DL-VizLab palette review sheet

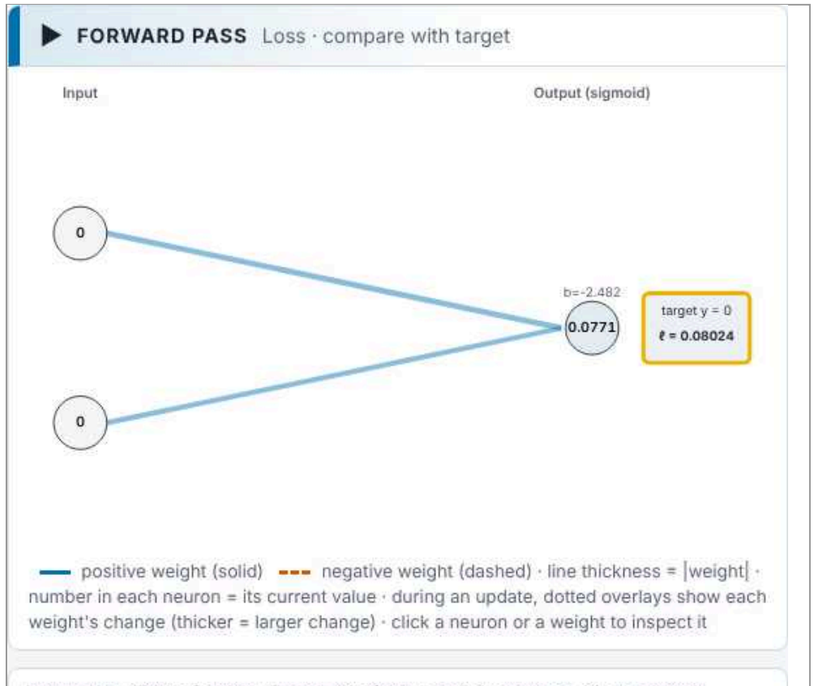
Every experiment's central views (network or structure view, explanation, charts and panels) in the light and dark themes, captured from the first example after the halfway point of the run (so the loss and gradient charts carry their series): once at the last layer of its forward pass and once one layer into its backward pass. Tall screens are split across pages. Each capture is shown as rendered, and with protanopia and deuteranopia simulated by the Machado, Oliveira & Fernandes (2009) matrices at severity 1.0 in linear RGB — the same simulation as `tests/unit/cvd.test.ts`.

The measured margins (ΔE in OKLab × 100 ≥ 15 for every meaning-carrying pair) are in `reports/cvd.json`. This sheet is for a person to judge legibility: whether the forward (blue) and backward (wine; coral in dark mode) phases, the loss-curve series and the signed weight colours are easy to tell apart, and whether any text on a coloured background is hard to read.

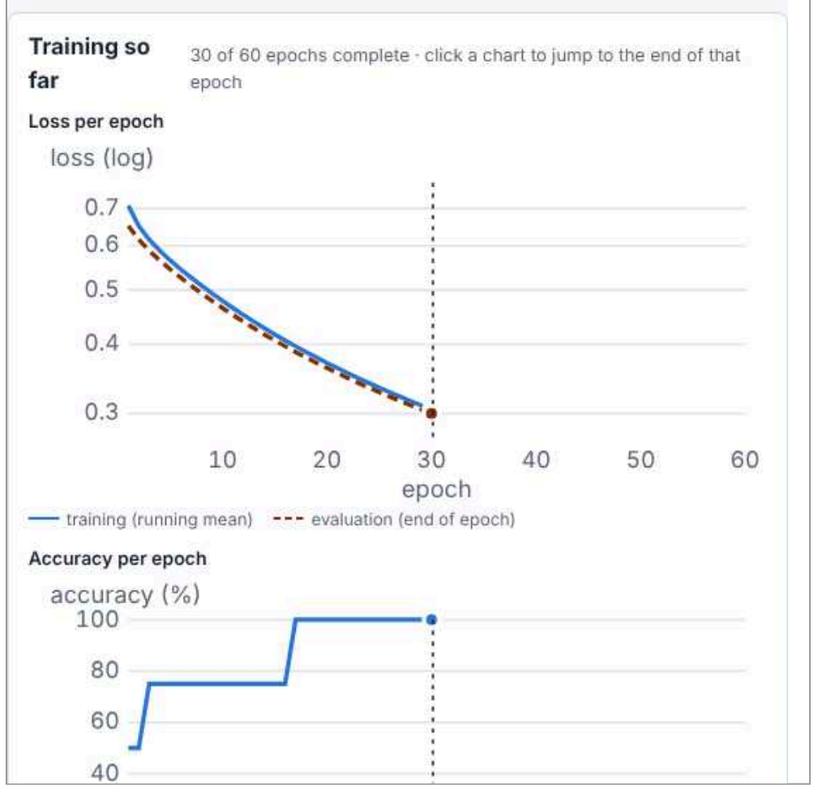
Generated by `scripts/gen-palette-sheet.ts` from the built app; regenerate after any palette change.

1. A single neuron — light theme, forward phase (part 1 of 2)

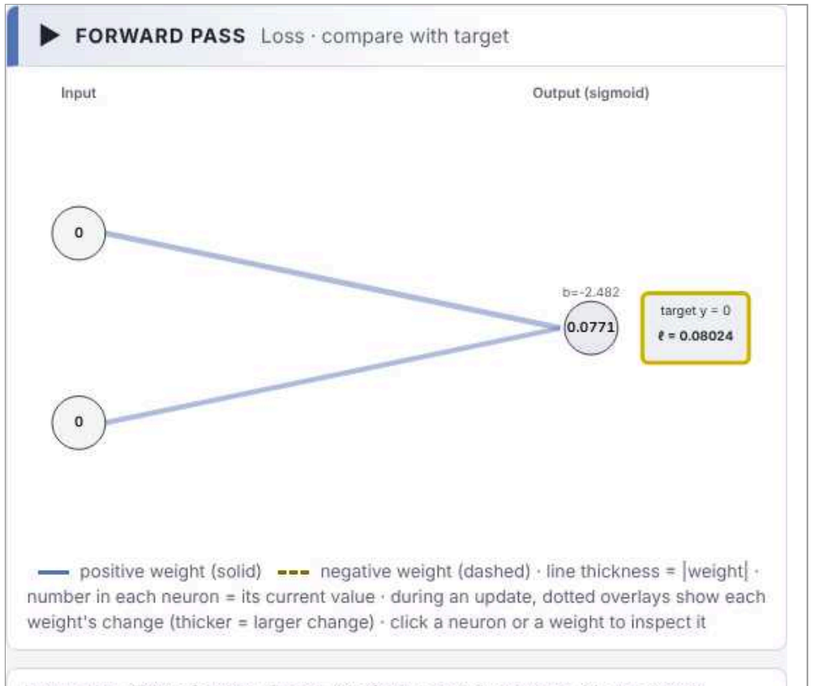
As rendered



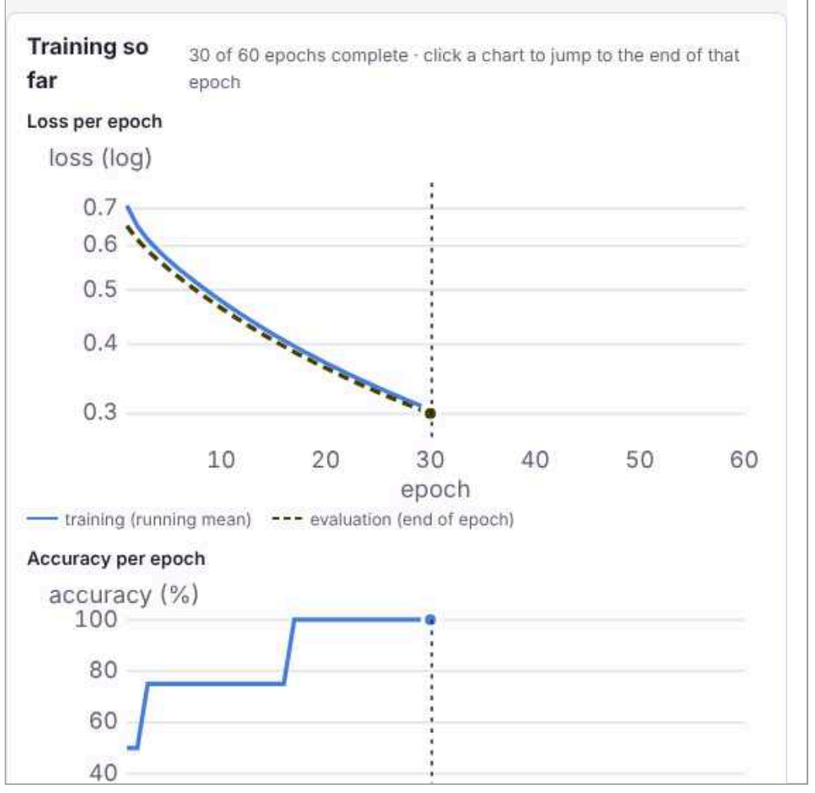
Instruction 1720 of 3,422. Epoch 31 of 60, batch 1, example 1 in the batch. Phase: loss. Input x = [0, 0], target y = [0]. Prediction 0.0771. Example loss 0.08024.



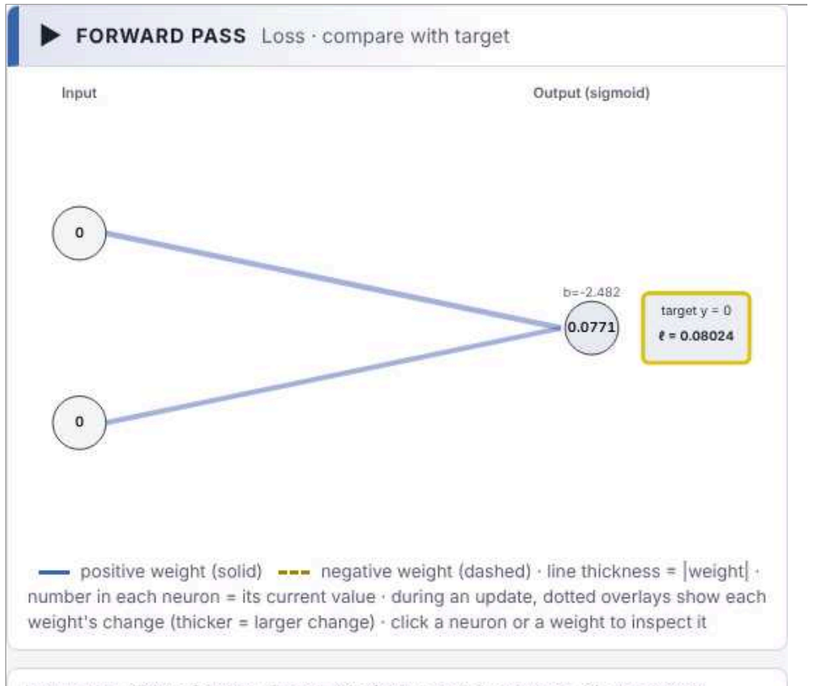
Simulated protanopia



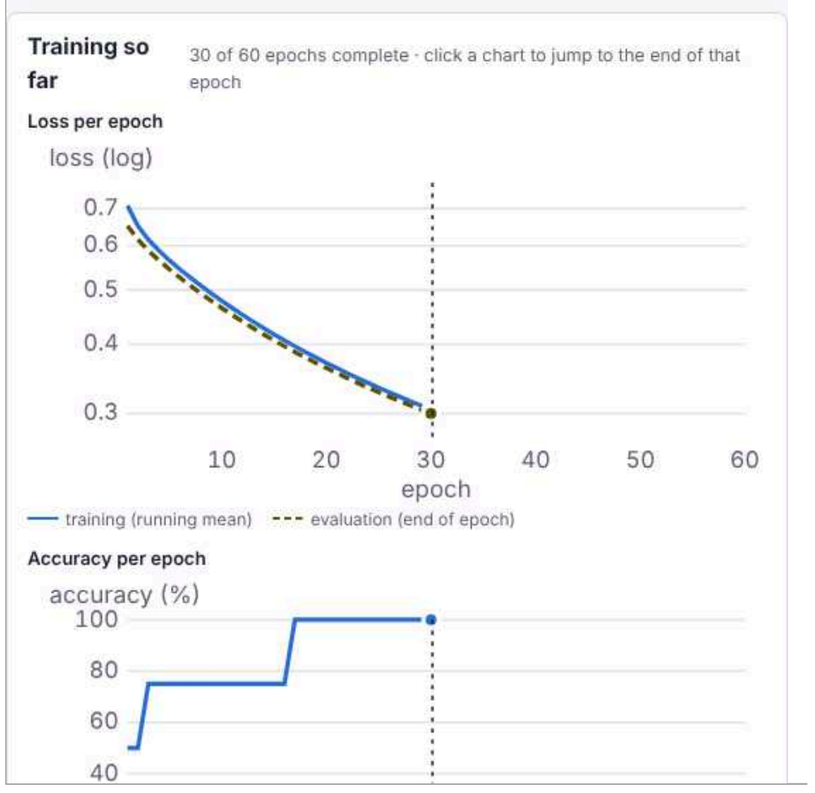
Instruction 1720 of 3,422. Epoch 31 of 60, batch 1, example 1 in the batch. Phase: loss. Input x = [0, 0], target y = [0]. Prediction 0.0771. Example loss 0.08024.



Simulated deuteranopia

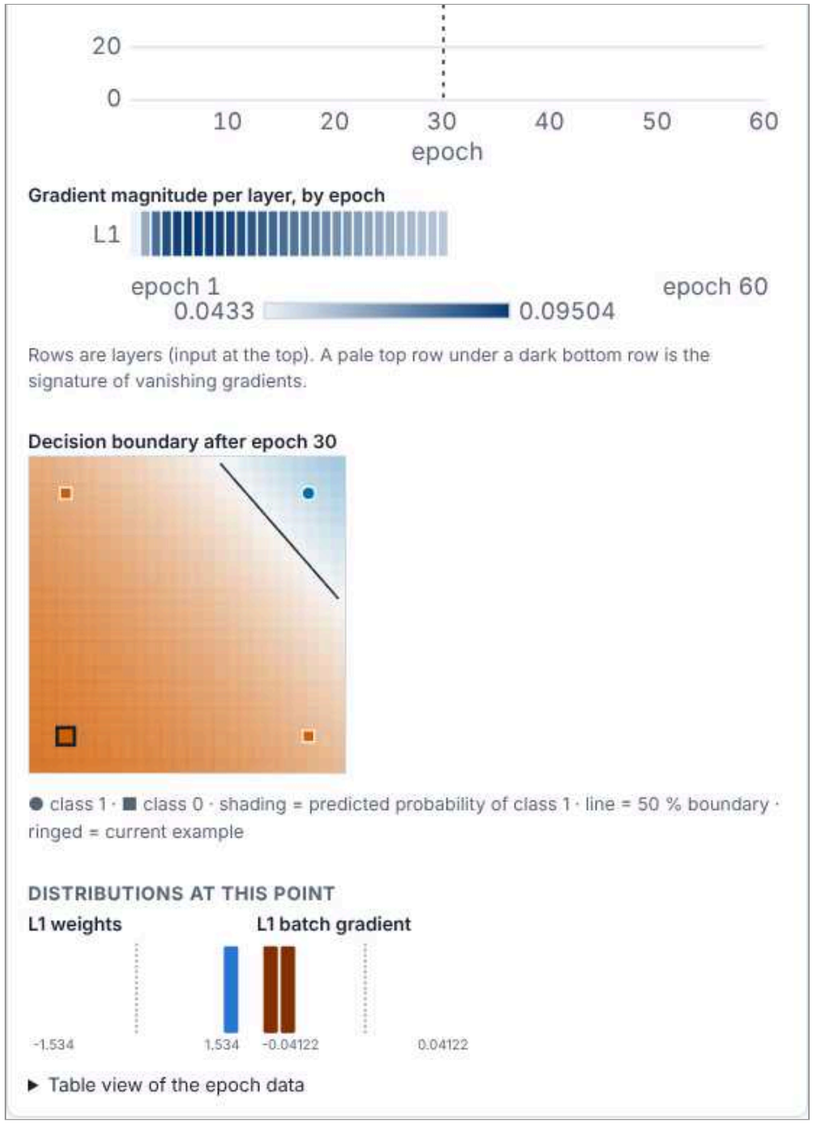


Instruction 1720 of 3,422. Epoch 31 of 60, batch 1, example 1 in the batch. Phase: loss. Input x = [0, 0], target y = [0]. Prediction 0.0771. Example loss 0.08024.

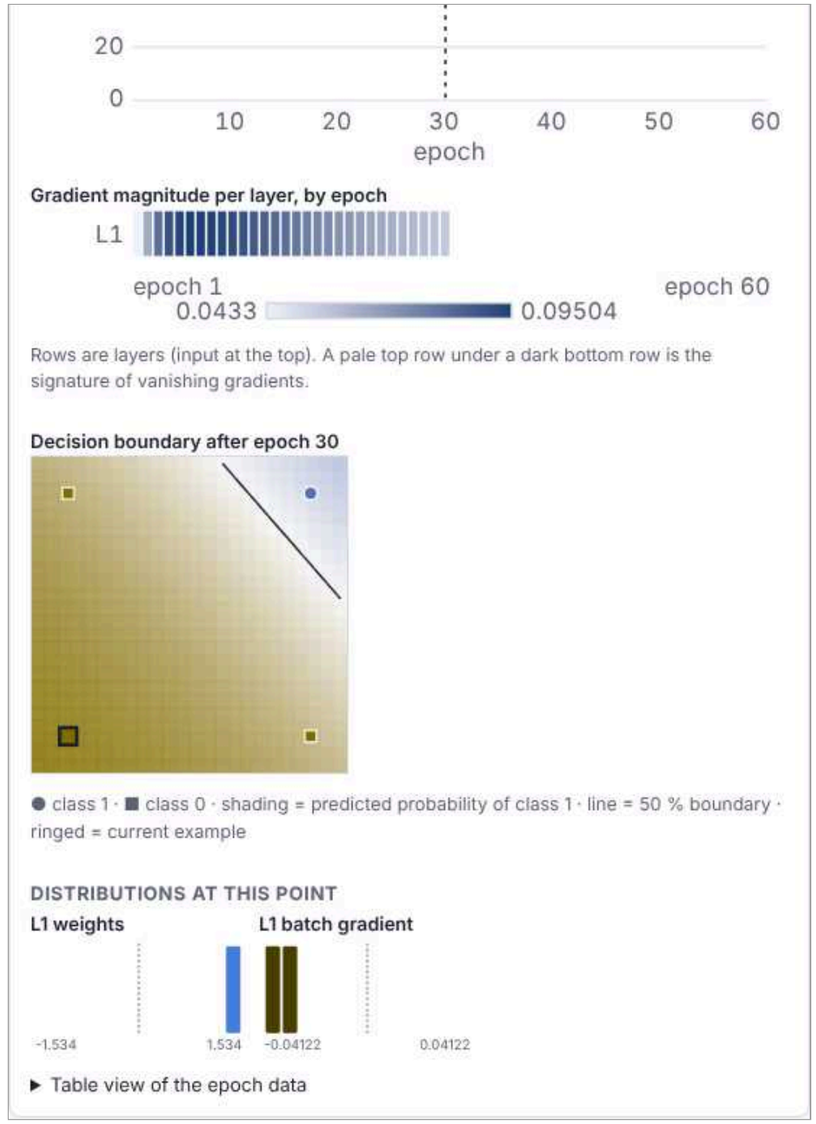


1. A single neuron — light theme, forward phase (part 2 of 2)

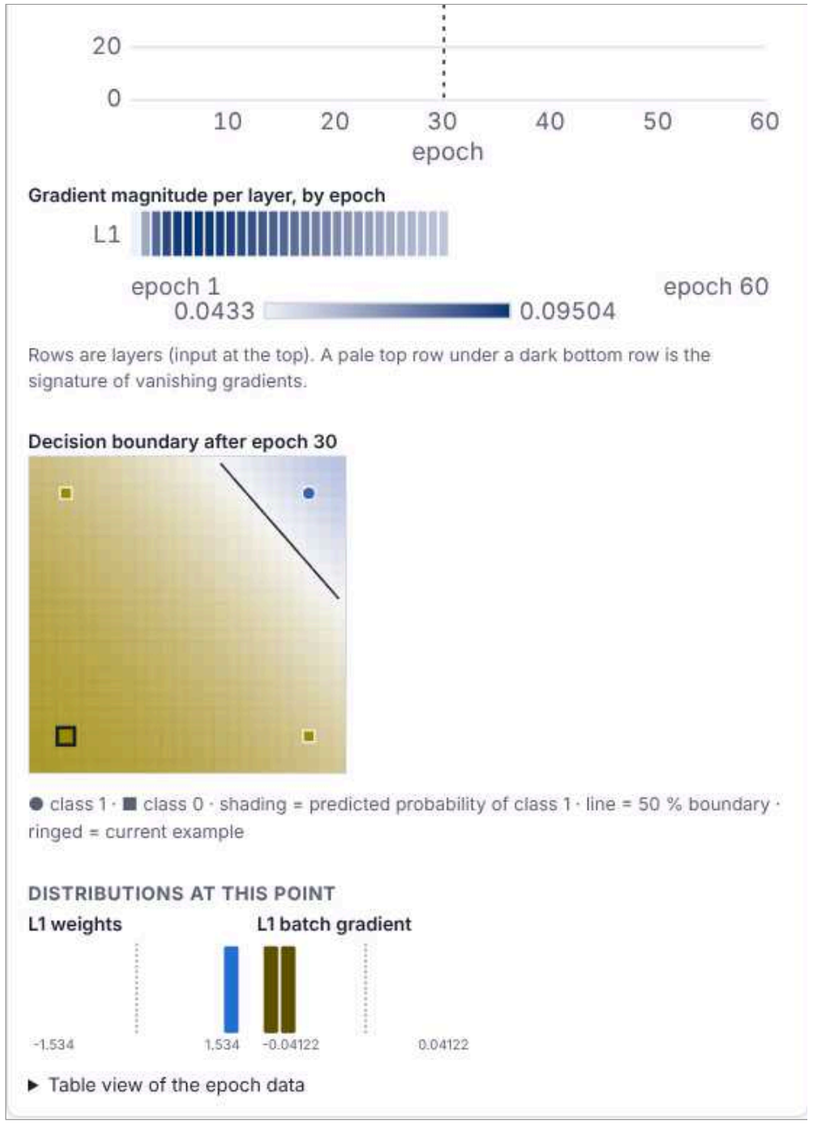
As rendered



Simulated protanopia

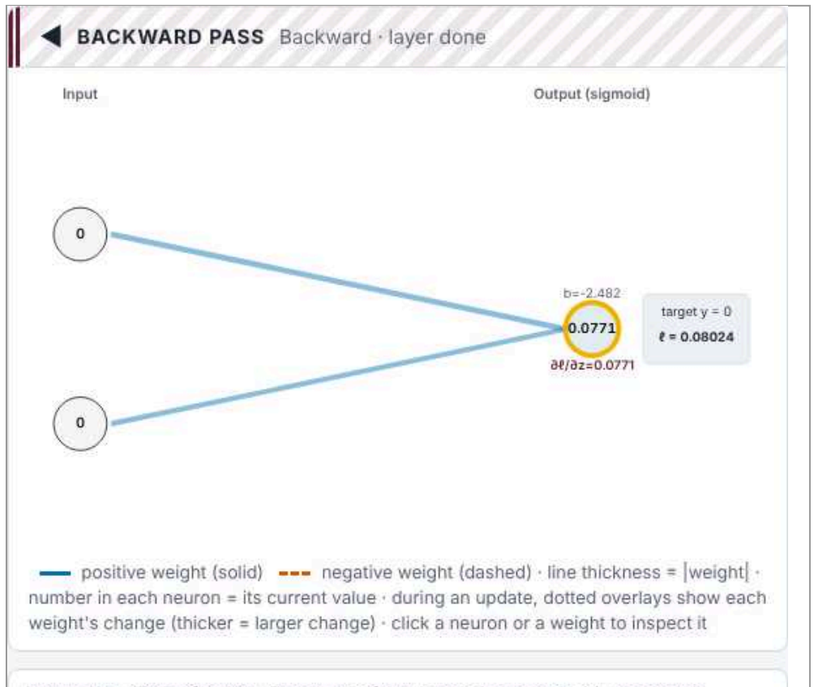


Simulated deuteranopia

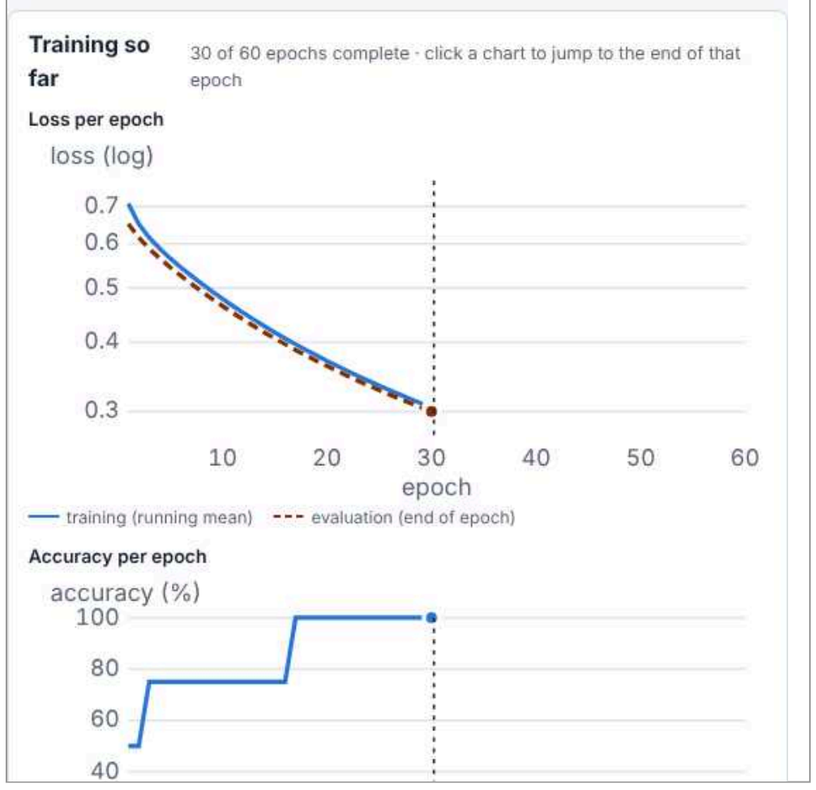


1. A single neuron — light theme, backward phase (part 1 of 2)

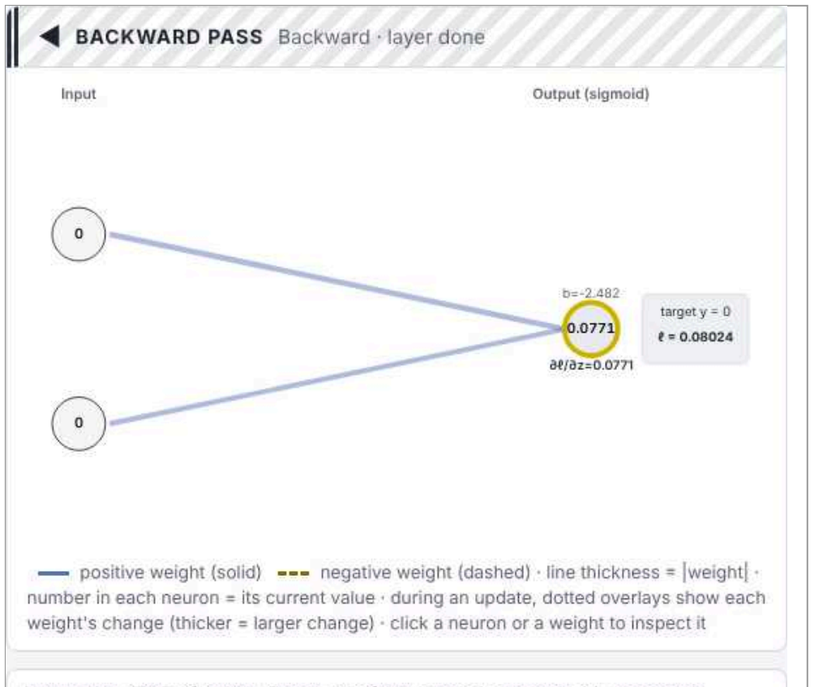
As rendered



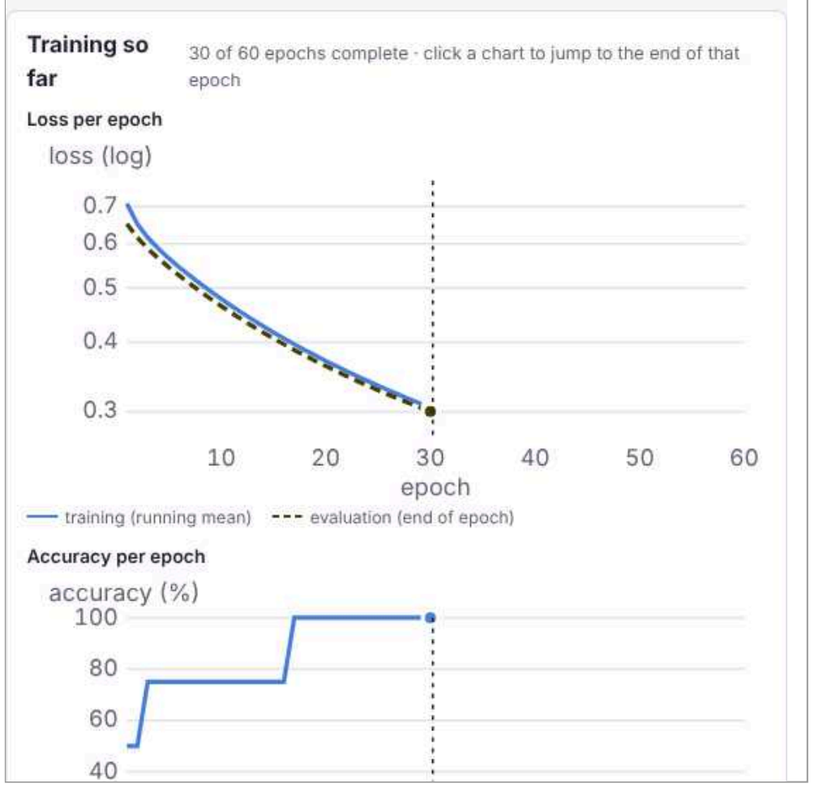
Instruction 1725 of 3,422. Epoch 31 of 60, batch 1, example 1 in the batch. Phase: backward pass, layer 1. Input x = [0, 0], target y = [0]. Prediction 0.0771. Example loss 0.08024.



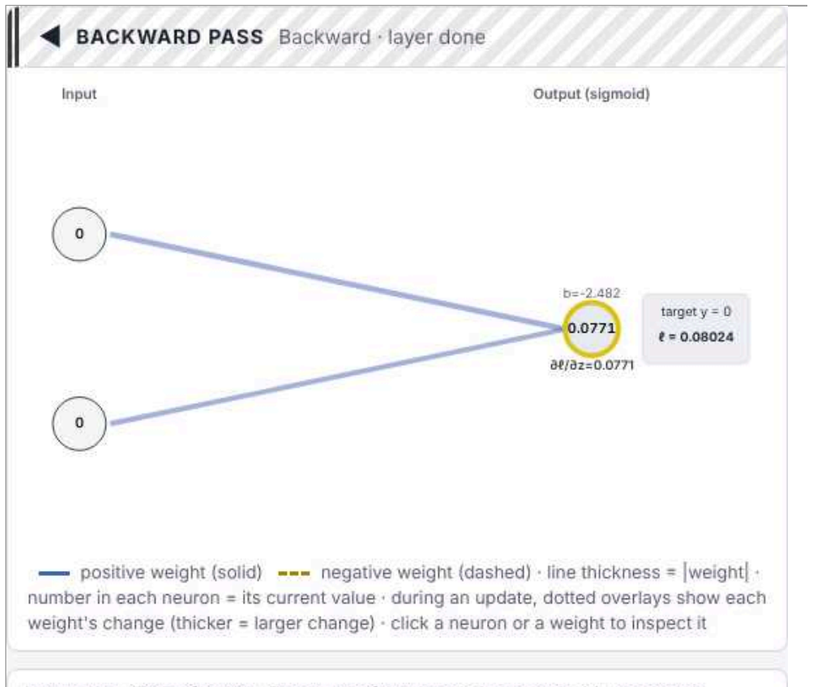
Simulated protanopia



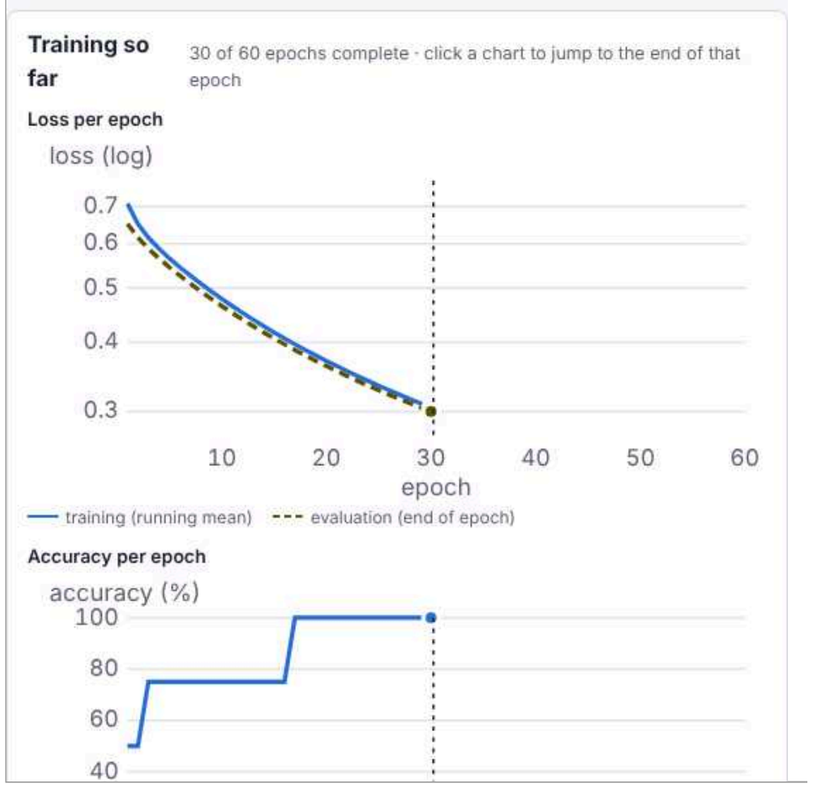
Instruction 1725 of 3,422. Epoch 31 of 60, batch 1, example 1 in the batch. Phase: backward pass, layer 1. Input x = [0, 0], target y = [0]. Prediction 0.0771. Example loss 0.08024.



Simulated deuteranopia

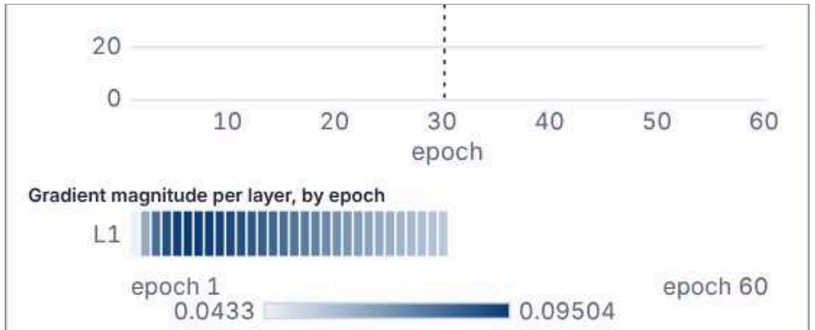


Instruction 1725 of 3,422. Epoch 31 of 60, batch 1, example 1 in the batch. Phase: backward pass, layer 1. Input x = [0, 0], target y = [0]. Prediction 0.0771. Example loss 0.08024.



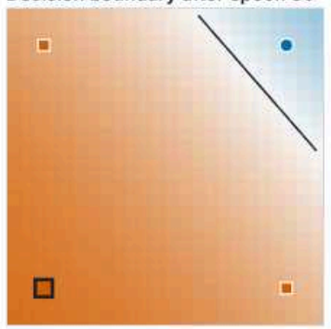
1. A single neuron — light theme, backward phase (part 2 of 2)

As rendered



Rows are layers (input at the top). A pale top row under a dark bottom row is the signature of vanishing gradients.

Decision boundary after epoch 30



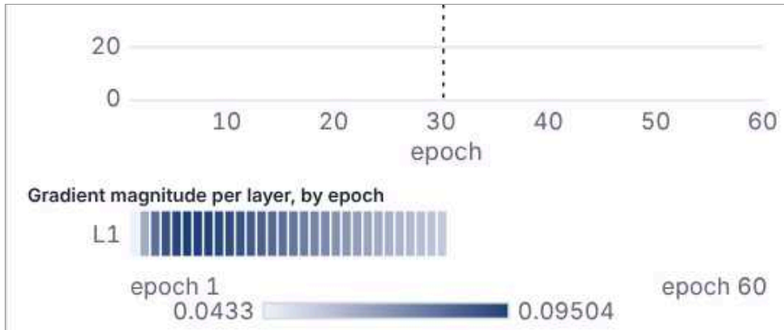
● class 1 · ■ class 0 · shading = predicted probability of class 1 · line = 50 % boundary · ringed = current example

DISTRIBUTIONS AT THIS POINT



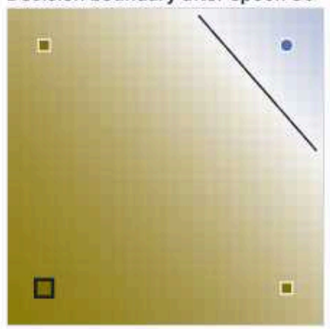
▶ Table view of the epoch data

Simulated protanopia



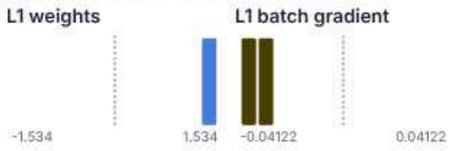
Rows are layers (input at the top). A pale top row under a dark bottom row is the signature of vanishing gradients.

Decision boundary after epoch 30



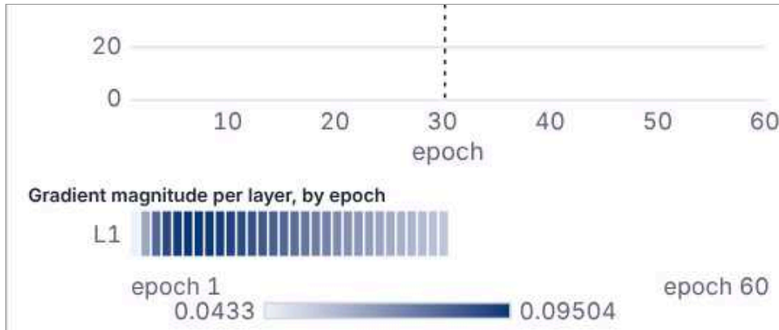
● class 1 · ■ class 0 · shading = predicted probability of class 1 · line = 50 % boundary · ringed = current example

DISTRIBUTIONS AT THIS POINT



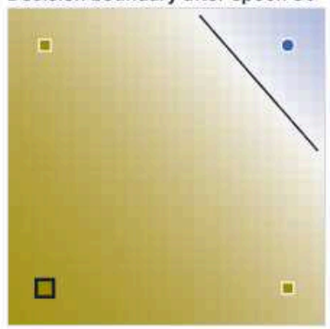
▶ Table view of the epoch data

Simulated deuteranopia



Rows are layers (input at the top). A pale top row under a dark bottom row is the signature of vanishing gradients.

Decision boundary after epoch 30



● class 1 · ■ class 0 · shading = predicted probability of class 1 · line = 50 % boundary · ringed = current example

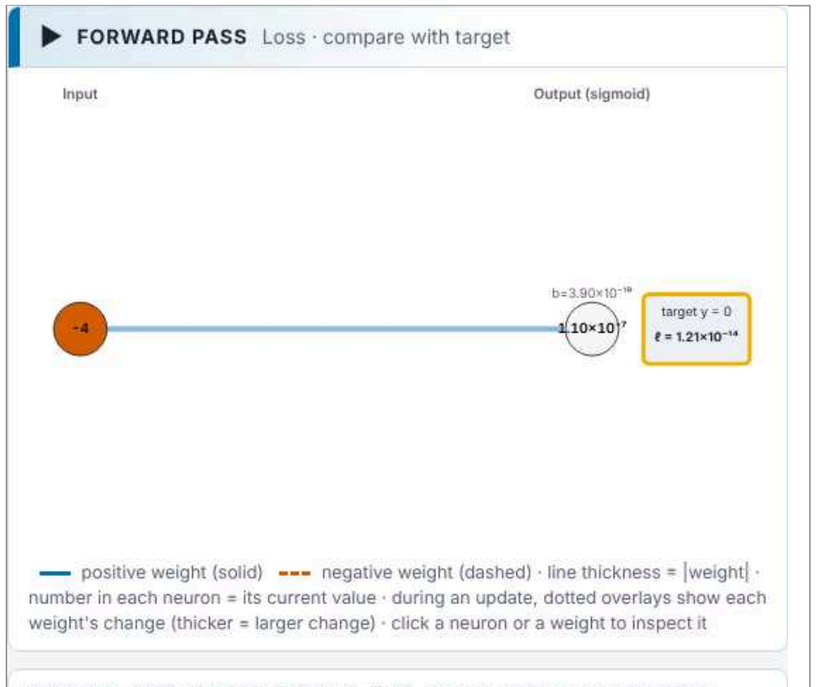
DISTRIBUTIONS AT THIS POINT



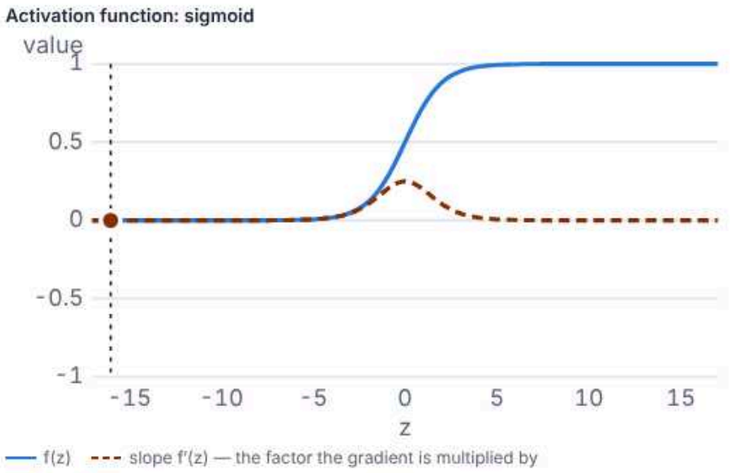
▶ Table view of the epoch data

2. Activation functions — light theme, forward phase (part 1 of 2)

As rendered



Instruction 2470 of 4,922. Epoch 21 of 40, batch 1, example 1 in the batch. Phase: loss. Input x = [-4], target y = [0]. Prediction 1.10×10^{-7}. Example loss 1.21×10^{-14}.



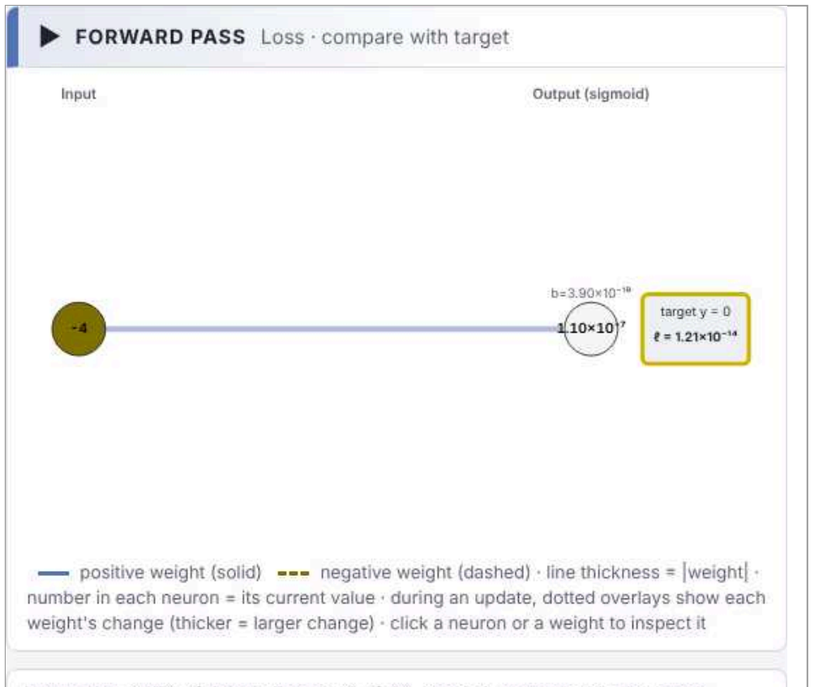
Current example: z = -16.02, f(z) = 1.10×10^{-7}, f′(z) = 1.10×10^{-7} — saturated: almost no gradient will pass back.

Training so far 20 of 40 epochs complete · click a chart to jump to the end of that epoch

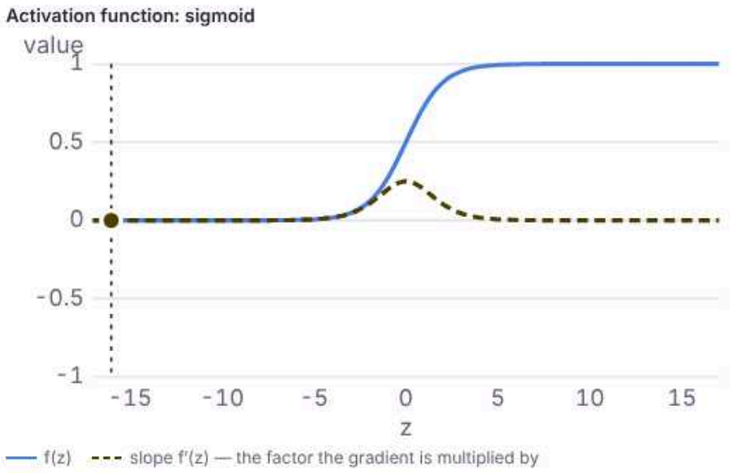
Loss per epoch

loss (log)

Simulated protanopia



Instruction 2470 of 4,922. Epoch 21 of 40, batch 1, example 1 in the batch. Phase: loss. Input x = [-4], target y = [0]. Prediction 1.10×10^{-7}. Example loss 1.21×10^{-14}.



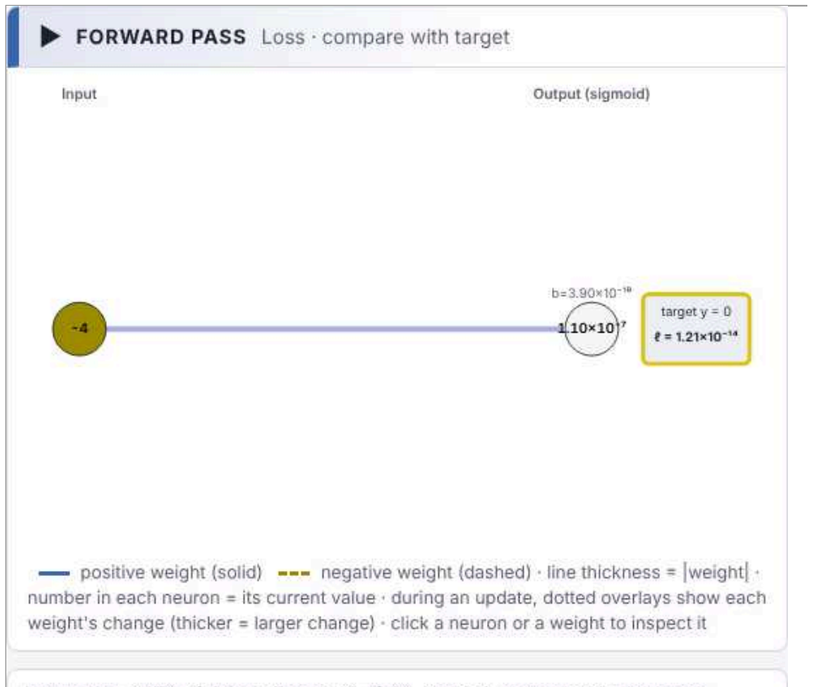
Current example: z = -16.02, f(z) = 1.10×10^{-7}, f′(z) = 1.10×10^{-7} — saturated: almost no gradient will pass back.

Training so far 20 of 40 epochs complete · click a chart to jump to the end of that epoch

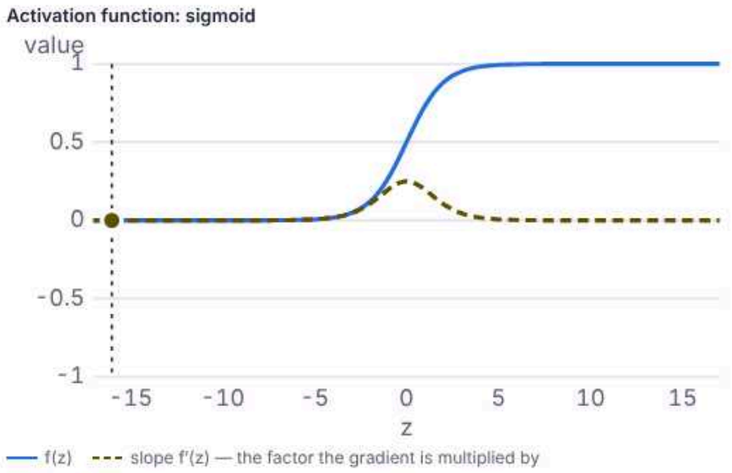
Loss per epoch

loss (log)

Simulated deuteranopia



Instruction 2470 of 4,922. Epoch 21 of 40, batch 1, example 1 in the batch. Phase: loss. Input x = [-4], target y = [0]. Prediction 1.10×10^{-7}. Example loss 1.21×10^{-14}.



Current example: z = -16.02, f(z) = 1.10×10^{-7}, f′(z) = 1.10×10^{-7} — saturated: almost no gradient will pass back.

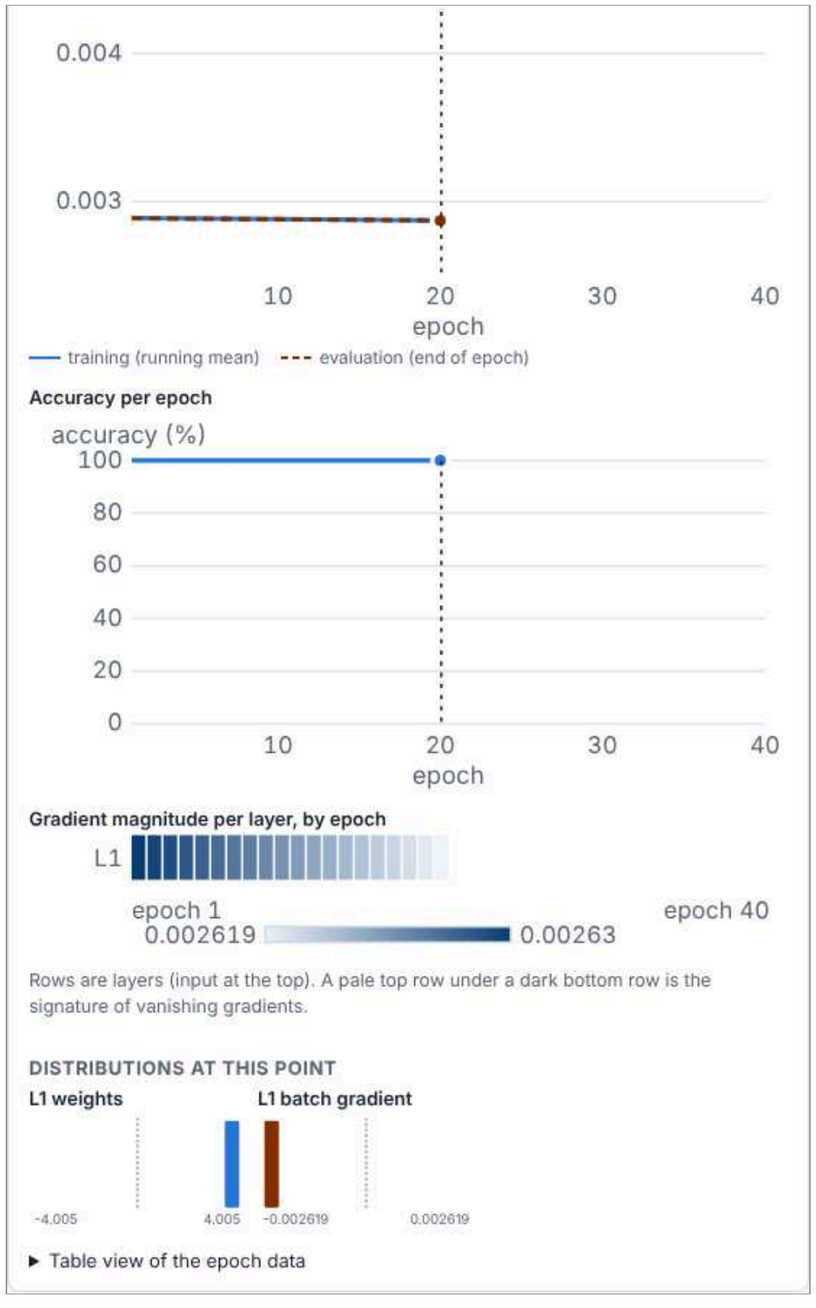
Training so far 20 of 40 epochs complete · click a chart to jump to the end of that epoch

Loss per epoch

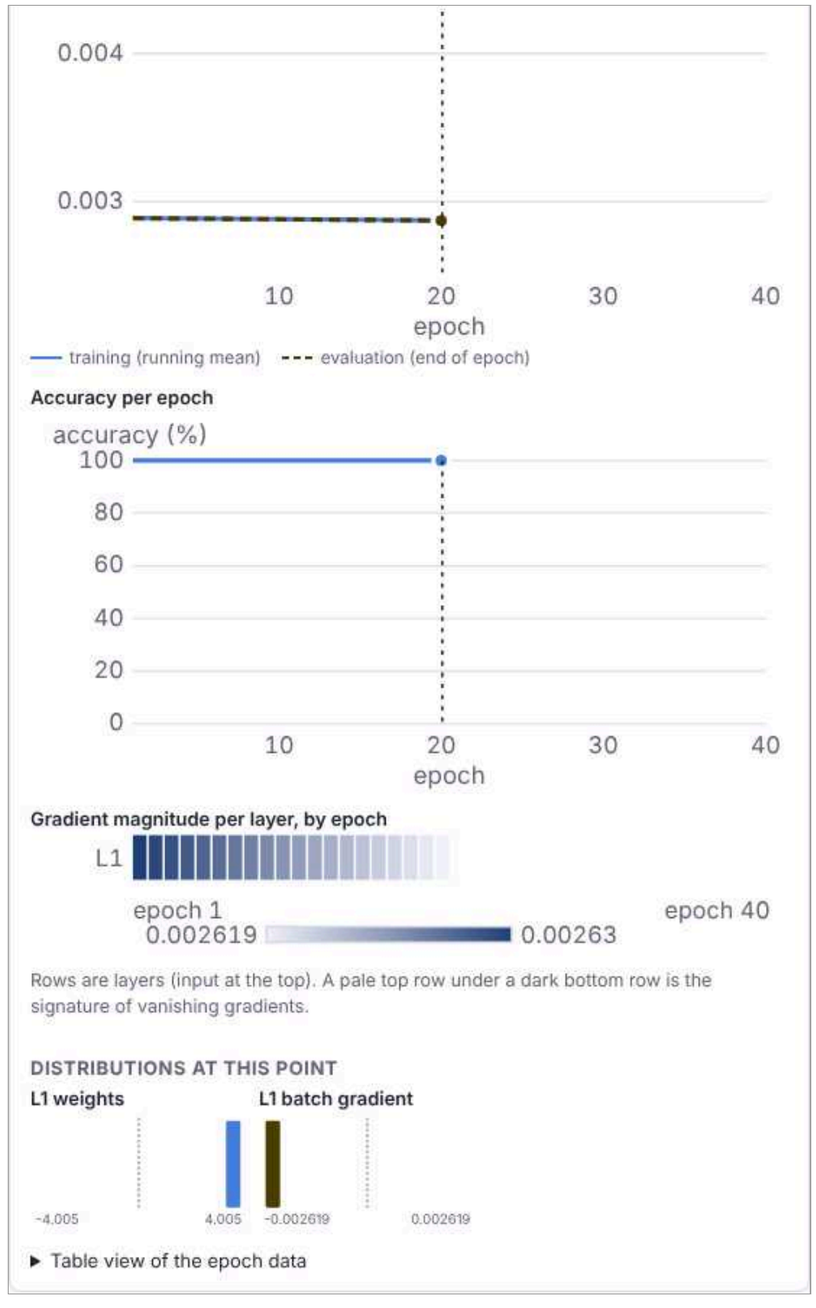
loss (log)

2. Activation functions — light theme, forward phase (part 2 of 2)

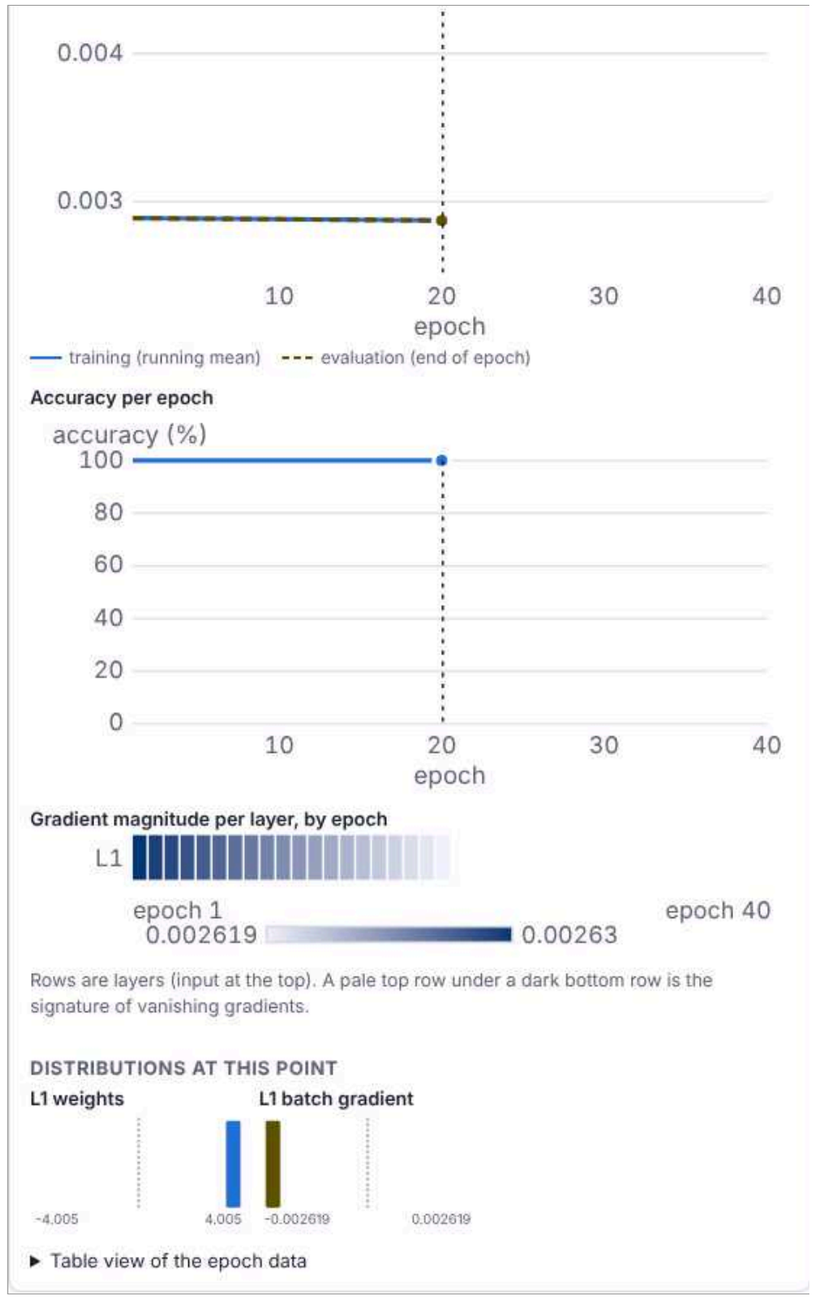
As rendered



Simulated protanopia

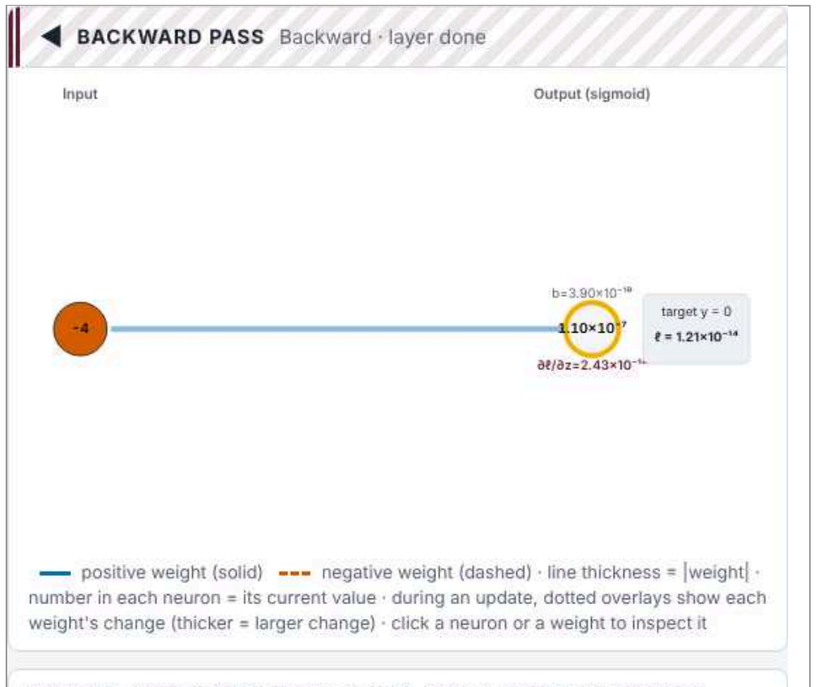


Simulated deuteranopia

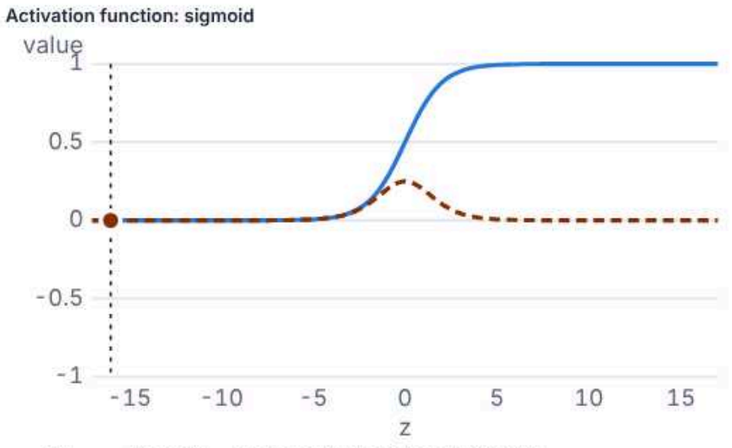


2. Activation functions — light theme, backward phase (part 1 of 2)

As rendered



Instruction 2475 of 4,922. Epoch 21 of 40, batch 1, example 1 in the batch. Phase: backward pass, layer 1. Input x = [-4], target y = [0]. Prediction 1.10×10^{-7}. Example loss 1.21×10^{-14}.



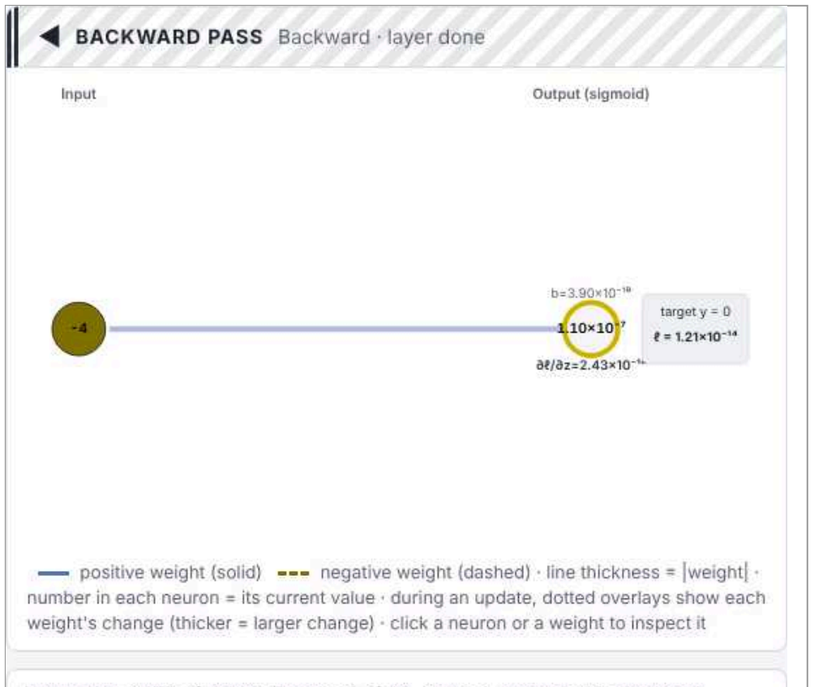
Current example: z = -16.02, f(z) = 1.10×10^{-7}, f′(z) = 1.10×10^{-7} — saturated: almost no gradient will pass back.

Training so far 20 of 40 epochs complete · click a chart to jump to the end of that epoch

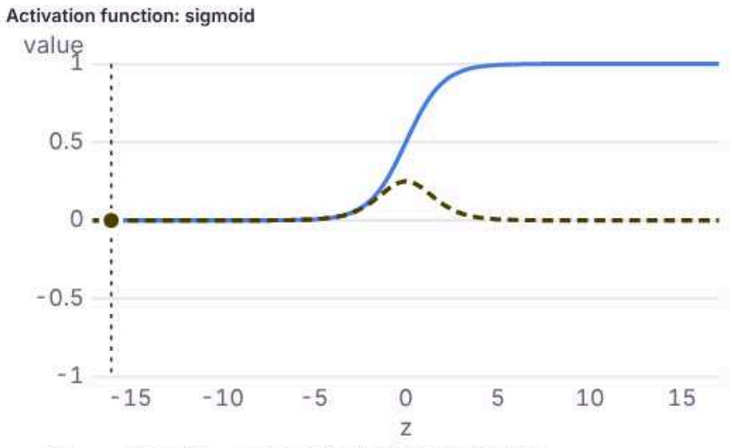
Loss per epoch

loss (log)

Simulated protanopia



Instruction 2475 of 4,922. Epoch 21 of 40, batch 1, example 1 in the batch. Phase: backward pass, layer 1. Input x = [-4], target y = [0]. Prediction 1.10×10^{-7}. Example loss 1.21×10^{-14}.



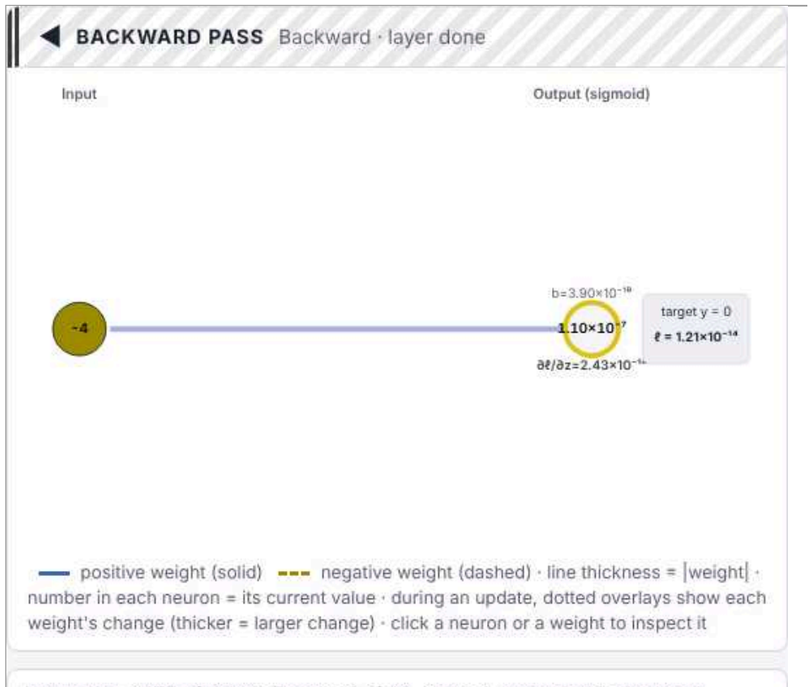
Current example: z = -16.02, f(z) = 1.10×10^{-7}, f′(z) = 1.10×10^{-7} — saturated: almost no gradient will pass back.

Training so far 20 of 40 epochs complete · click a chart to jump to the end of that epoch

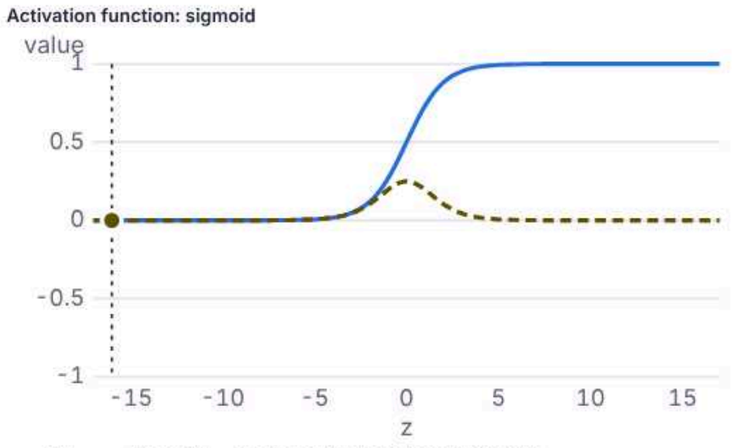
Loss per epoch

loss (log)

Simulated deuteranopia



Instruction 2475 of 4,922. Epoch 21 of 40, batch 1, example 1 in the batch. Phase: backward pass, layer 1. Input x = [-4], target y = [0]. Prediction 1.10×10^{-7}. Example loss 1.21×10^{-14}.



Current example: z = -16.02, f(z) = 1.10×10^{-7}, f′(z) = 1.10×10^{-7} — saturated: almost no gradient will pass back.

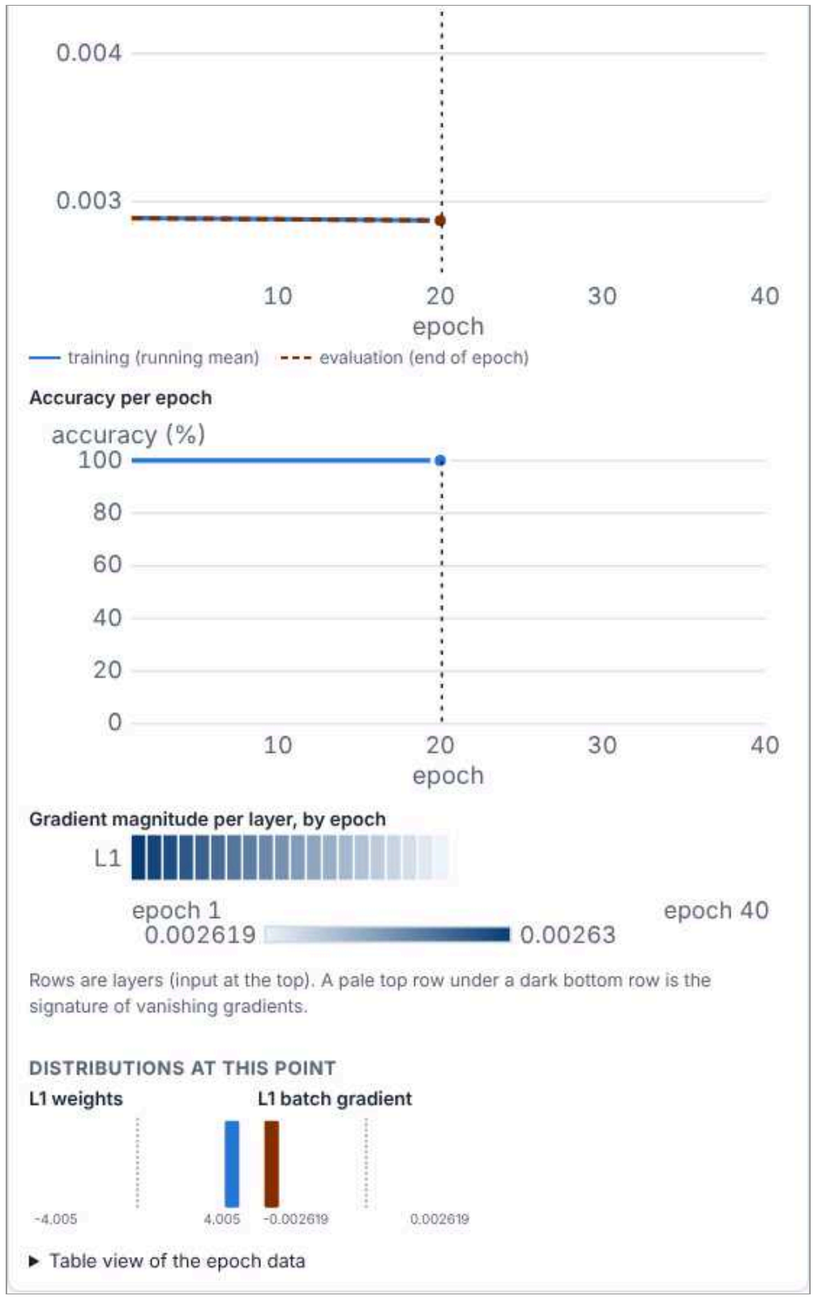
Training so far 20 of 40 epochs complete · click a chart to jump to the end of that epoch

Loss per epoch

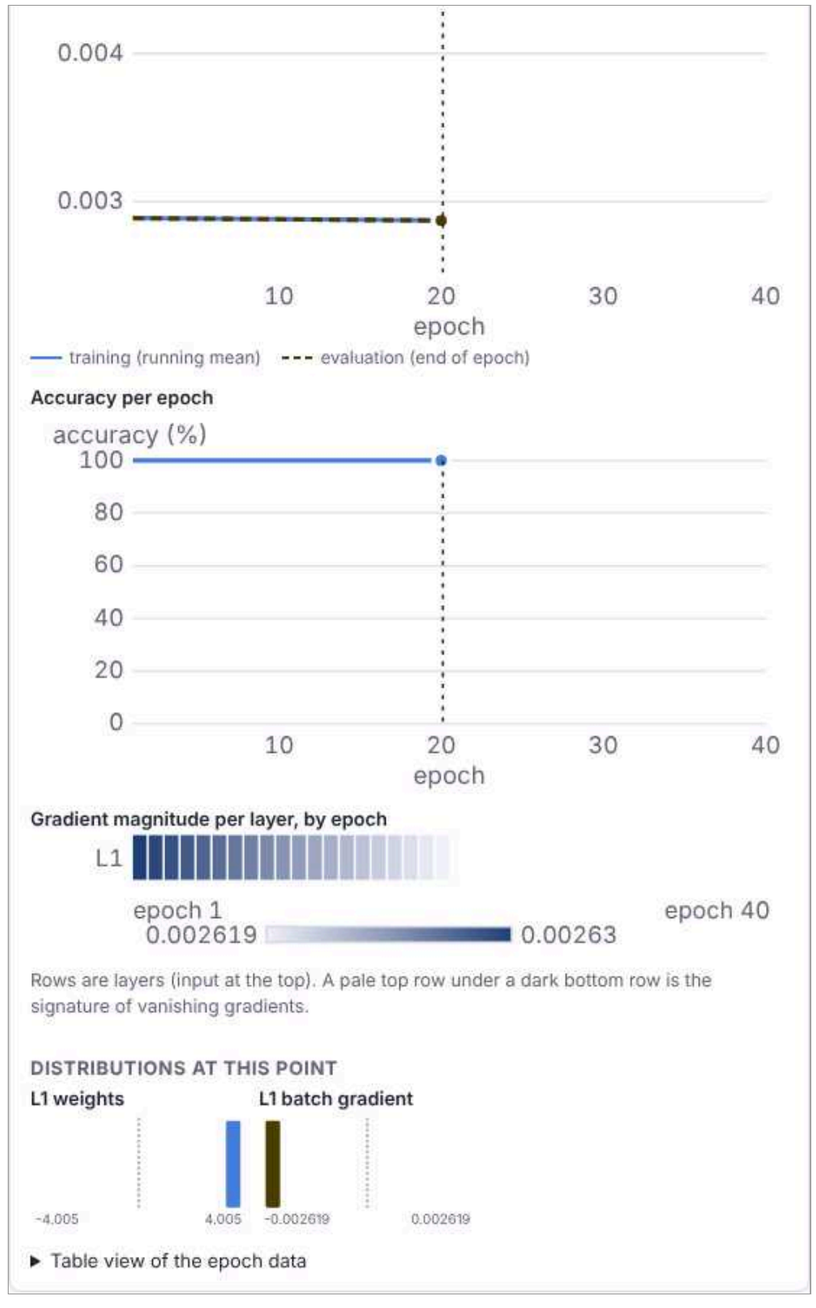
loss (log)

2. Activation functions — light theme, backward phase (part 2 of 2)

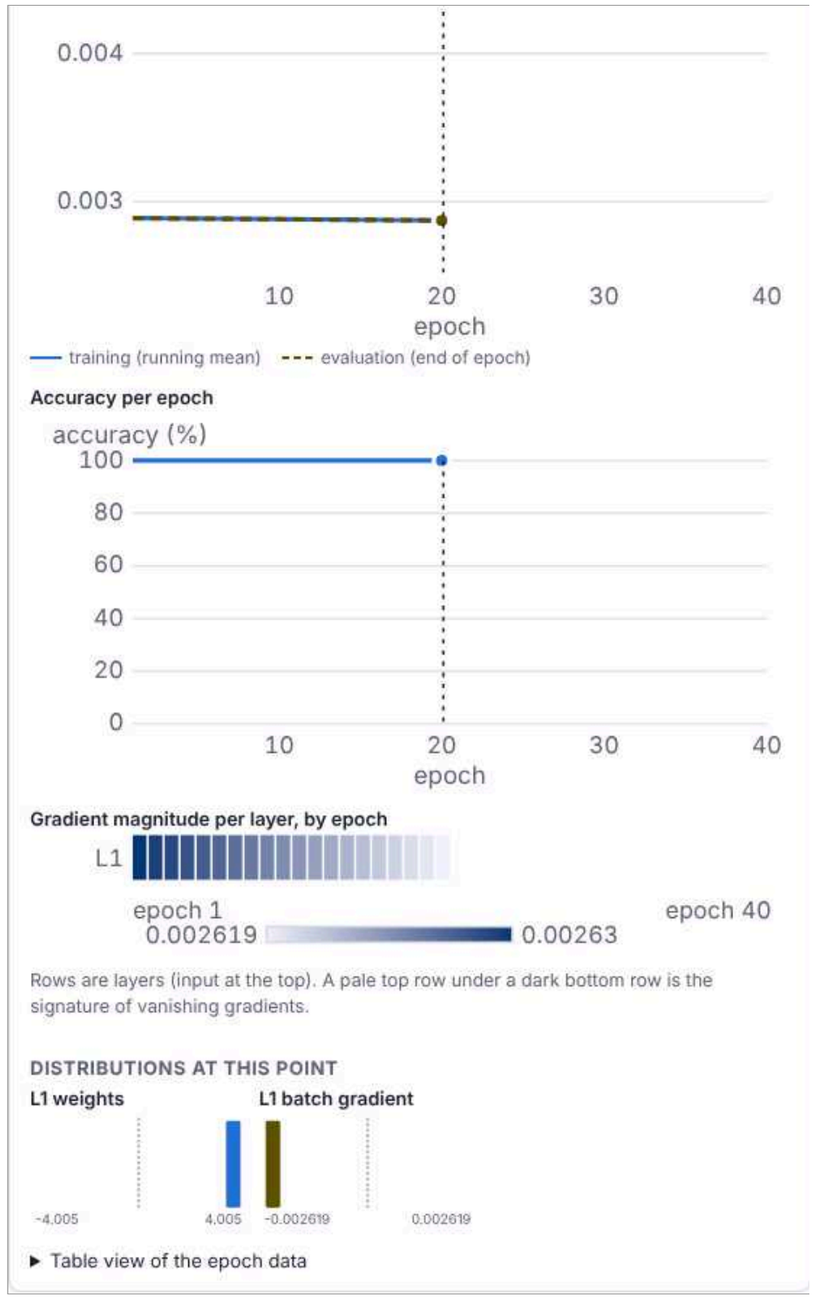
As rendered



Simulated protanopia

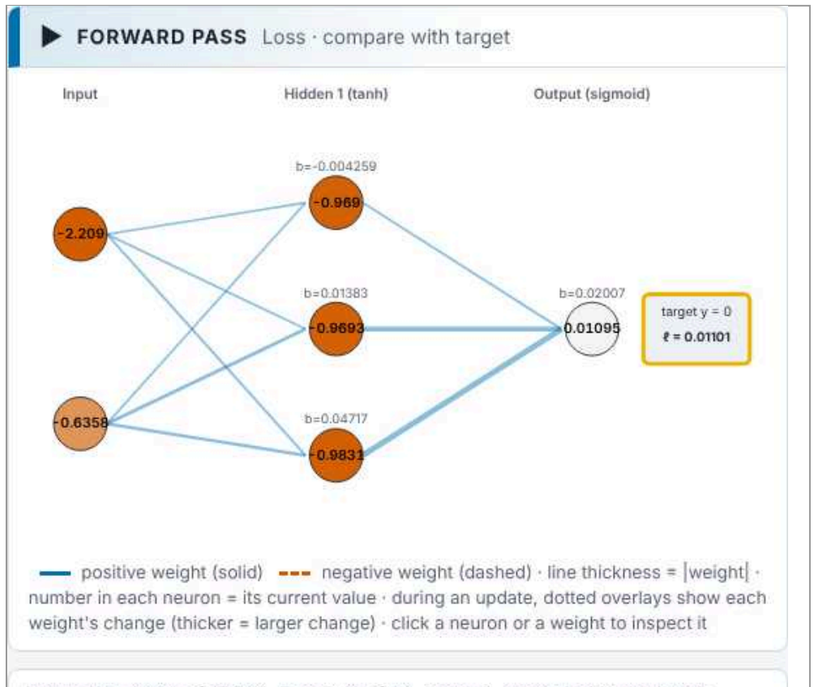


Simulated deuteranopia

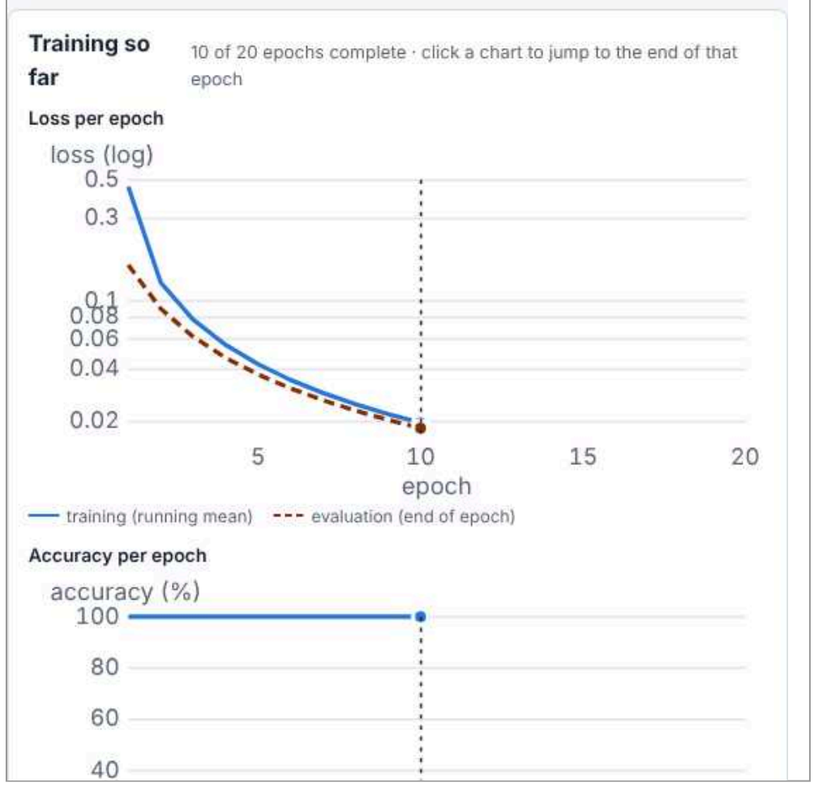


3. Forward propagation — light theme, forward phase (part 1 of 2)

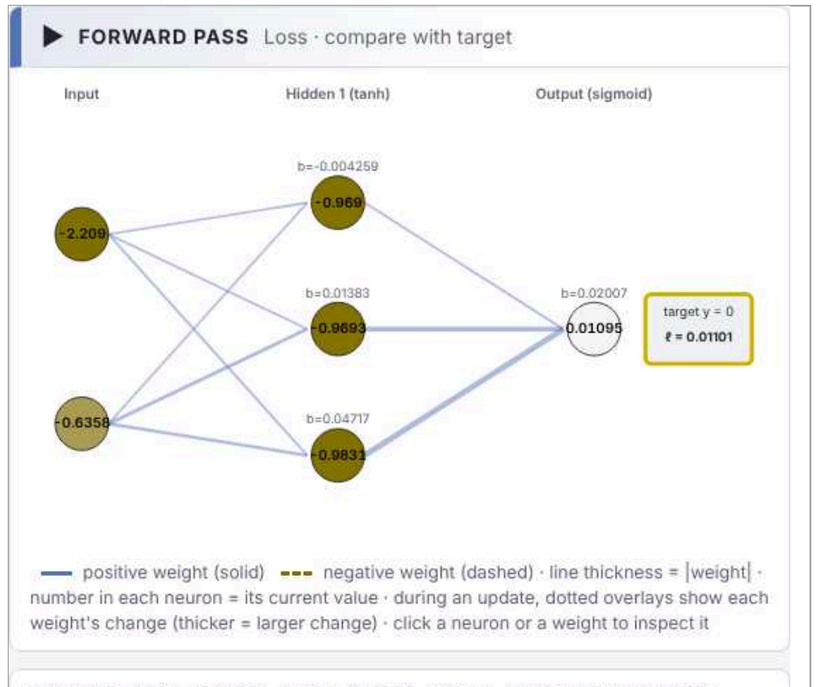
As rendered



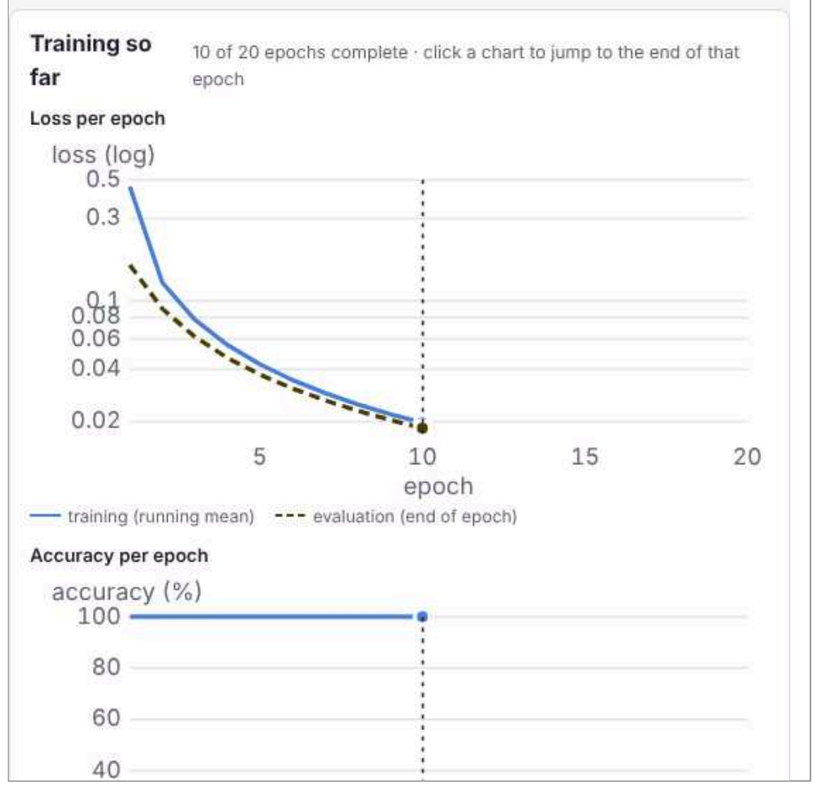
Instruction 8806 of 17,584. Epoch 11 of 20, batch 1, example 1 in the batch. Phase: loss. Input x = [-2.209, -0.6358], target y = [0]. Prediction 0.01095. Example loss 0.01101.



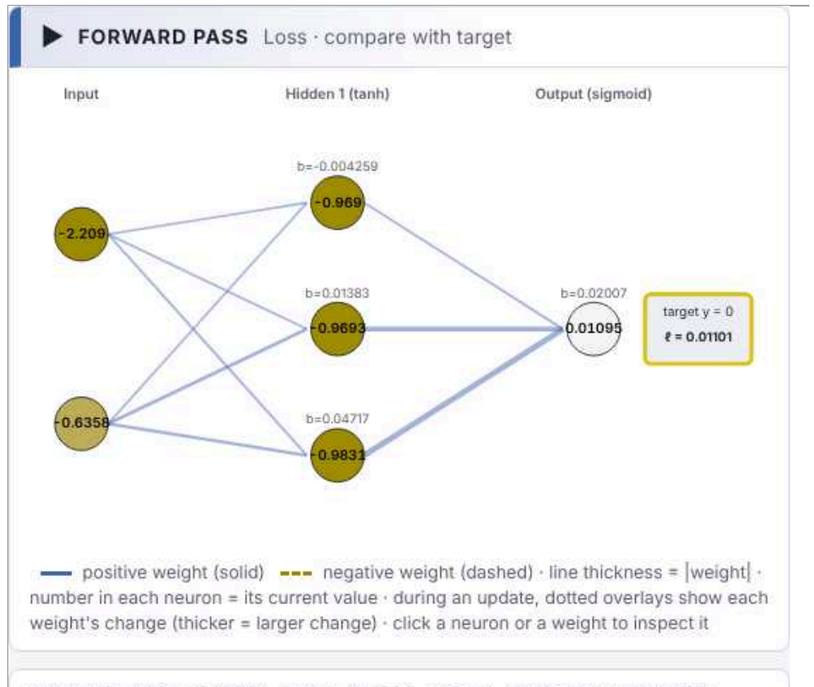
Simulated protanopia



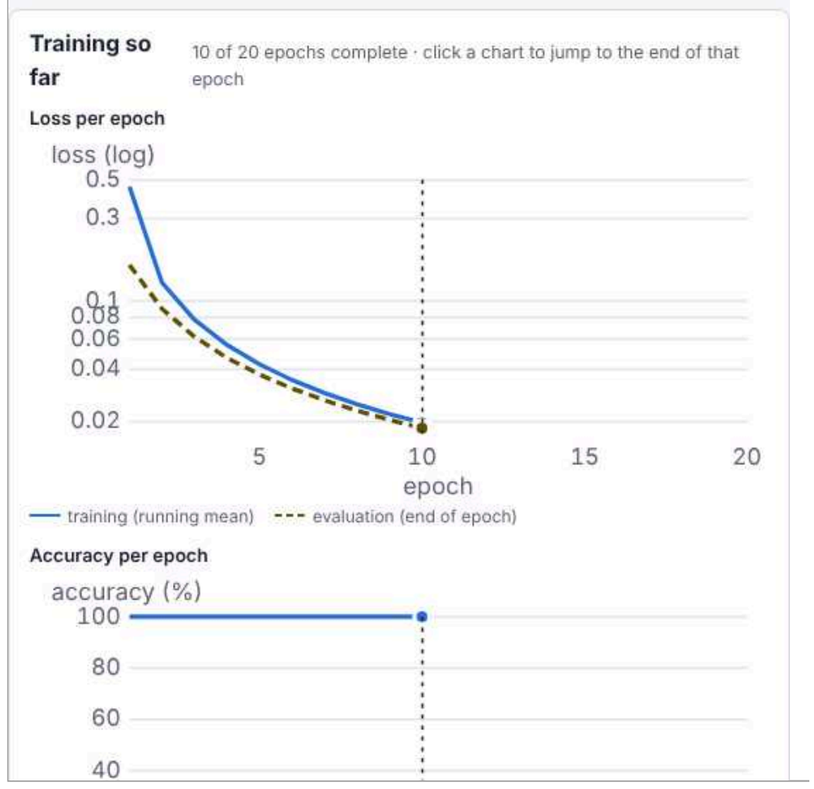
Instruction 8806 of 17,584. Epoch 11 of 20, batch 1, example 1 in the batch. Phase: loss. Input x = [-2.209, -0.6358], target y = [0]. Prediction 0.01095. Example loss 0.01101.



Simulated deuteranopia

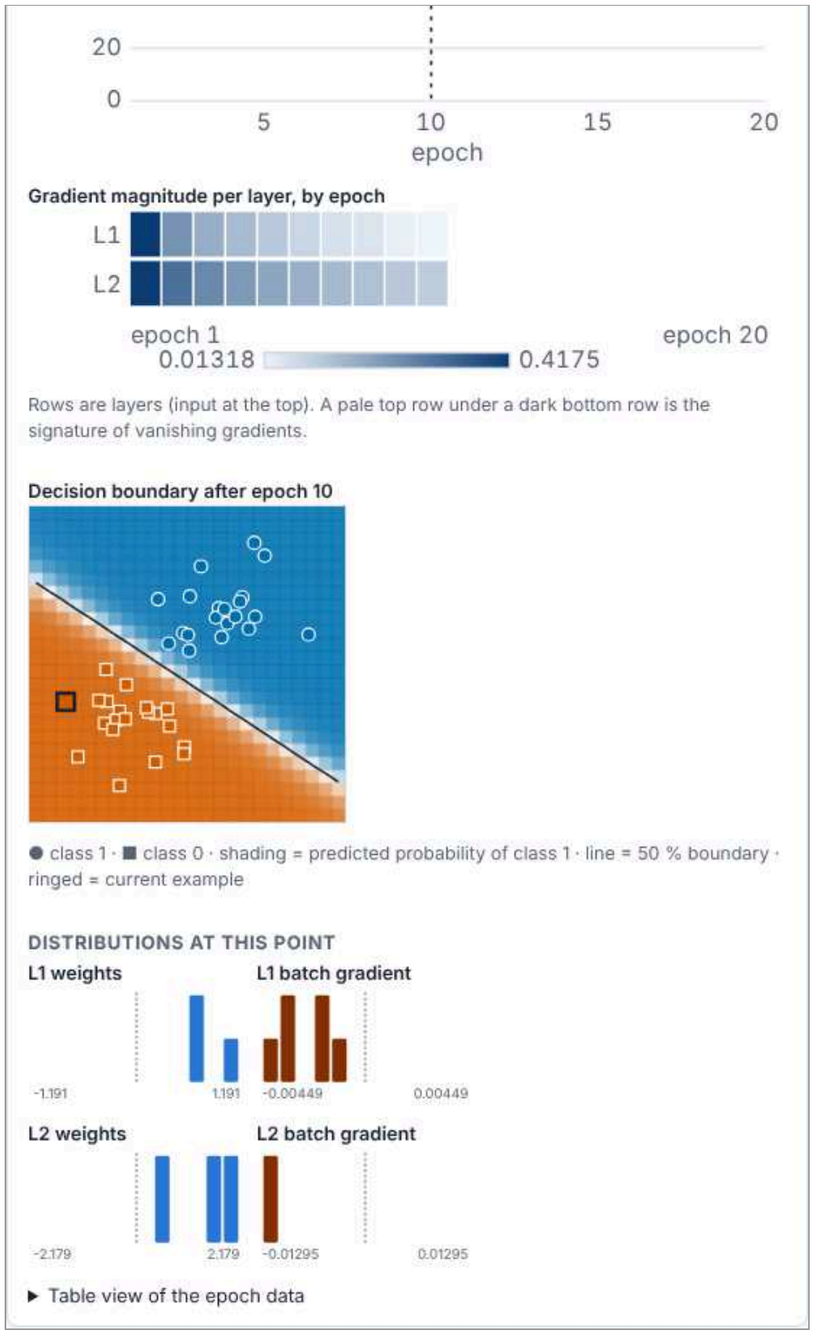


Instruction 8806 of 17,584. Epoch 11 of 20, batch 1, example 1 in the batch. Phase: loss. Input x = [-2.209, -0.6358], target y = [0]. Prediction 0.01095. Example loss 0.01101.

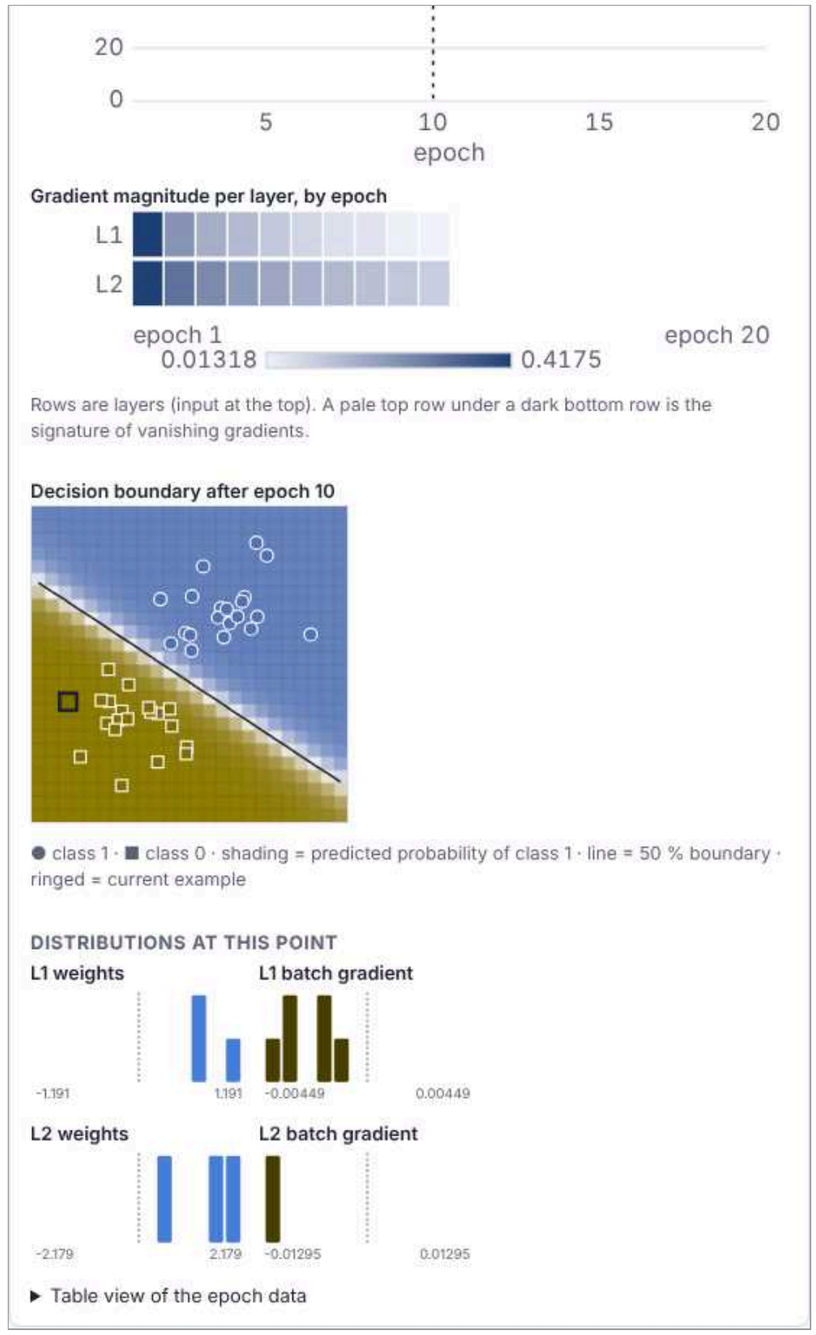


3. Forward propagation — light theme, forward phase (part 2 of 2)

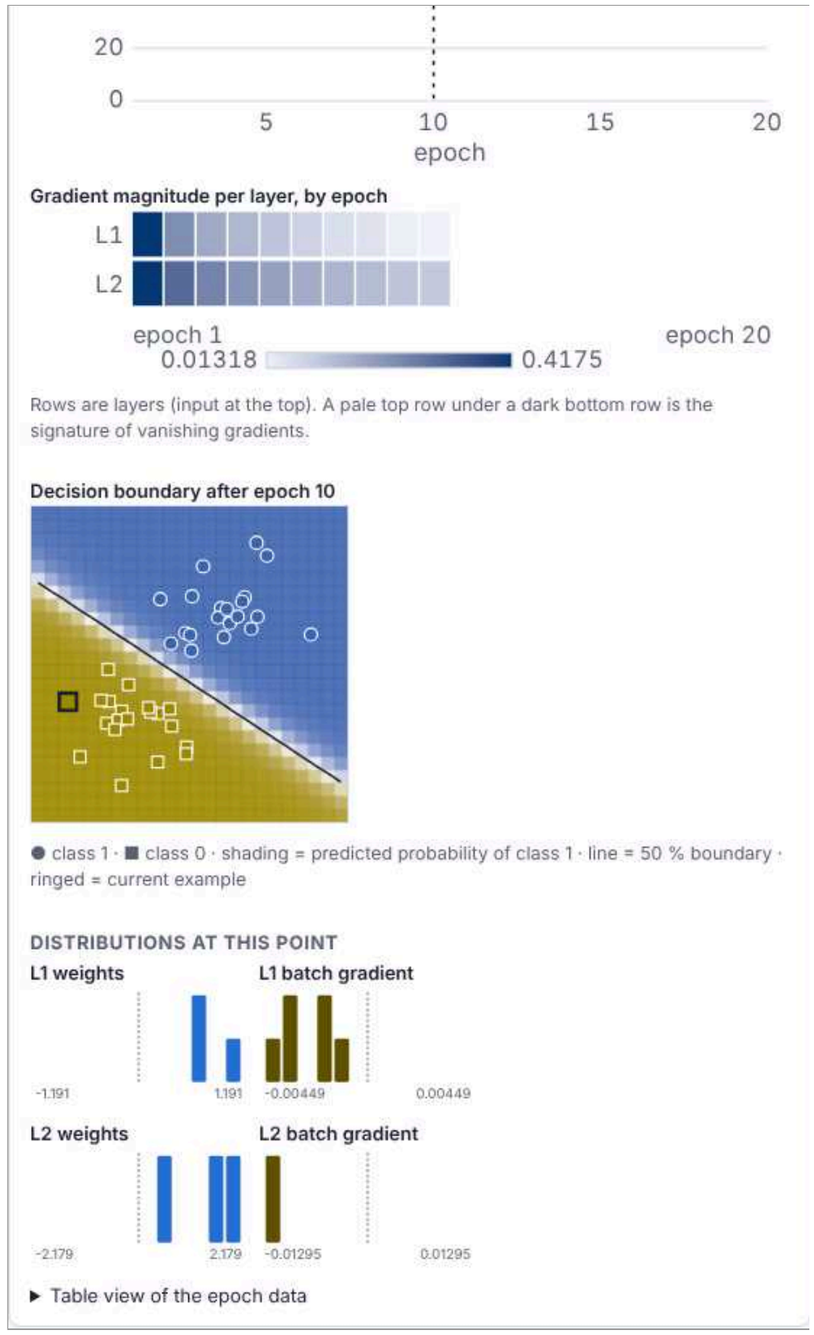
As rendered



Simulated protanopia

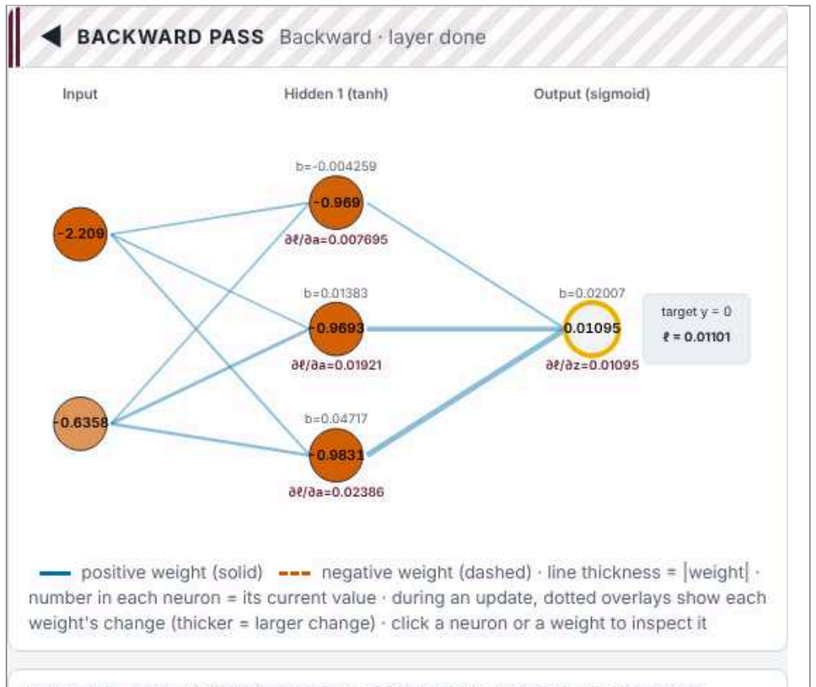


Simulated deuteranopia

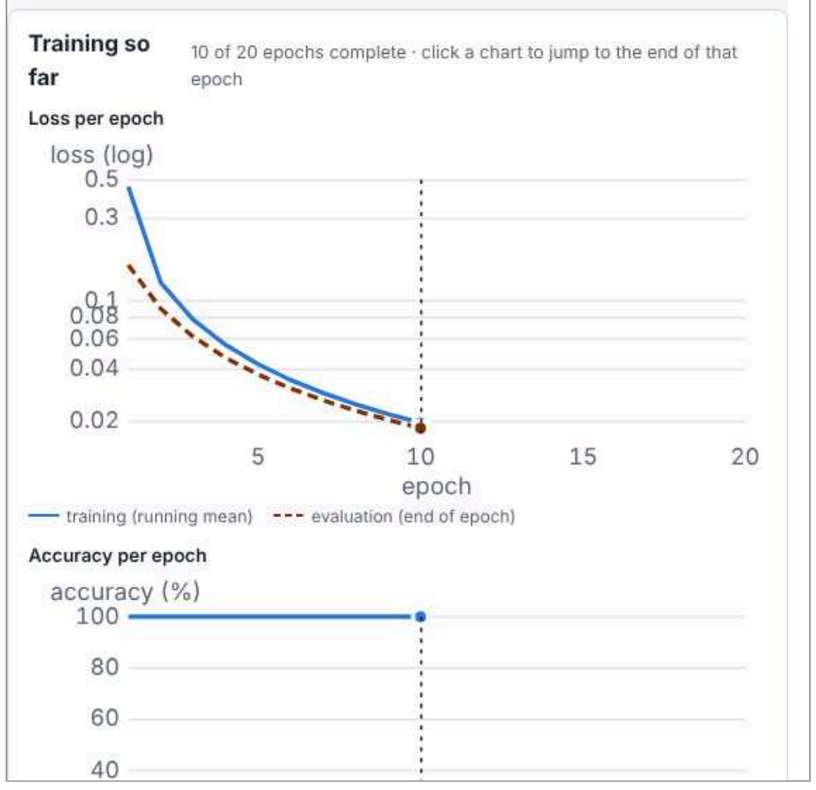


3. Forward propagation — light theme, backward phase (part 1 of 2)

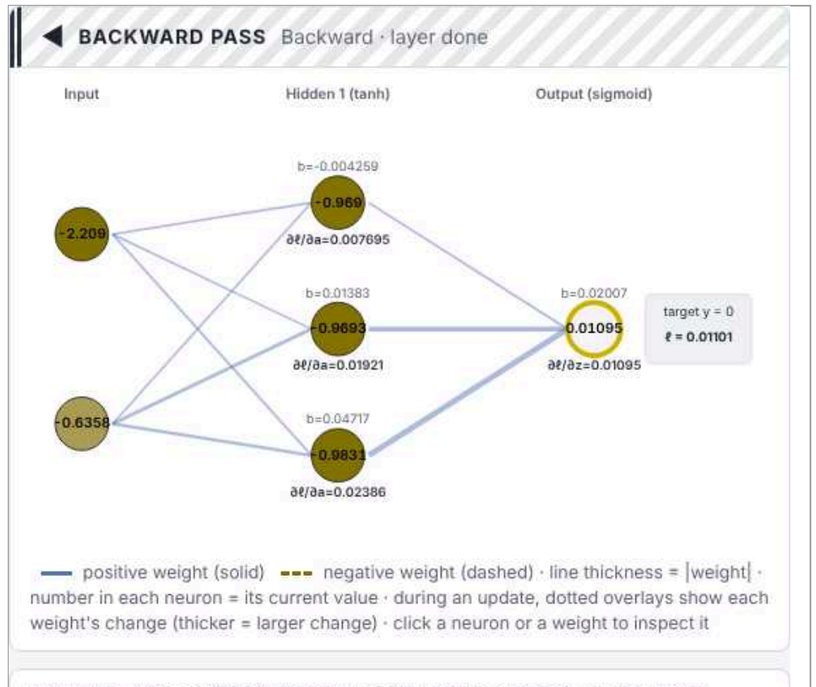
As rendered



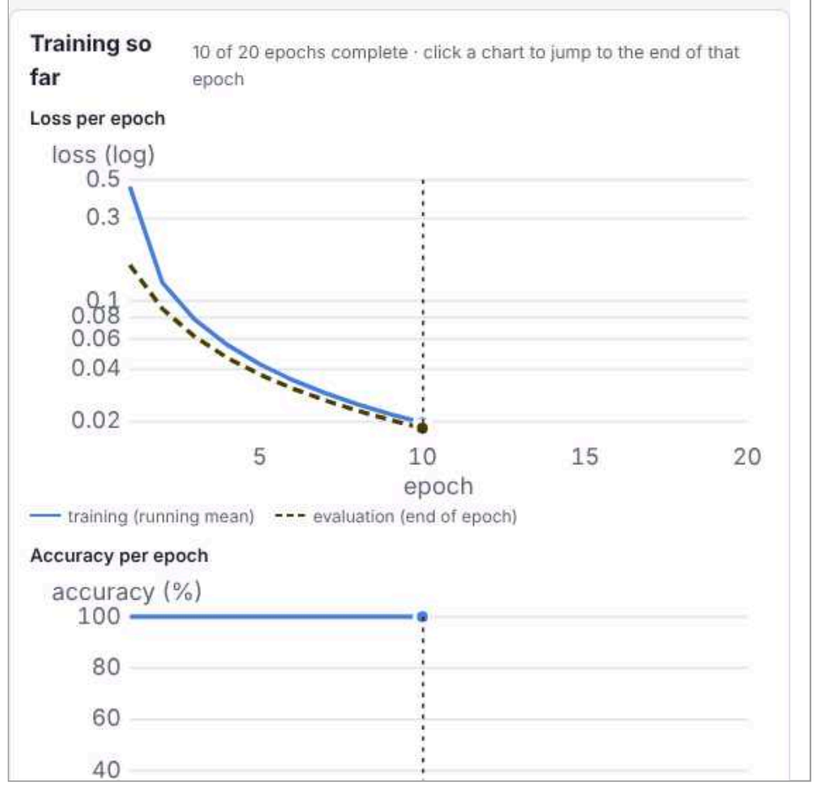
Instruction 8812 of 17,584. Epoch 11 of 20, batch 1, example 1 in the batch. Phase: backward pass, layer 2. Input x = [-2.209, -0.6358], target y = [0]. Prediction 0.01095. Example loss 0.01101.



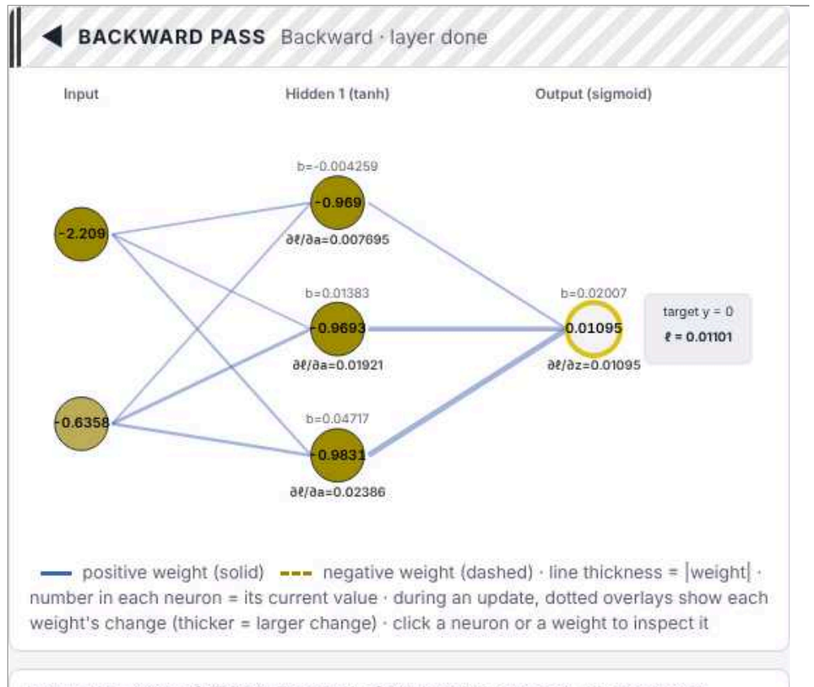
Simulated protanopia



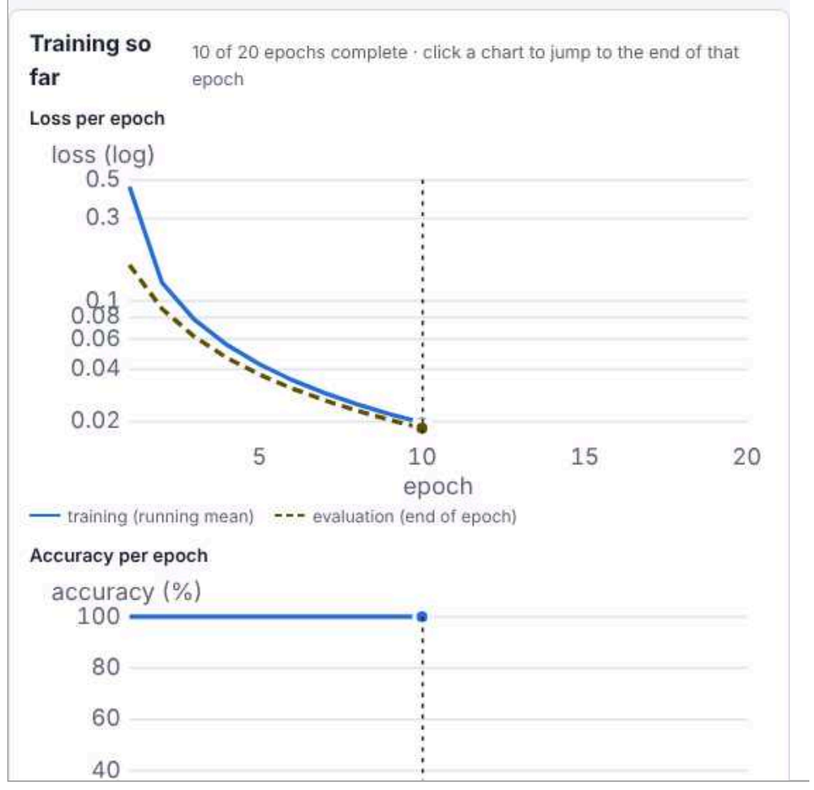
Instruction 8812 of 17,584. Epoch 11 of 20, batch 1, example 1 in the batch. Phase: backward pass, layer 2. Input x = [-2.209, -0.6358], target y = [0]. Prediction 0.01095. Example loss 0.01101.



Simulated deuteranopia

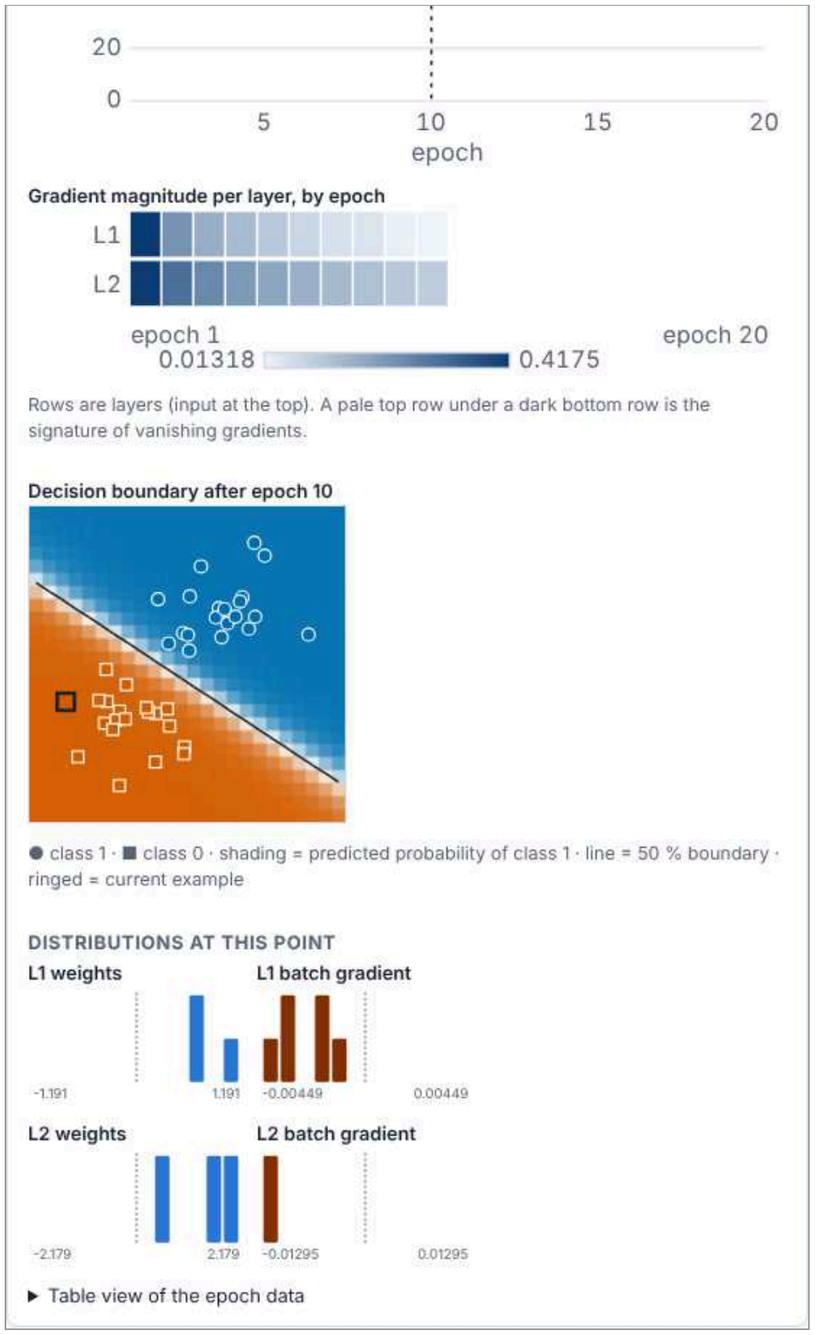


Instruction 8812 of 17,584. Epoch 11 of 20, batch 1, example 1 in the batch. Phase: backward pass, layer 2. Input x = [-2.209, -0.6358], target y = [0]. Prediction 0.01095. Example loss 0.01101.

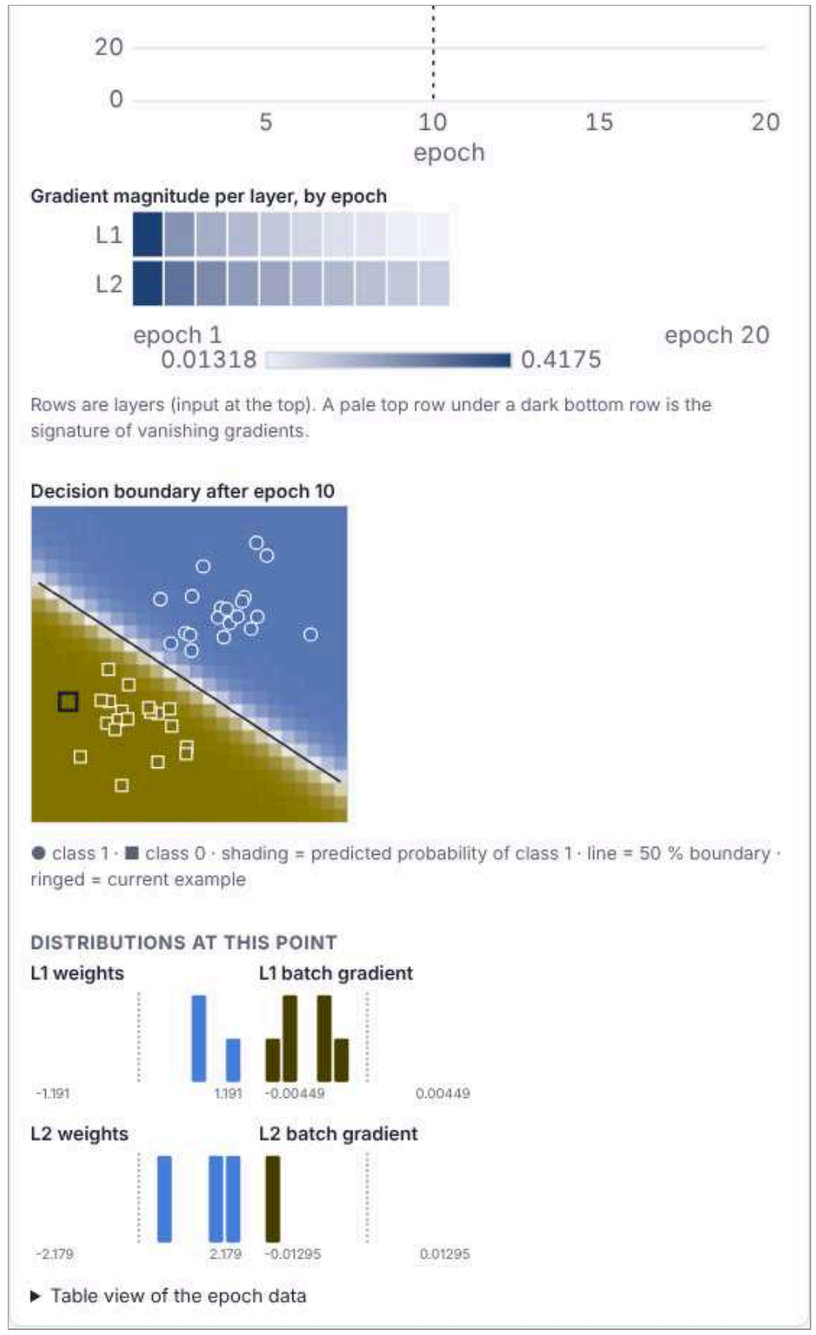


3. Forward propagation — light theme, backward phase (part 2 of 2)

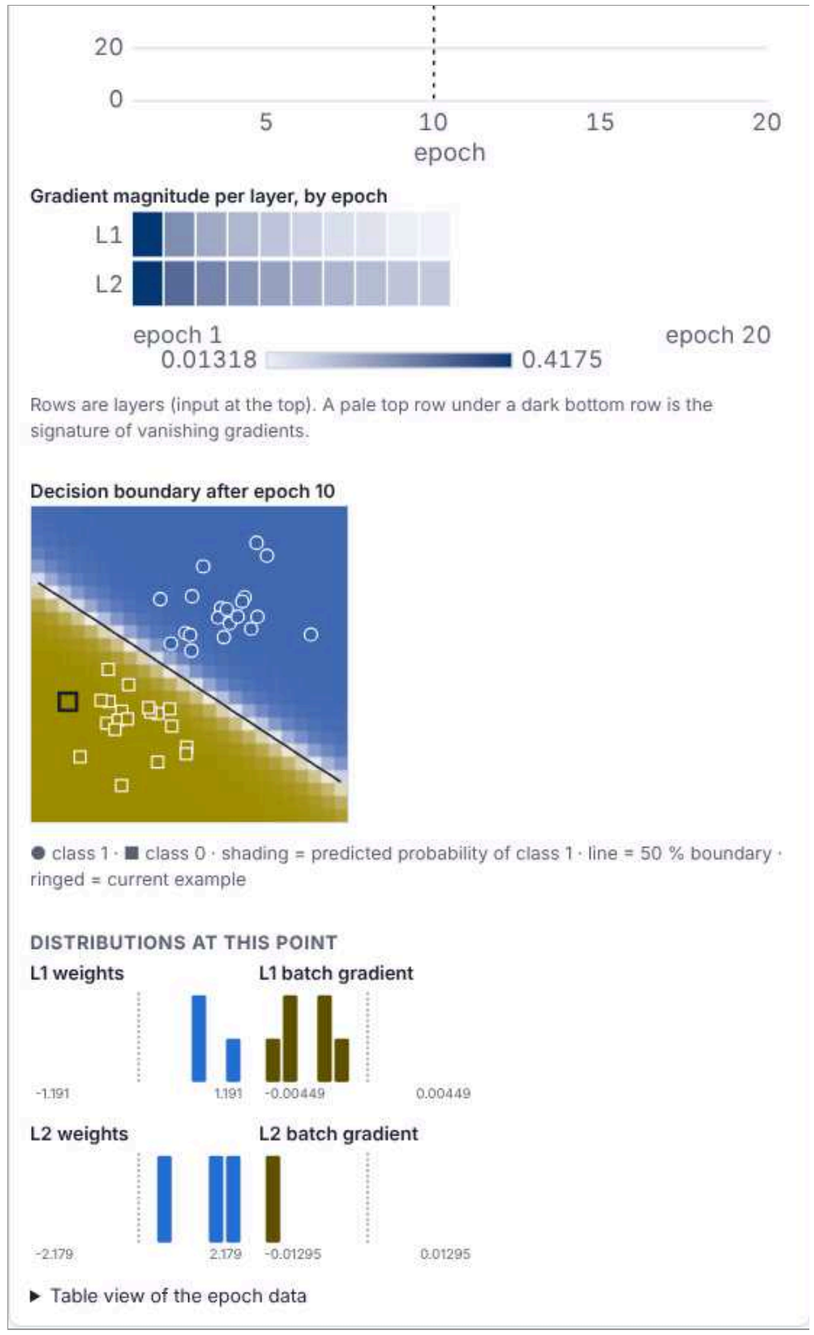
As rendered



Simulated protanopia

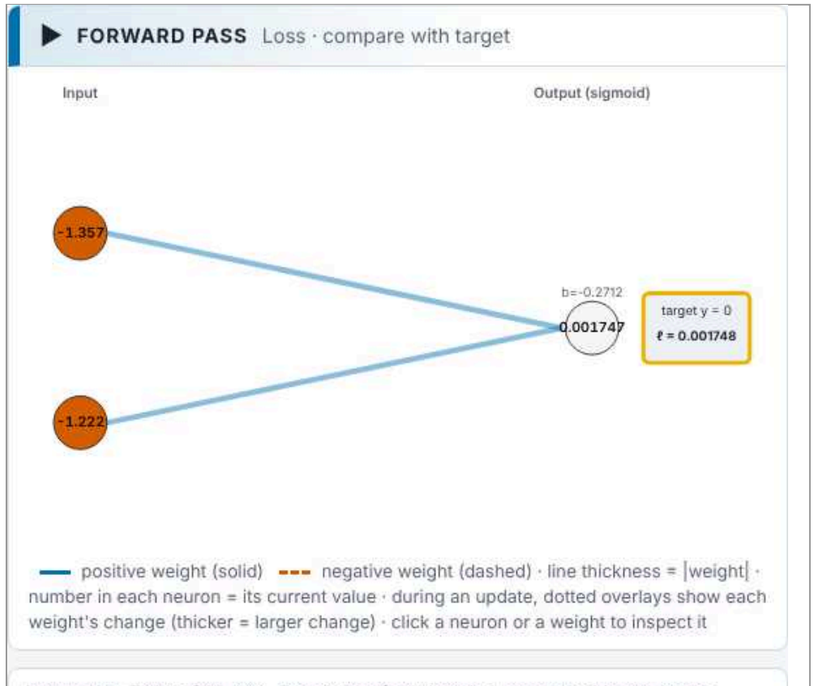


Simulated deuteranopia

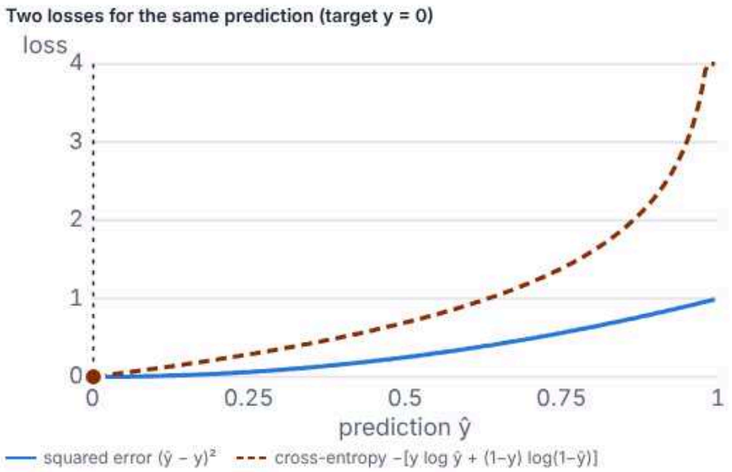


4. Loss functions — light theme, forward phase (part 1 of 3)

As rendered



Instruction 7210 of 14,402. Epoch 16 of 30, batch 1, example 1 in the batch. Phase: loss. Input x = [-1.357, -1.222], target y = [0]. Prediction 0.001747. Example loss 0.001748.



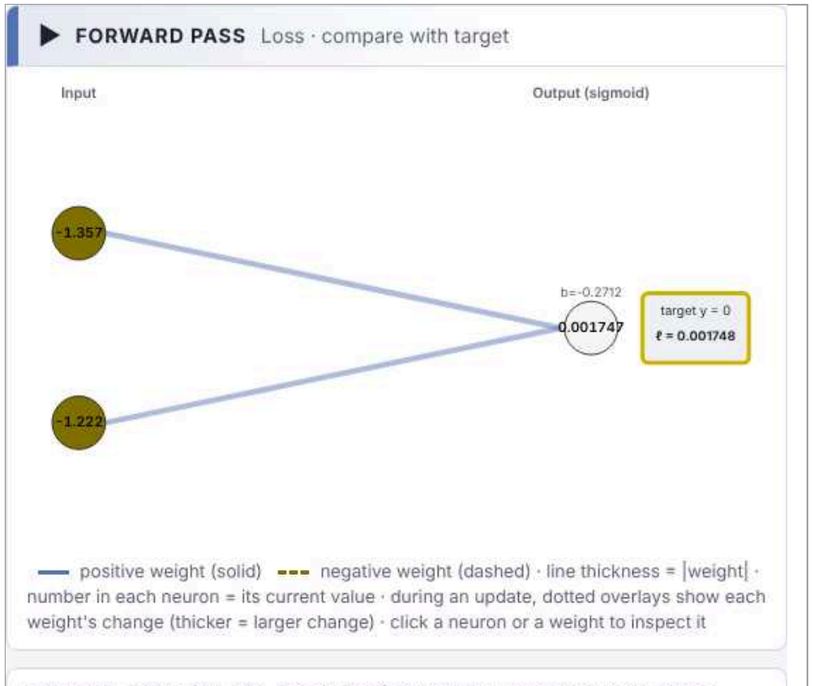
Current ŷ = 0.001747: squared error 3.05×10^{-6}, cross-entropy 0.001748. The run trains with cross-entropy.

Training so far 15 of 30 epochs complete · click a chart to jump to the end of that epoch

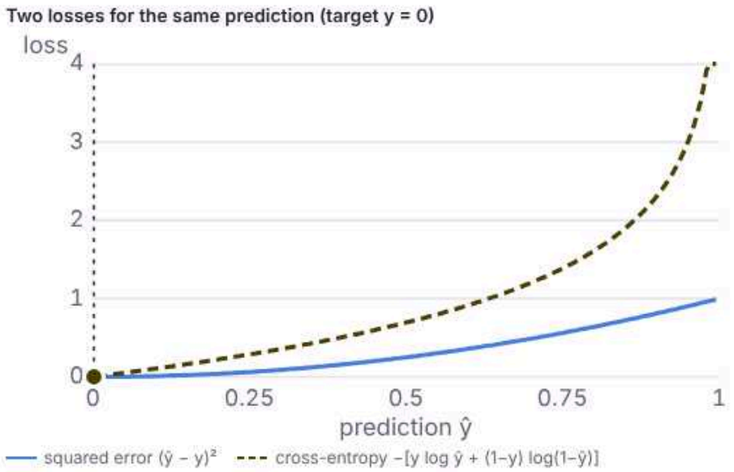
Loss per epoch

loss (log)

Simulated protanopia



Instruction 7210 of 14,402. Epoch 16 of 30, batch 1, example 1 in the batch. Phase: loss. Input x = [-1.357, -1.222], target y = [0]. Prediction 0.001747. Example loss 0.001748.



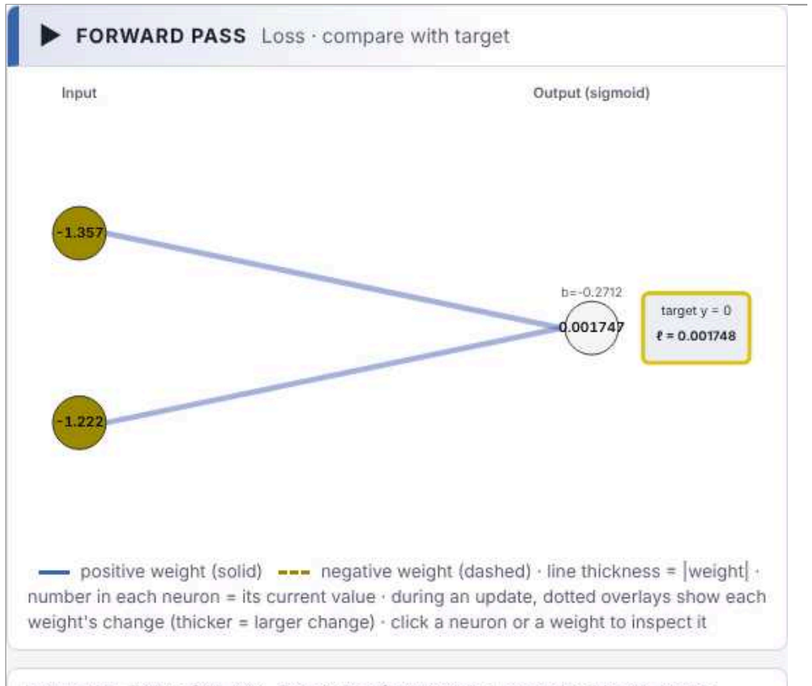
Current ŷ = 0.001747: squared error 3.05×10^{-6}, cross-entropy 0.001748. The run trains with cross-entropy.

Training so far 15 of 30 epochs complete · click a chart to jump to the end of that epoch

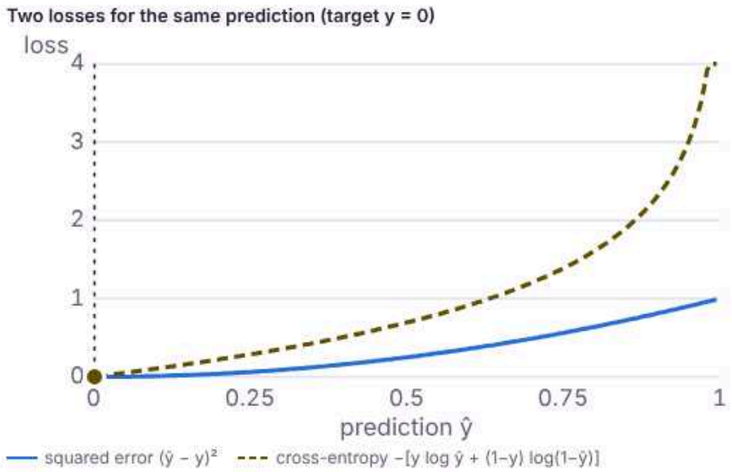
Loss per epoch

loss (log)

Simulated deuteranopia



Instruction 7210 of 14,402. Epoch 16 of 30, batch 1, example 1 in the batch. Phase: loss. Input x = [-1.357, -1.222], target y = [0]. Prediction 0.001747. Example loss 0.001748.



Current ŷ = 0.001747: squared error 3.05×10^{-6}, cross-entropy 0.001748. The run trains with cross-entropy.

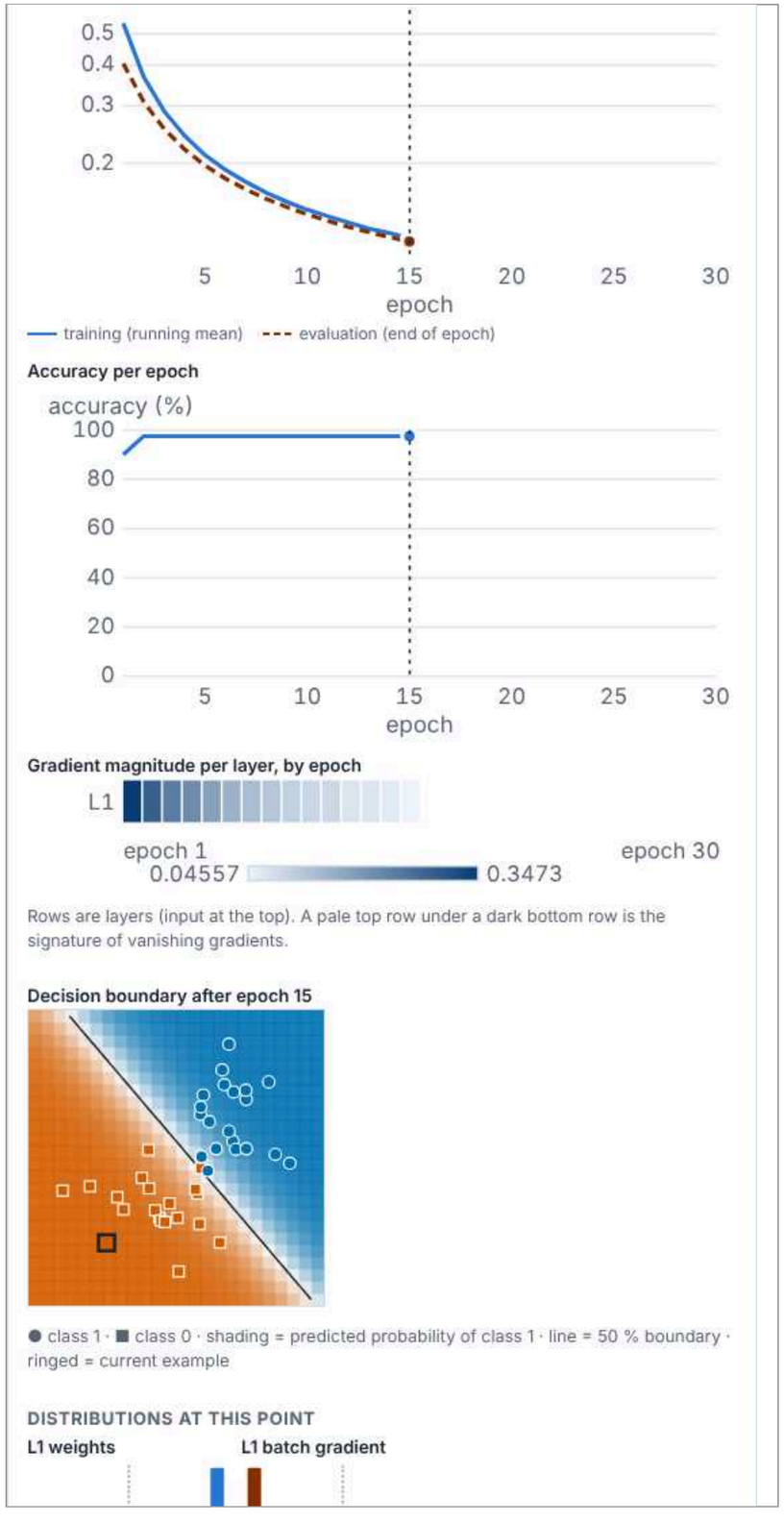
Training so far 15 of 30 epochs complete · click a chart to jump to the end of that epoch

Loss per epoch

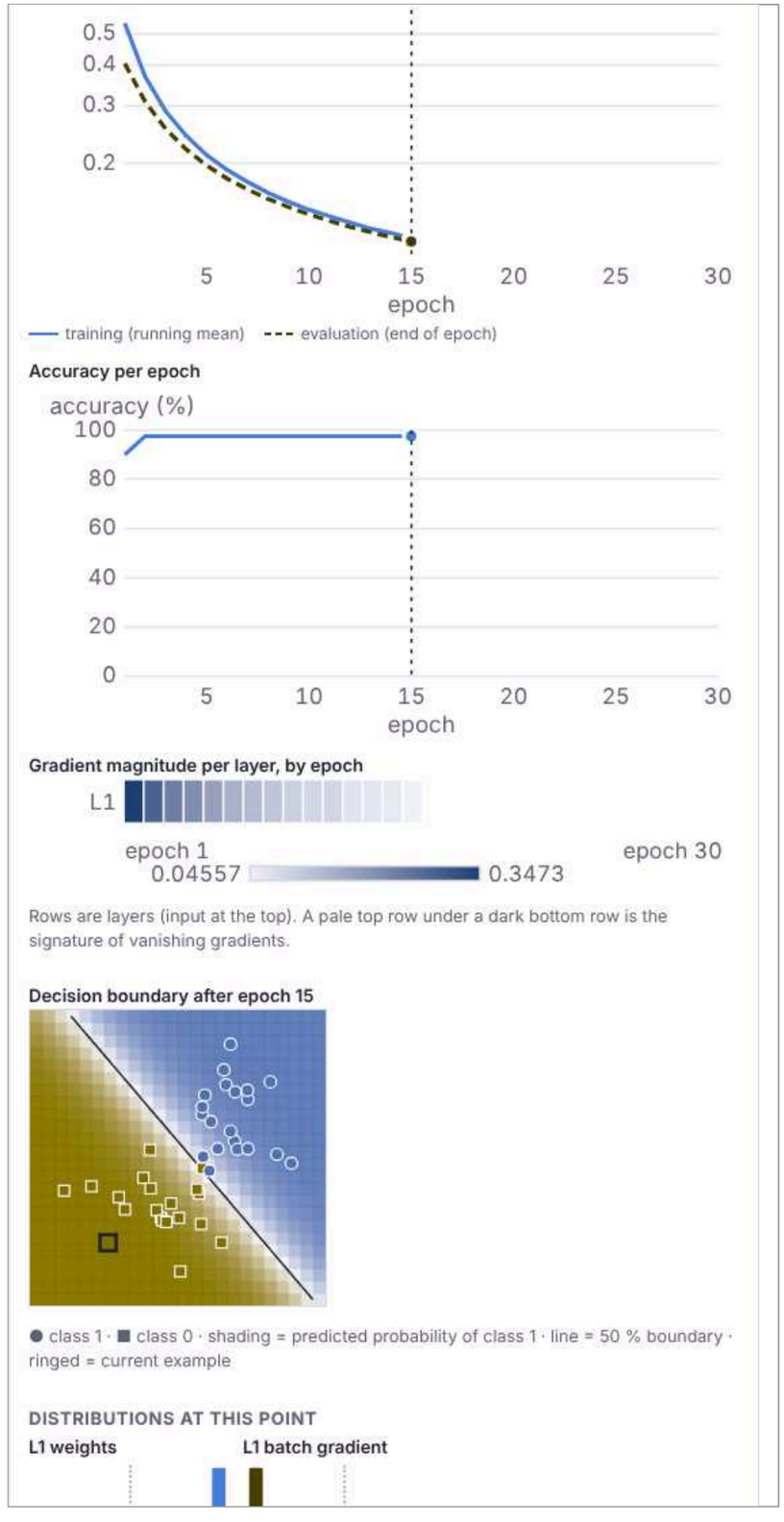
loss (log)

4. Loss functions — light theme, forward phase (part 2 of 3)

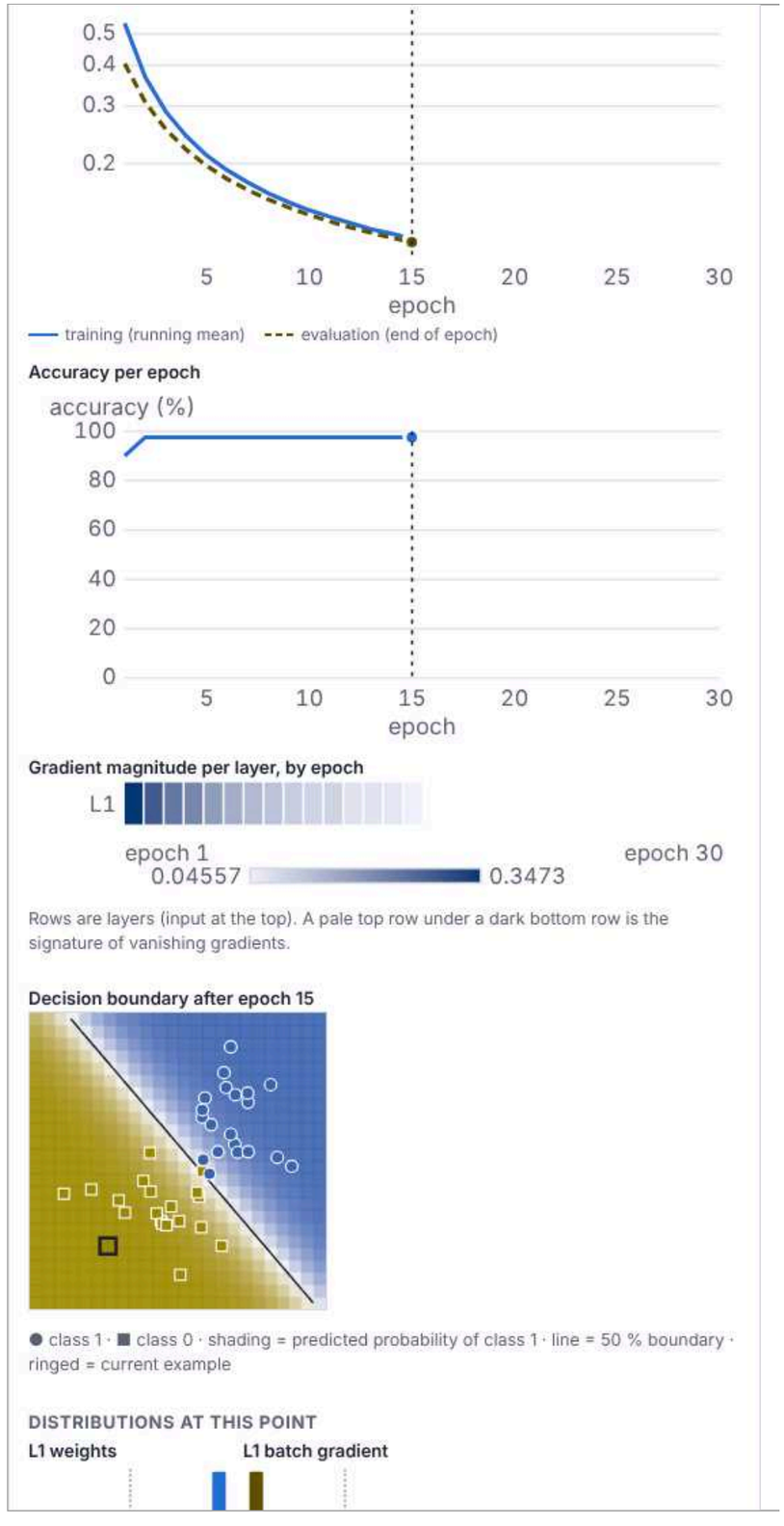
As rendered



Simulated protanopia



Simulated deuteranopia



4. Loss functions — light theme, forward phase (part 3 of 3)

As rendered



Simulated protanopia

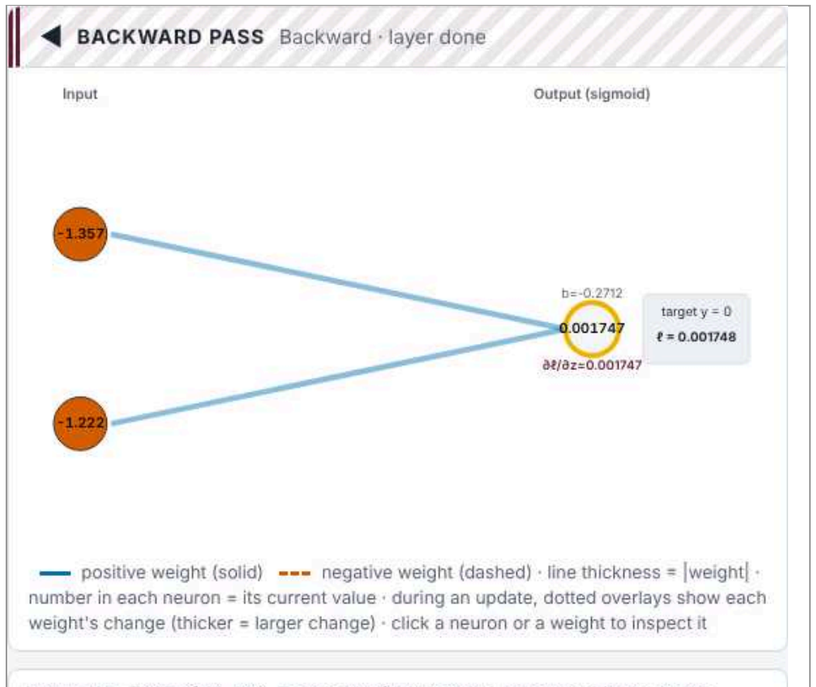


Simulated deuteranopia

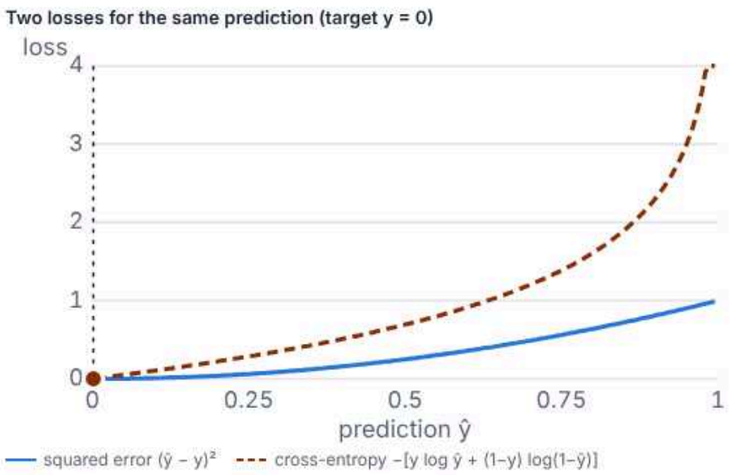


4. Loss functions — light theme, backward phase (part 1 of 3)

As rendered



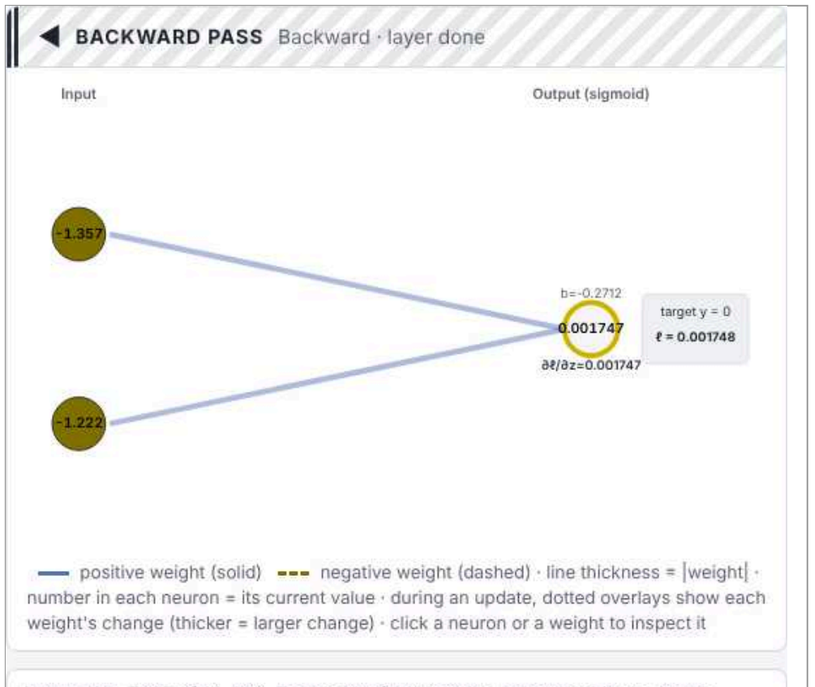
Instruction 7215 of 14,402. Epoch 16 of 30, batch 1, example 1 in the batch. Phase: backward pass, layer 1. Input x = [-1.357, -1.222], target y = [0]. Prediction 0.001747. Example loss 0.001748.



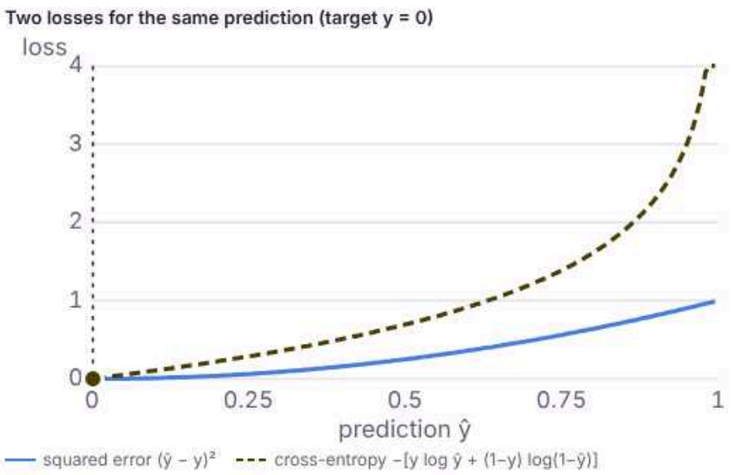
Current ŷ = 0.001747: squared error 3.05×10^{-6}, cross-entropy 0.001748. The run trains with cross-entropy.



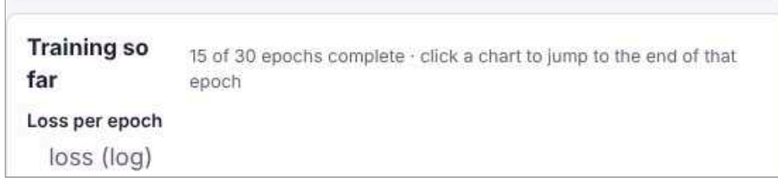
Simulated protanopia



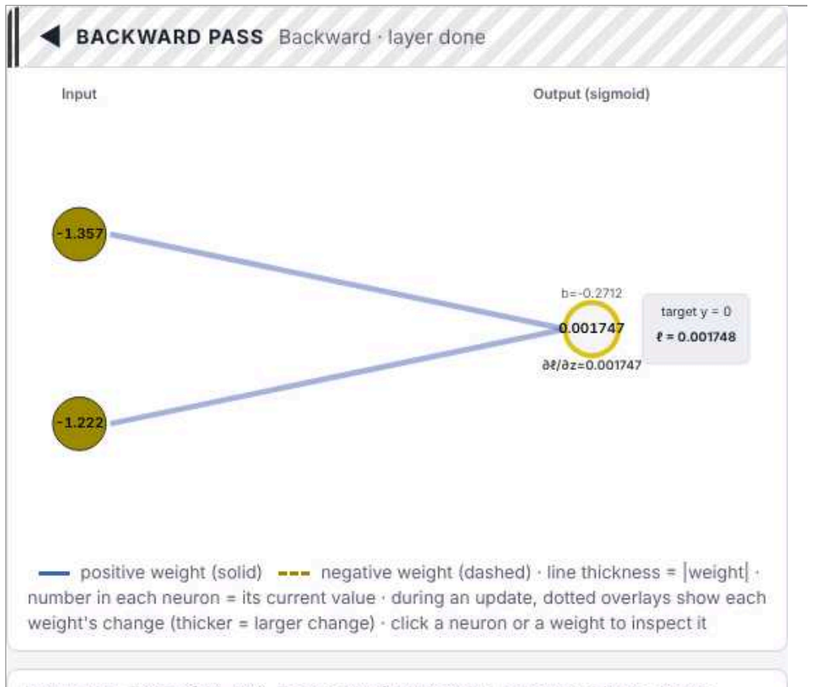
Instruction 7215 of 14,402. Epoch 16 of 30, batch 1, example 1 in the batch. Phase: backward pass, layer 1. Input x = [-1.357, -1.222], target y = [0]. Prediction 0.001747. Example loss 0.001748.



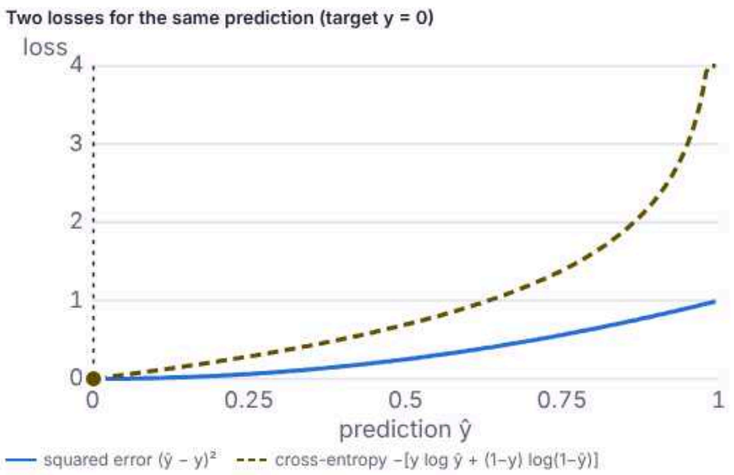
Current ŷ = 0.001747: squared error 3.05×10^{-6}, cross-entropy 0.001748. The run trains with cross-entropy.



Simulated deuteranopia



Instruction 7215 of 14,402. Epoch 16 of 30, batch 1, example 1 in the batch. Phase: backward pass, layer 1. Input x = [-1.357, -1.222], target y = [0]. Prediction 0.001747. Example loss 0.001748.

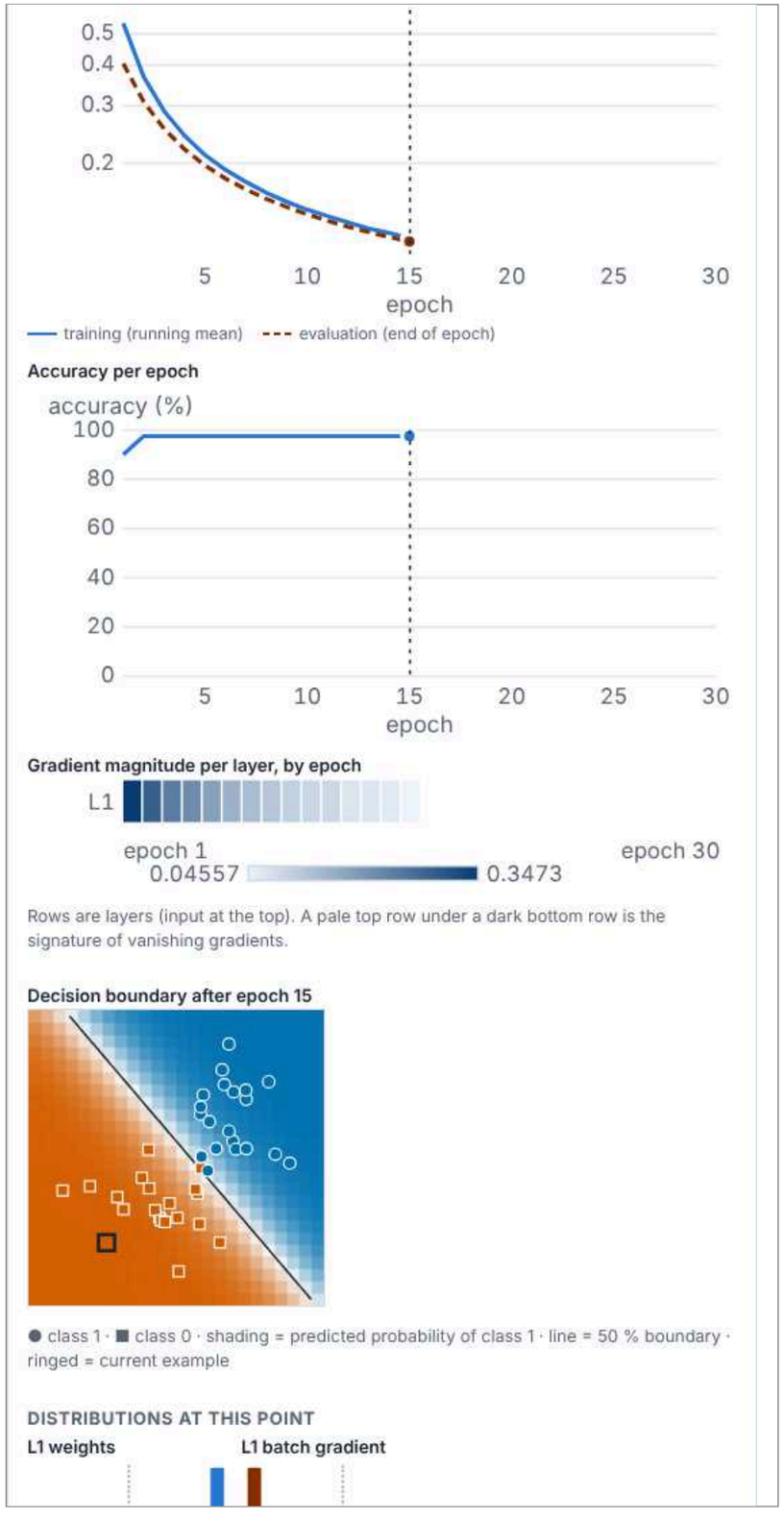


Current ŷ = 0.001747: squared error 3.05×10^{-6}, cross-entropy 0.001748. The run trains with cross-entropy.

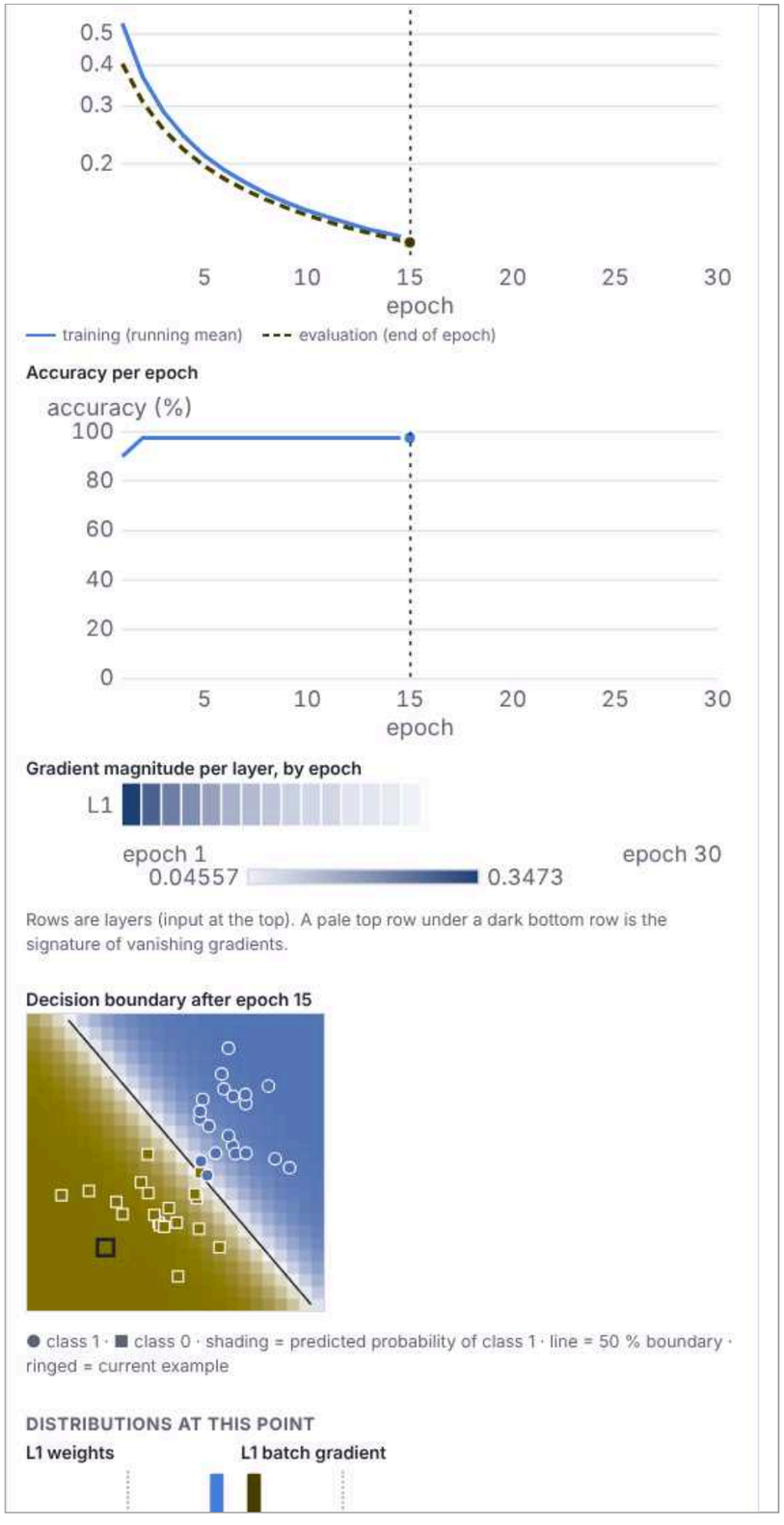


4. Loss functions — light theme, backward phase (part 2 of 3)

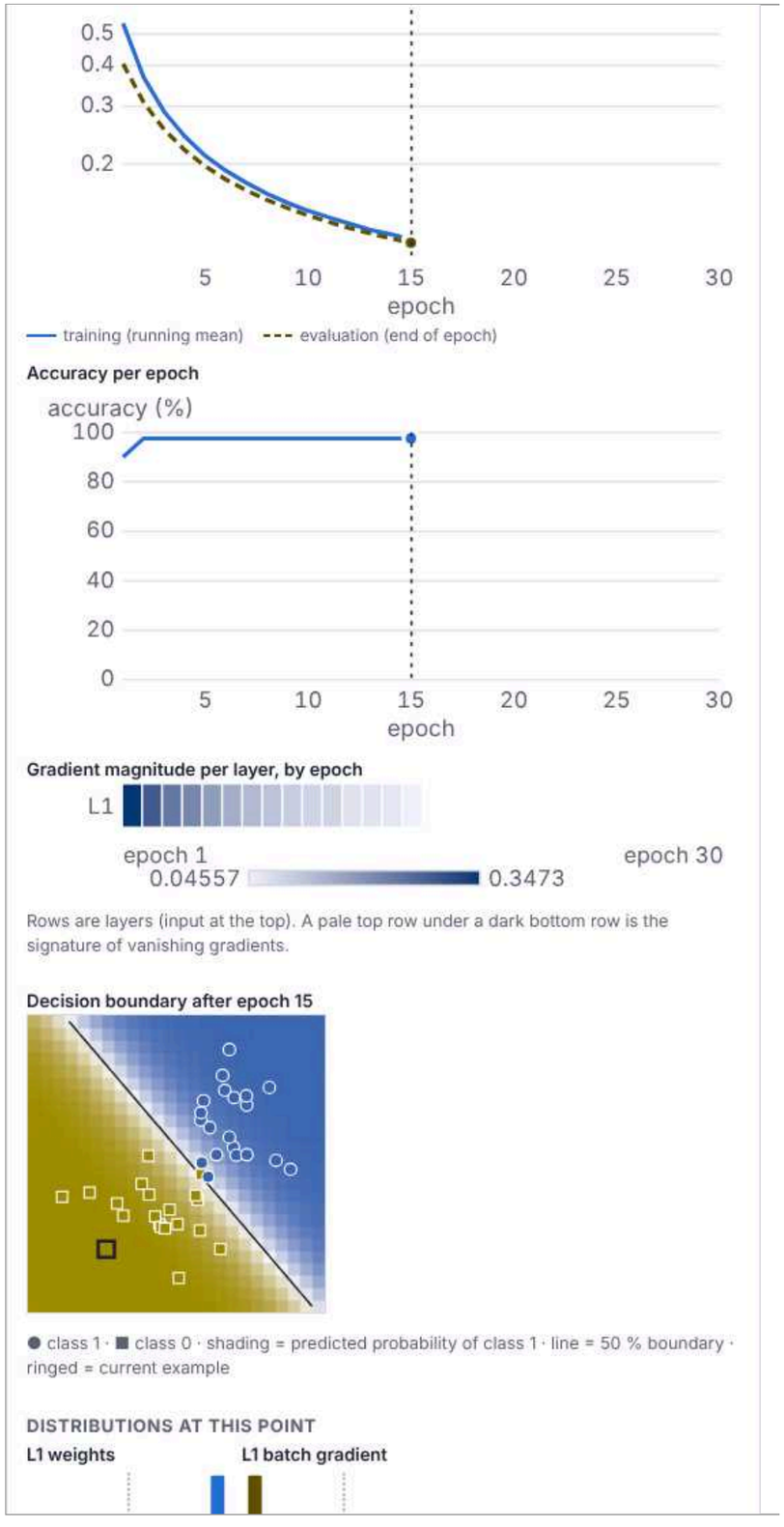
As rendered



Simulated protanopia



Simulated deuteranopia



4. Loss functions — light theme, backward phase (part 3 of 3)

As rendered



▶ Table view of the epoch data

Simulated protanopia



▶ Table view of the epoch data

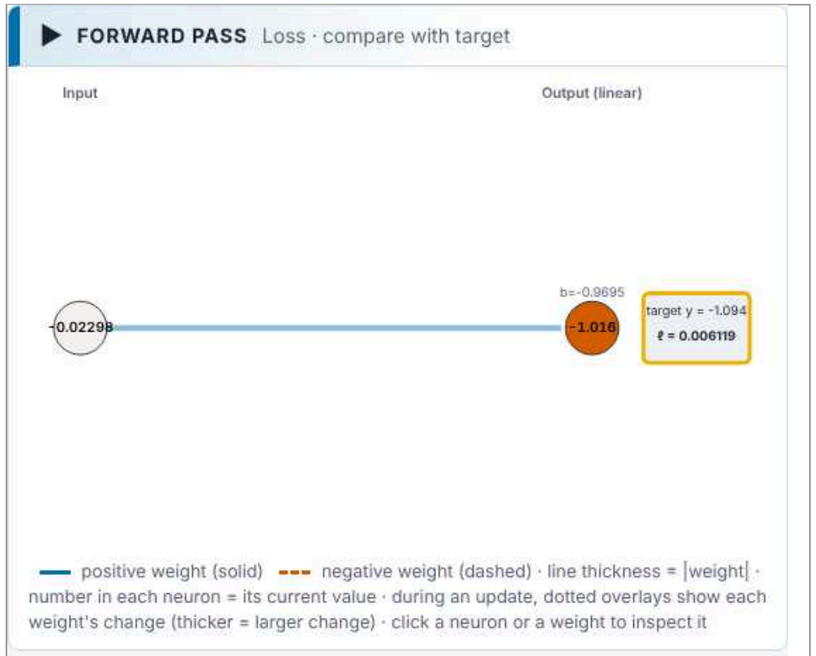
Simulated deuteranopia



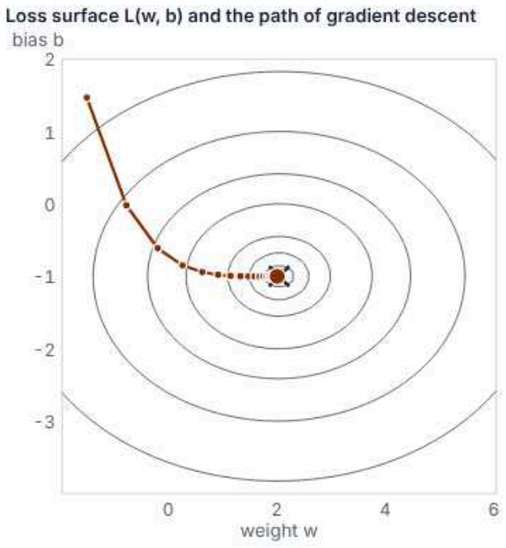
▶ Table view of the epoch data

5. Gradient descent — light theme, forward phase (part 1 of 2)

As rendered



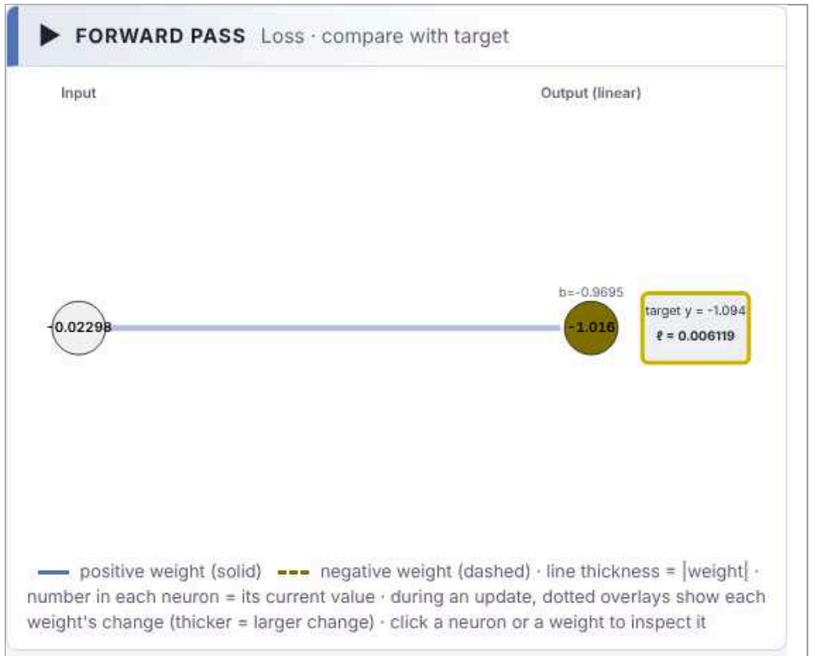
Instruction 11270 of 22,522. Epoch 21 of 40, batch 1, example 1 in the batch. Phase: loss. Input x = [-0.02298], target y = [-1.094]. Prediction -1.016. Example loss 0.006119.



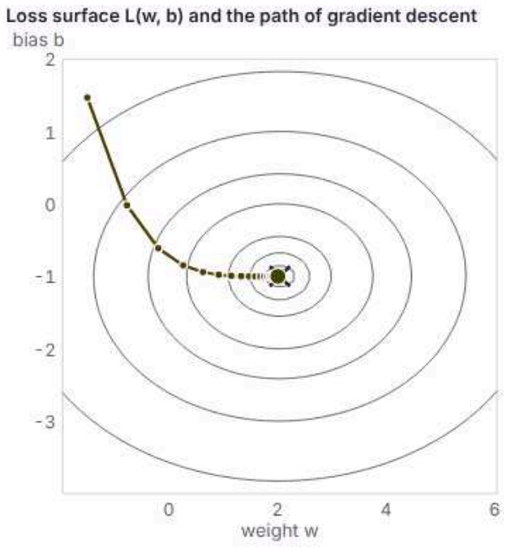
Contours are levels of the loss over all 50 points; × marks the minimum (least-squares fit). Dots are the parameters after each update so far: now w = 2.008, b = -0.9695.

Training so far 20 of 40 epochs complete · click a chart to jump to the end of that epoch

Simulated protanopia



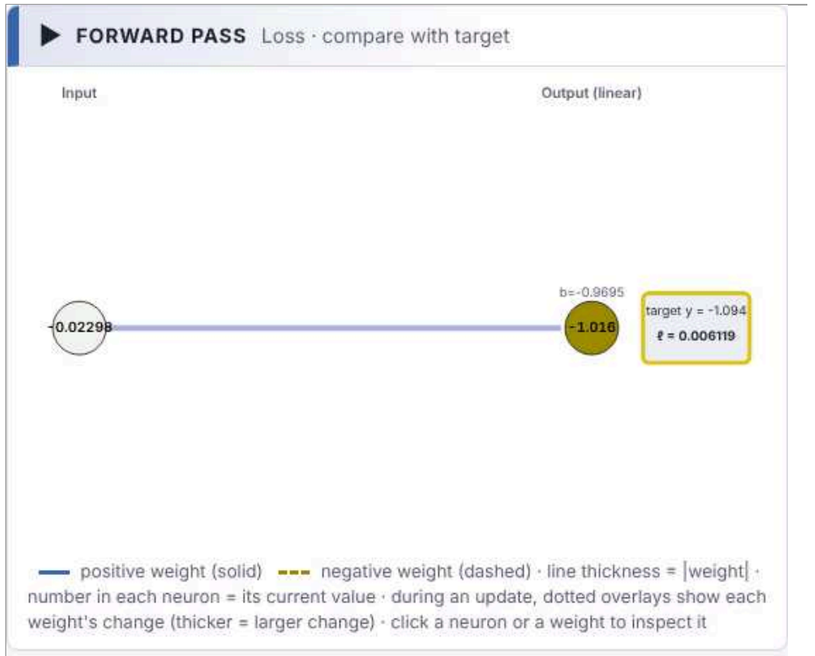
Instruction 11270 of 22,522. Epoch 21 of 40, batch 1, example 1 in the batch. Phase: loss. Input x = [-0.02298], target y = [-1.094]. Prediction -1.016. Example loss 0.006119.



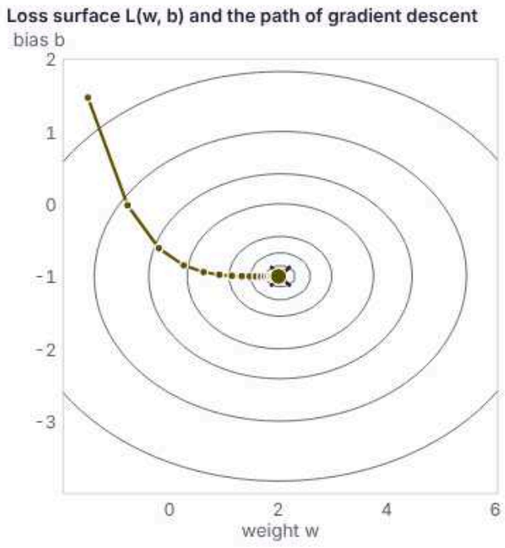
Contours are levels of the loss over all 50 points; × marks the minimum (least-squares fit). Dots are the parameters after each update so far: now w = 2.008, b = -0.9695.

Training so far 20 of 40 epochs complete · click a chart to jump to the end of that epoch

Simulated deuteranopia



Instruction 11270 of 22,522. Epoch 21 of 40, batch 1, example 1 in the batch. Phase: loss. Input x = [-0.02298], target y = [-1.094]. Prediction -1.016. Example loss 0.006119.

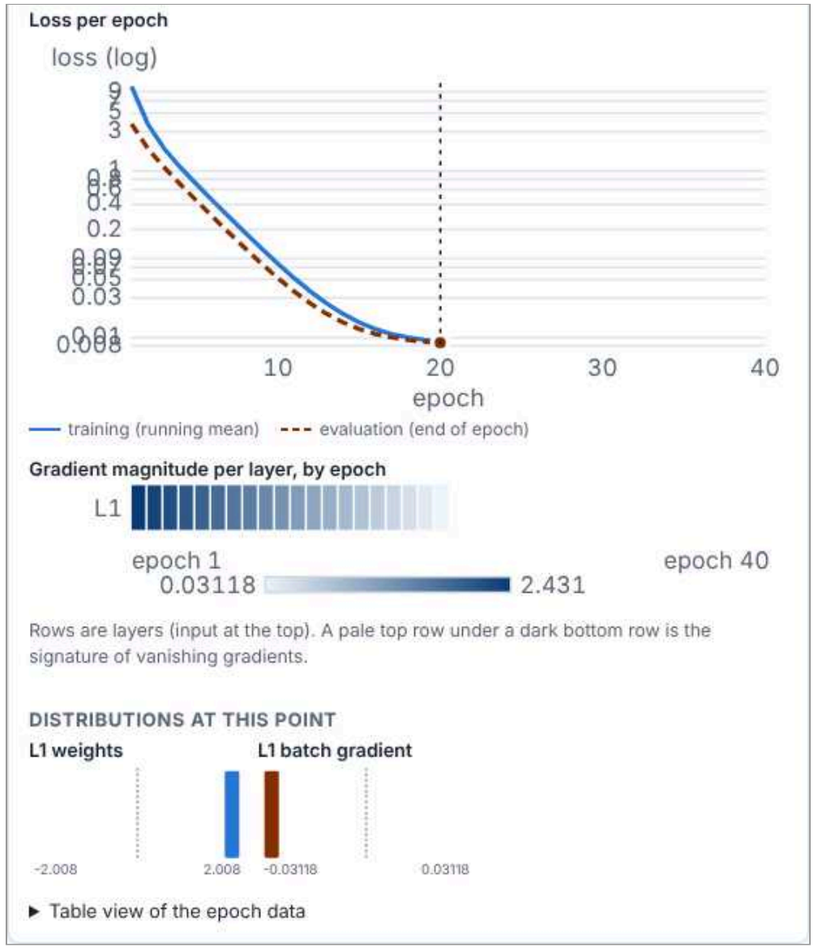


Contours are levels of the loss over all 50 points; × marks the minimum (least-squares fit). Dots are the parameters after each update so far: now w = 2.008, b = -0.9695.

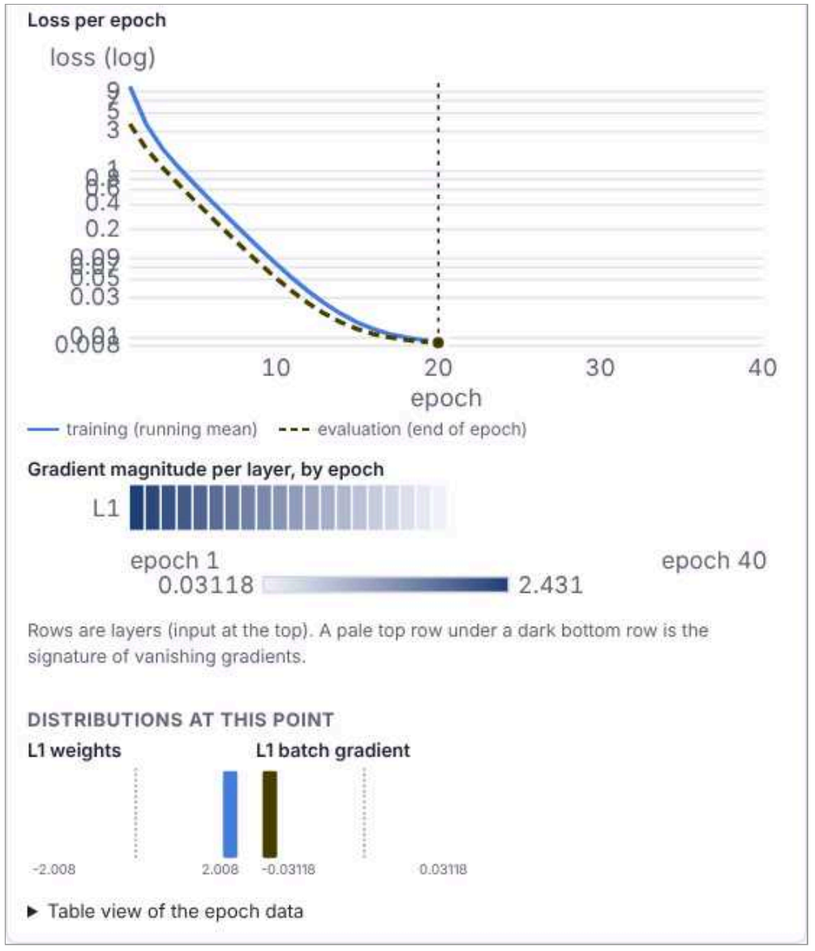
Training so far 20 of 40 epochs complete · click a chart to jump to the end of that epoch

5. Gradient descent — light theme, forward phase (part 2 of 2)

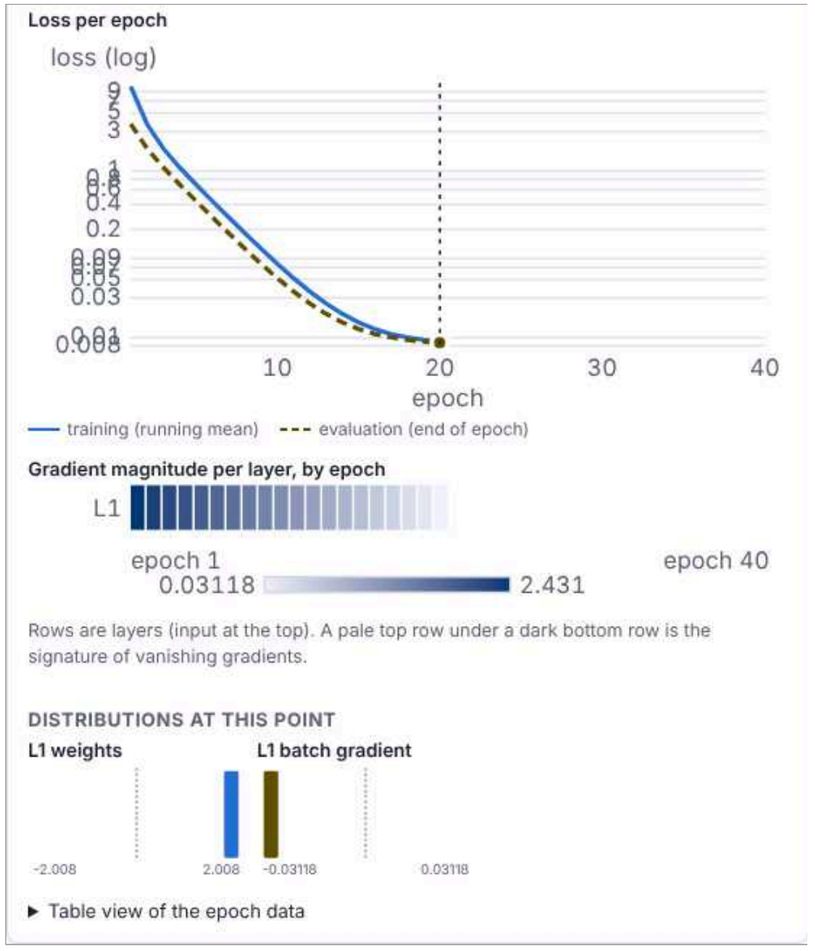
As rendered



Simulated protanopia



Simulated deuteranopia

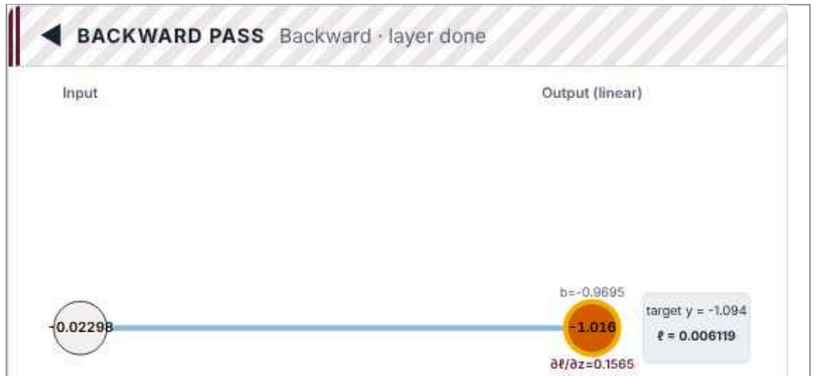


5. Gradient descent — light theme, backward phase (part 1 of 2)

As rendered

Simulated protanopia

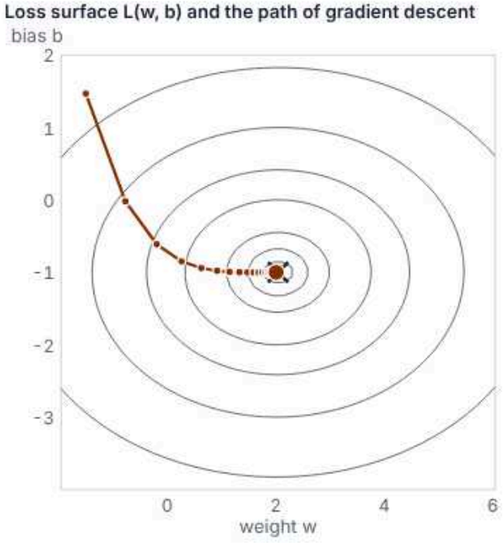
Simulated deuteranopia



positive weight (solid) negative weight (dashed) · line thickness = |weight| · number in each neuron = its current value · during an update, dotted overlays show each weight's change (thicker = larger change) · click a neuron or a weight to inspect it

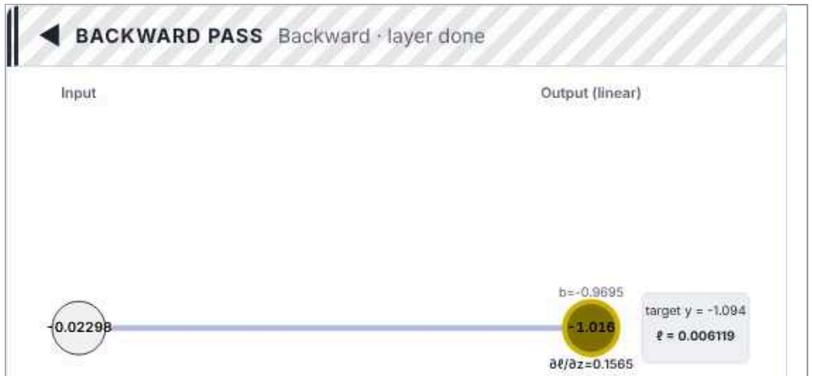
Instruction 11275 of 22,522. Epoch 21 of 40, batch 1, example 1 in the batch. Phase: backward pass, layer 1. Input x = [-0.02298], target y = [-1.094]. Prediction -1.016. Example loss 0.006119.

Loss surface L(w, b) and the path of gradient descent



Contours are levels of the loss over all 50 points; × marks the minimum (least-squares fit). Dots are the parameters after each update so far: now w = 2.008, b = -0.9695.

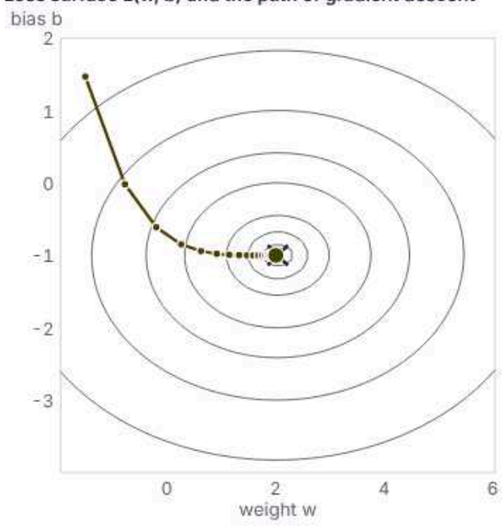
Training so far 20 of 40 epochs complete · click a chart to jump to the end of that epoch



positive weight (solid) negative weight (dashed) · line thickness = |weight| · number in each neuron = its current value · during an update, dotted overlays show each weight's change (thicker = larger change) · click a neuron or a weight to inspect it

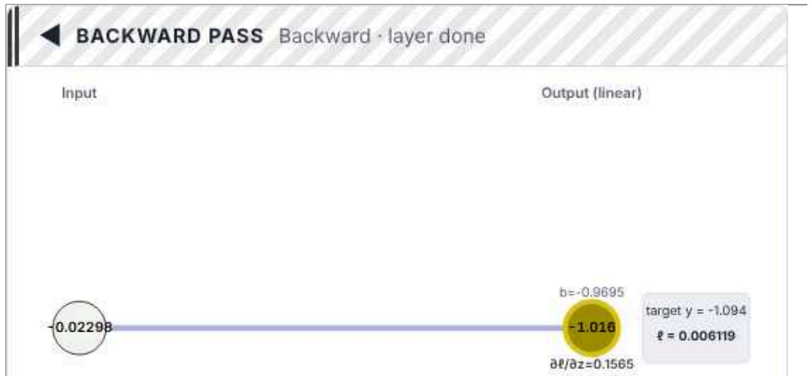
Instruction 11275 of 22,522. Epoch 21 of 40, batch 1, example 1 in the batch. Phase: backward pass, layer 1. Input x = [-0.02298], target y = [-1.094]. Prediction -1.016. Example loss 0.006119.

Loss surface L(w, b) and the path of gradient descent



Contours are levels of the loss over all 50 points; × marks the minimum (least-squares fit). Dots are the parameters after each update so far: now w = 2.008, b = -0.9695.

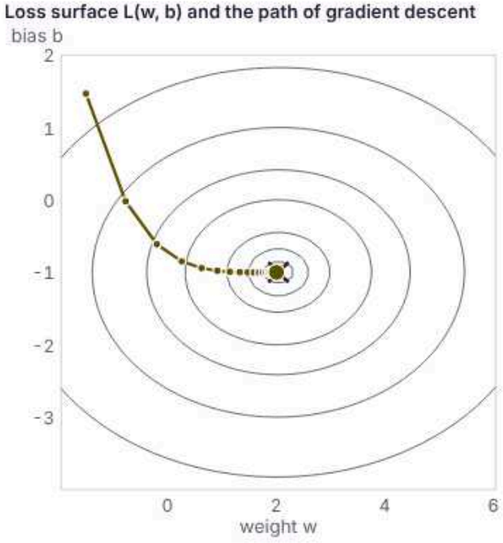
Training so far 20 of 40 epochs complete · click a chart to jump to the end of that epoch



positive weight (solid) negative weight (dashed) · line thickness = |weight| · number in each neuron = its current value · during an update, dotted overlays show each weight's change (thicker = larger change) · click a neuron or a weight to inspect it

Instruction 11275 of 22,522. Epoch 21 of 40, batch 1, example 1 in the batch. Phase: backward pass, layer 1. Input x = [-0.02298], target y = [-1.094]. Prediction -1.016. Example loss 0.006119.

Loss surface L(w, b) and the path of gradient descent

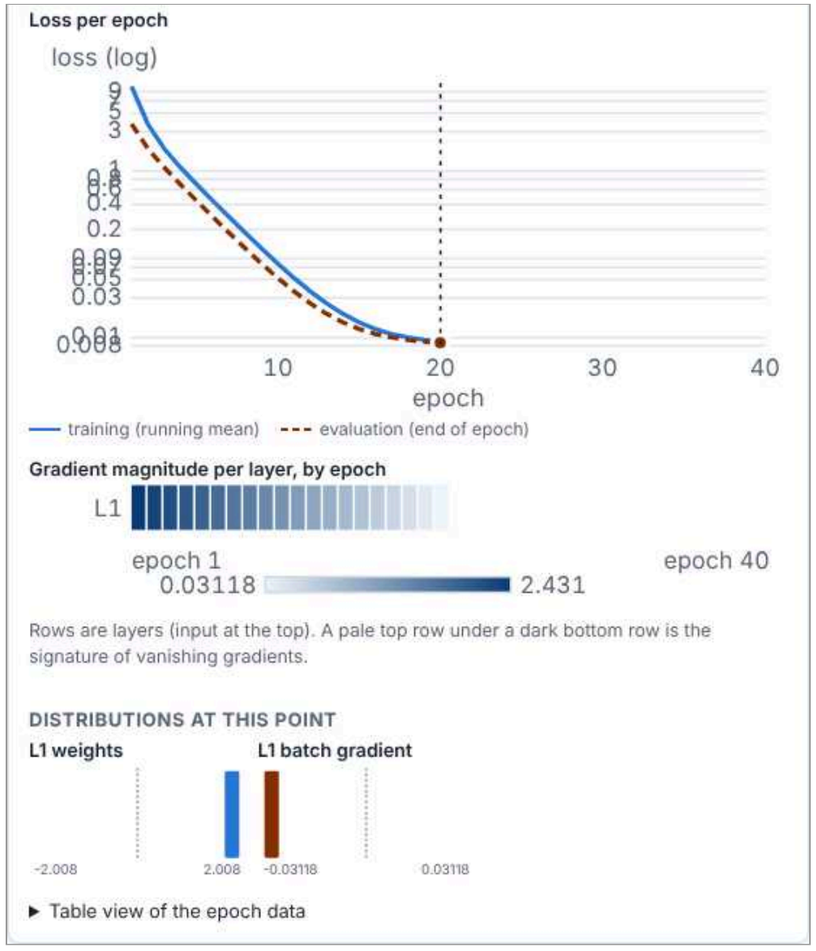


Contours are levels of the loss over all 50 points; × marks the minimum (least-squares fit). Dots are the parameters after each update so far: now w = 2.008, b = -0.9695.

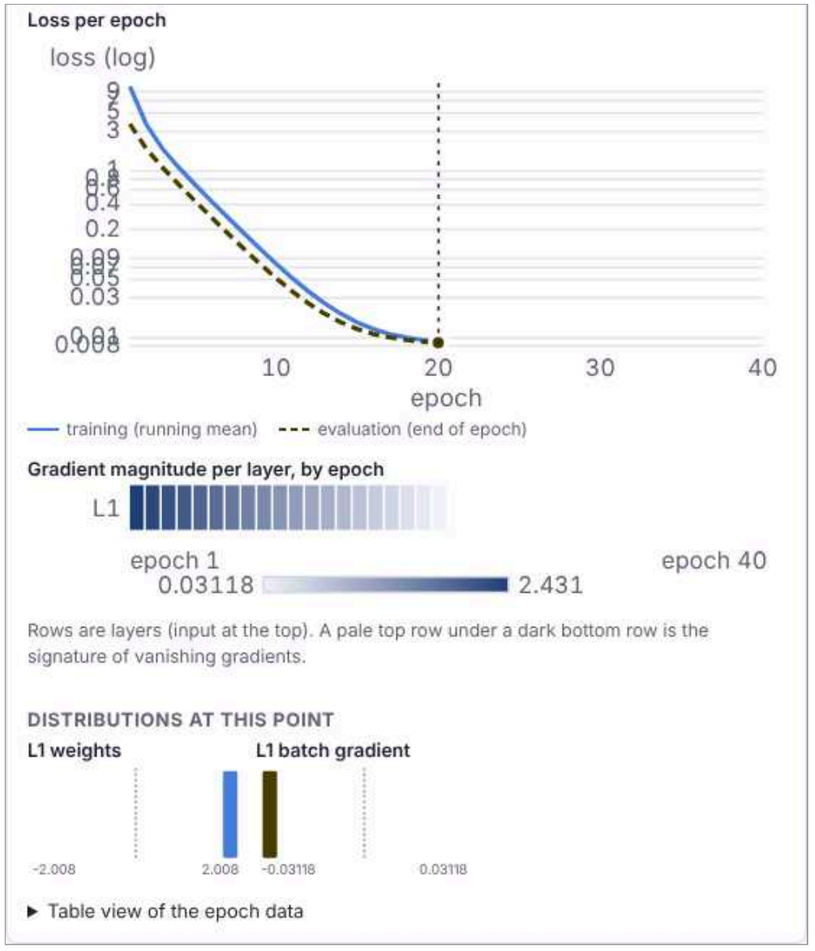
Training so far 20 of 40 epochs complete · click a chart to jump to the end of that epoch

5. Gradient descent — light theme, backward phase (part 2 of 2)

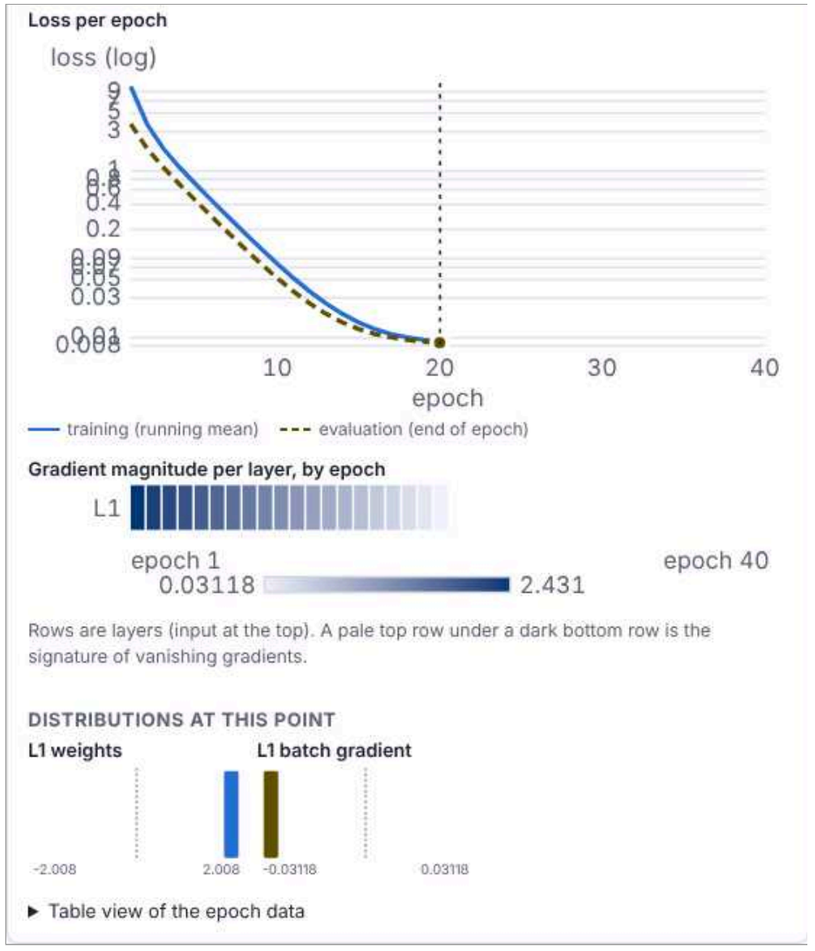
As rendered



Simulated protanopia

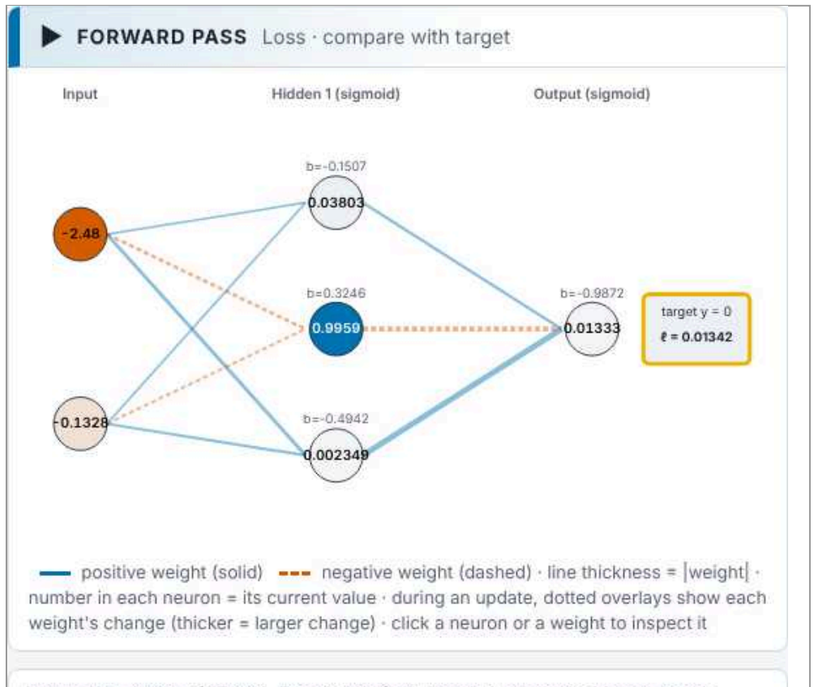


Simulated deuteranopia

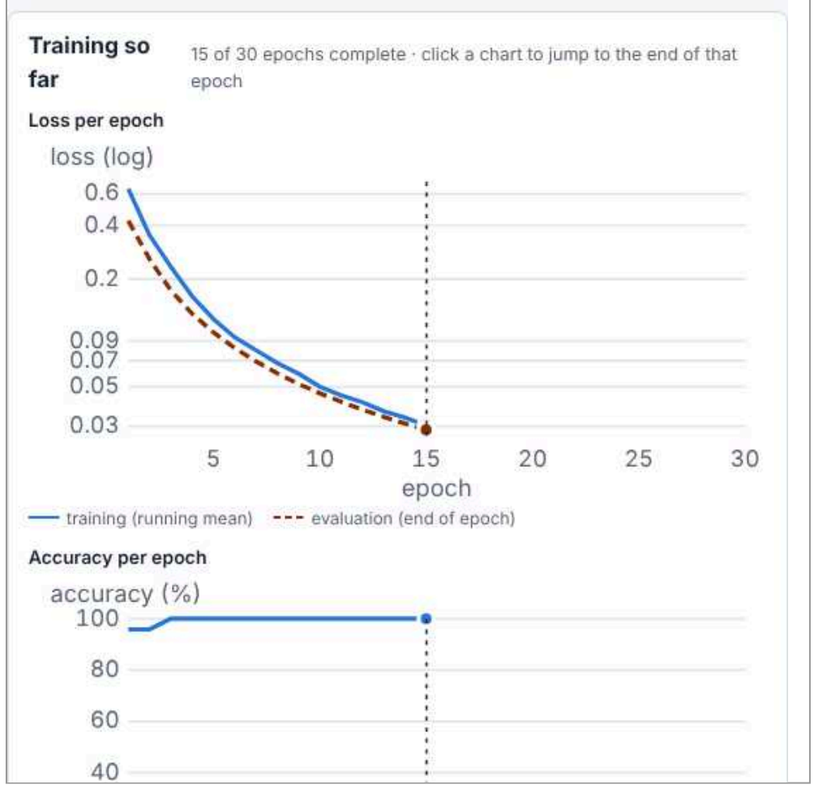


6. Back-propagation — light theme, forward phase (part 1 of 2)

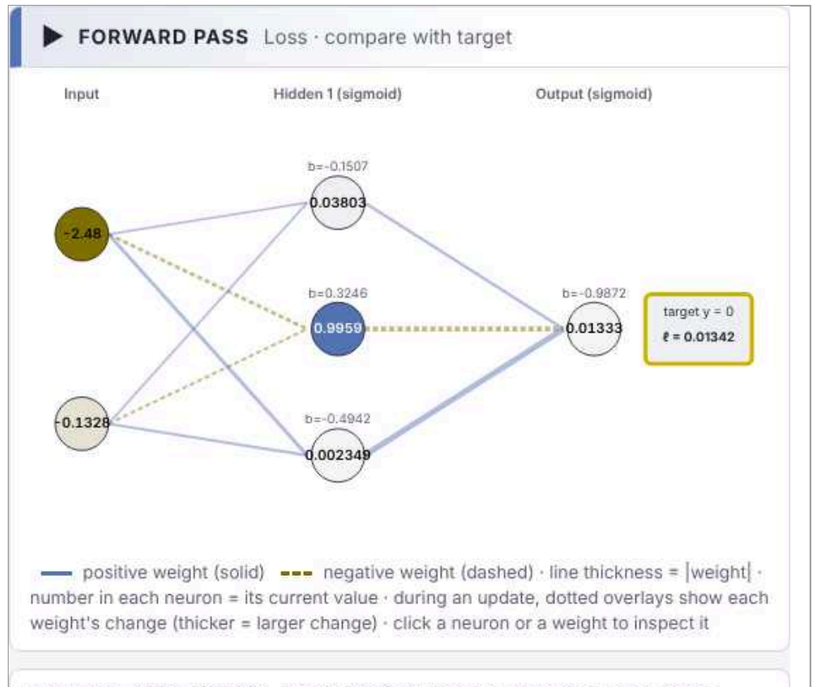
As rendered



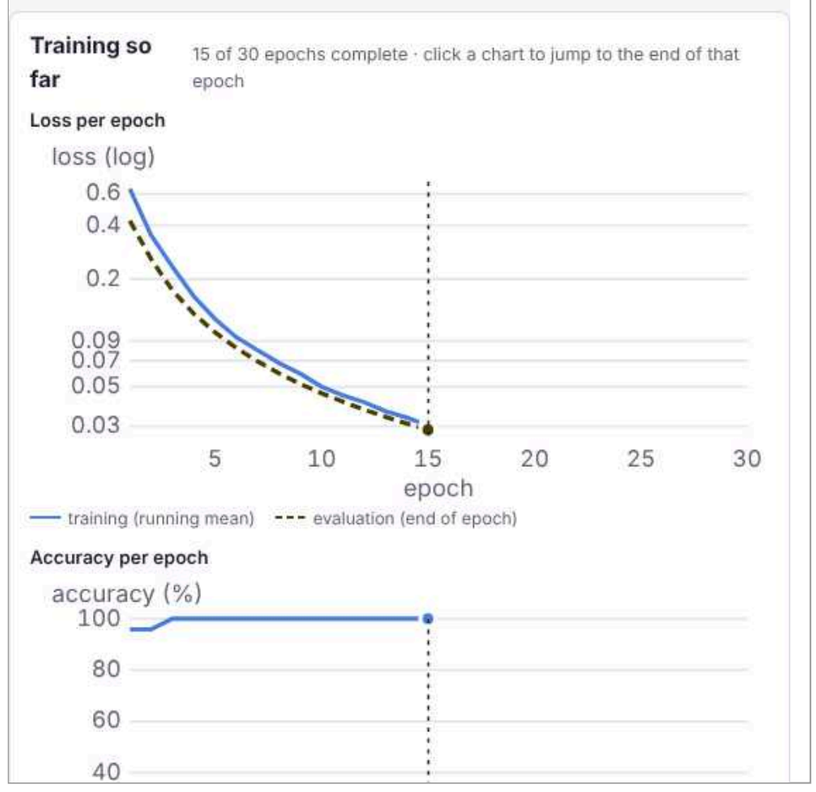
Instruction 8626 of 17,224. Epoch 16 of 30, batch 1, example 1 in the batch. Phase: loss. Input x = [-2.48, -0.1328], target y = [0]. Prediction 0.01333. Example loss 0.01342.



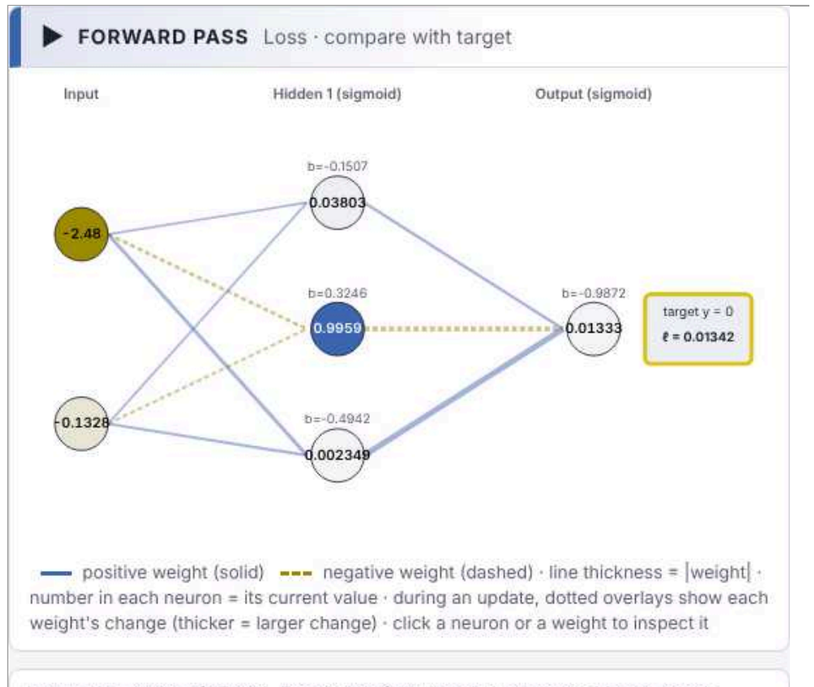
Simulated protanopia



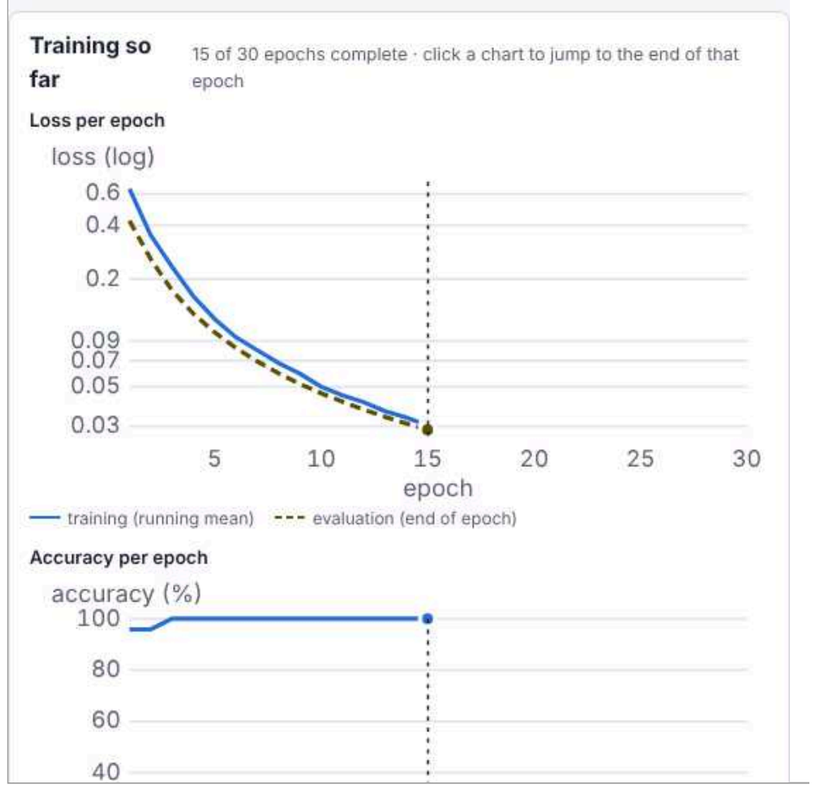
Instruction 8626 of 17,224. Epoch 16 of 30, batch 1, example 1 in the batch. Phase: loss. Input x = [-2.48, -0.1328], target y = [0]. Prediction 0.01333. Example loss 0.01342.



Simulated deuteranopia

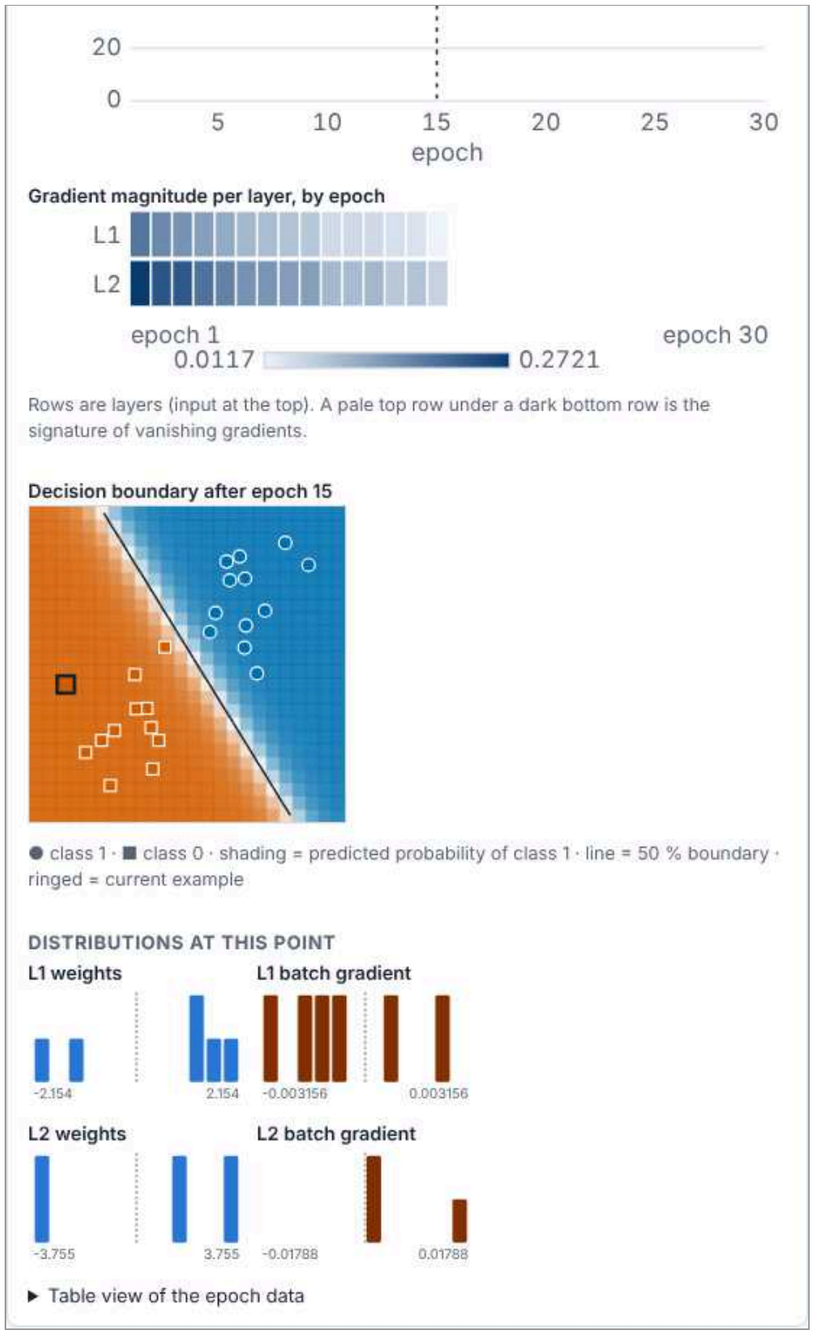


Instruction 8626 of 17,224. Epoch 16 of 30, batch 1, example 1 in the batch. Phase: loss. Input x = [-2.48, -0.1328], target y = [0]. Prediction 0.01333. Example loss 0.01342.



6. Back-propagation — light theme, forward phase (part 2 of 2)

As rendered



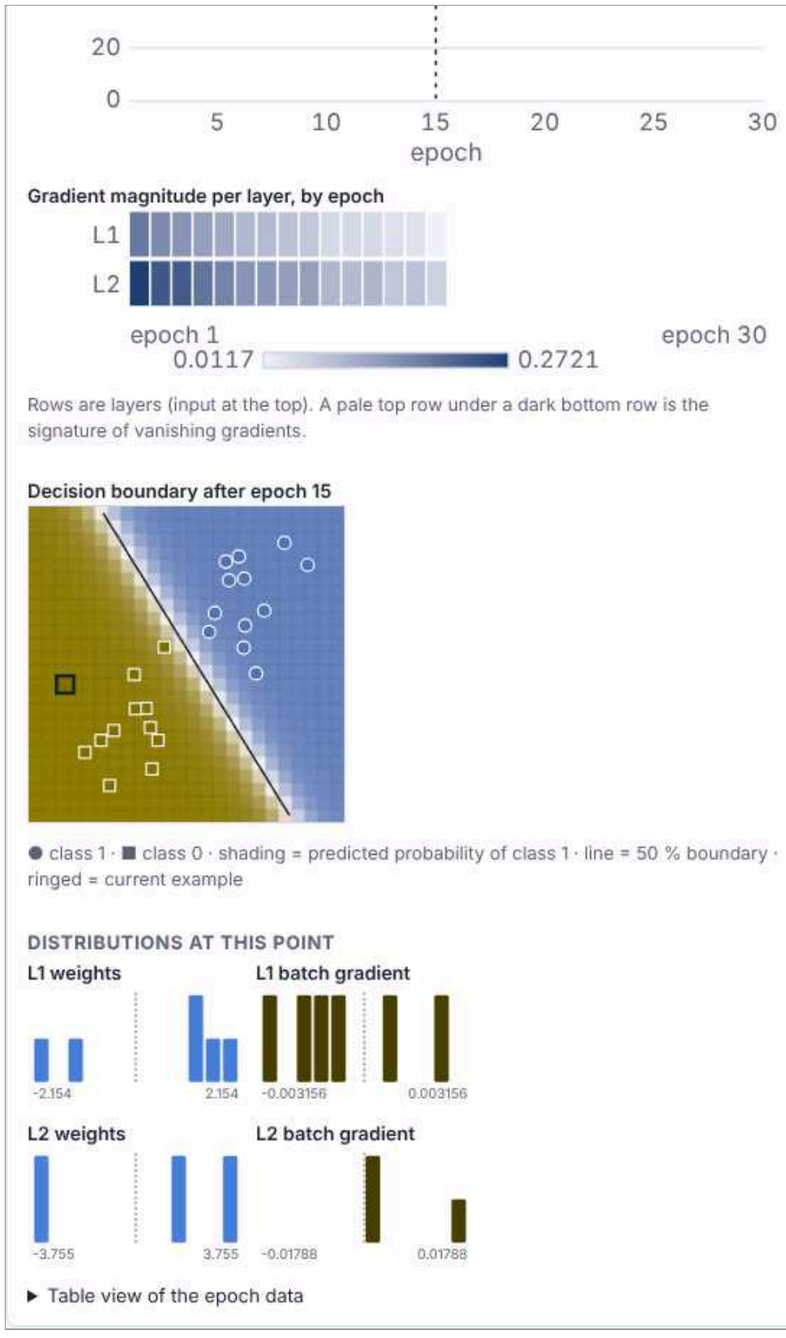
Rows are layers (input at the top). A pale top row under a dark bottom row is the signature of vanishing gradients.

● class 1 · ■ class 0 · shading = predicted probability of class 1 · line = 50 % boundary · ringed = current example

DISTRIBUTIONS AT THIS POINT

▸ Table view of the epoch data

Simulated protanopia



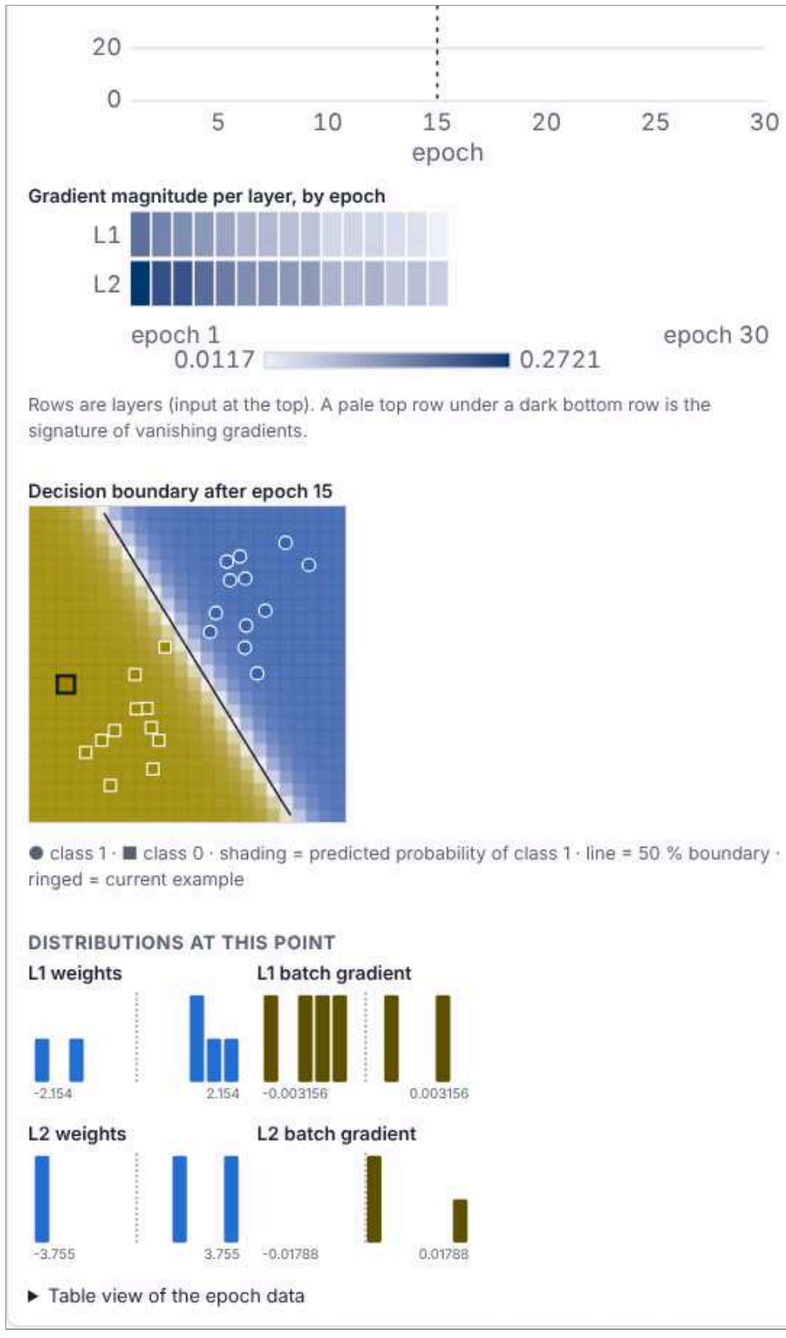
Rows are layers (input at the top). A pale top row under a dark bottom row is the signature of vanishing gradients.

● class 1 · ■ class 0 · shading = predicted probability of class 1 · line = 50 % boundary · ringed = current example

DISTRIBUTIONS AT THIS POINT

▸ Table view of the epoch data

Simulated deuteranopia



Rows are layers (input at the top). A pale top row under a dark bottom row is the signature of vanishing gradients.

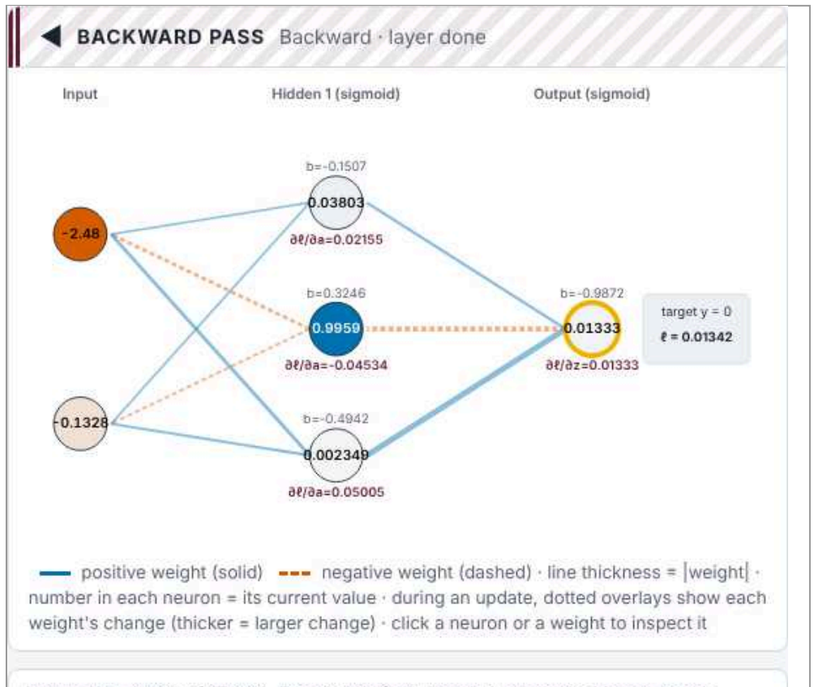
● class 1 · ■ class 0 · shading = predicted probability of class 1 · line = 50 % boundary · ringed = current example

DISTRIBUTIONS AT THIS POINT

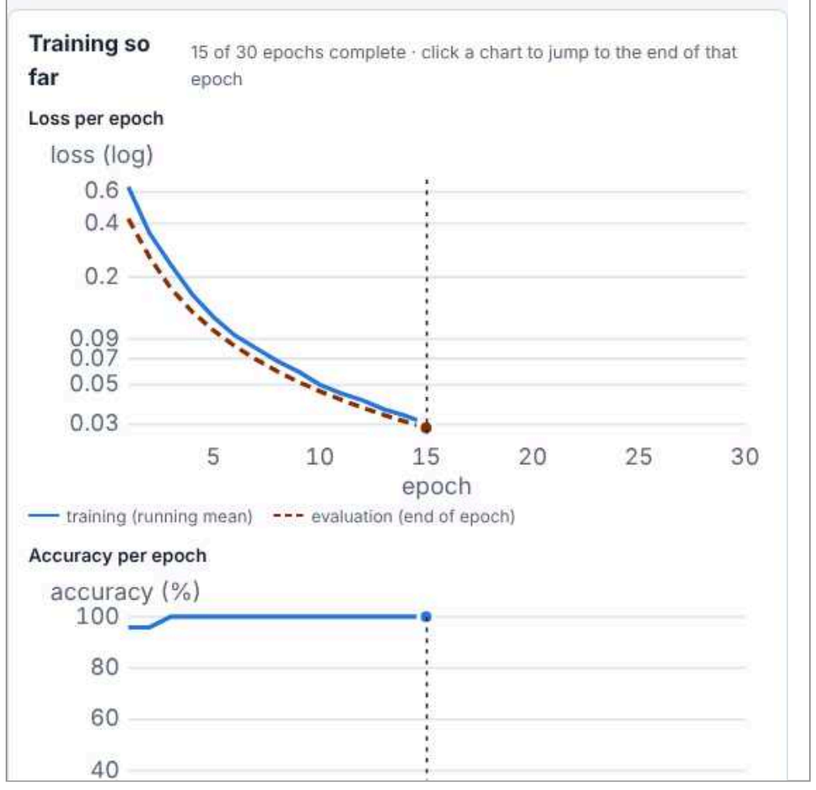
▸ Table view of the epoch data

6. Back-propagation — light theme, backward phase (part 1 of 2)

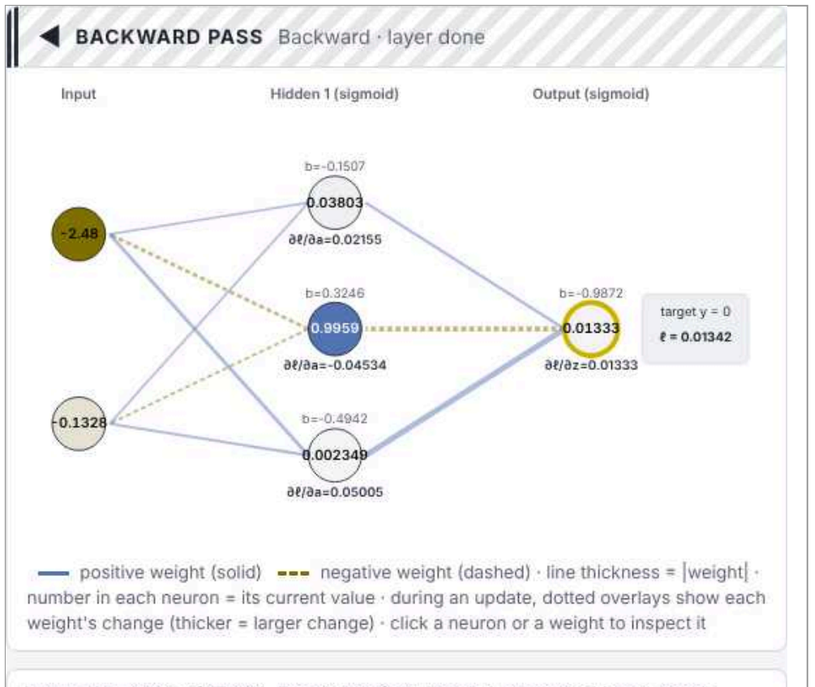
As rendered



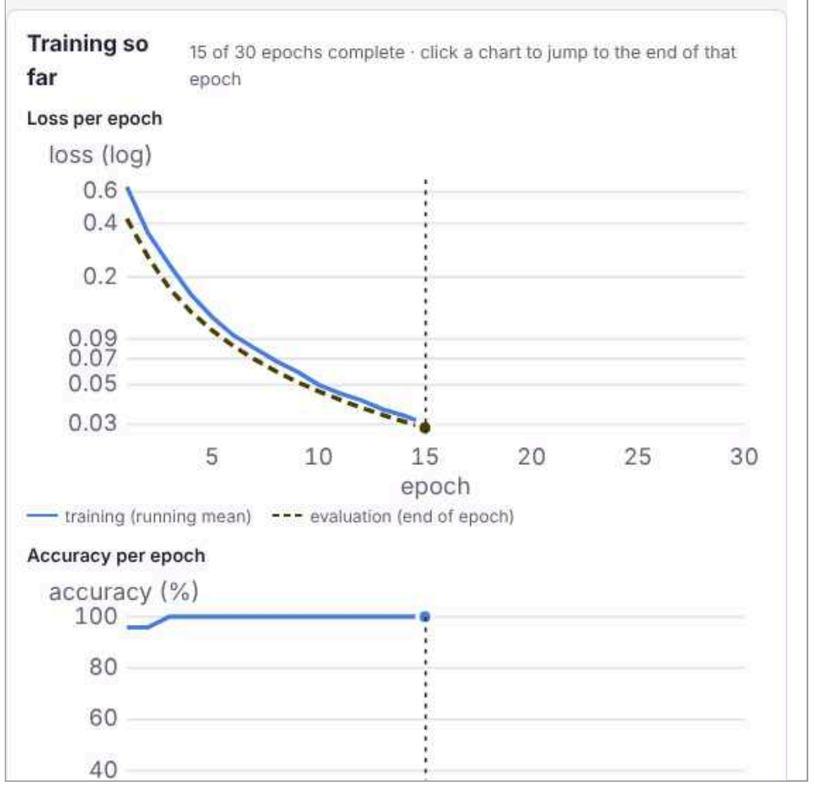
Instruction 8632 of 17,224. Epoch 16 of 30, batch 1, example 1 in the batch. Phase: backward pass, layer 2. Input x = [-2.48, -0.1328], target y = [0]. Prediction 0.01333. Example loss 0.01342.



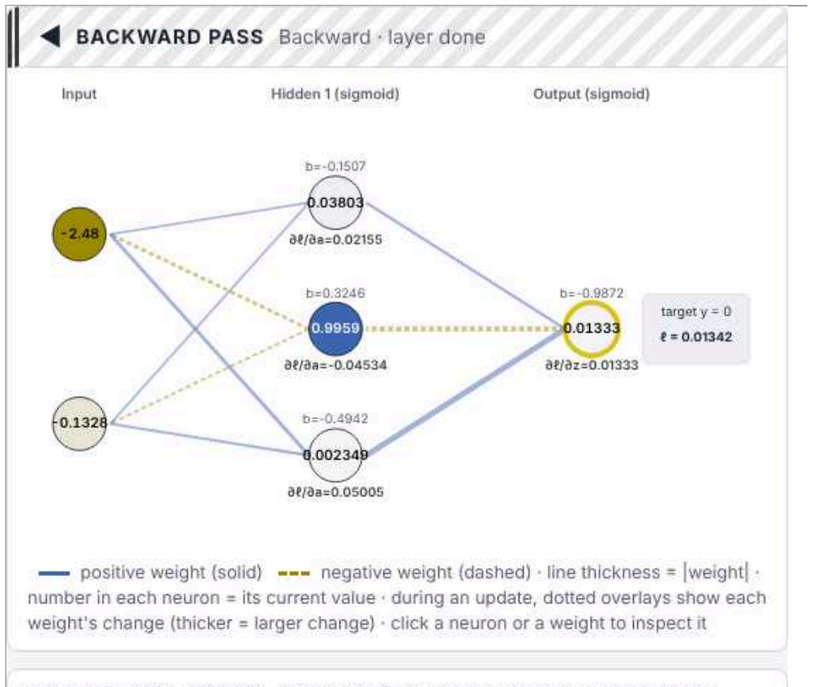
Simulated protanopia



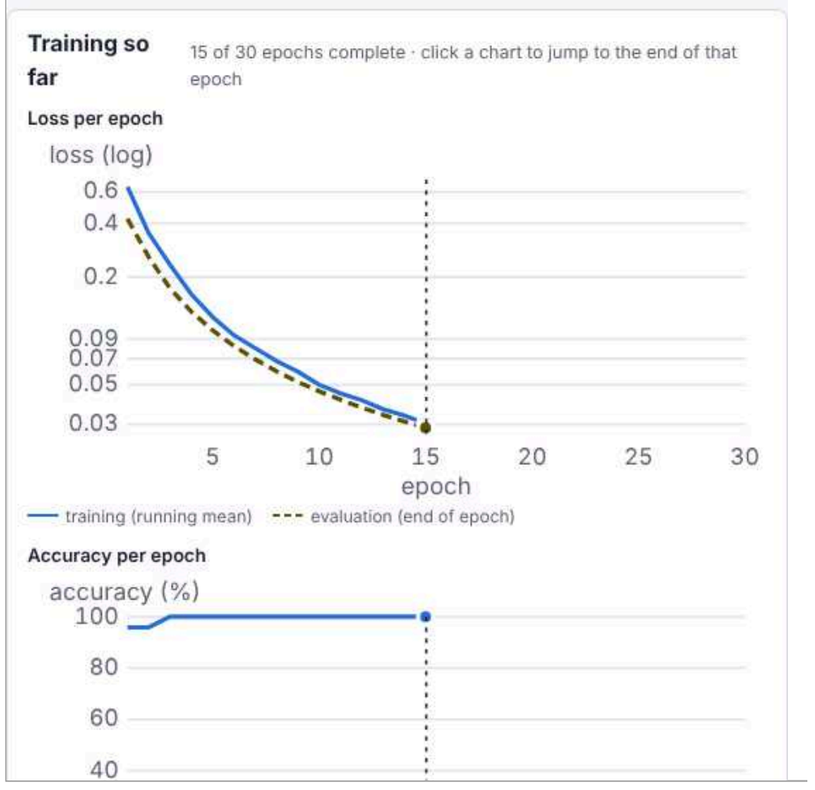
Instruction 8632 of 17,224. Epoch 16 of 30, batch 1, example 1 in the batch. Phase: backward pass, layer 2. Input x = [-2.48, -0.1328], target y = [0]. Prediction 0.01333. Example loss 0.01342.



Simulated deuteranopia

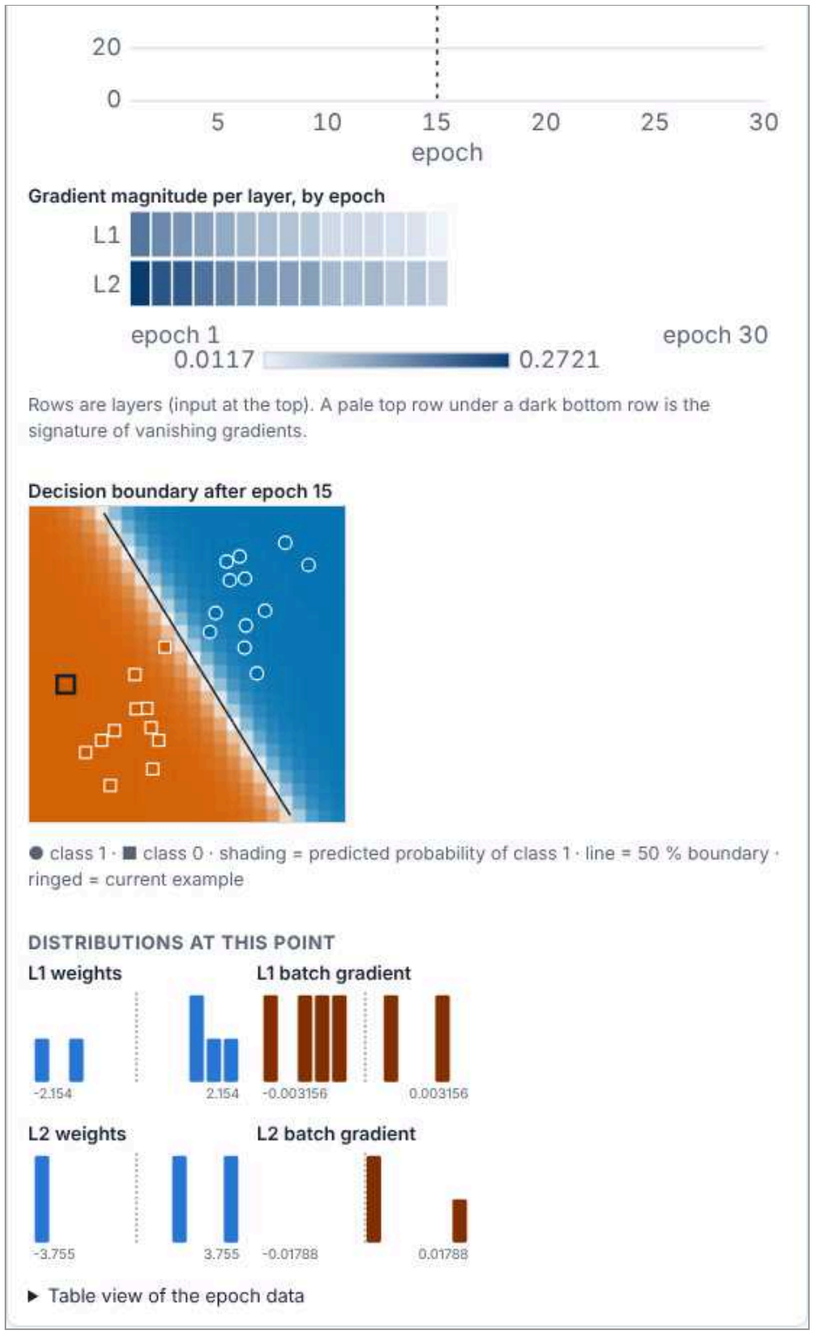


Instruction 8632 of 17,224. Epoch 16 of 30, batch 1, example 1 in the batch. Phase: backward pass, layer 2. Input x = [-2.48, -0.1328], target y = [0]. Prediction 0.01333. Example loss 0.01342.

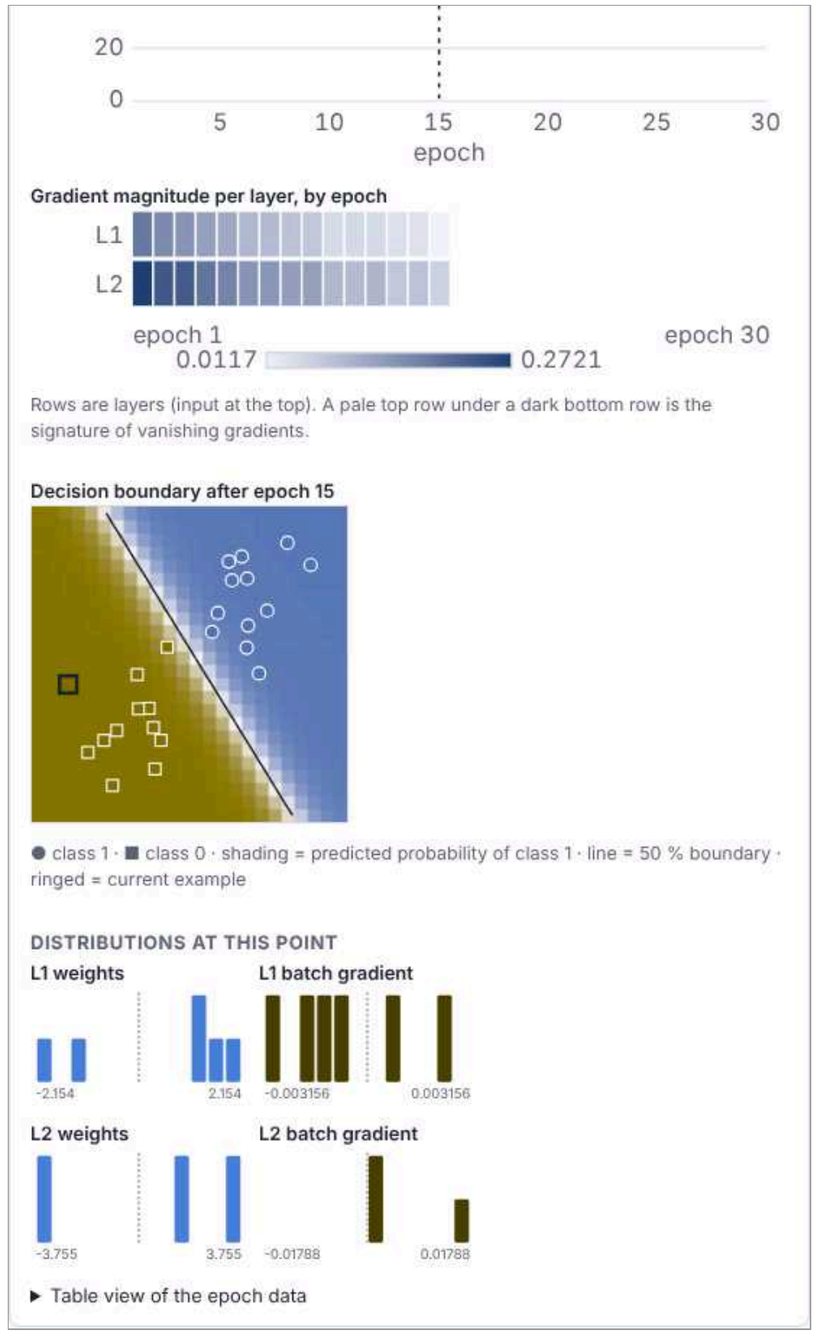


6. Back-propagation — light theme, backward phase (part 2 of 2)

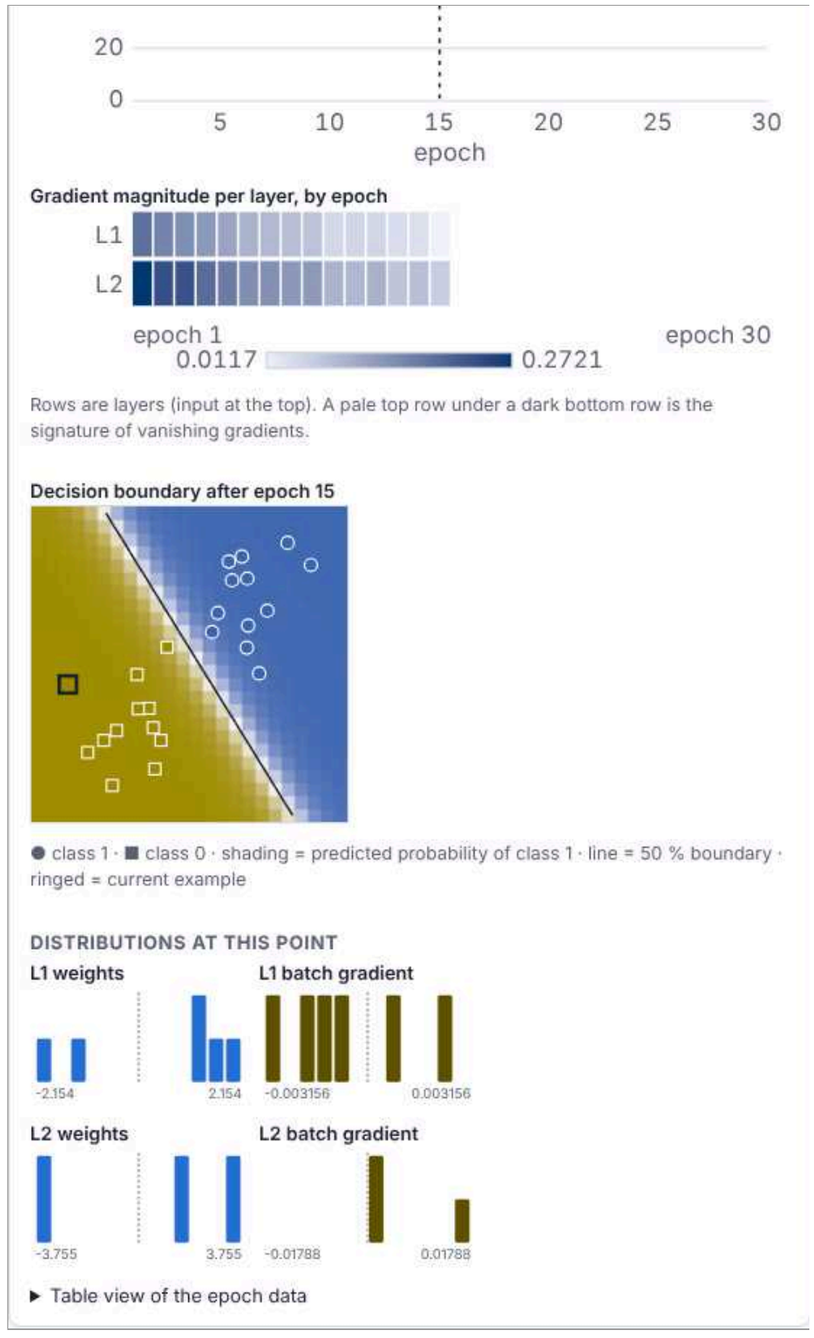
As rendered



Simulated protanopia

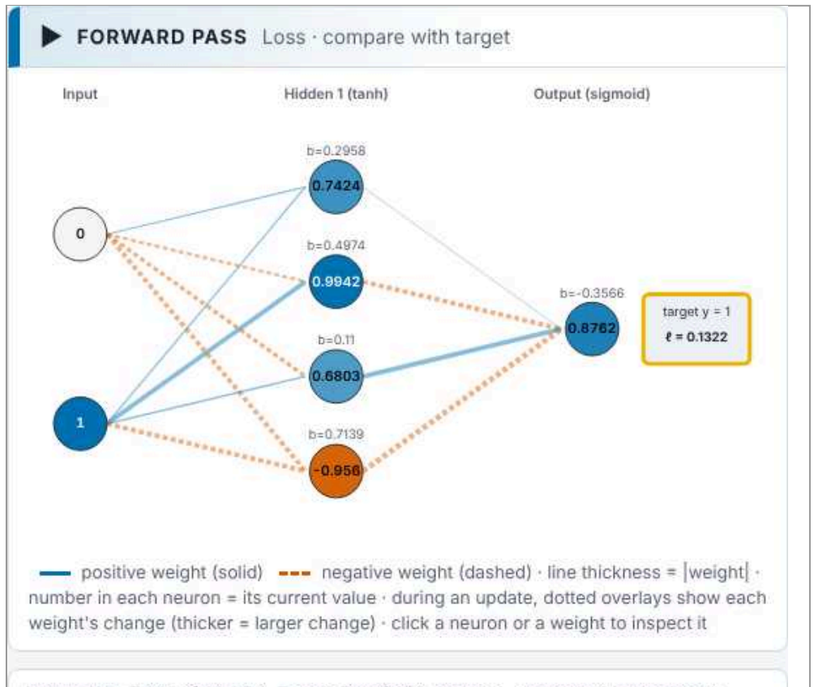


Simulated deuteranopia

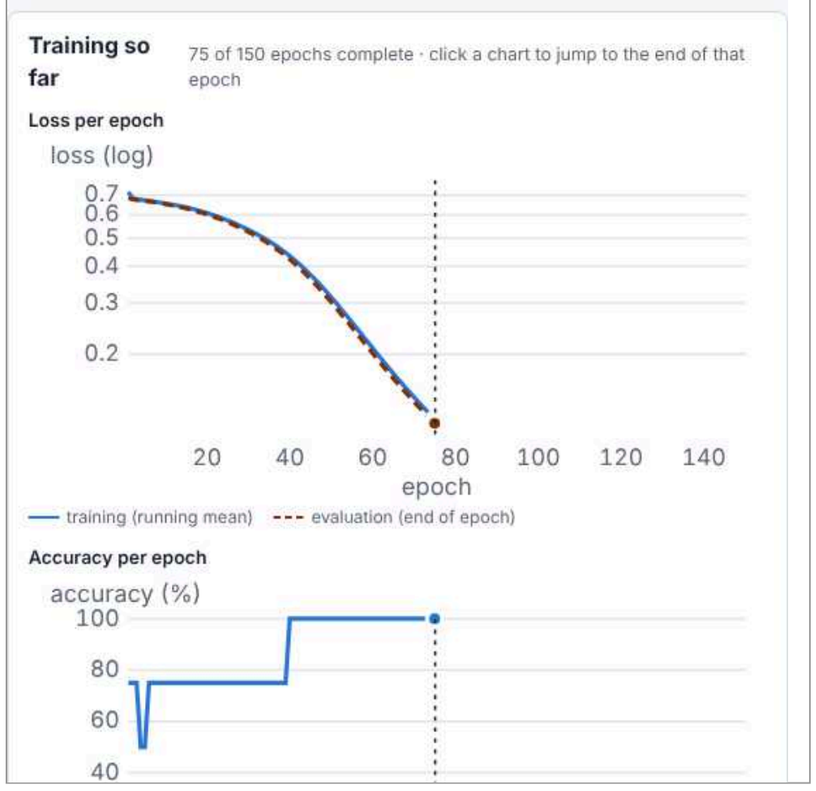


7. XOR needs a hidden layer — light theme, forward phase (part 1 of 2)

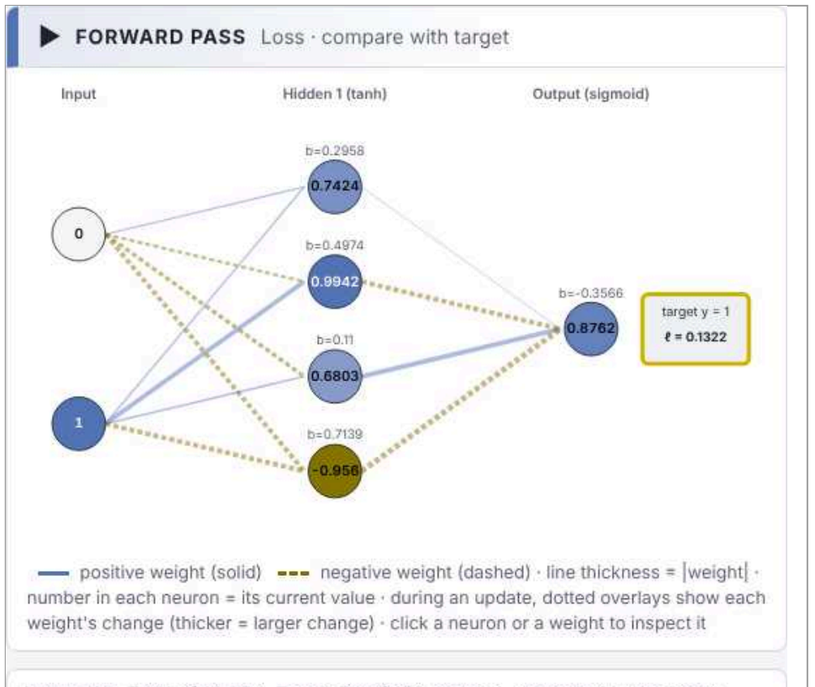
As rendered



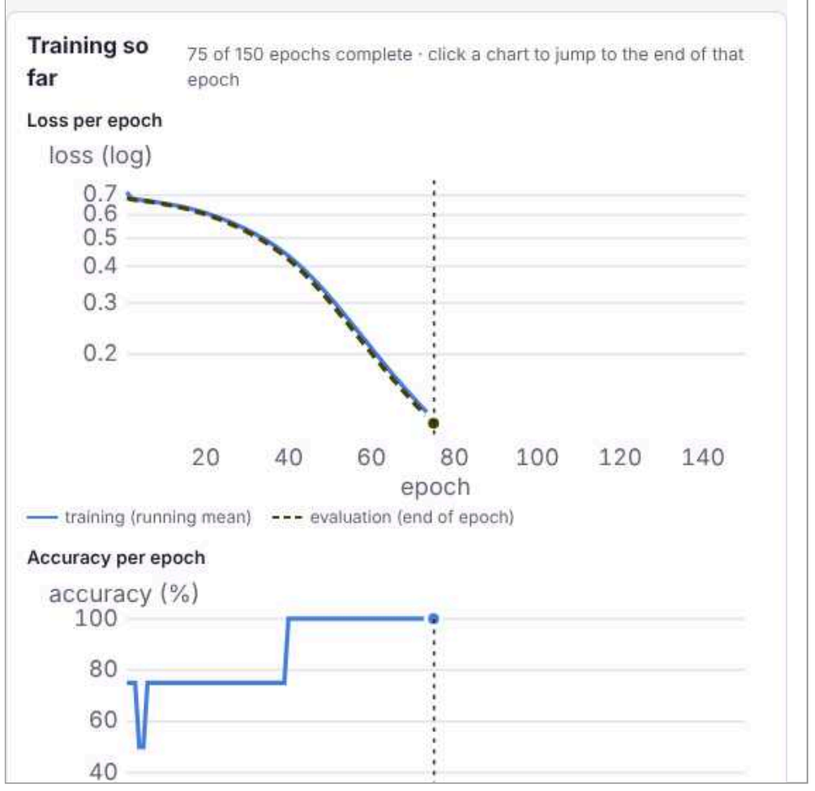
Instruction 7441 of 14,854. Epoch 76 of 150, batch 1, example 1 in the batch. Phase: loss. Input x = [0, 1], target y = [1]. Prediction 0.8762. Example loss 0.1322.



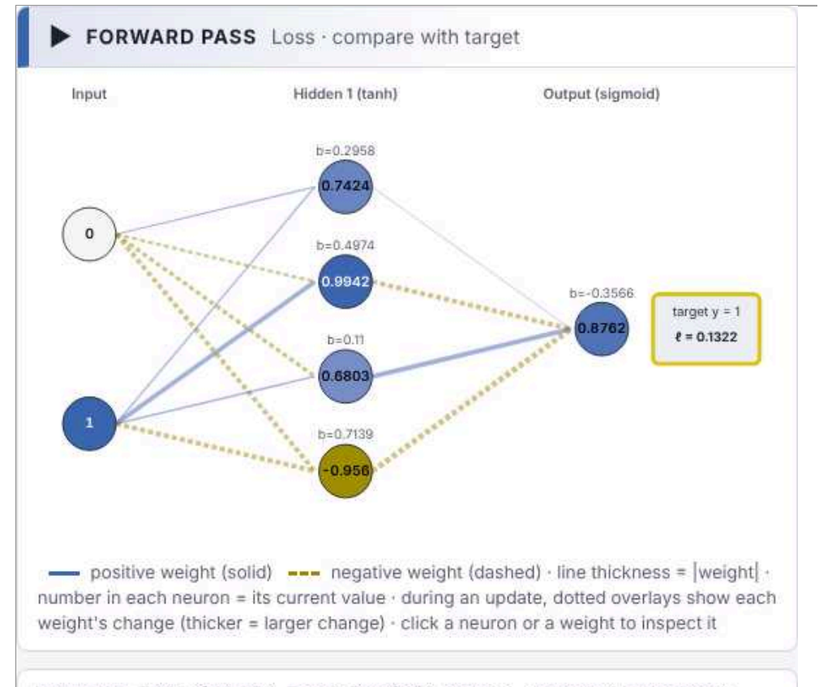
Simulated protanopia



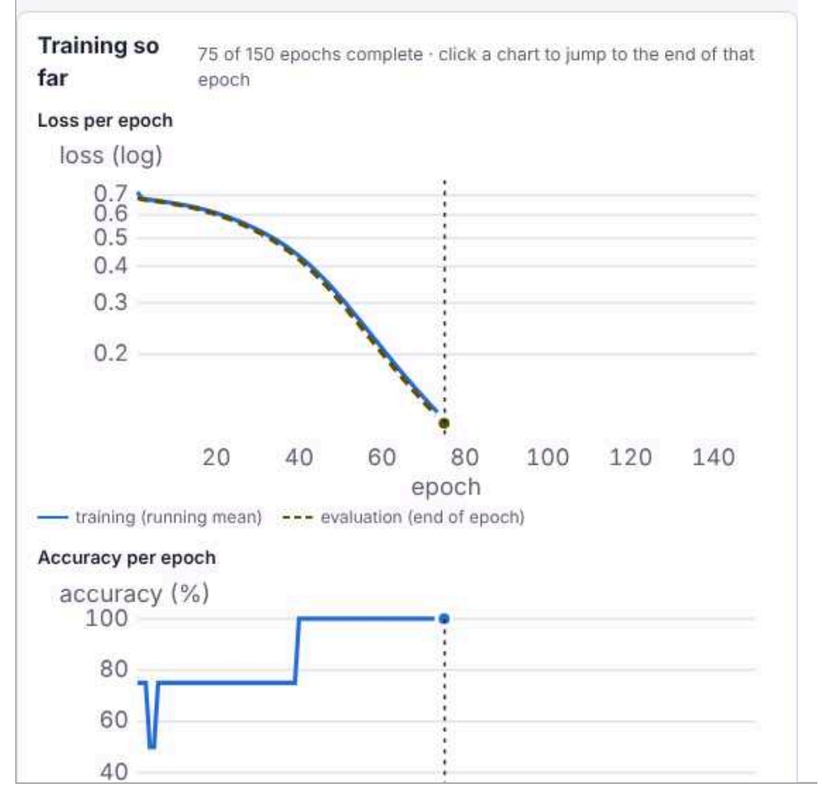
Instruction 7441 of 14,854. Epoch 76 of 150, batch 1, example 1 in the batch. Phase: loss. Input x = [0, 1], target y = [1]. Prediction 0.8762. Example loss 0.1322.



Simulated deuteranopia

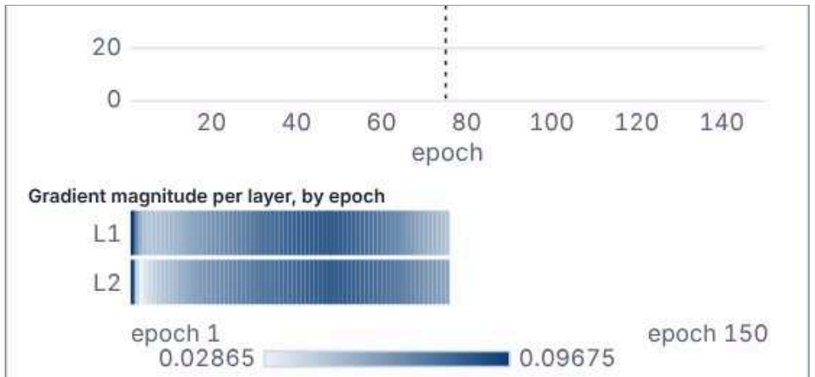


Instruction 7441 of 14,854. Epoch 76 of 150, batch 1, example 1 in the batch. Phase: loss. Input x = [0, 1], target y = [1]. Prediction 0.8762. Example loss 0.1322.

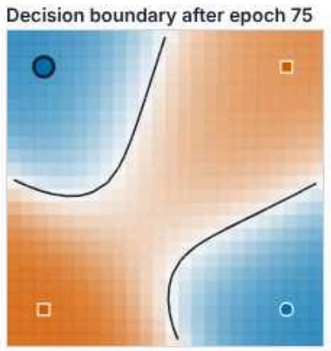


7. XOR needs a hidden layer — light theme, forward phase (part 2 of 2)

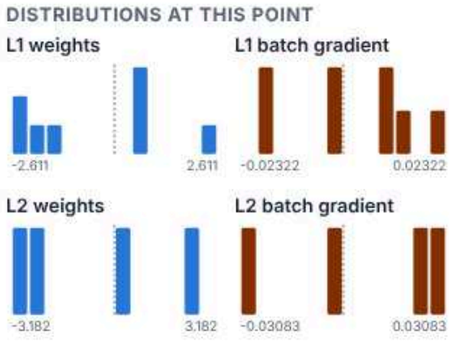
As rendered



Rows are layers (input at the top). A pale top row under a dark bottom row is the signature of vanishing gradients.

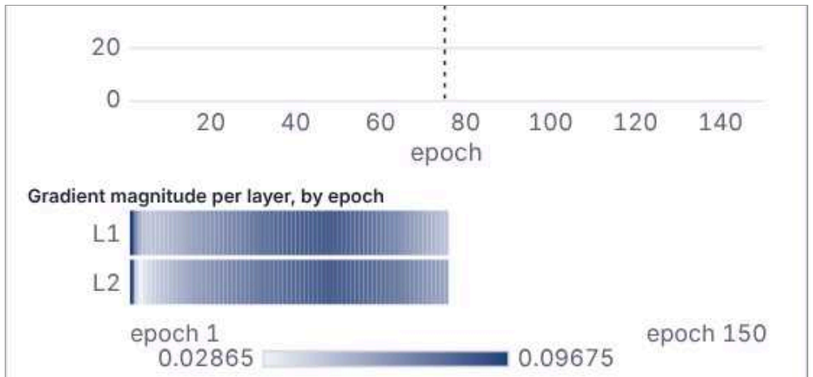


● class 1 · ■ class 0 · shading = predicted probability of class 1 · line = 50 % boundary · ringed = current example

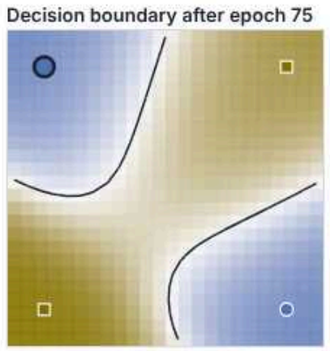


▸ Table view of the epoch data

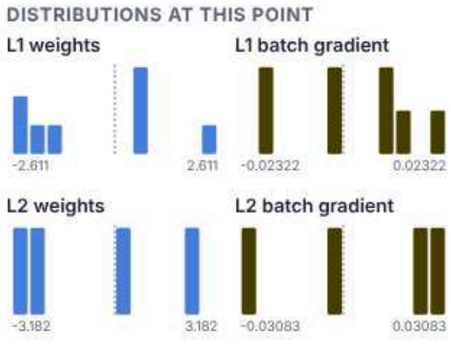
Simulated protanopia



Rows are layers (input at the top). A pale top row under a dark bottom row is the signature of vanishing gradients.

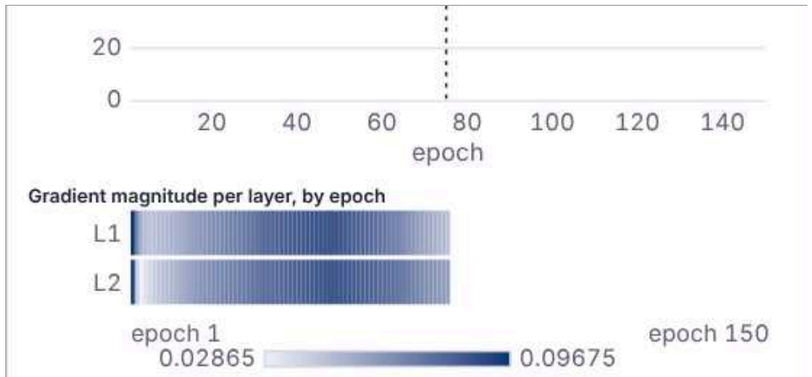


● class 1 · ■ class 0 · shading = predicted probability of class 1 · line = 50 % boundary · ringed = current example

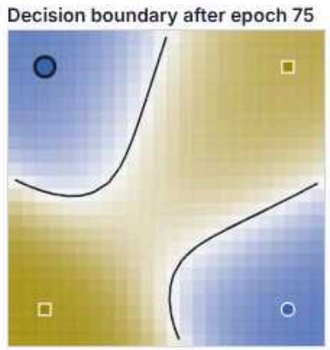


▸ Table view of the epoch data

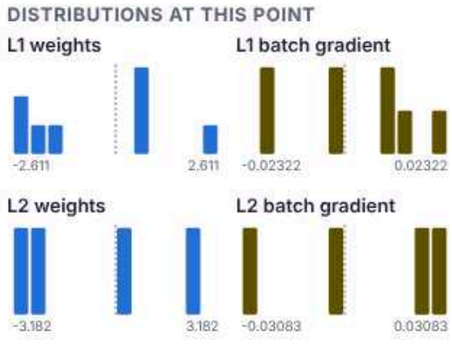
Simulated deuteranopia



Rows are layers (input at the top). A pale top row under a dark bottom row is the signature of vanishing gradients.



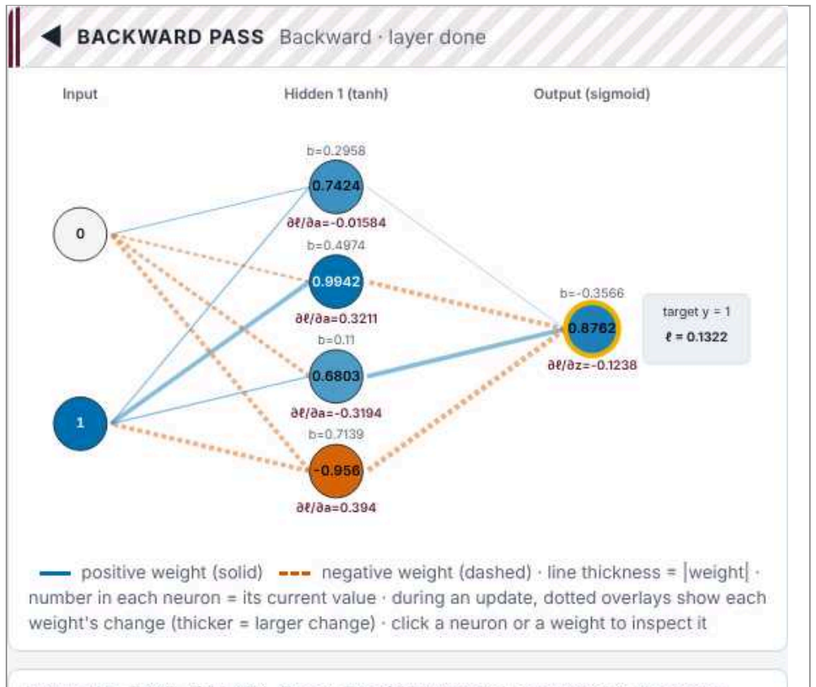
● class 1 · ■ class 0 · shading = predicted probability of class 1 · line = 50 % boundary · ringed = current example



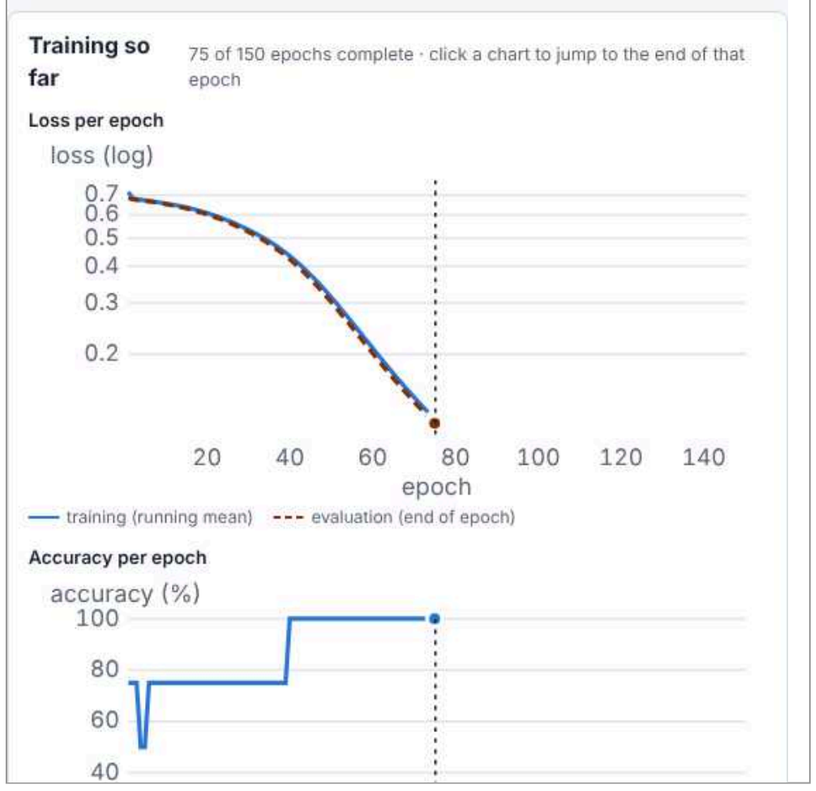
▸ Table view of the epoch data

7. XOR needs a hidden layer — light theme, backward phase (part 1 of 2)

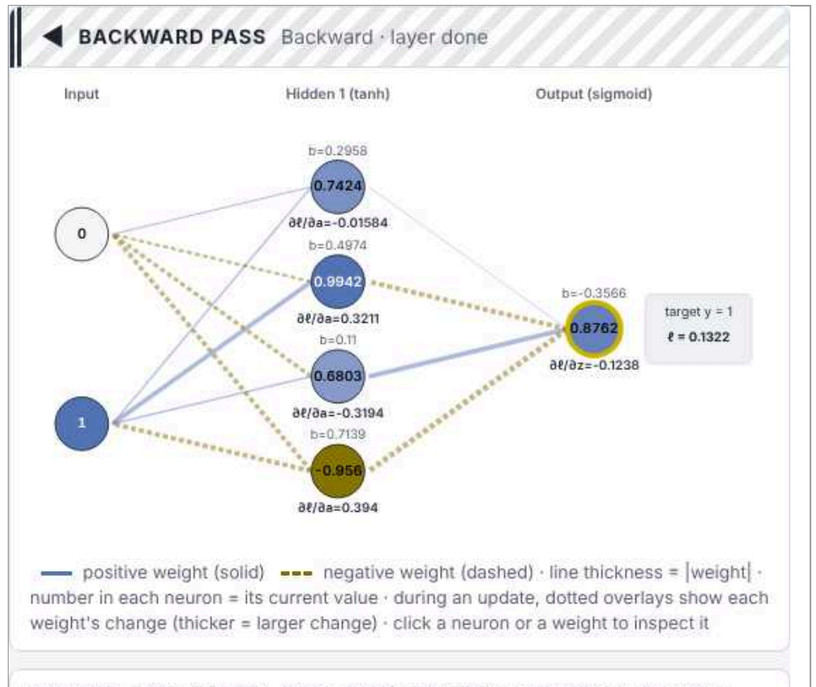
As rendered



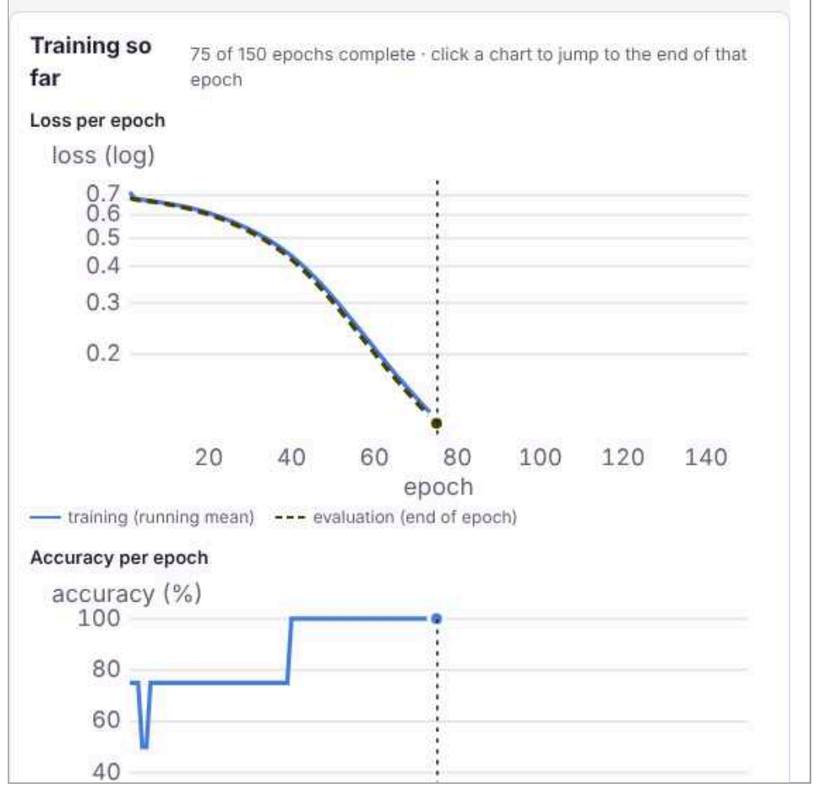
Instruction 7447 of 14,854. Epoch 76 of 150, batch 1, example 1 in the batch. Phase: backward pass, layer 2. Input x = [0, 1], target y = [1]. Prediction 0.8762. Example loss 0.1322.



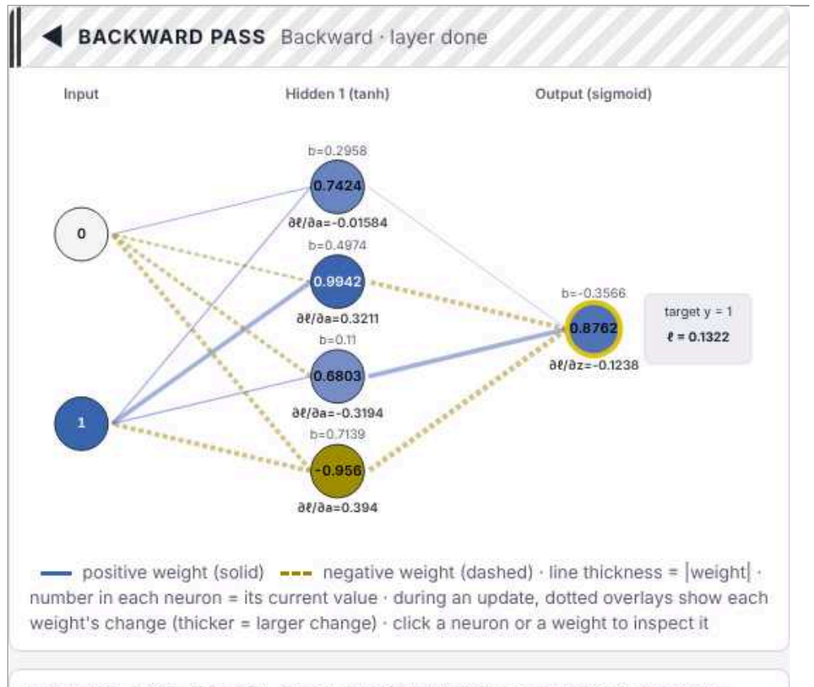
Simulated protanopia



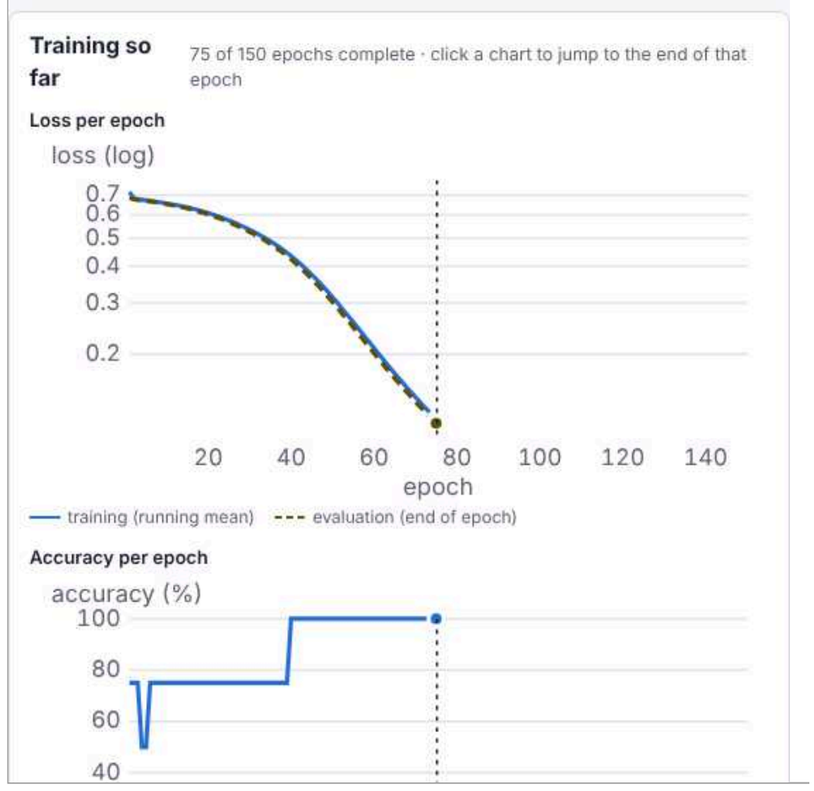
Instruction 7447 of 14,854. Epoch 76 of 150, batch 1, example 1 in the batch. Phase: backward pass, layer 2. Input x = [0, 1], target y = [1]. Prediction 0.8762. Example loss 0.1322.



Simulated deuteranopia

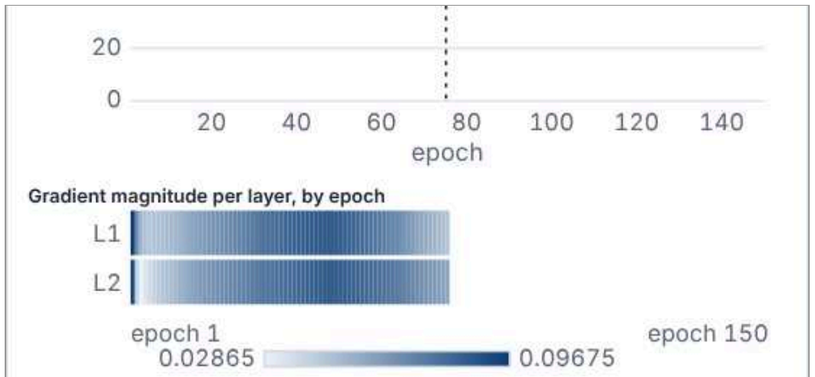


Instruction 7447 of 14,854. Epoch 76 of 150, batch 1, example 1 in the batch. Phase: backward pass, layer 2. Input x = [0, 1], target y = [1]. Prediction 0.8762. Example loss 0.1322.

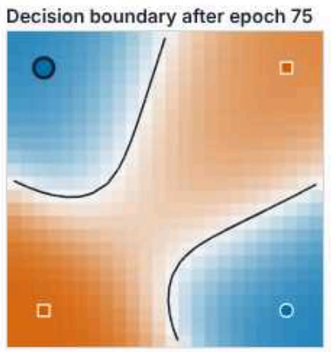


7. XOR needs a hidden layer — light theme, backward phase (part 2 of 2)

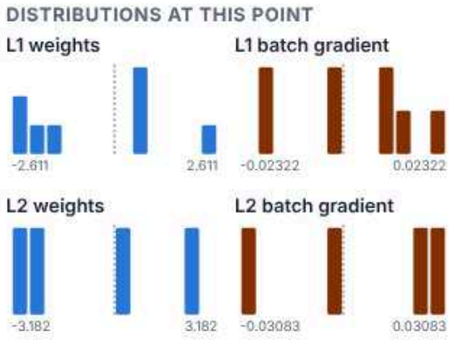
As rendered



Rows are layers (input at the top). A pale top row under a dark bottom row is the signature of vanishing gradients.

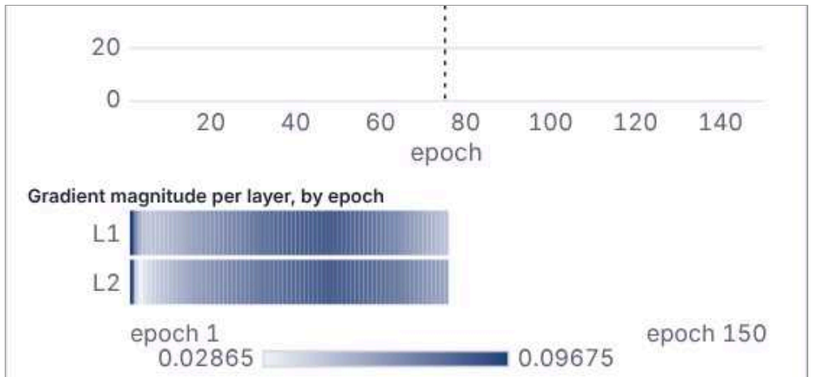


● class 1 · ■ class 0 · shading = predicted probability of class 1 · line = 50 % boundary · ringed = current example

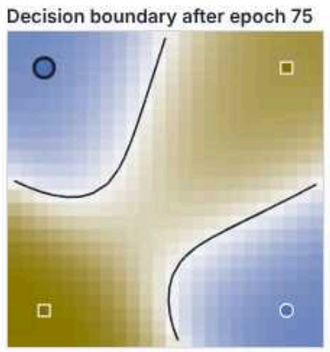


▸ Table view of the epoch data

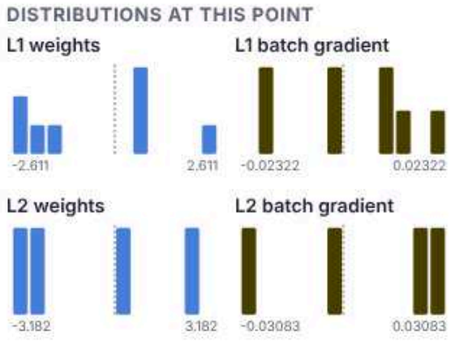
Simulated protanopia



Rows are layers (input at the top). A pale top row under a dark bottom row is the signature of vanishing gradients.

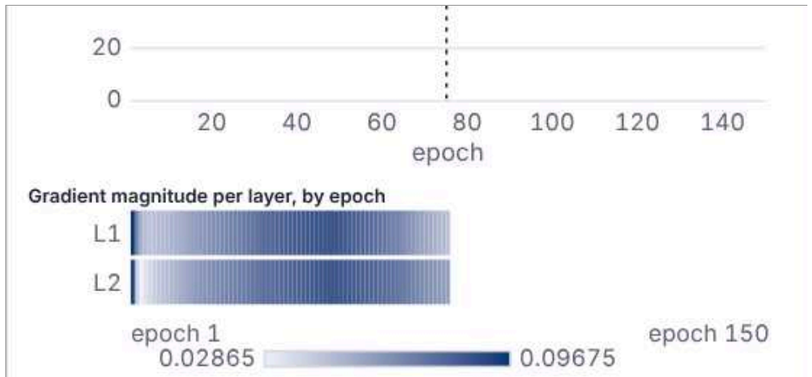


● class 1 · ■ class 0 · shading = predicted probability of class 1 · line = 50 % boundary · ringed = current example

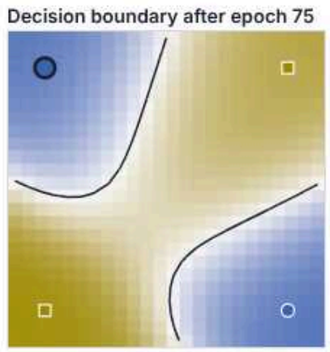


▸ Table view of the epoch data

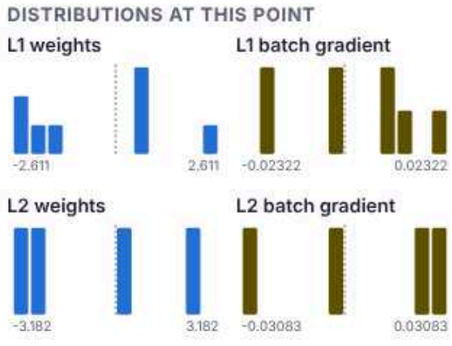
Simulated deuteranopia



Rows are layers (input at the top). A pale top row under a dark bottom row is the signature of vanishing gradients.



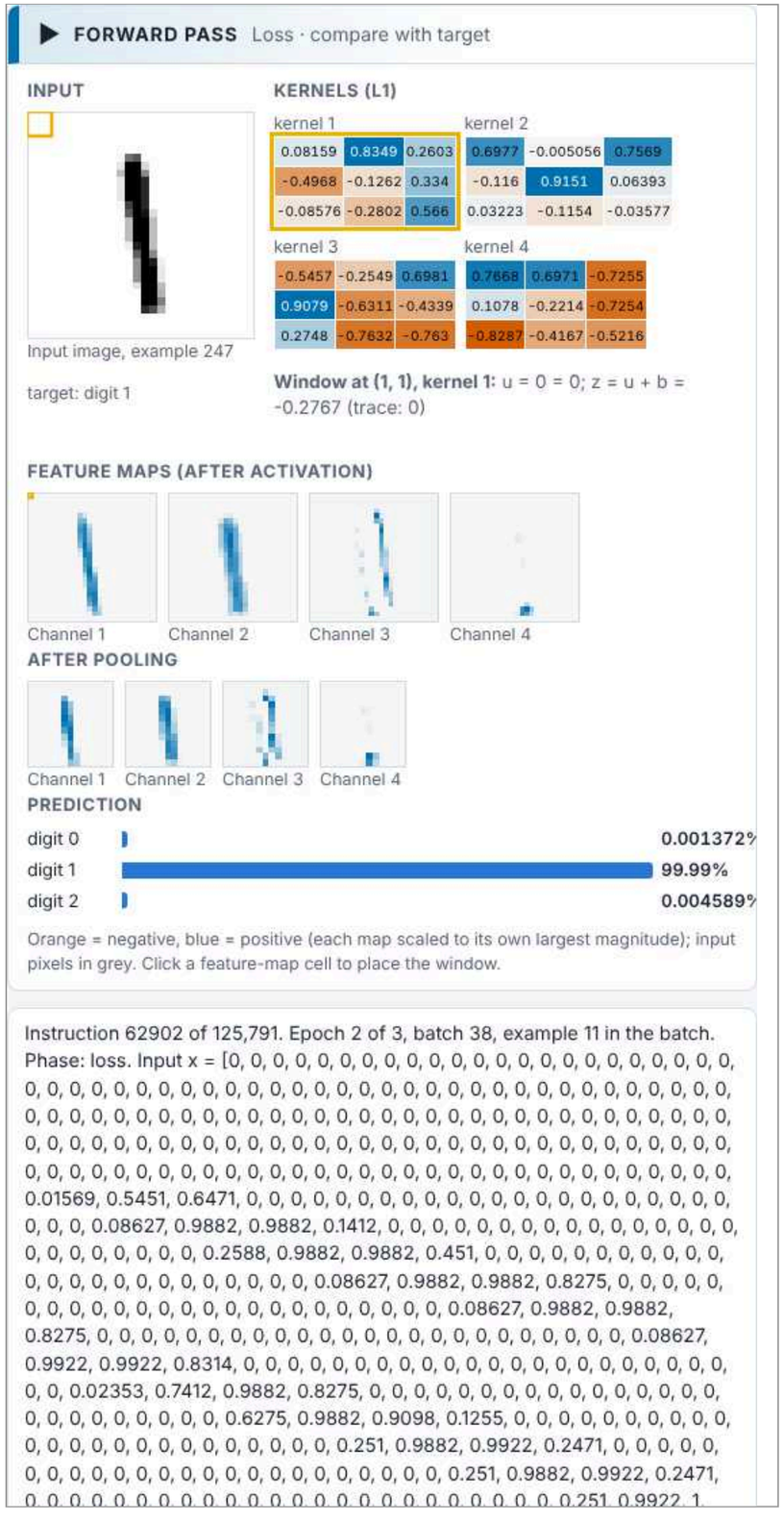
● class 1 · ■ class 0 · shading = predicted probability of class 1 · line = 50 % boundary · ringed = current example



▸ Table view of the epoch data

8. A CNN on handwritten digits — light theme, forward phase (part 1 of 3)

As rendered



Instruction 62902 of 125,791. Epoch 2 of 3, batch 38, example 11 in the batch. Phase: loss. Input x = [0, 0, 0,

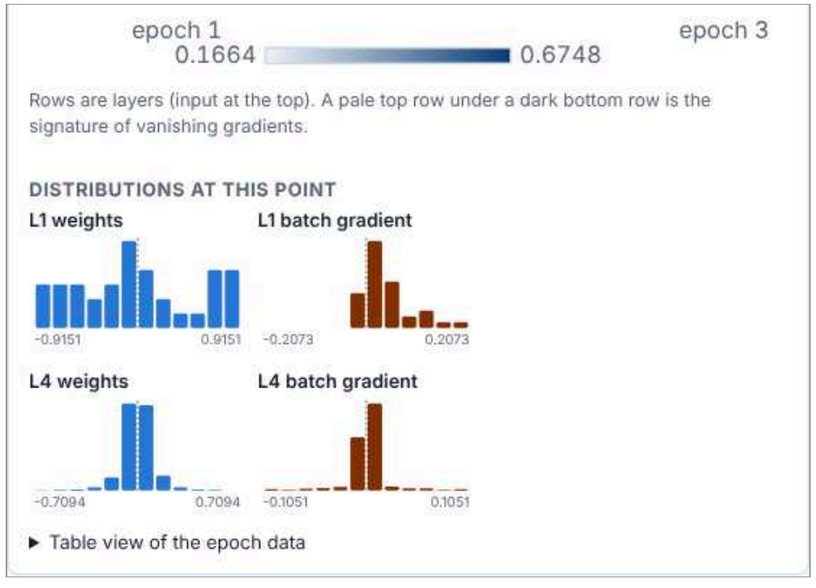
8. A CNN on handwritten digits — light theme, forward phase (part 2 of 3)

As rendered

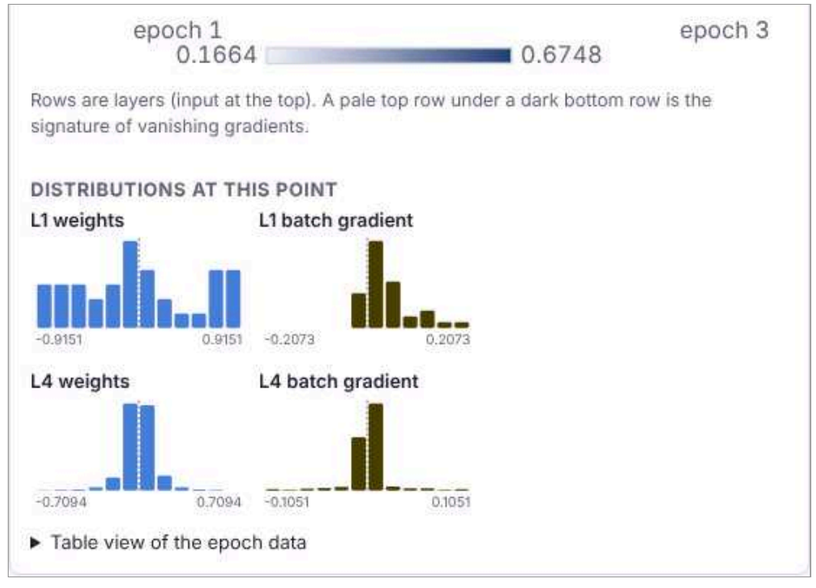
0.2471, 0, 0, 0

8. A CNN on handwritten digits — light theme, forward phase (part 3 of 3)

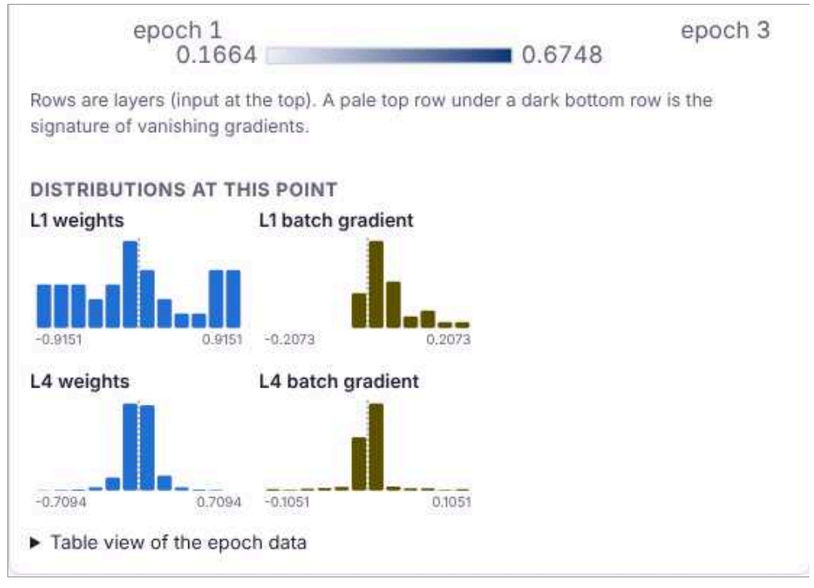
As rendered



Simulated protanopia

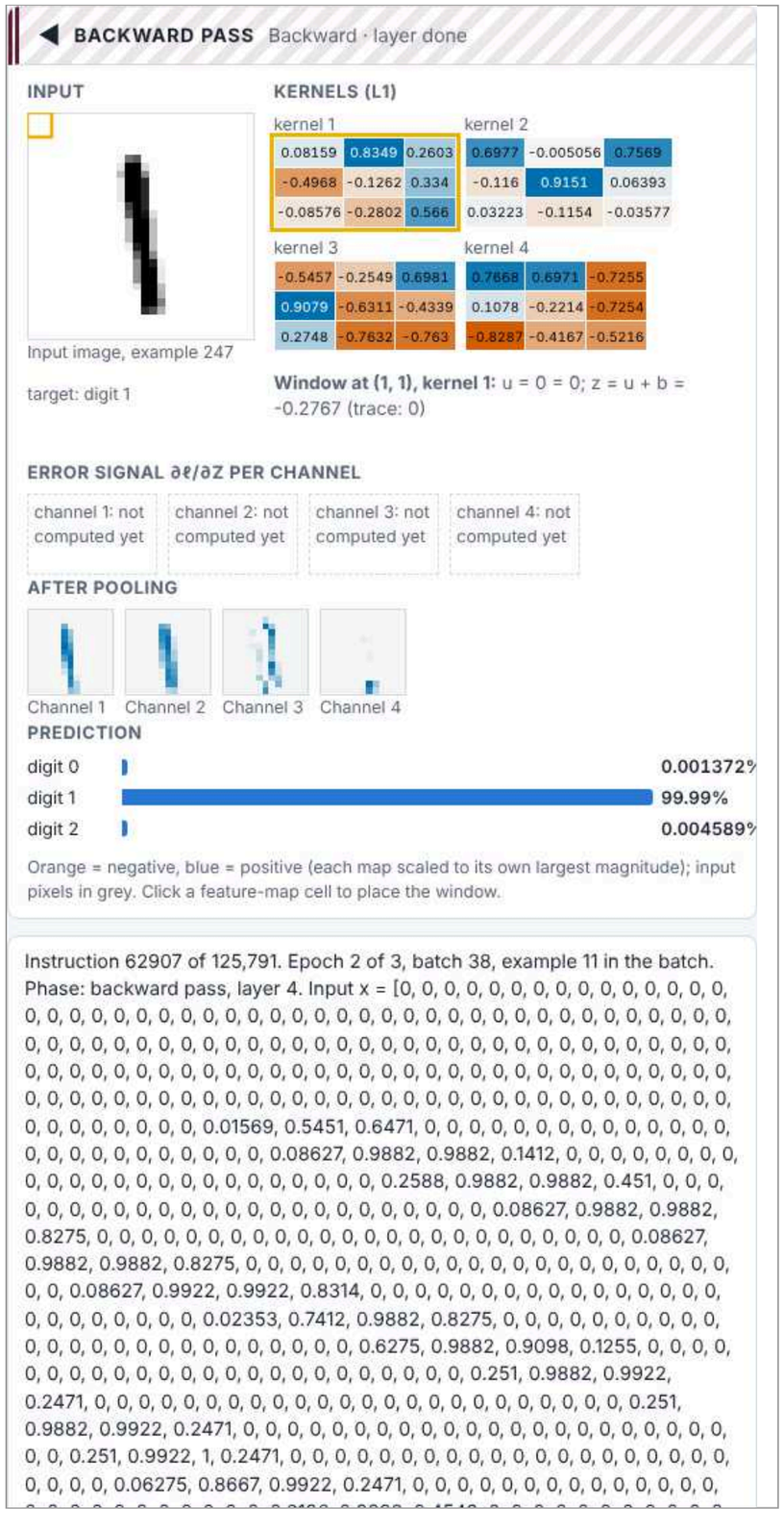


Simulated deuteranopia



8. A CNN on handwritten digits — light theme, backward phase (part 1 of 3)

As rendered



Instruction 62907 of 125,791. Epoch 2 of 3, batch 38, example 11 in the batch. Phase: backward pass, layer 4. Input x = [0, 0, 0,

8. A CNN on handwritten digits — light theme, backward phase (part 2 of 3)

As rendered

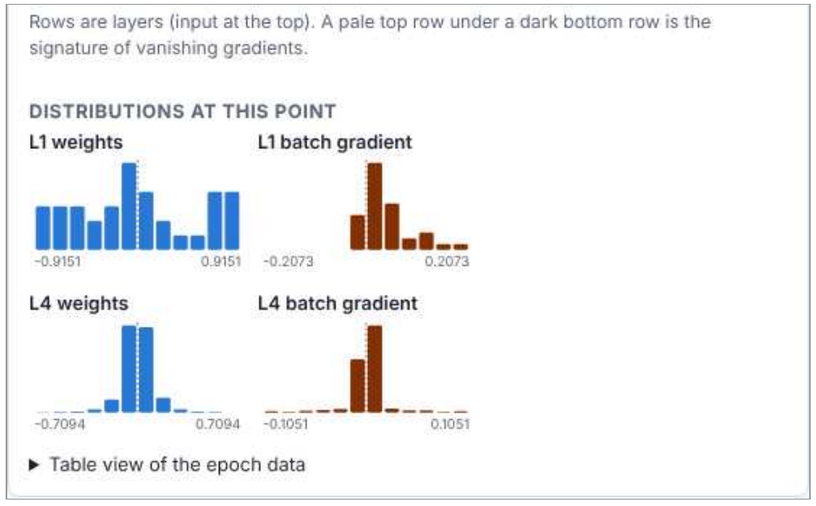
Simulated protanopia

Simulated deuteranopia

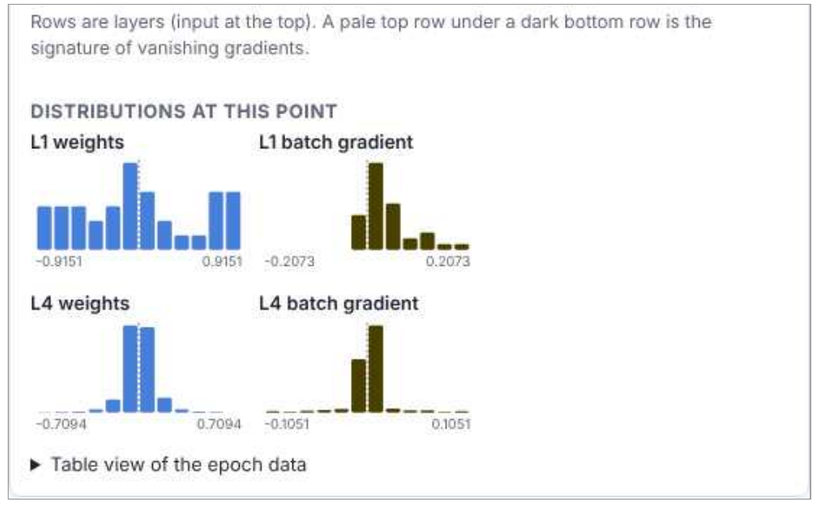
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0.6196, 0.9922, 0.4549, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0

8. A CNN on handwritten digits — light theme, backward phase (part 3 of 3)

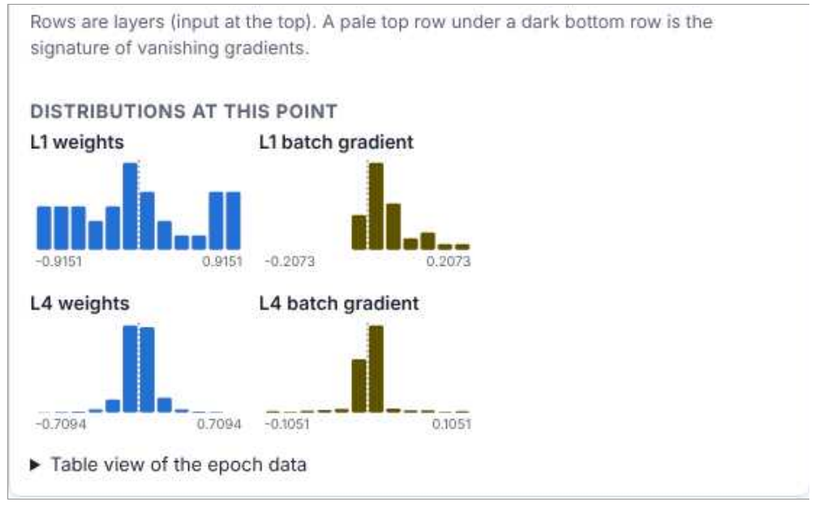
As rendered



Simulated protanopia

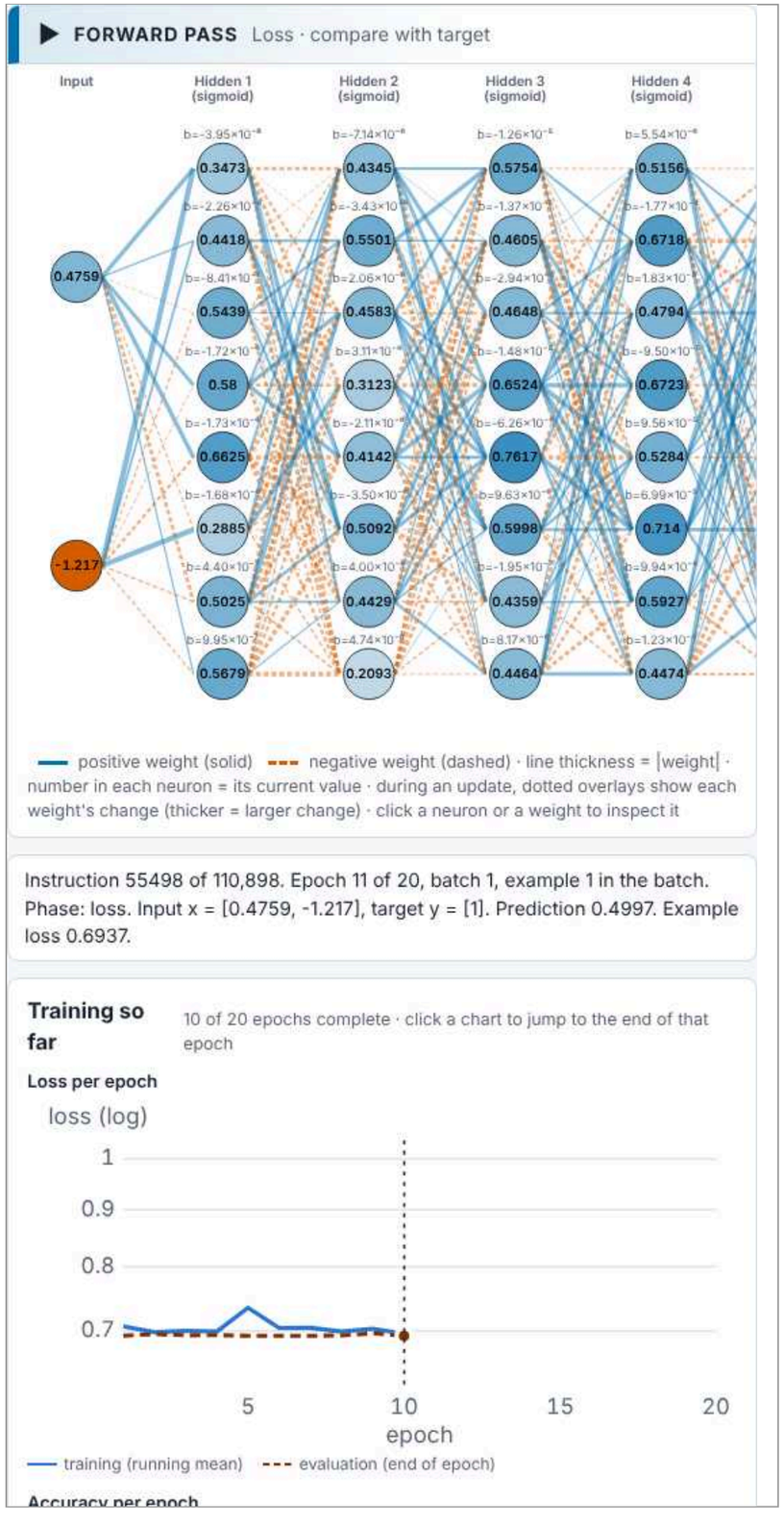


Simulated deuteranopia

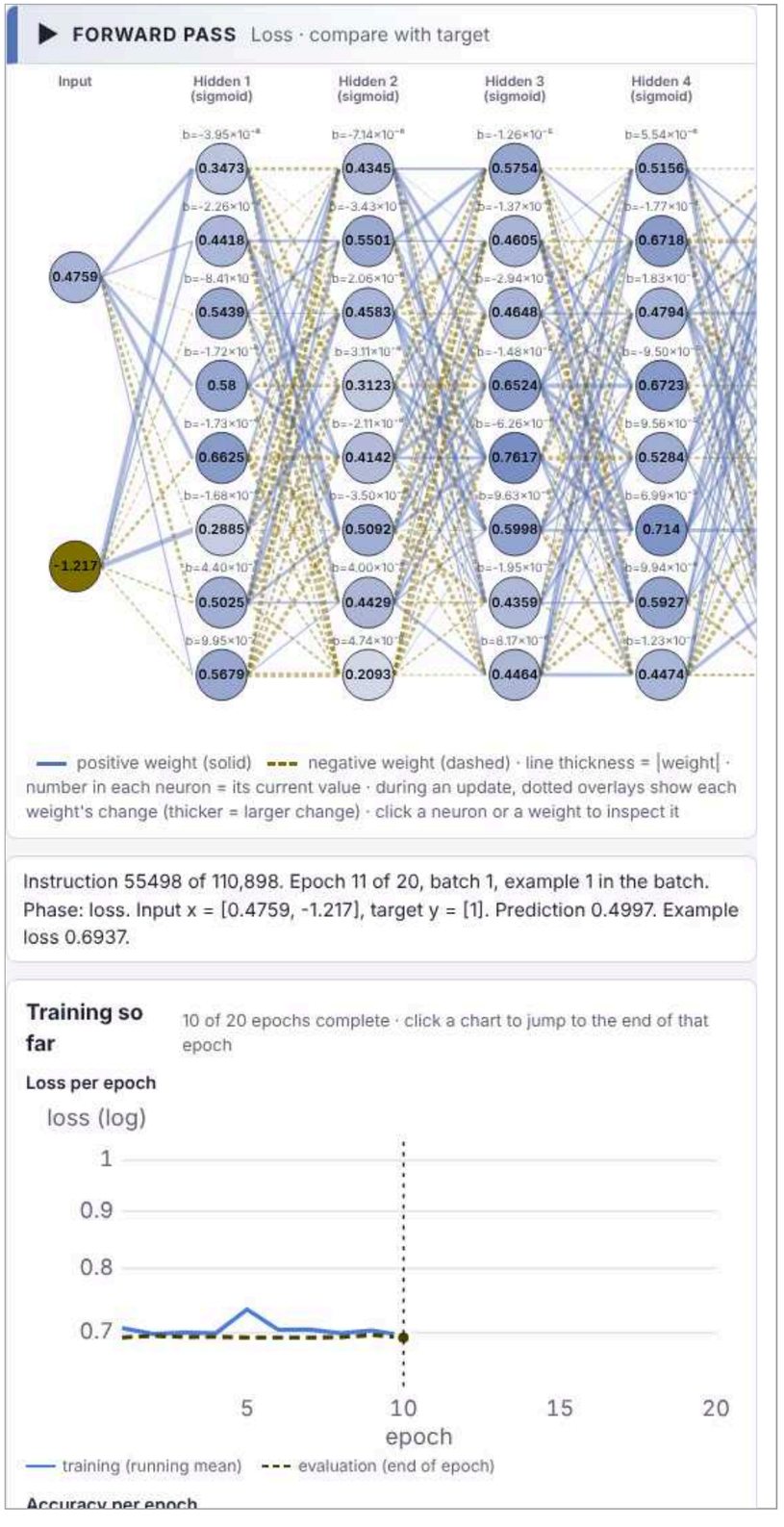


D1. Vanishing gradients — light theme, forward phase (part 1 of 3)

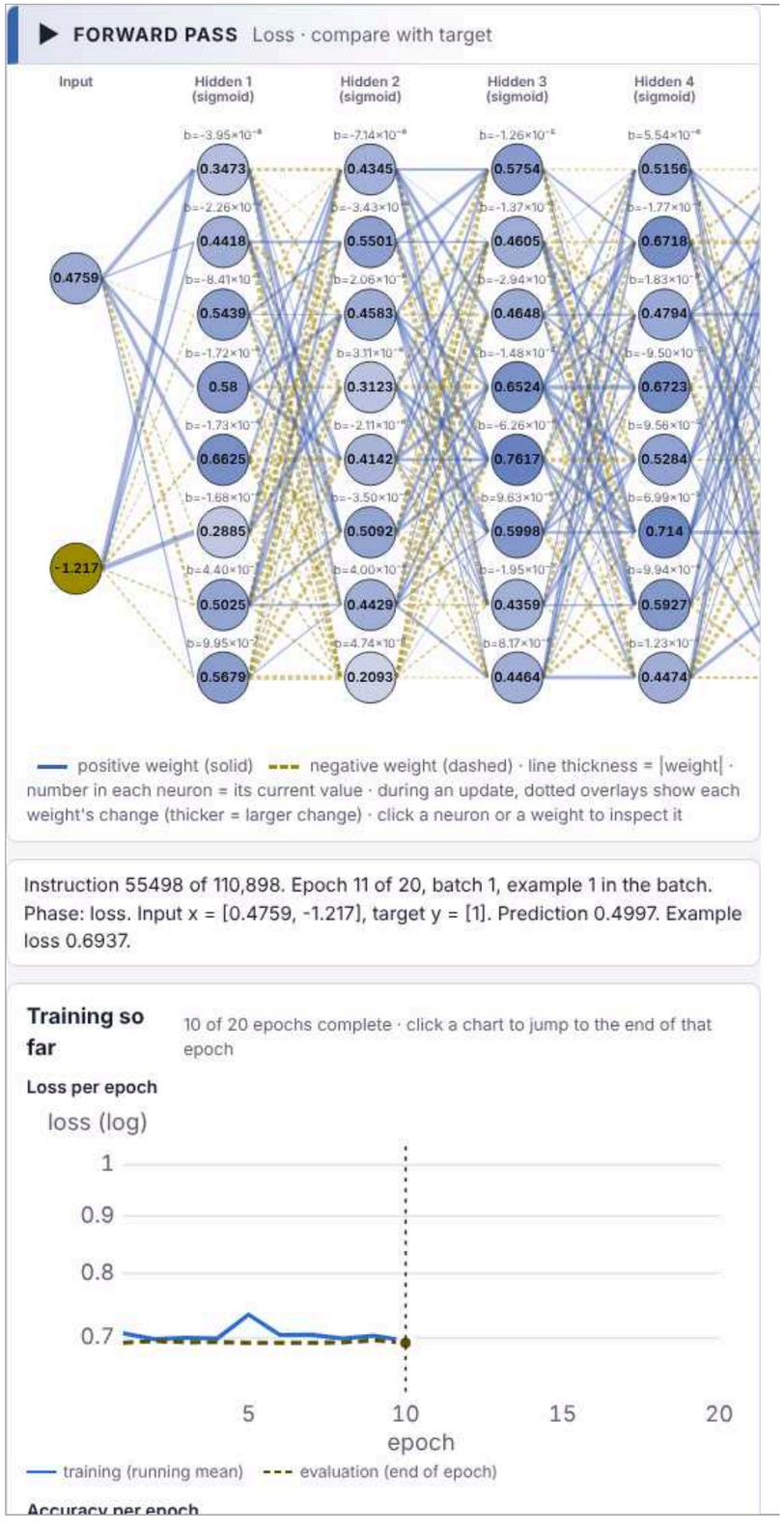
As rendered



Simulated protanopia

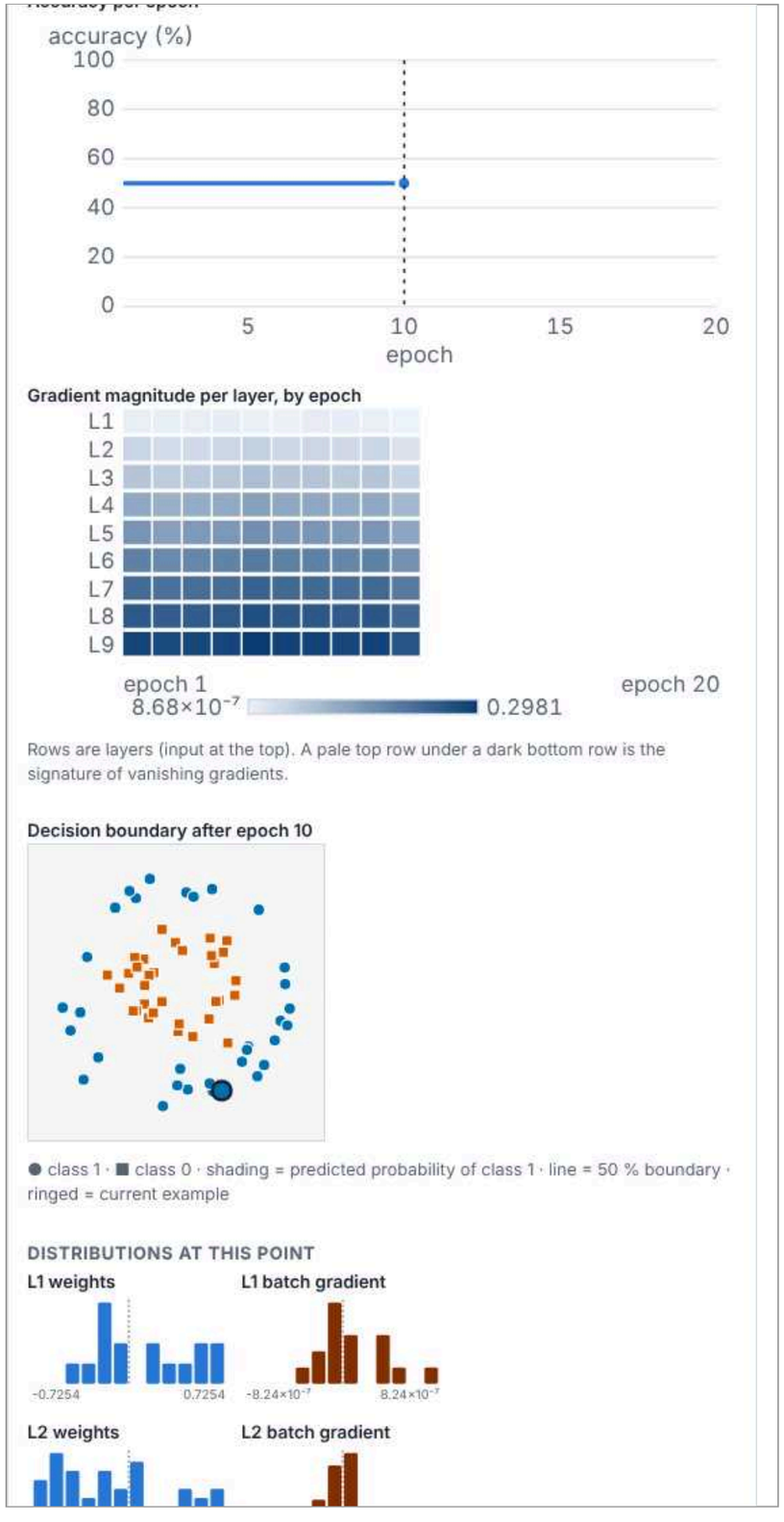


Simulated deuteranopia

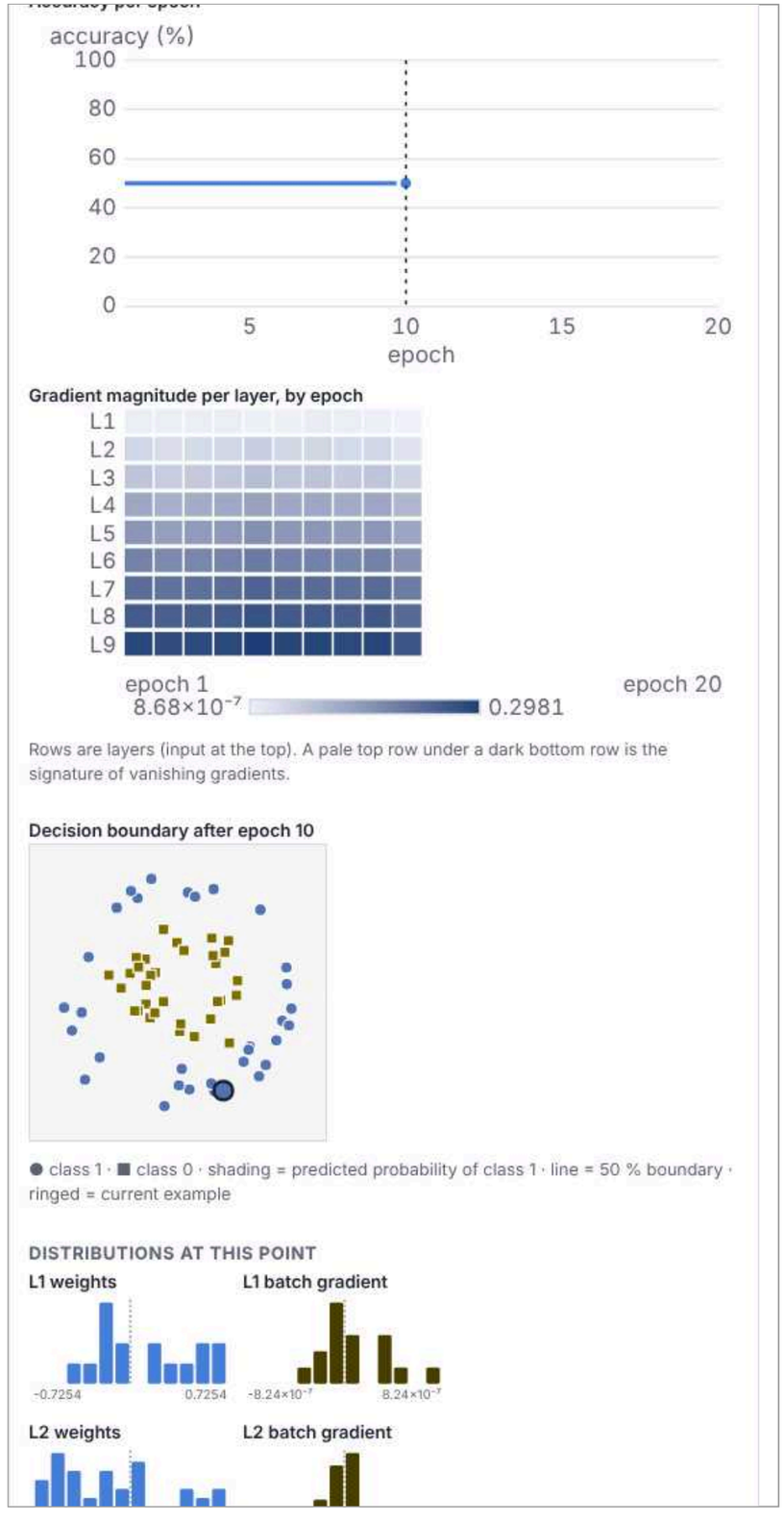


D1. Vanishing gradients — light theme, forward phase (part 2 of 3)

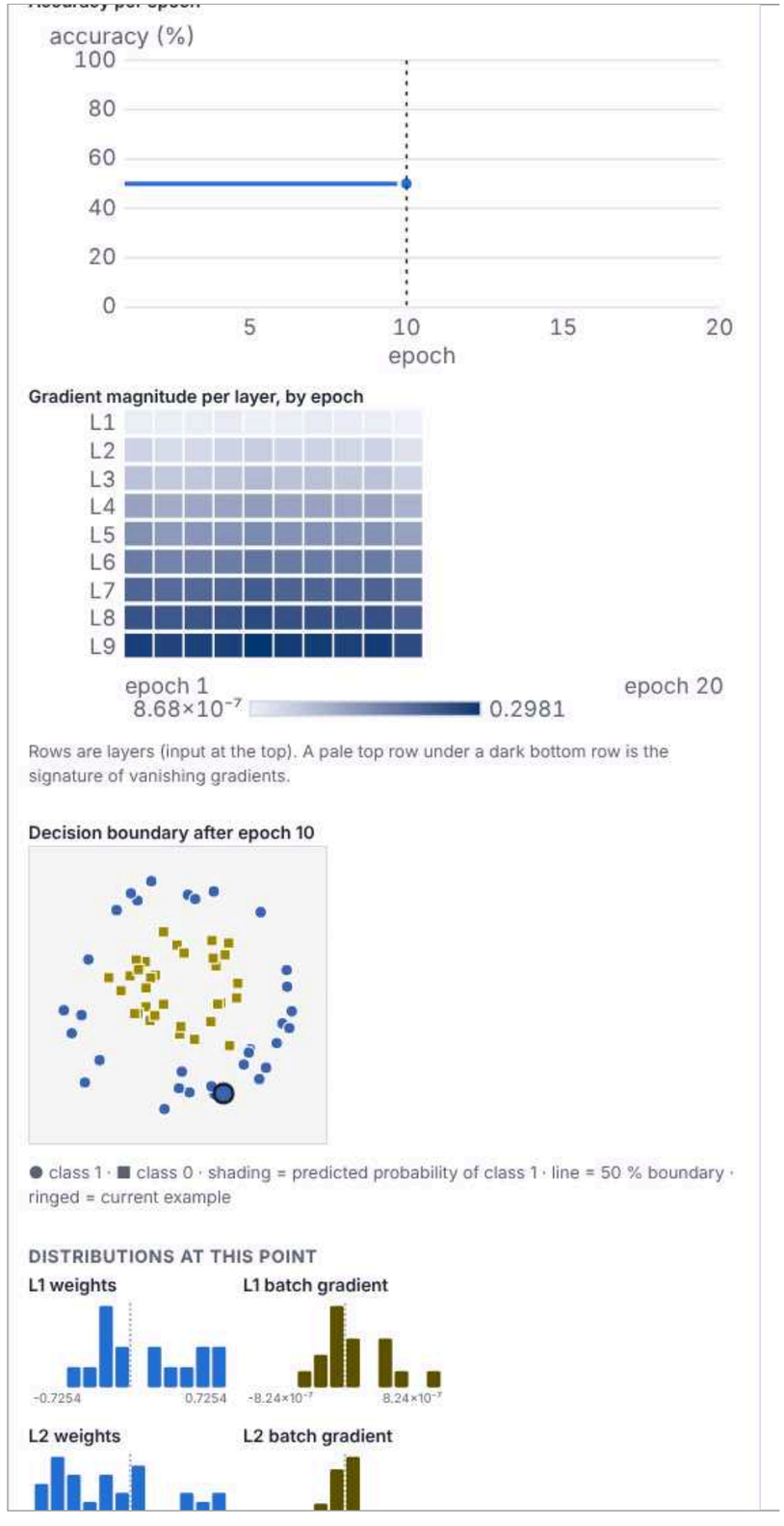
As rendered



Simulated protanopia

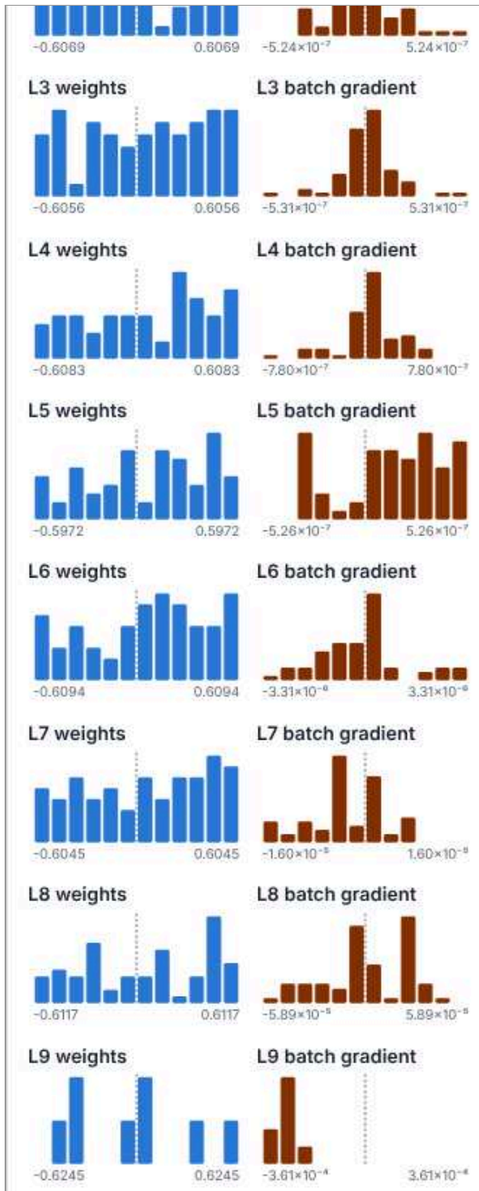


Simulated deuteranopia



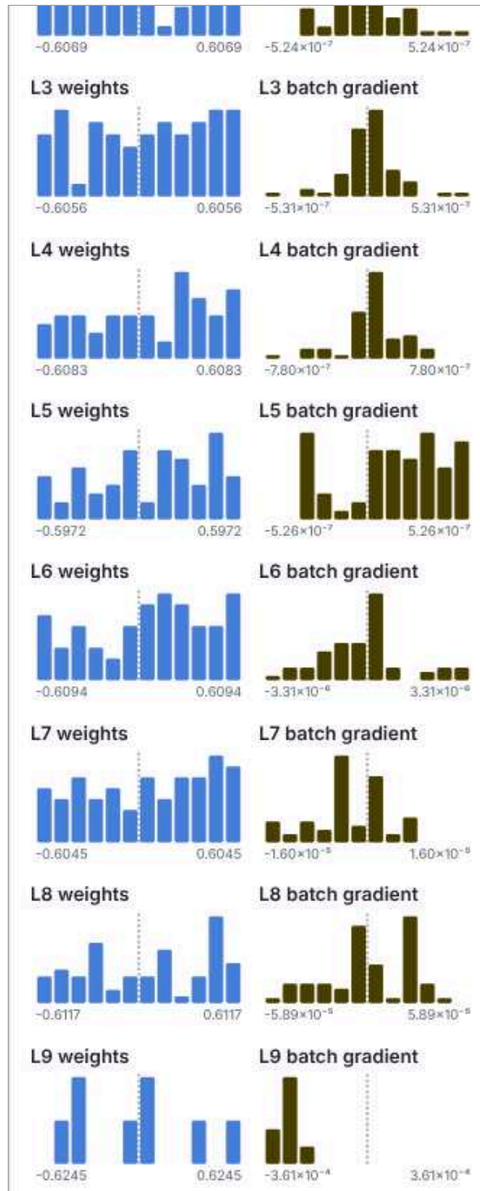
D1. Vanishing gradients — light theme, forward phase (part 3 of 3)

As rendered



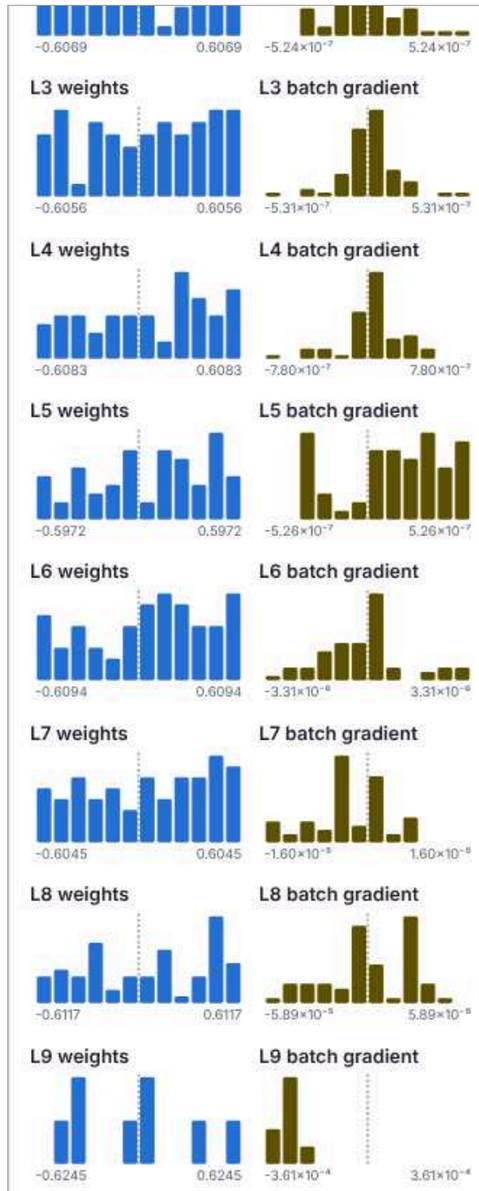
▶ Table view of the epoch data

Simulated protanopia



▶ Table view of the epoch data

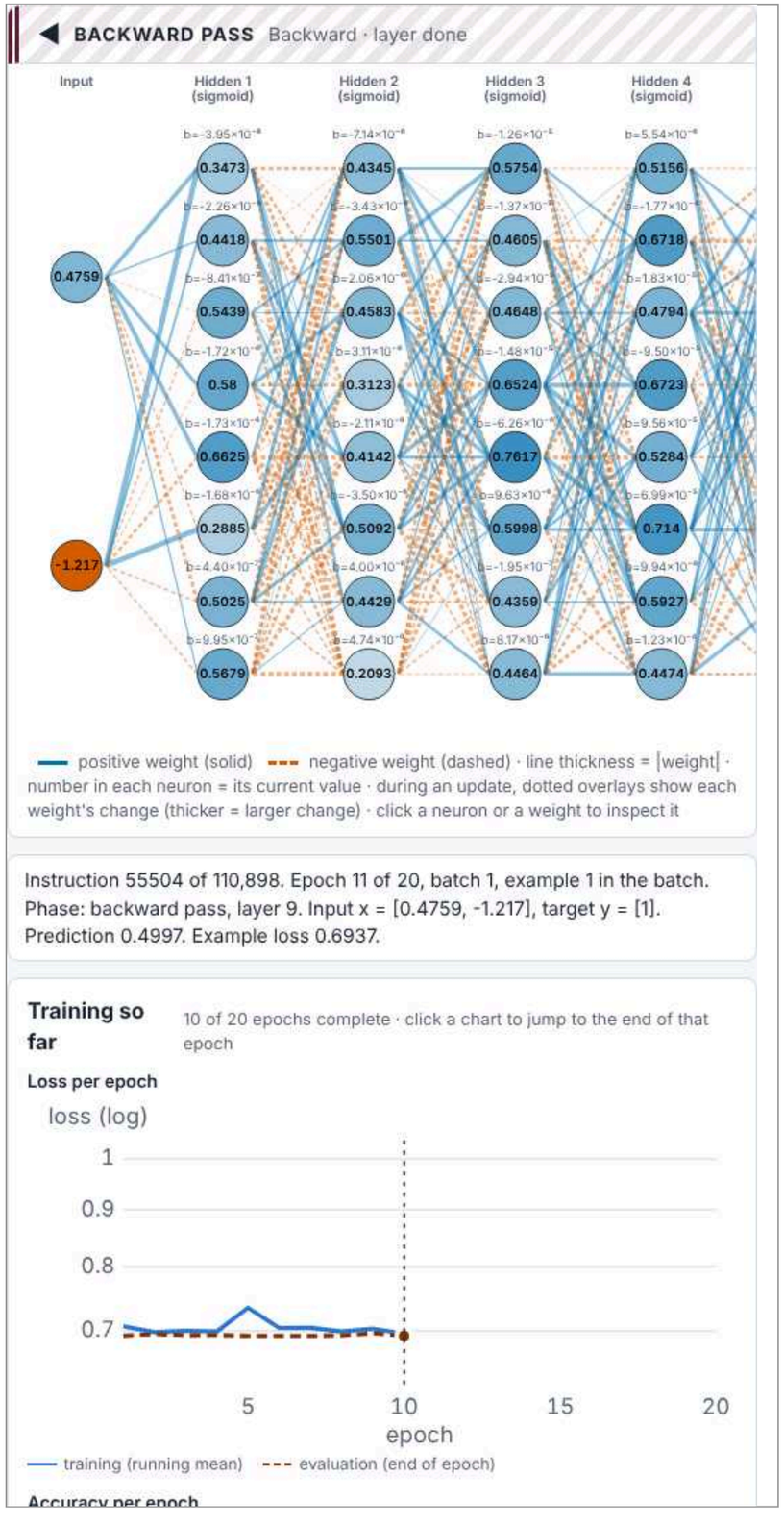
Simulated deuteranopia



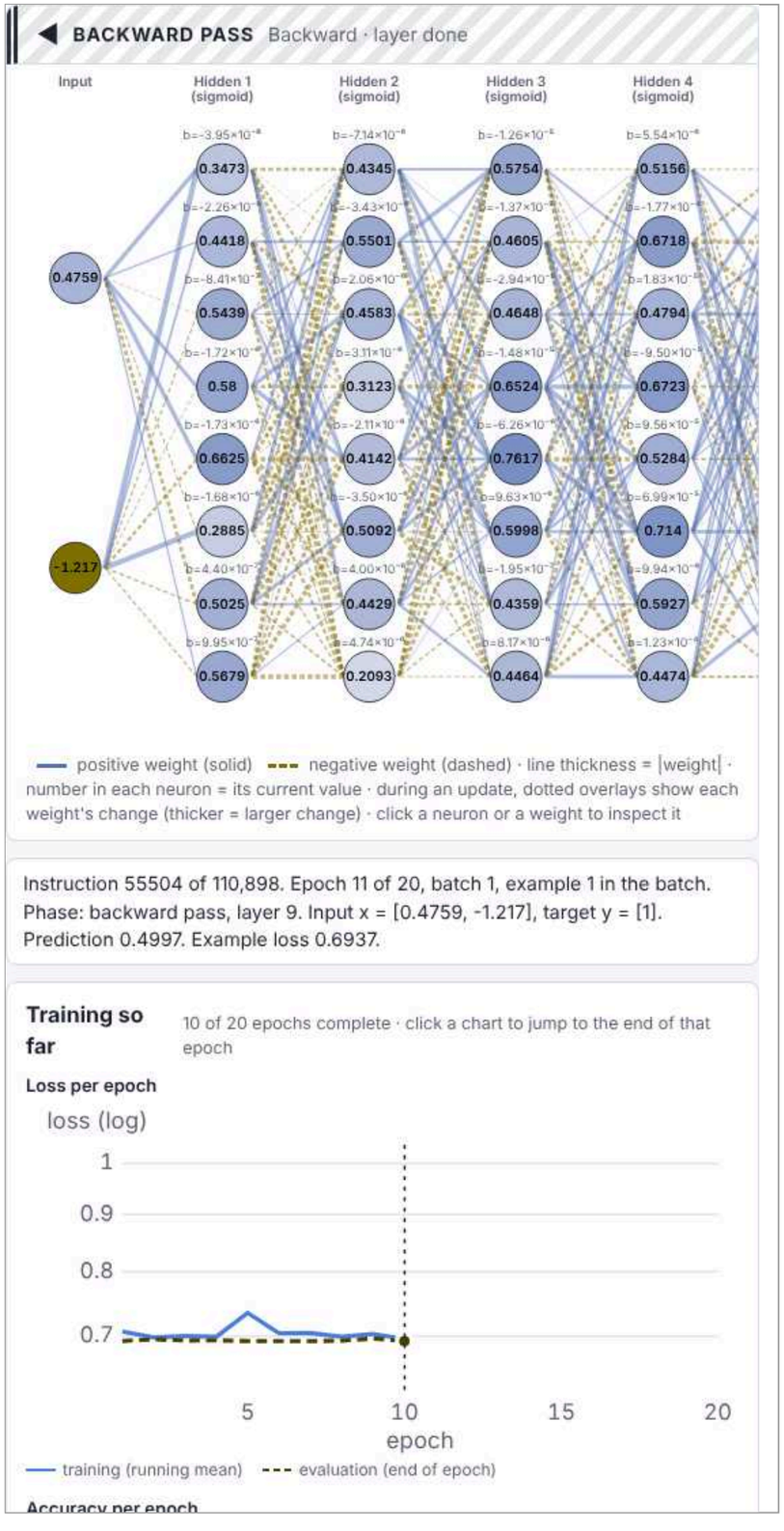
▶ Table view of the epoch data

D1. Vanishing gradients — light theme, backward phase (part 1 of 3)

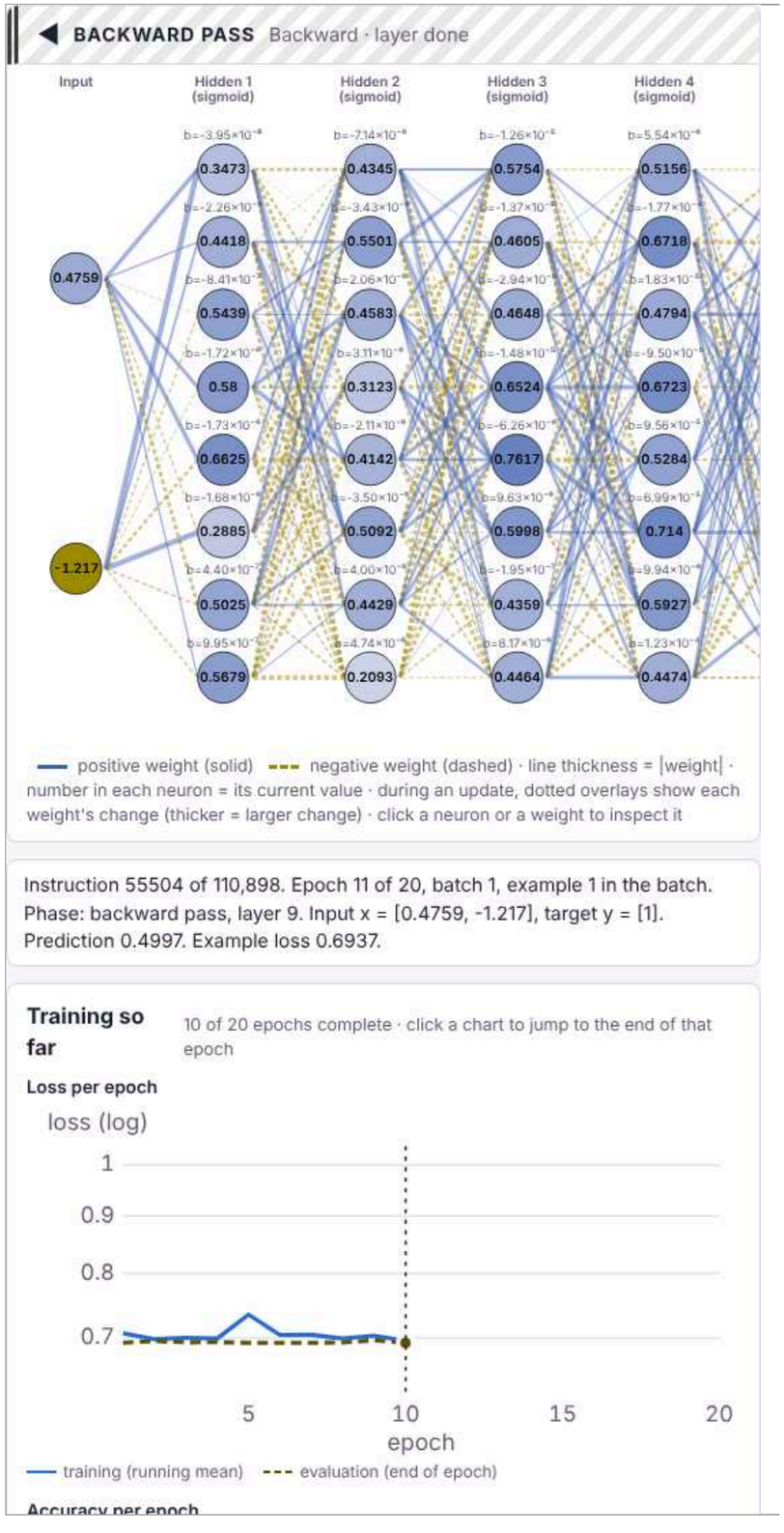
As rendered



Simulated protanopia

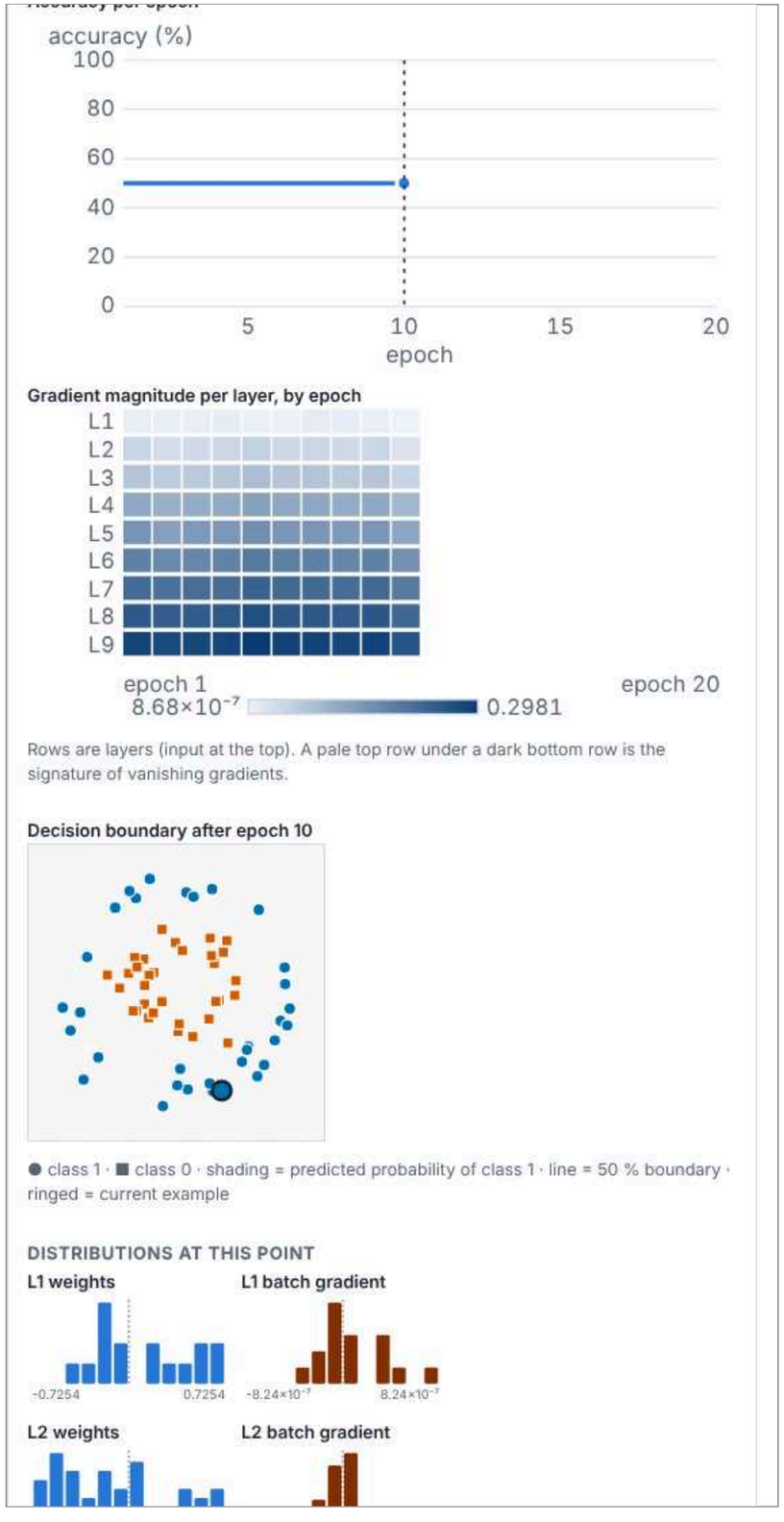


Simulated deuteranopia

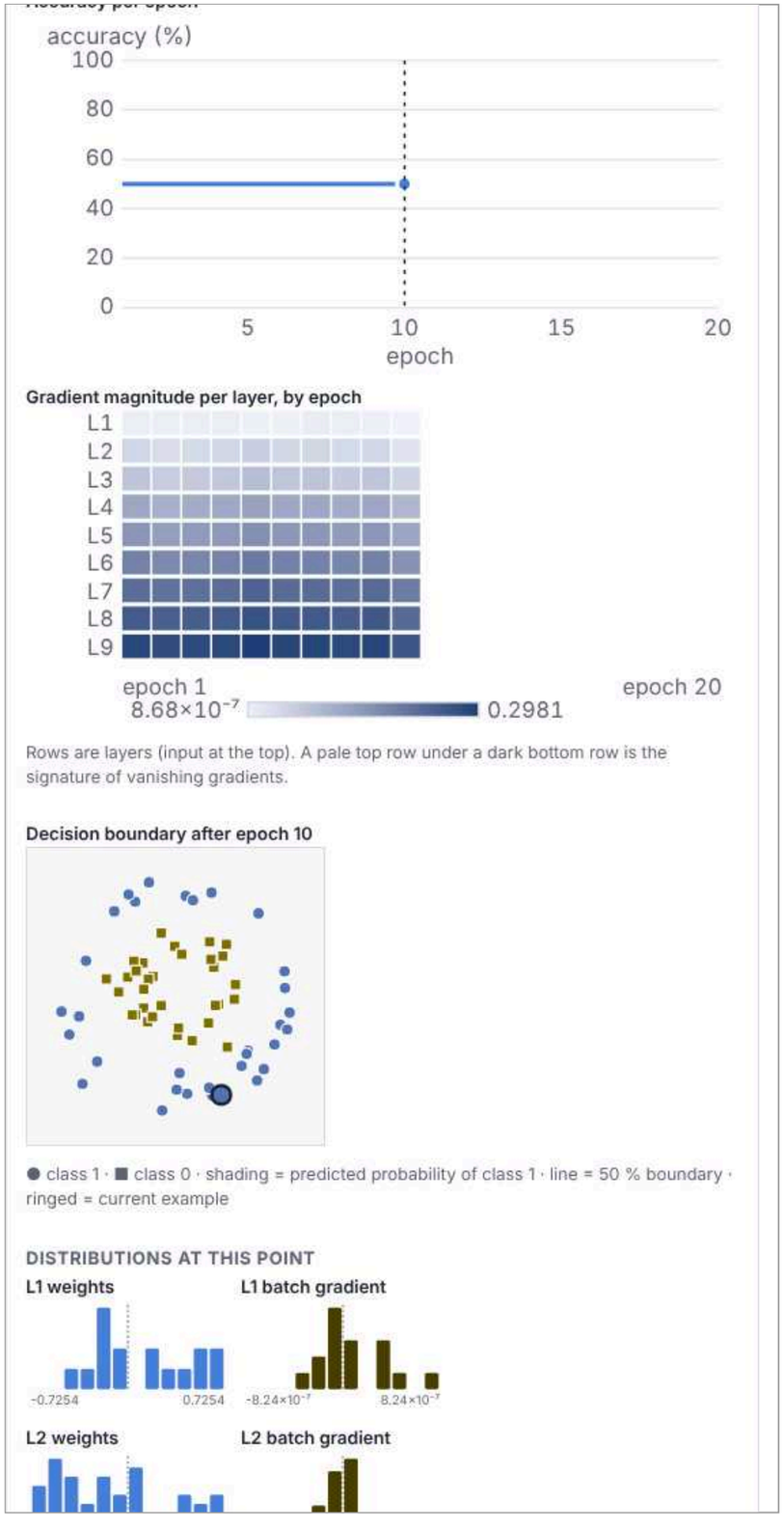


D1. Vanishing gradients — light theme, backward phase (part 2 of 3)

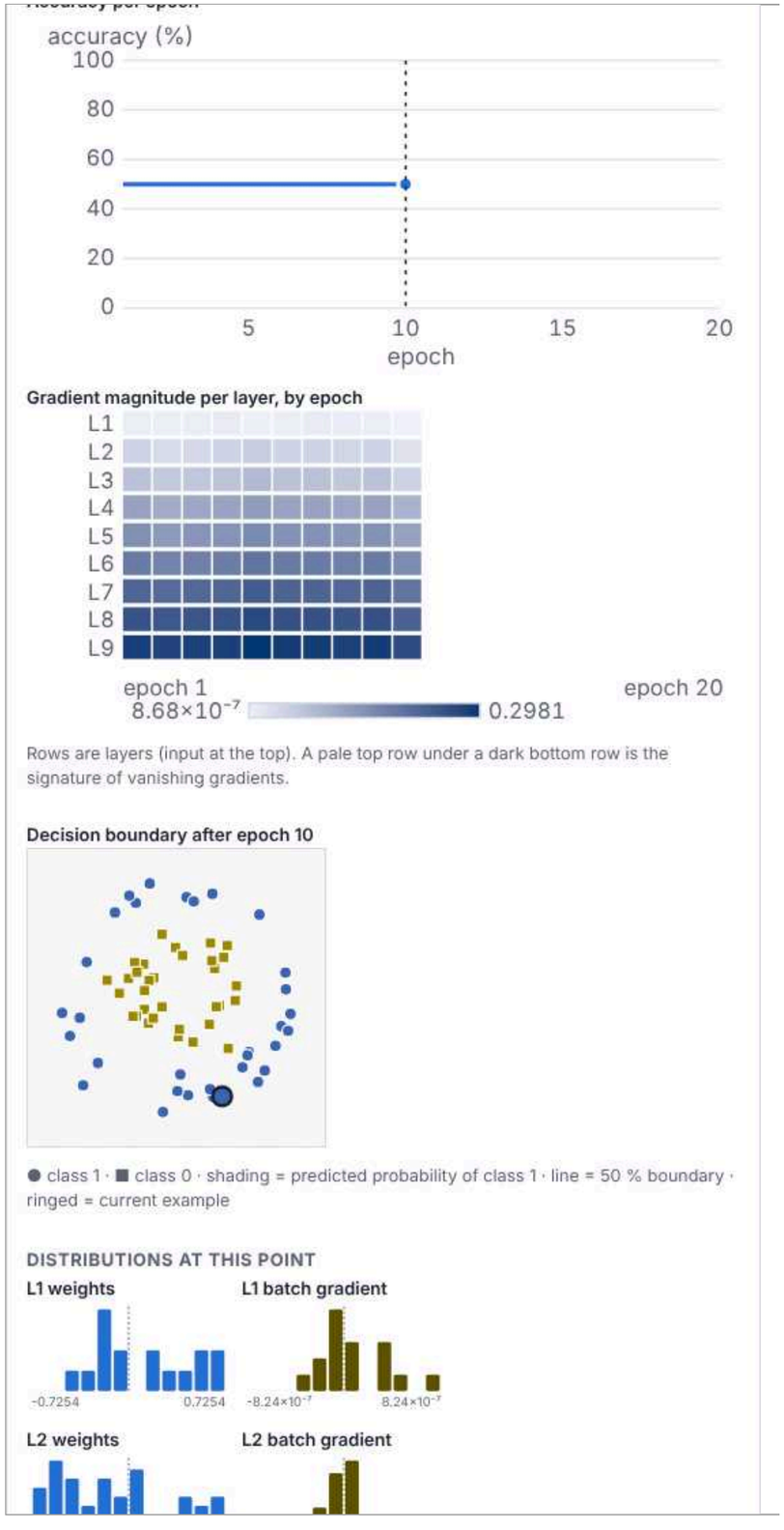
As rendered



Simulated protanopia

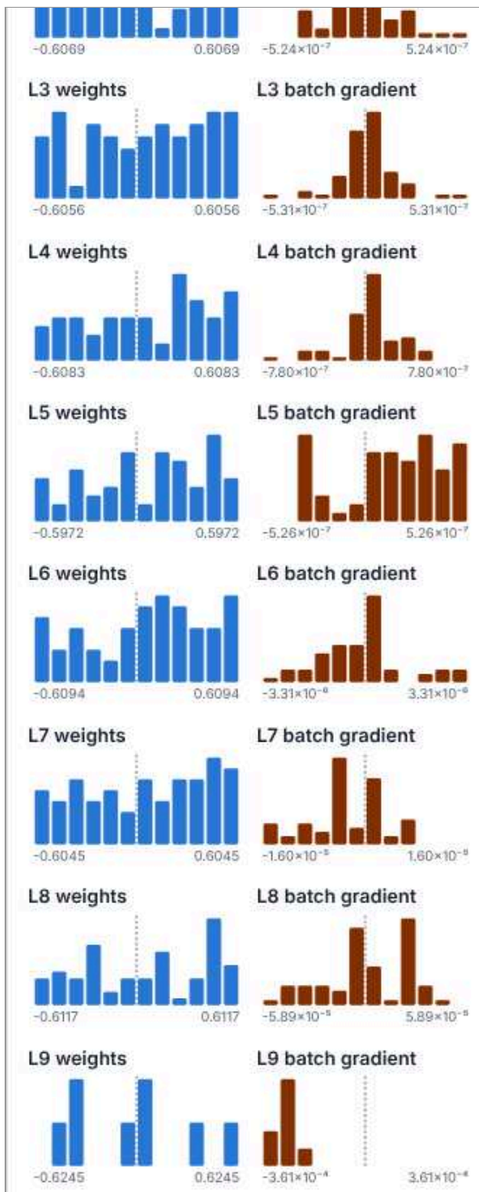


Simulated deuteranopia



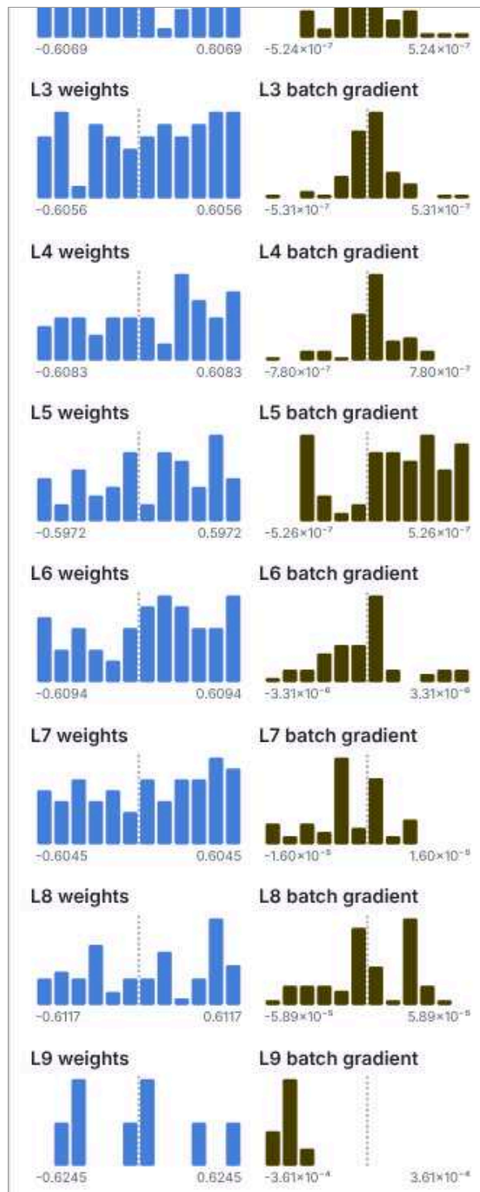
D1. Vanishing gradients — light theme, backward phase (part 3 of 3)

As rendered



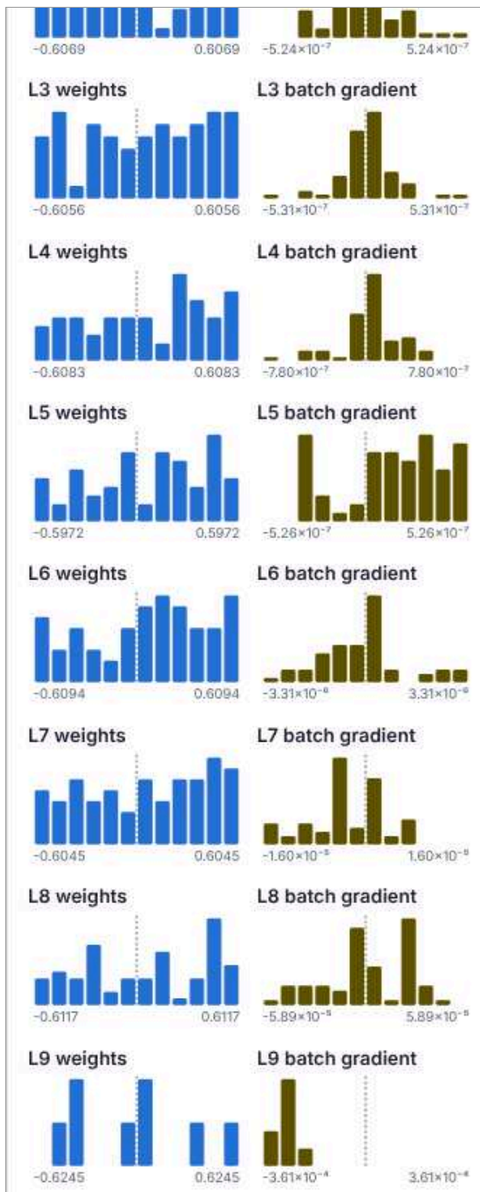
▶ Table view of the epoch data

Simulated protanopia



▶ Table view of the epoch data

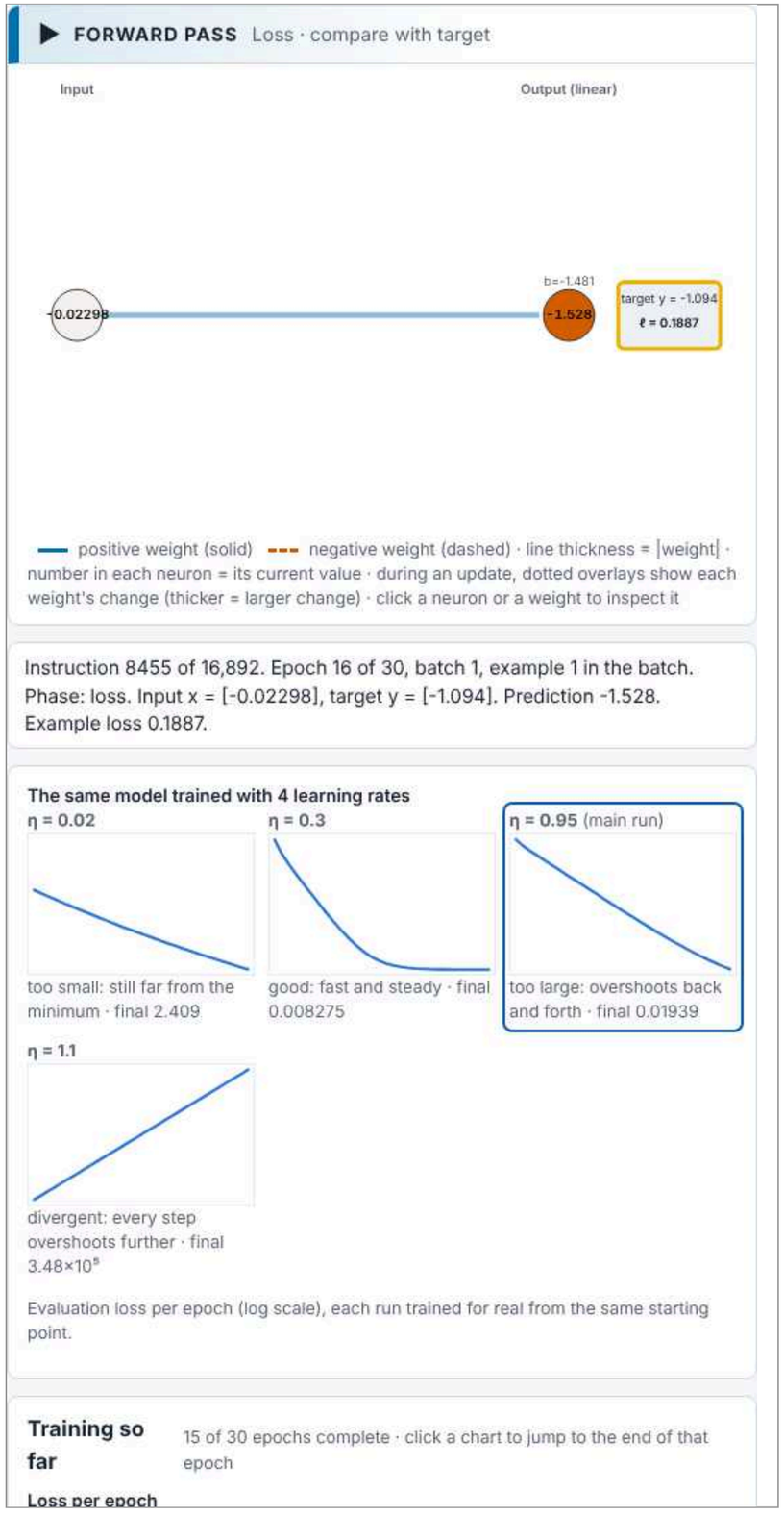
Simulated deuteranopia



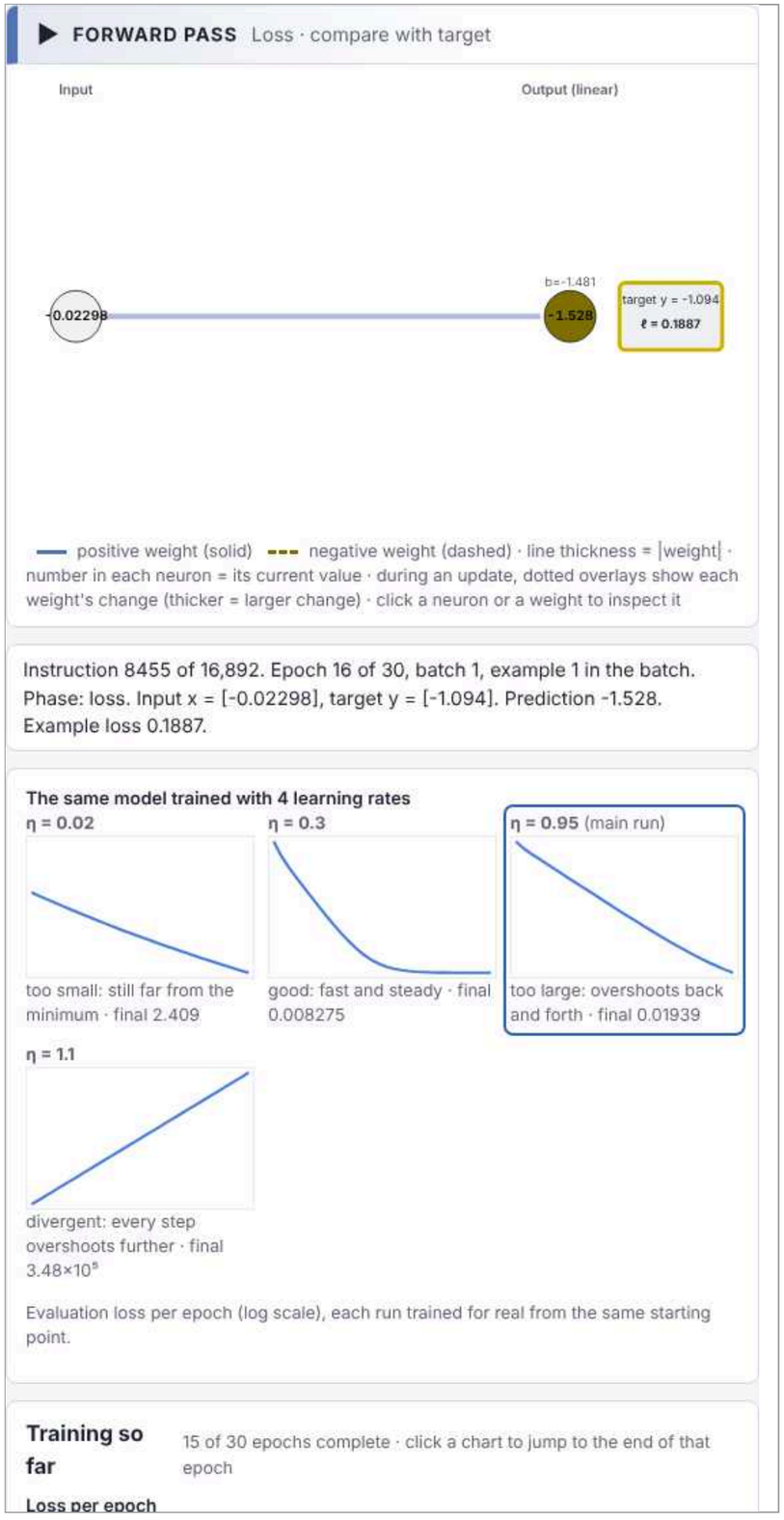
▶ Table view of the epoch data

D2. Learning-rate pathology — light theme, forward phase (part 1 of 2)

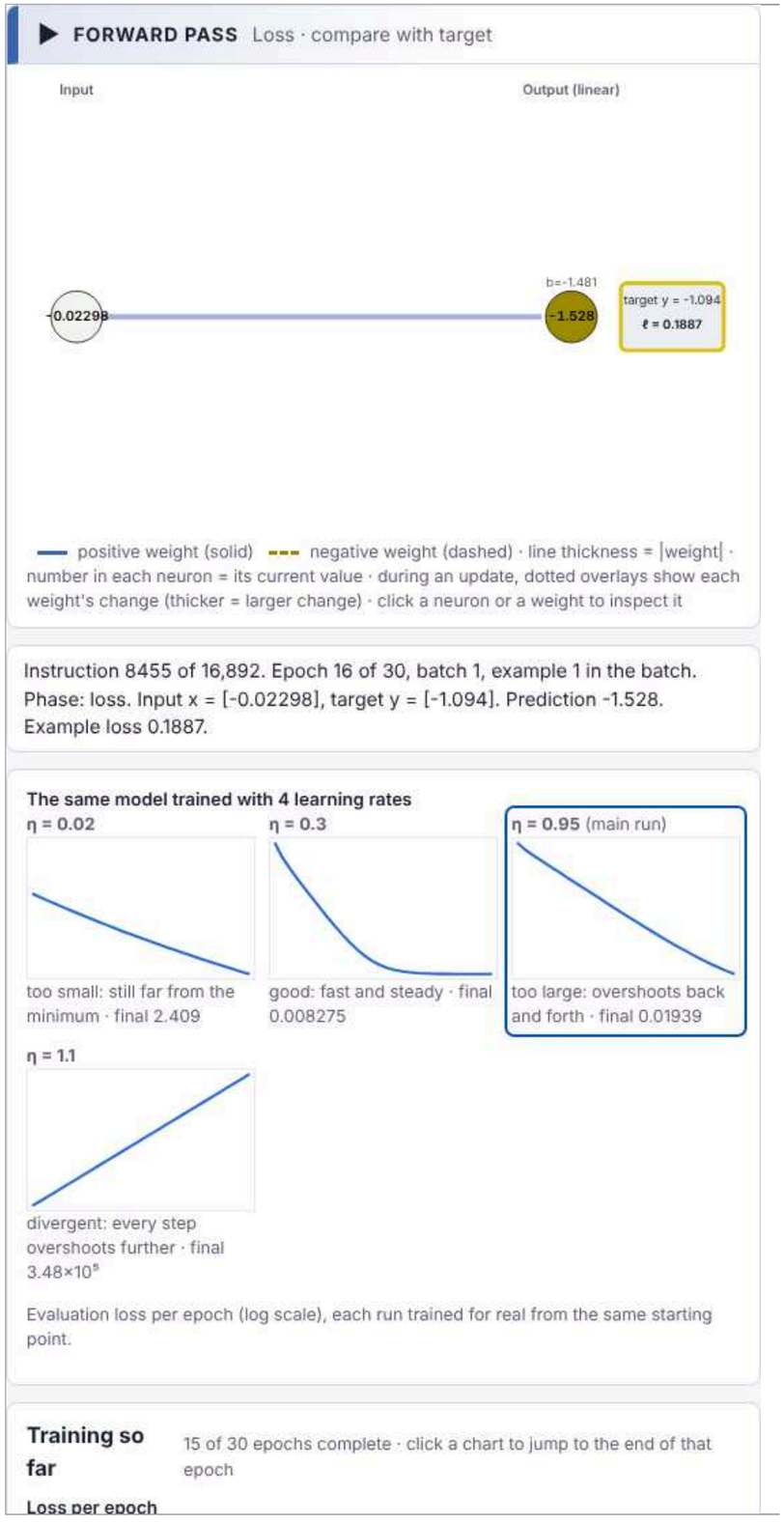
As rendered



Simulated protanopia

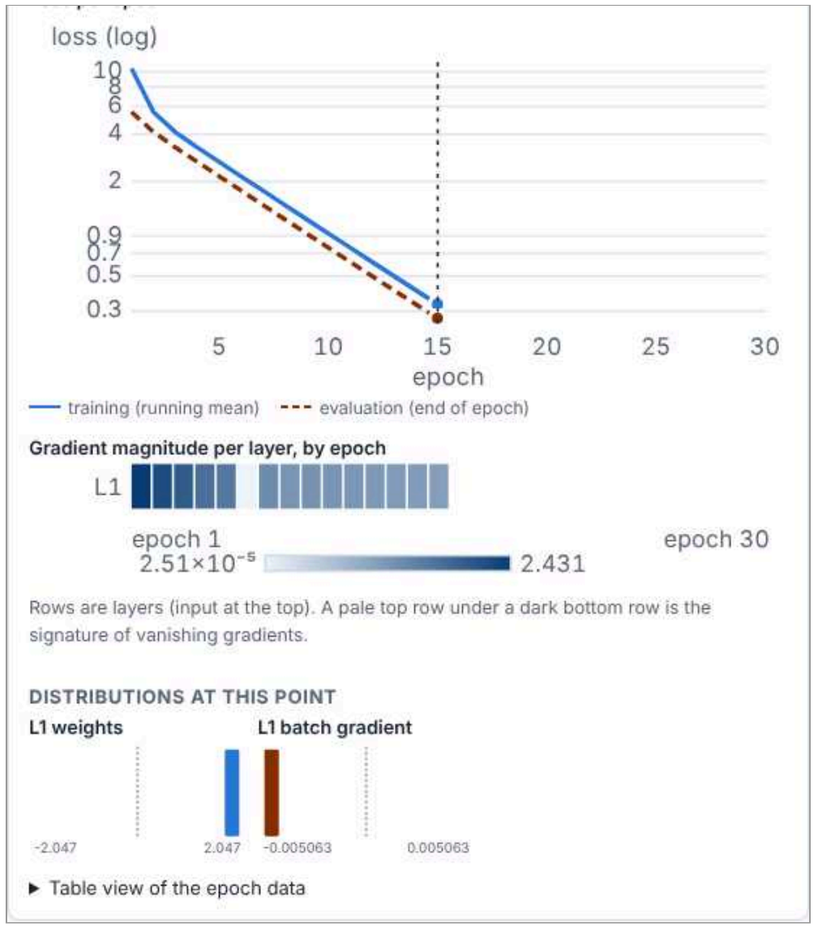


Simulated deuteranopia

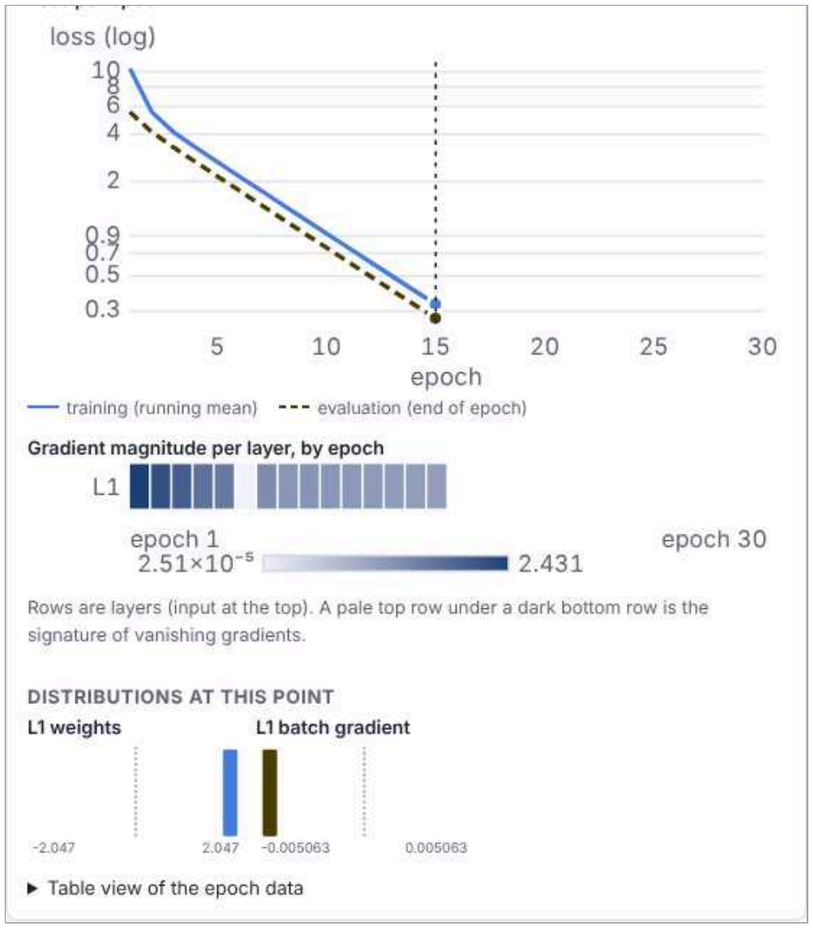


D2. Learning-rate pathology — light theme, forward phase (part 2 of 2)

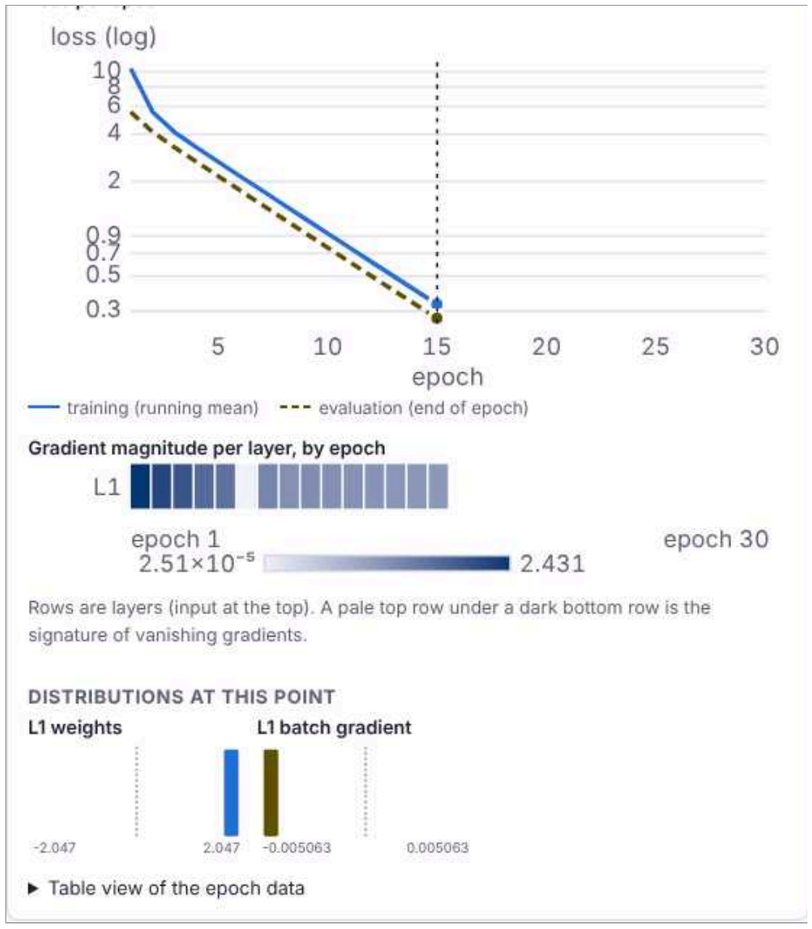
As rendered



Simulated protanopia

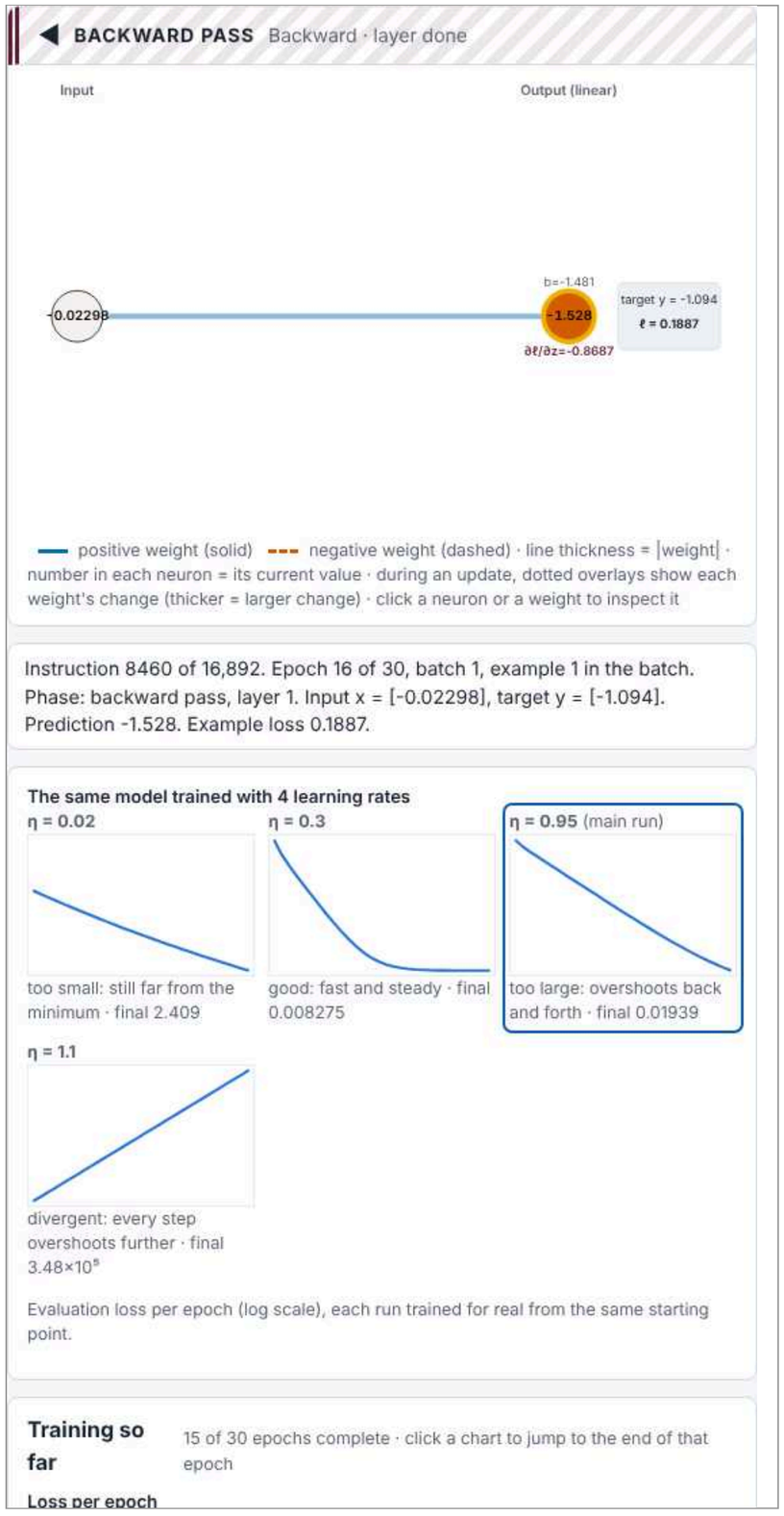


Simulated deuteranopia

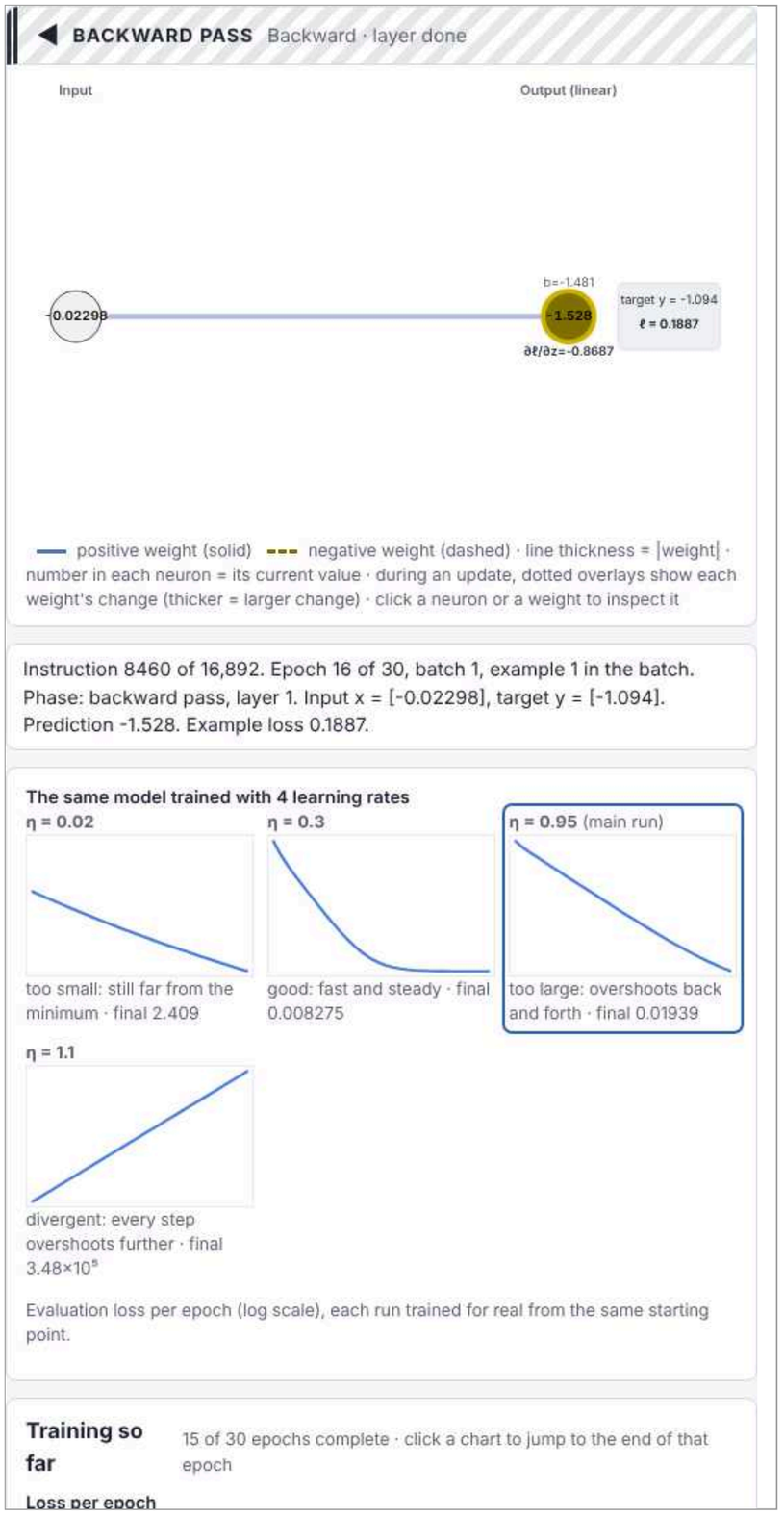


D2. Learning-rate pathology — light theme, backward phase (part 1 of 2)

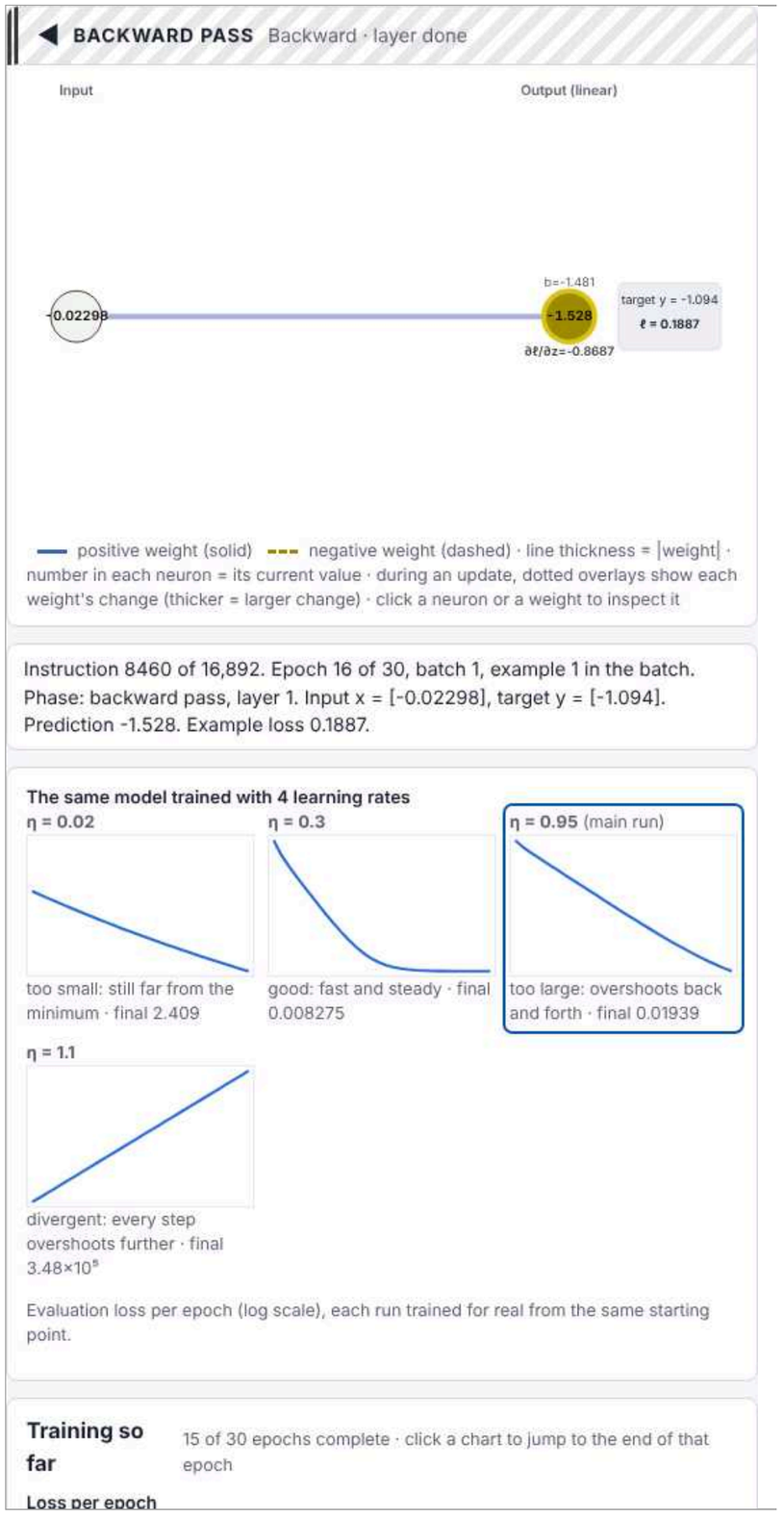
As rendered



Simulated protanopia

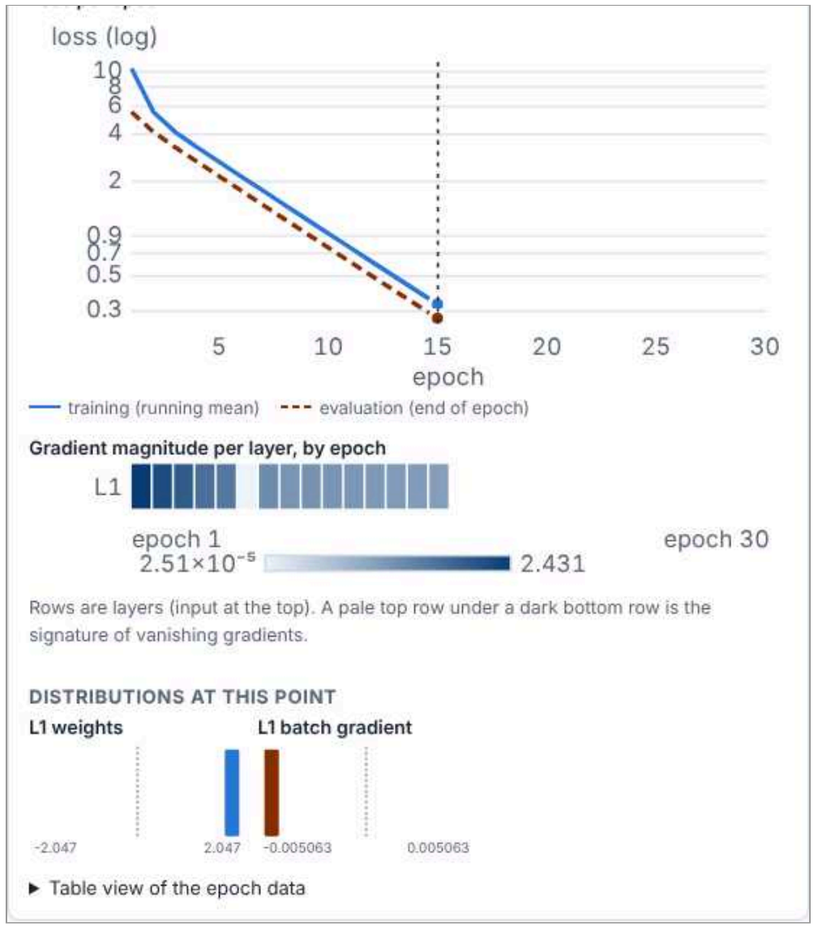


Simulated deuteranopia

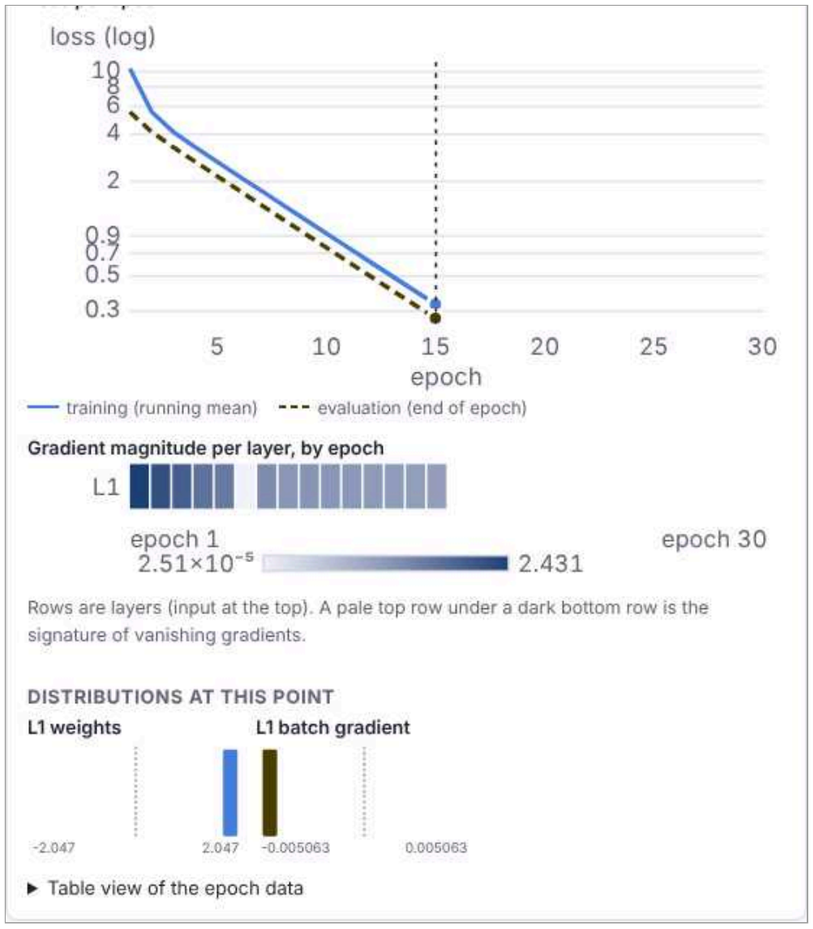


D2. Learning-rate pathology — light theme, backward phase (part 2 of 2)

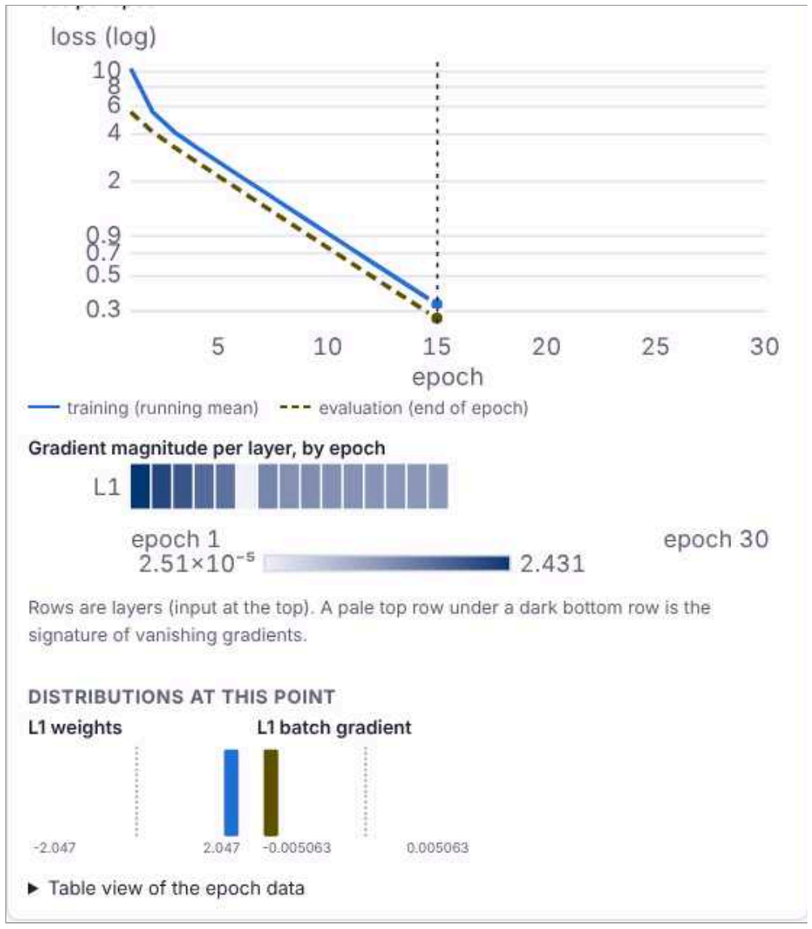
As rendered



Simulated protanopia

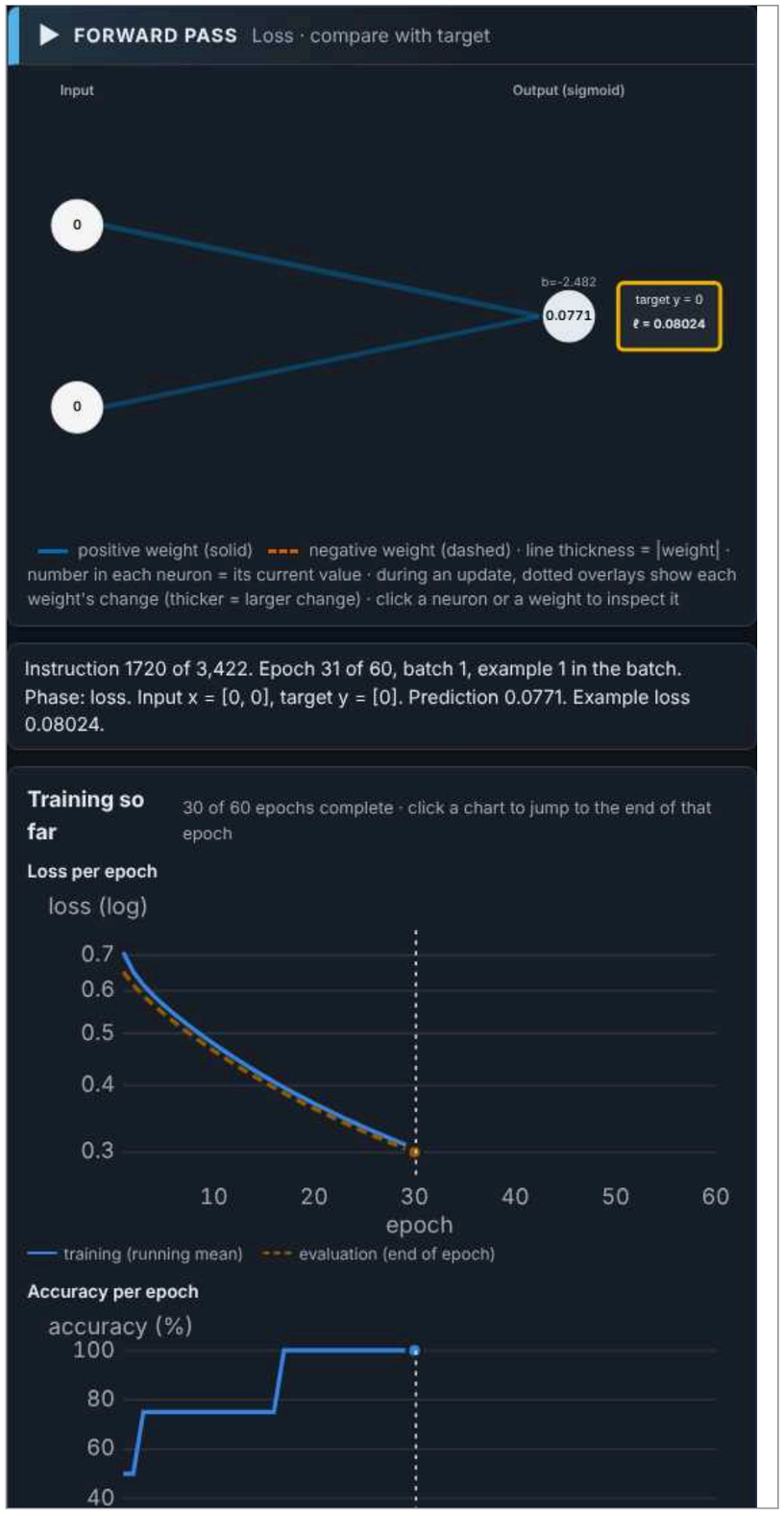


Simulated deuteranopia

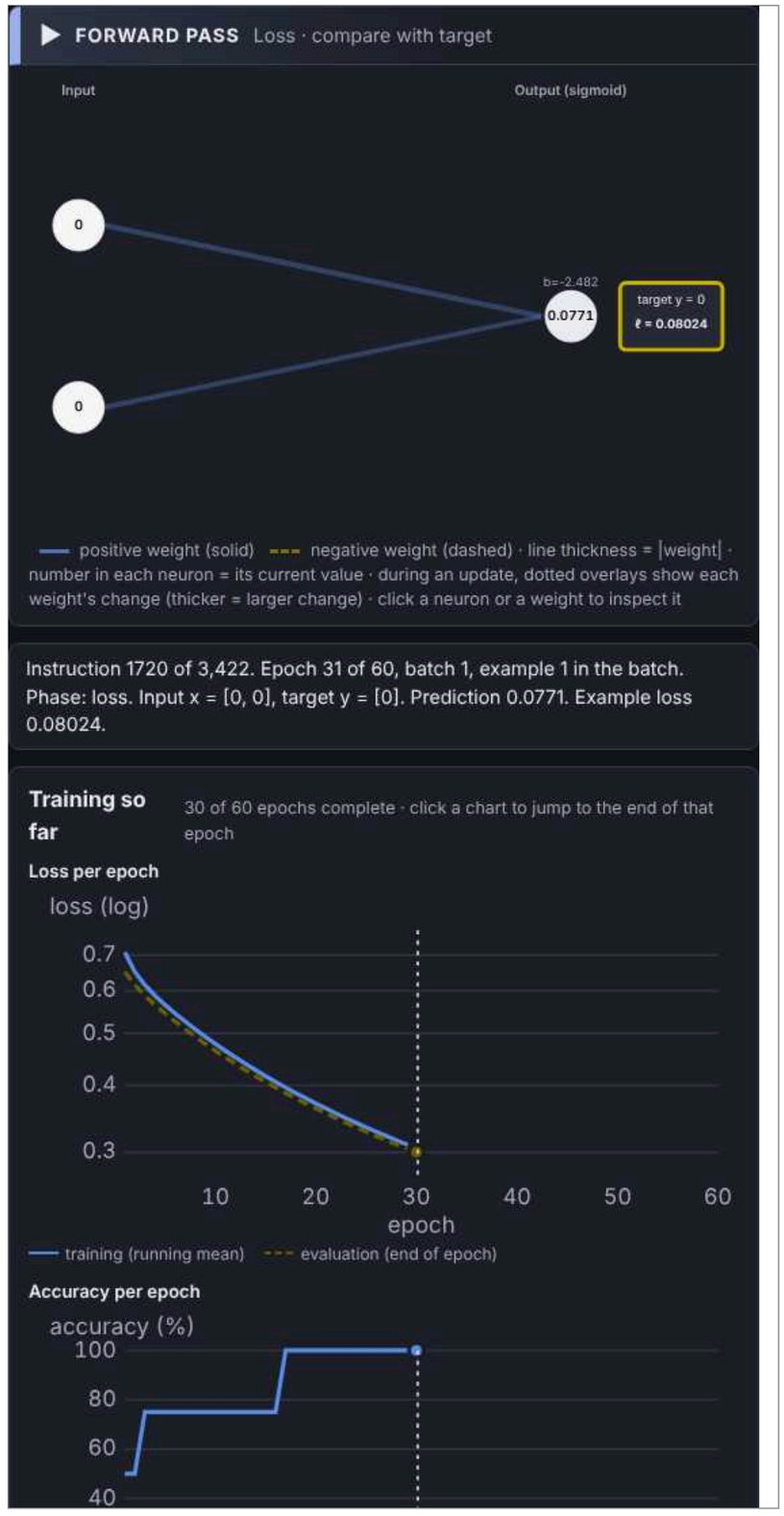


1. A single neuron — dark theme, forward phase (part 1 of 2)

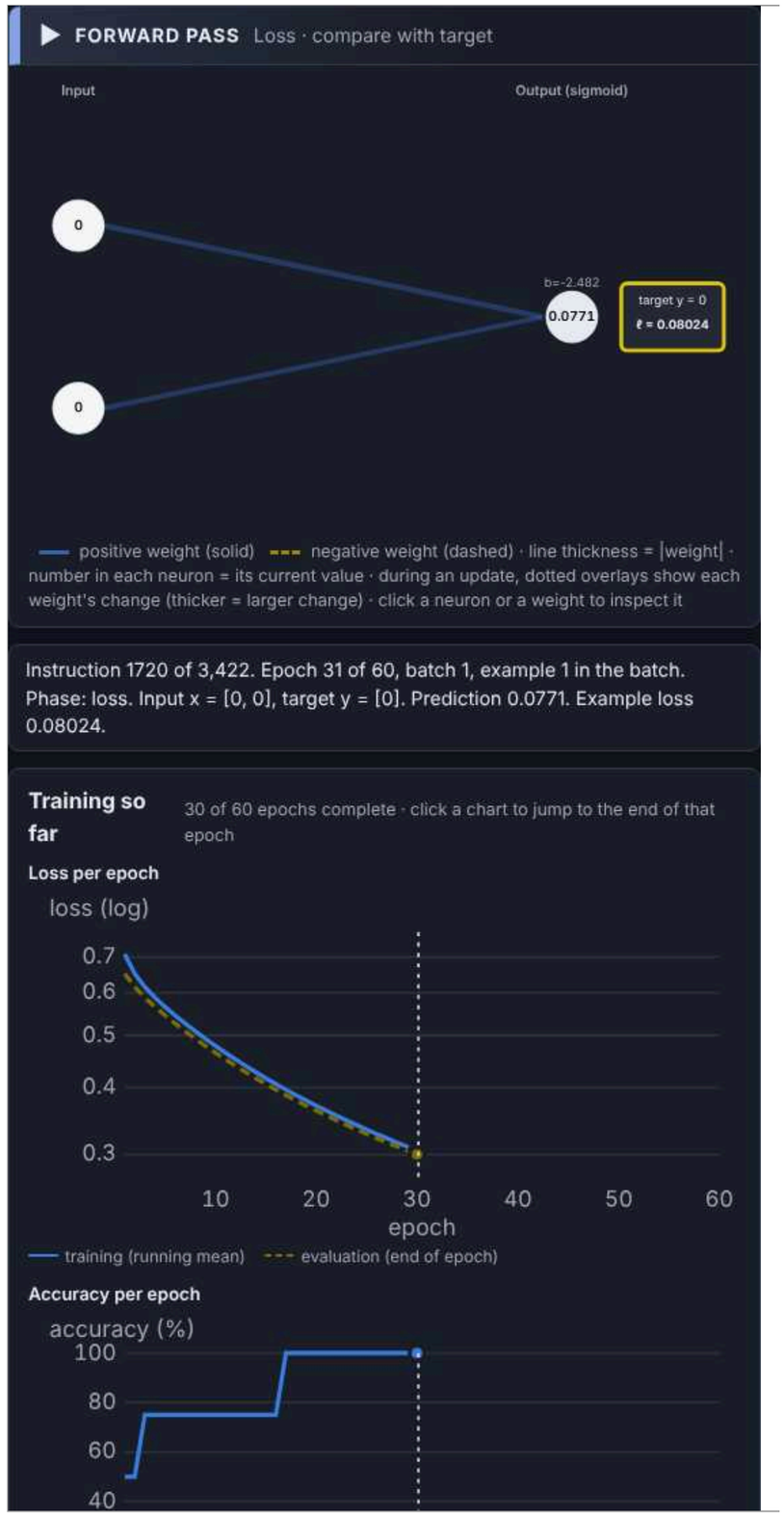
As rendered



Simulated protanopia

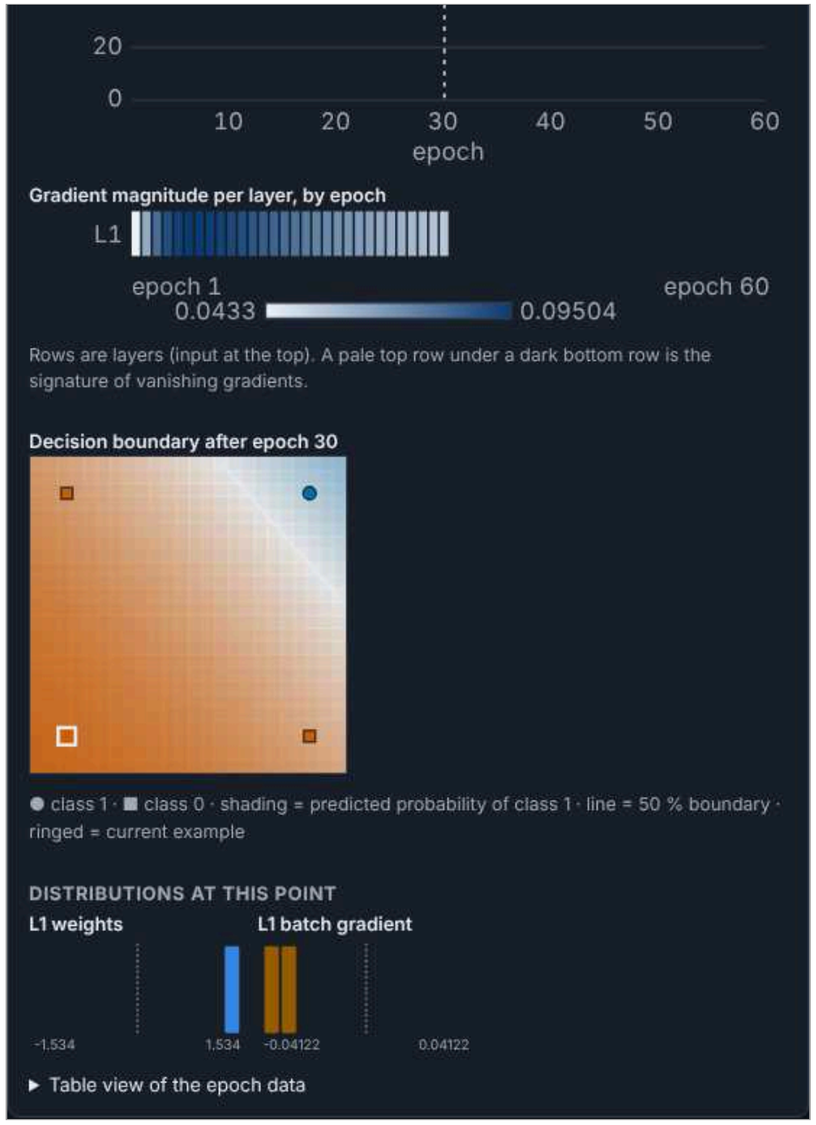


Simulated deuteranopia

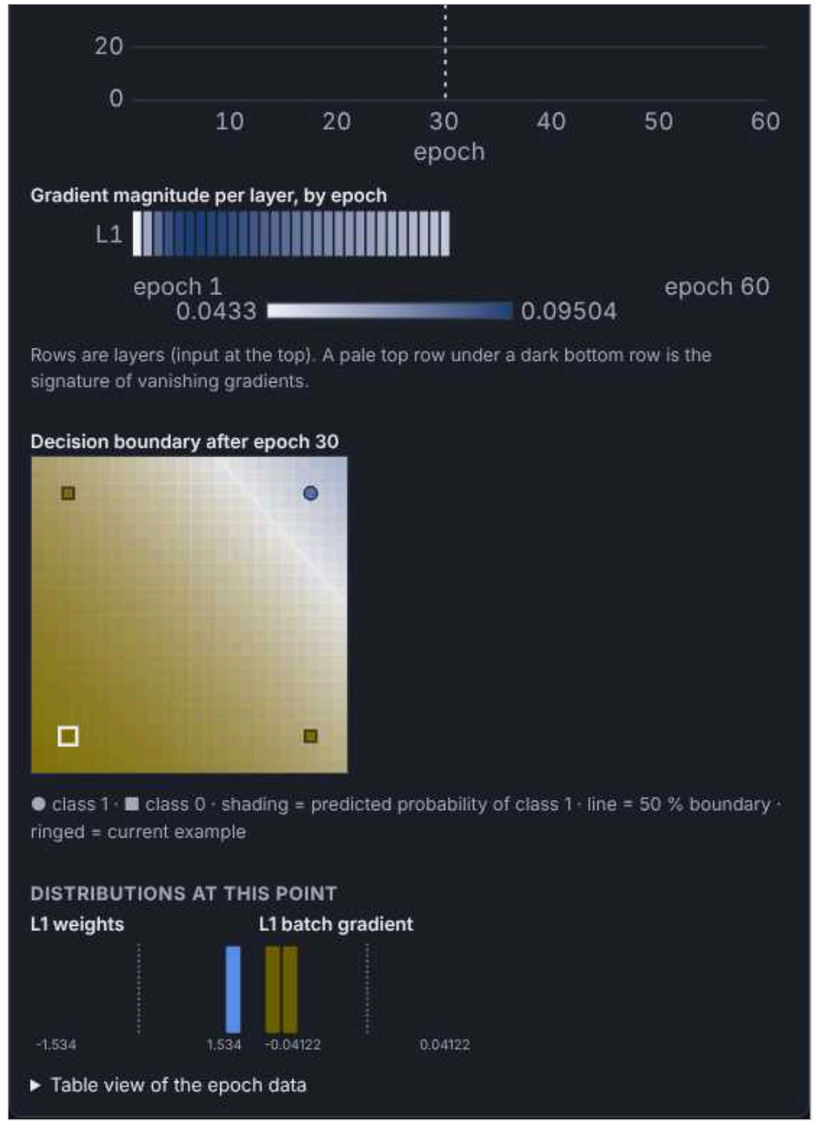


1. A single neuron — dark theme, forward phase (part 2 of 2)

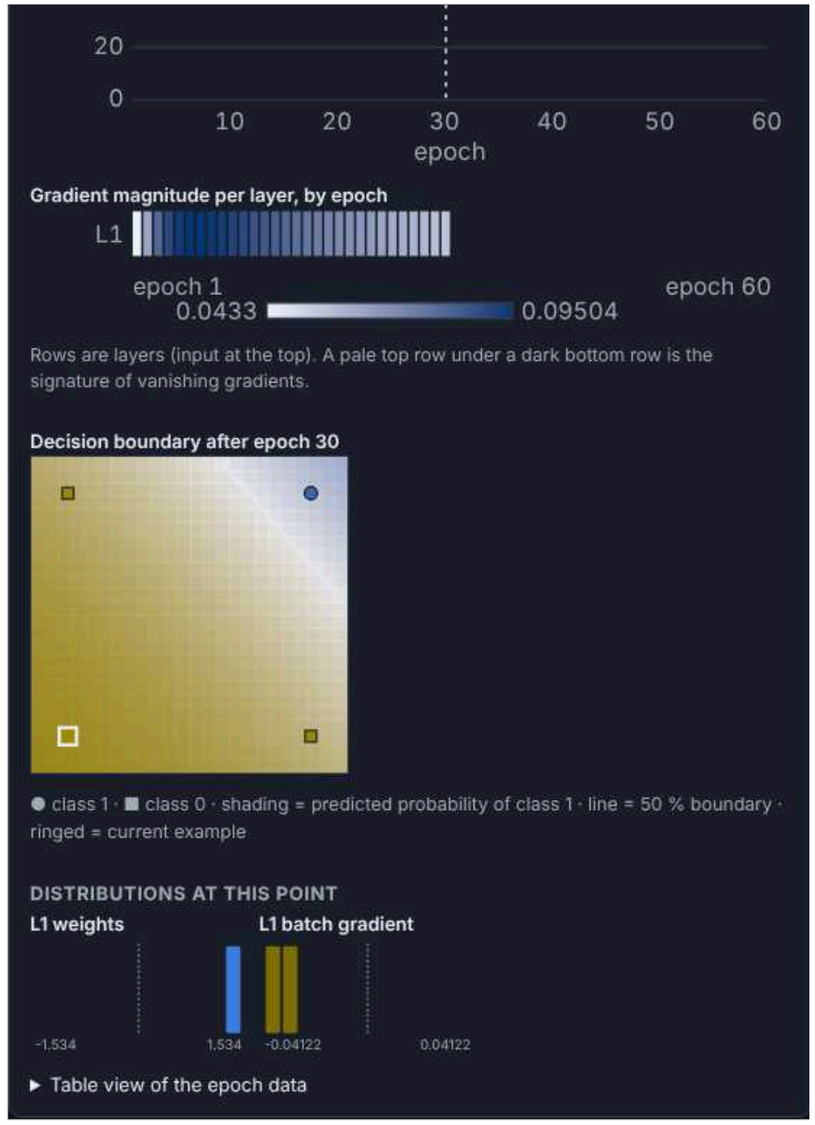
As rendered



Simulated protanopia

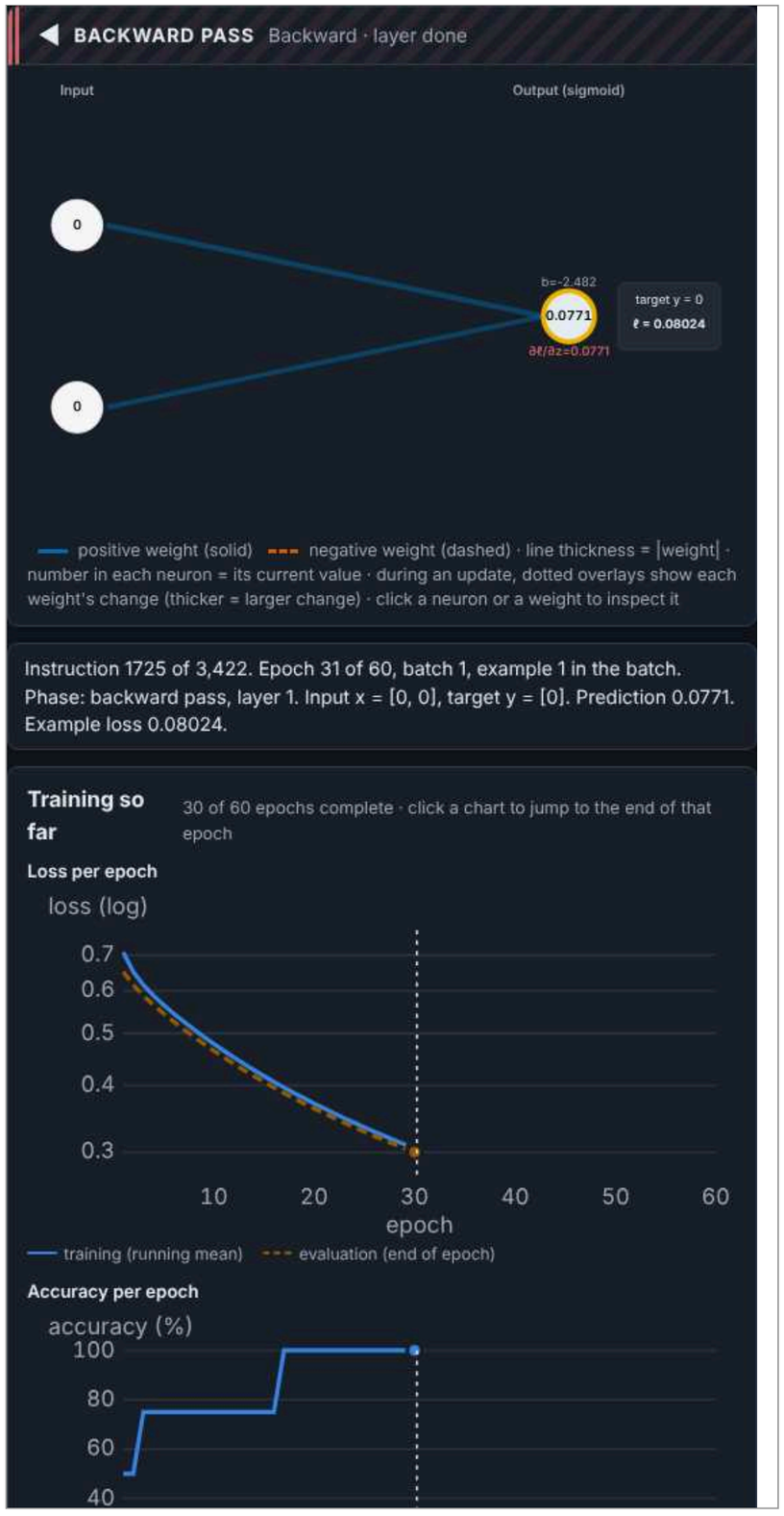


Simulated deuteranopia

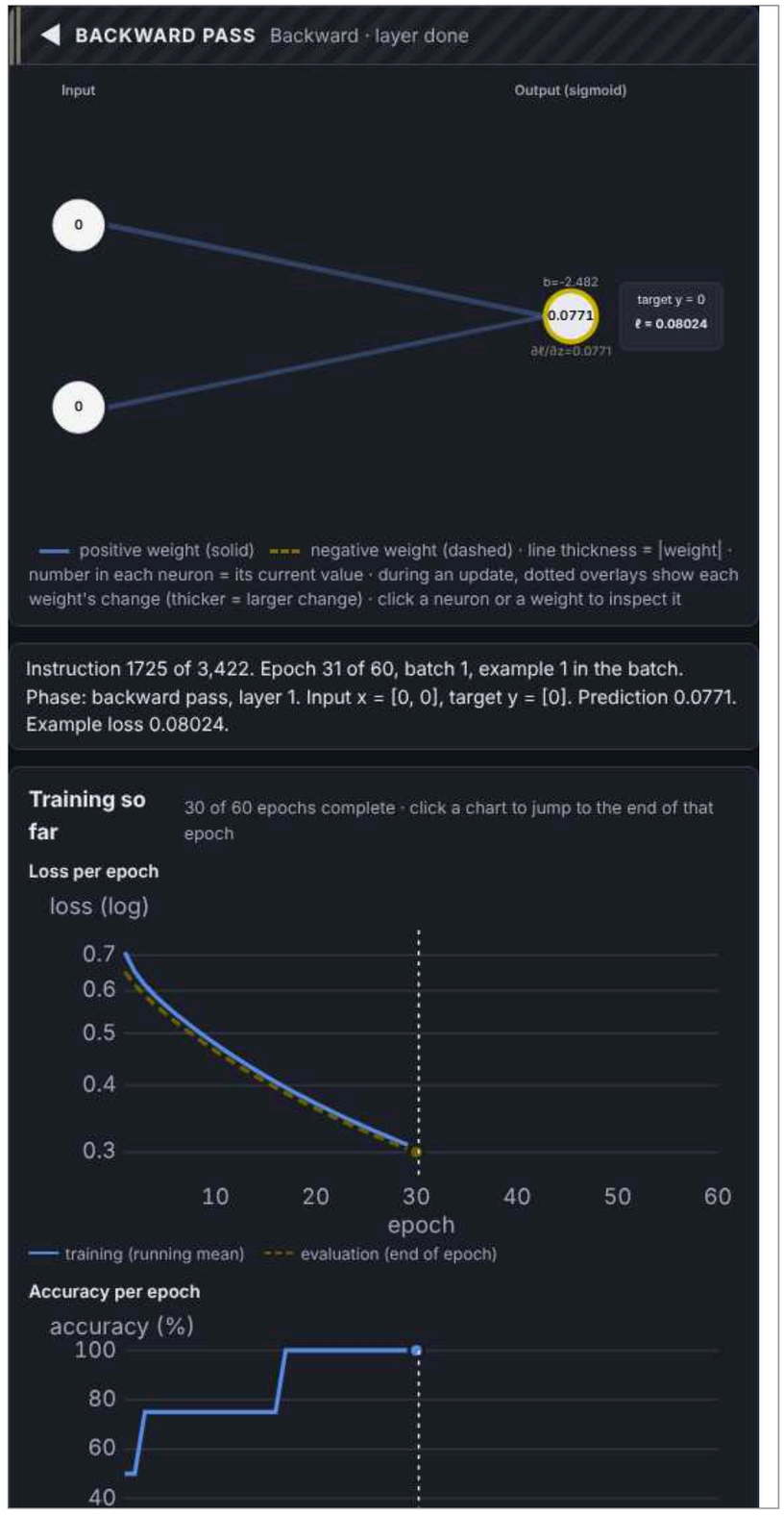


1. A single neuron — dark theme, backward phase (part 1 of 2)

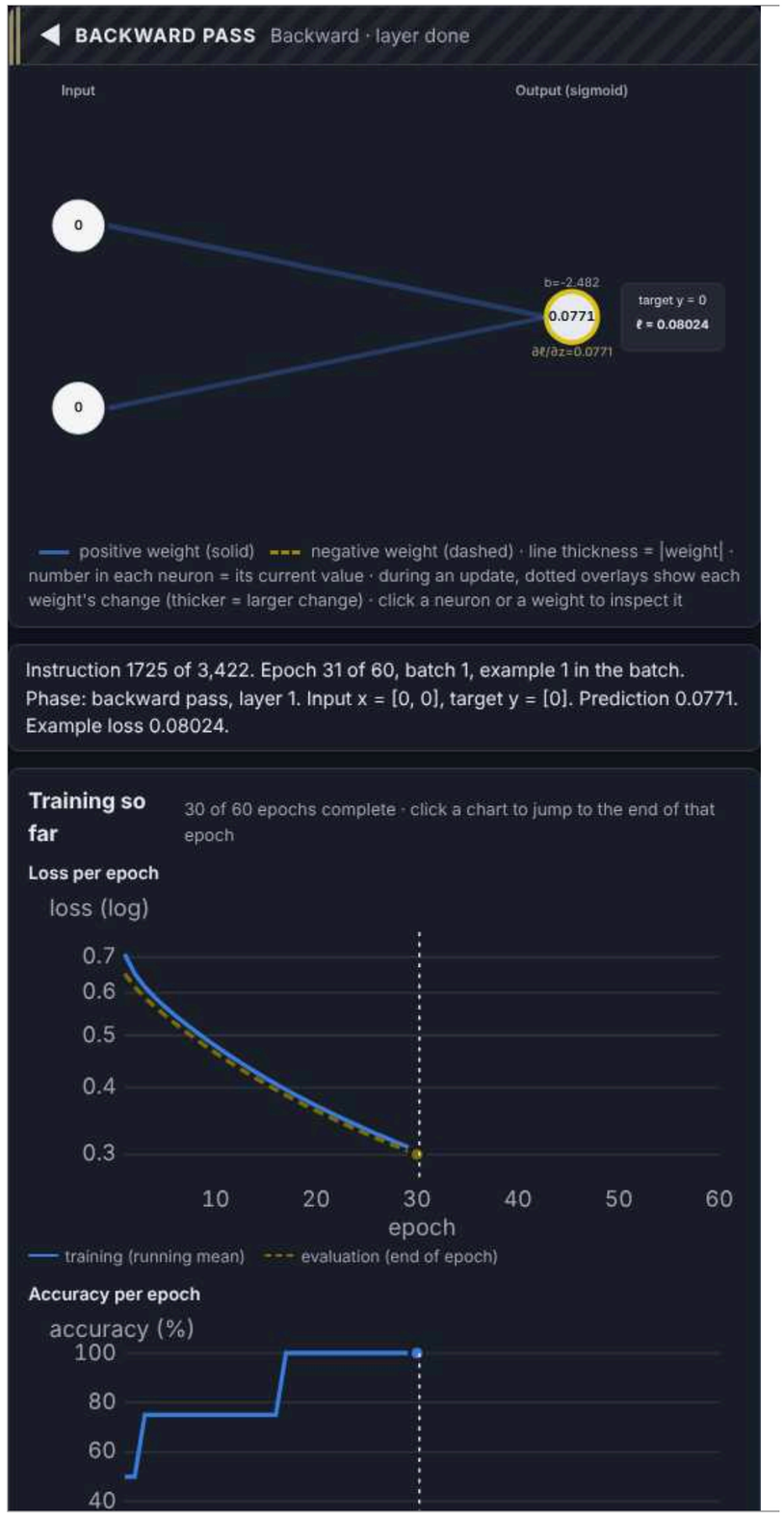
As rendered



Simulated protanopia

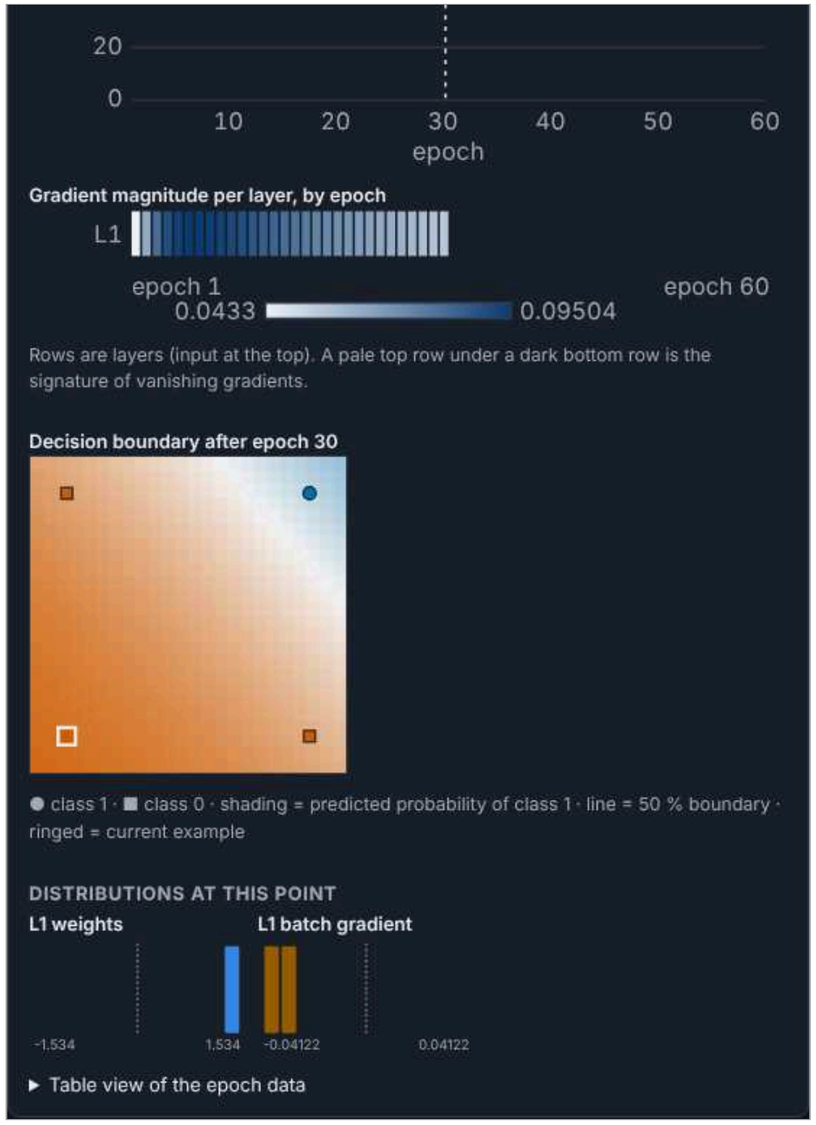


Simulated deuteranopia

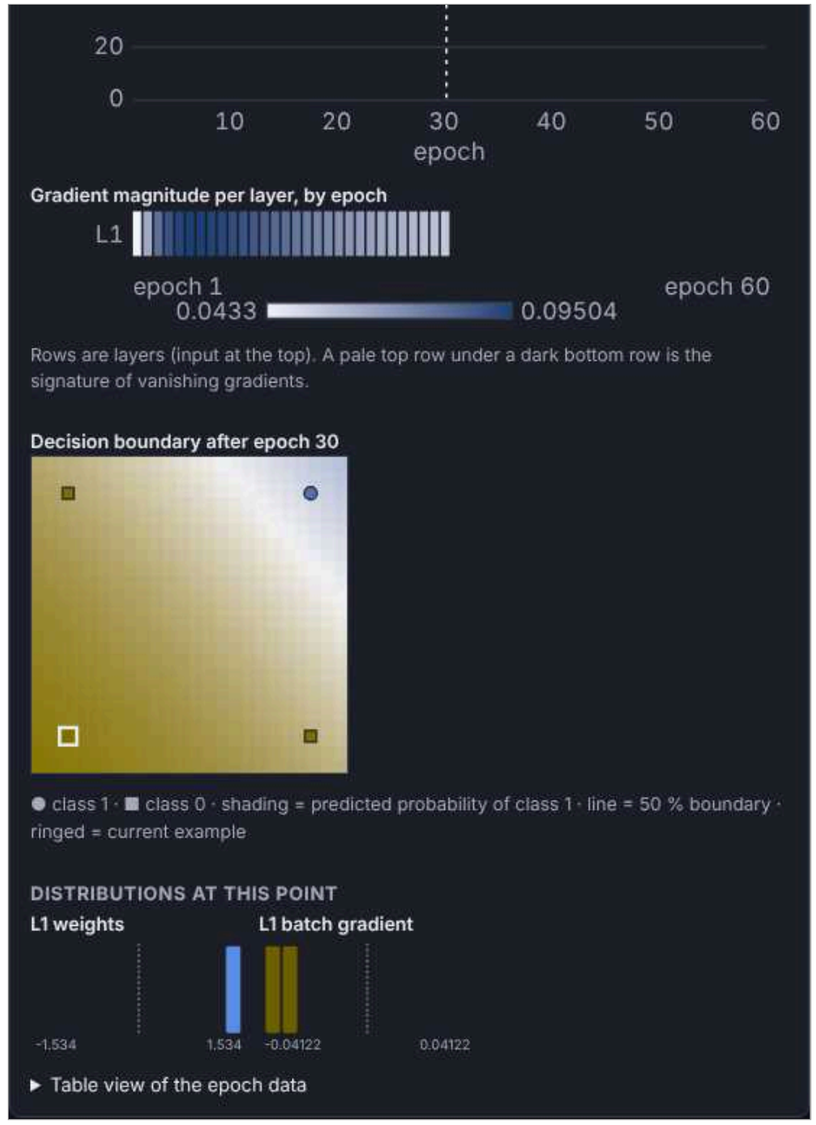


1. A single neuron — dark theme, backward phase (part 2 of 2)

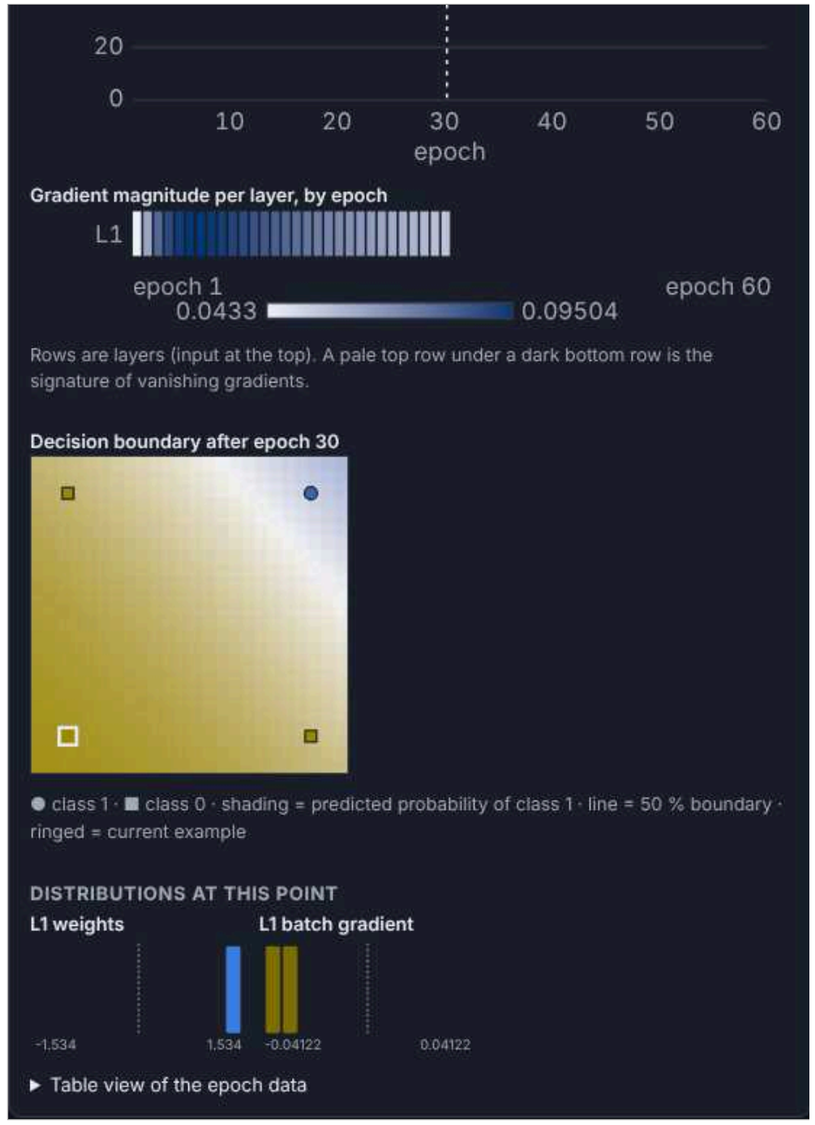
As rendered



Simulated protanopia

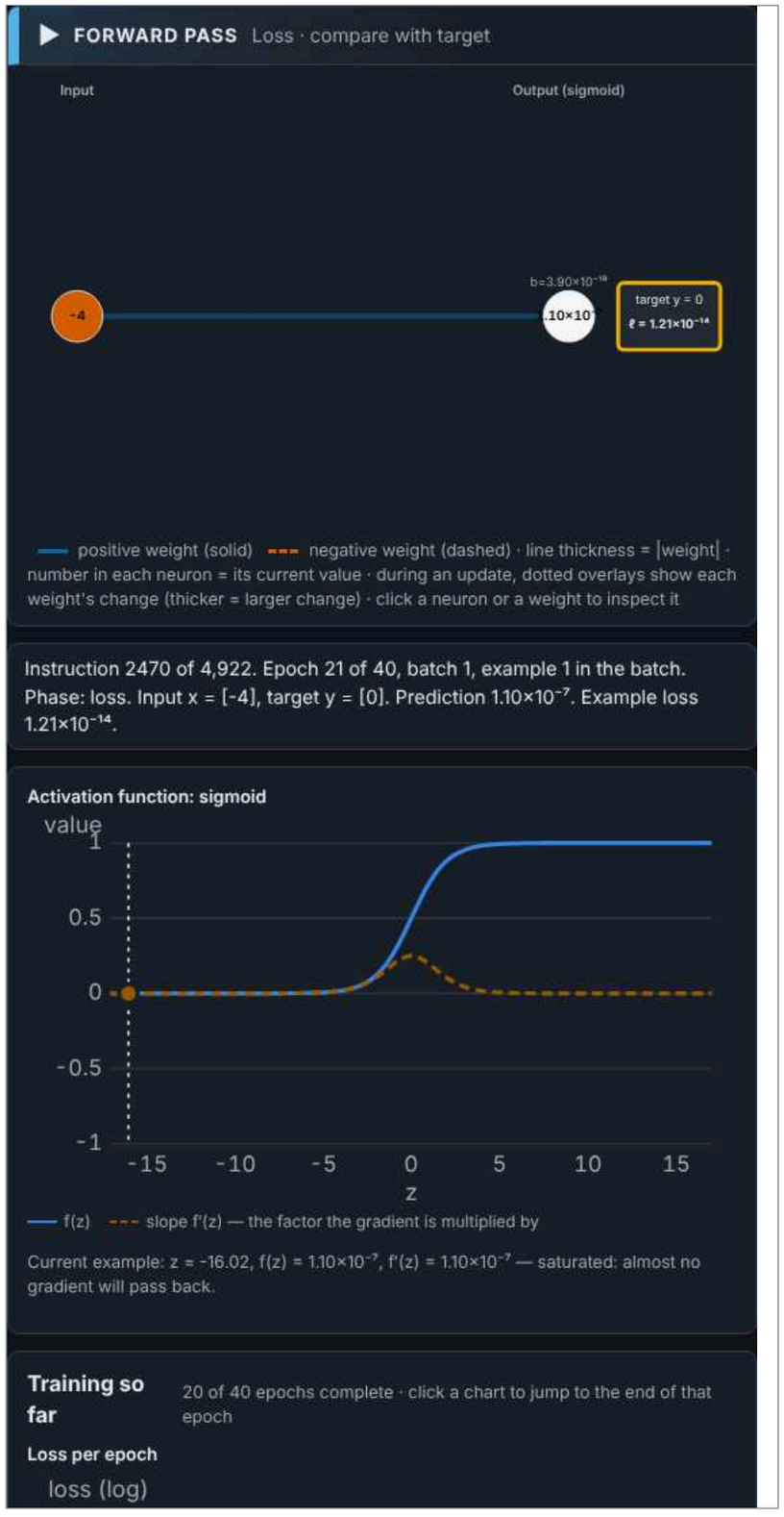


Simulated deuteranopia

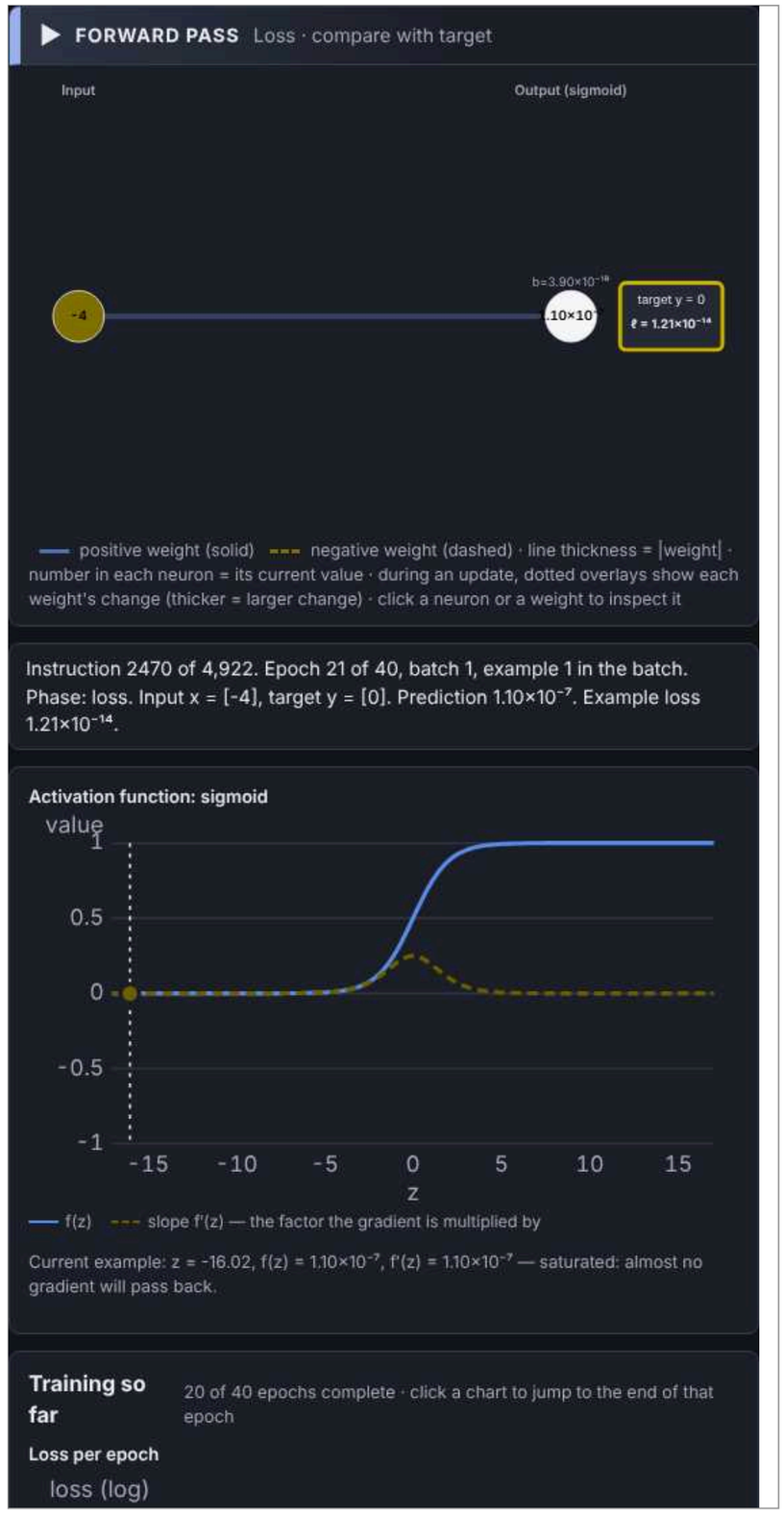


2. Activation functions — dark theme, forward phase (part 1 of 2)

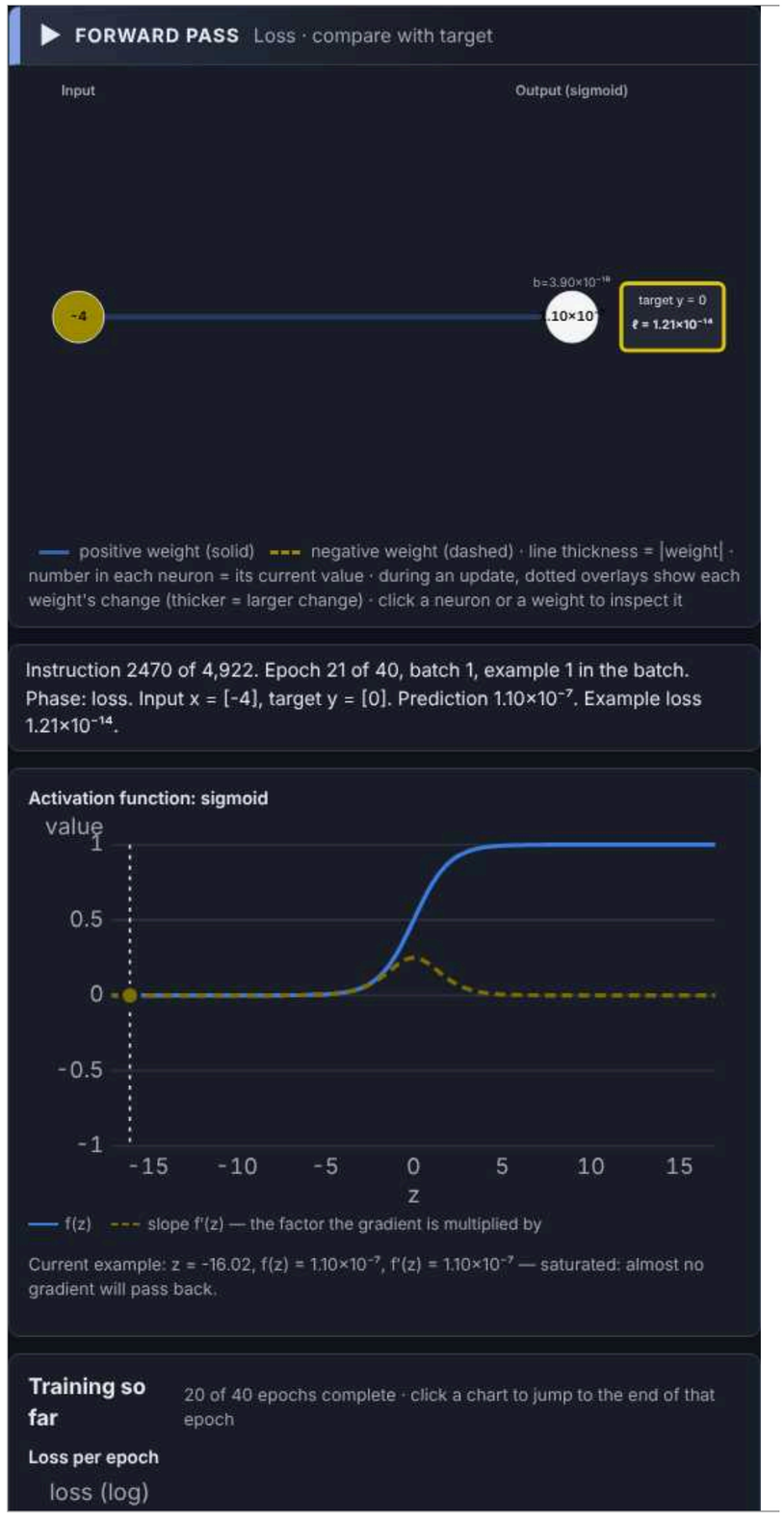
As rendered



Simulated protanopia

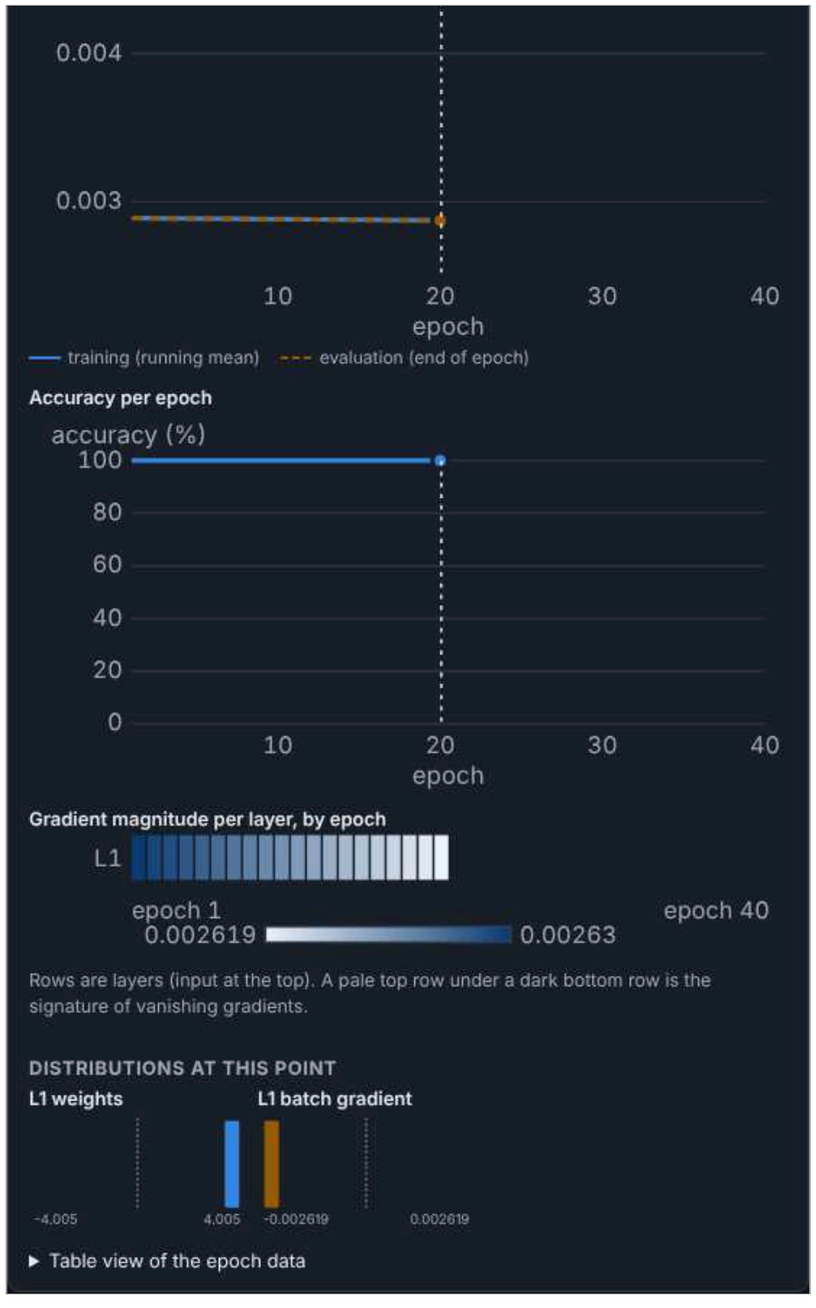


Simulated deuteranopia

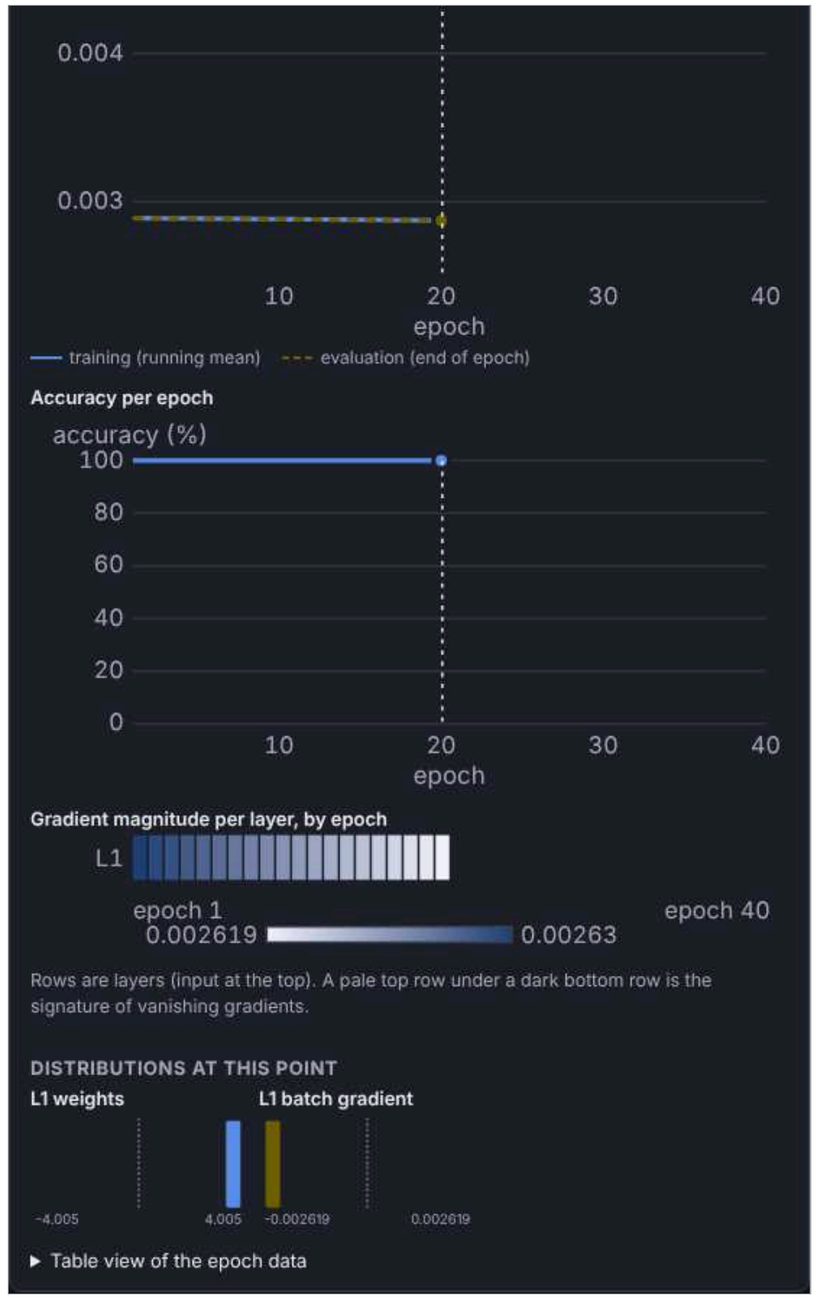


2. Activation functions — dark theme, forward phase (part 2 of 2)

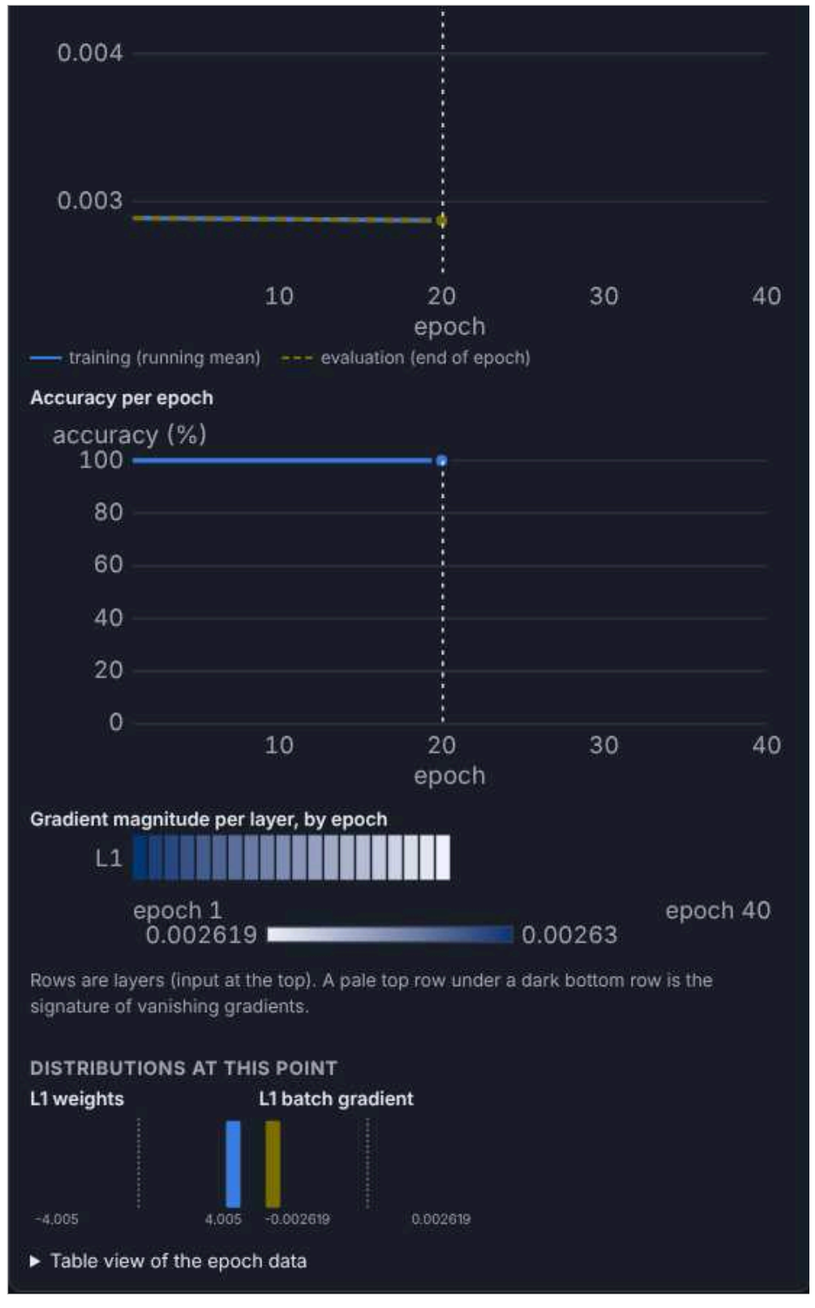
As rendered



Simulated protanopia

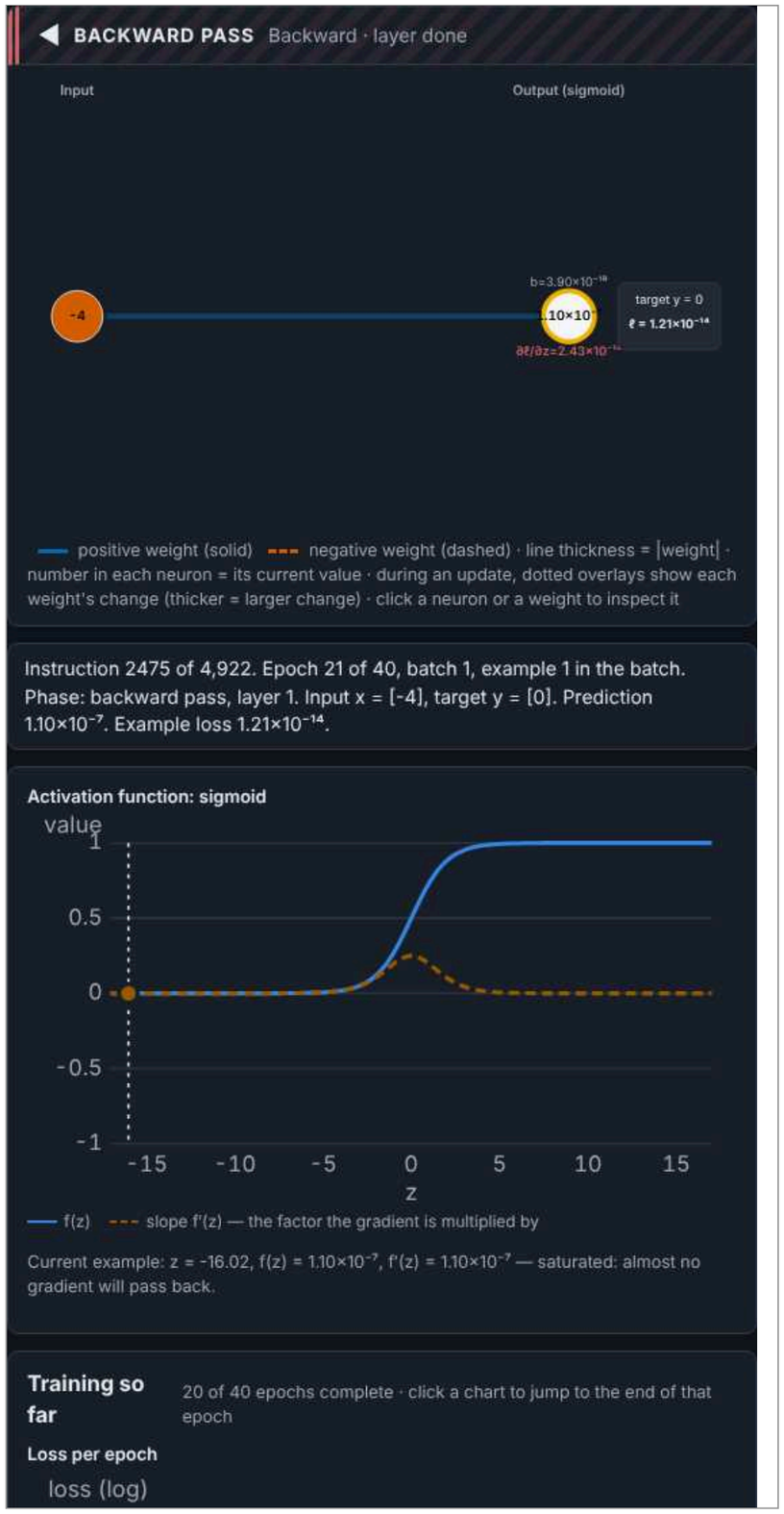


Simulated deuteranopia

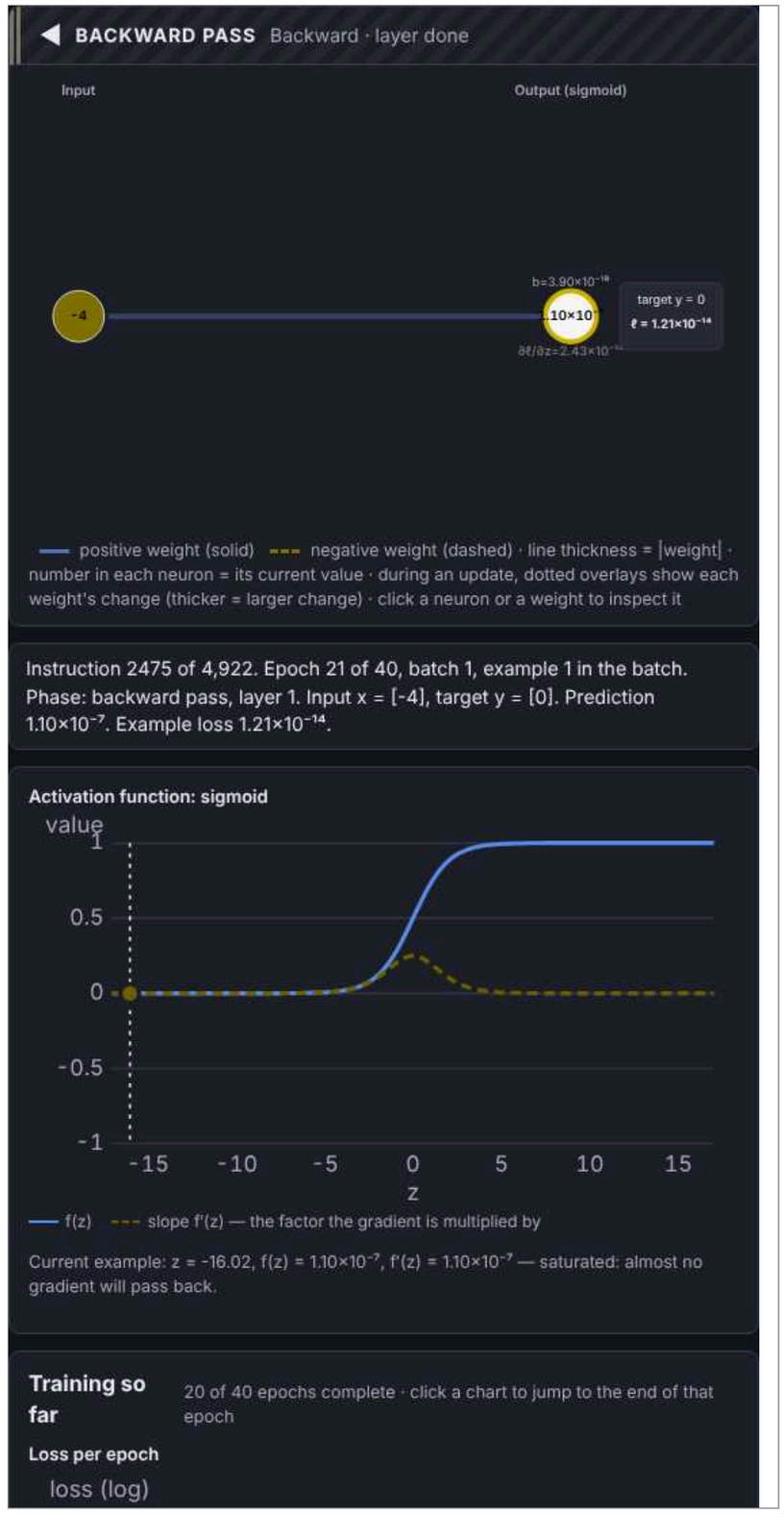


2. Activation functions — dark theme, backward phase (part 1 of 2)

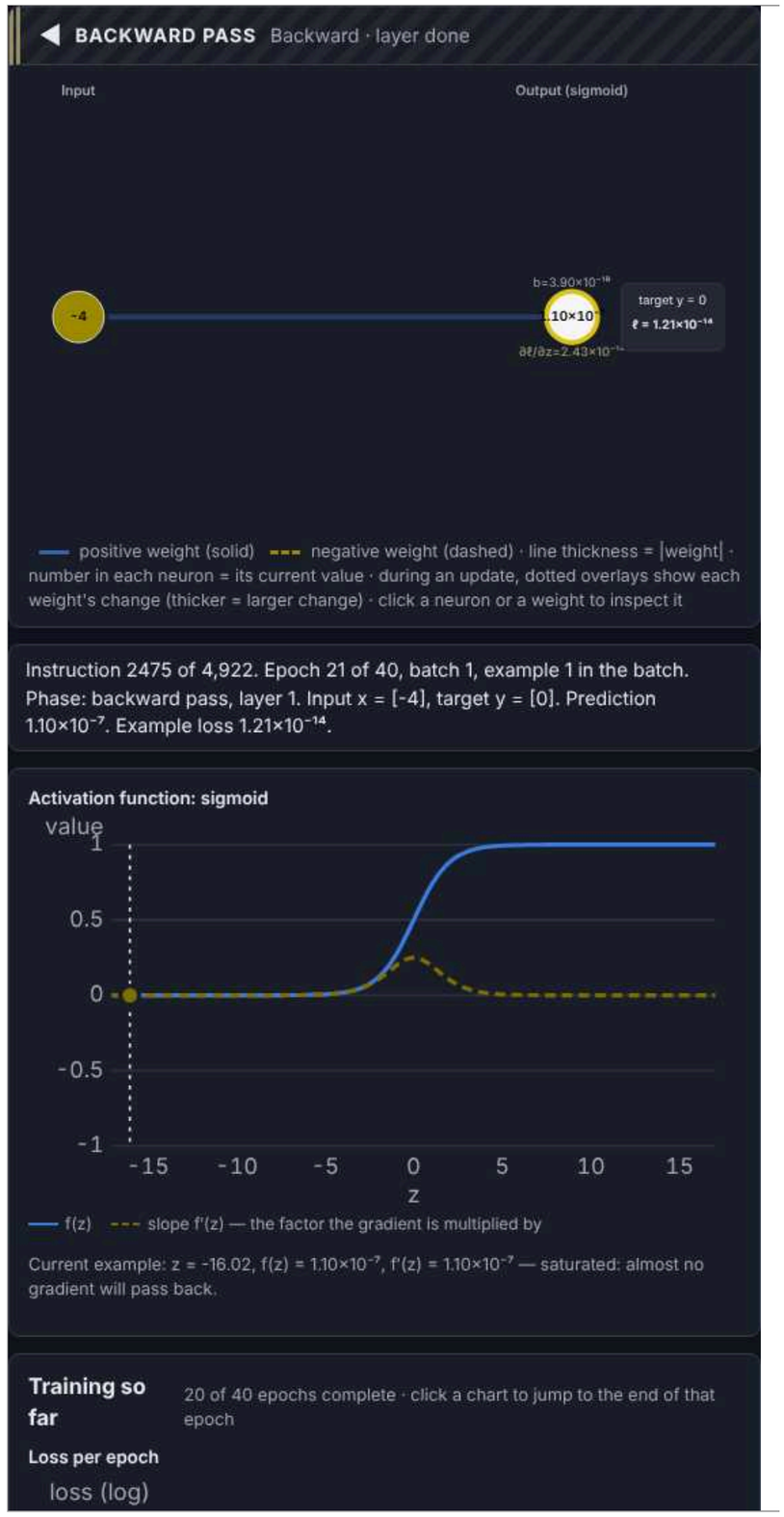
As rendered



Simulated protanopia

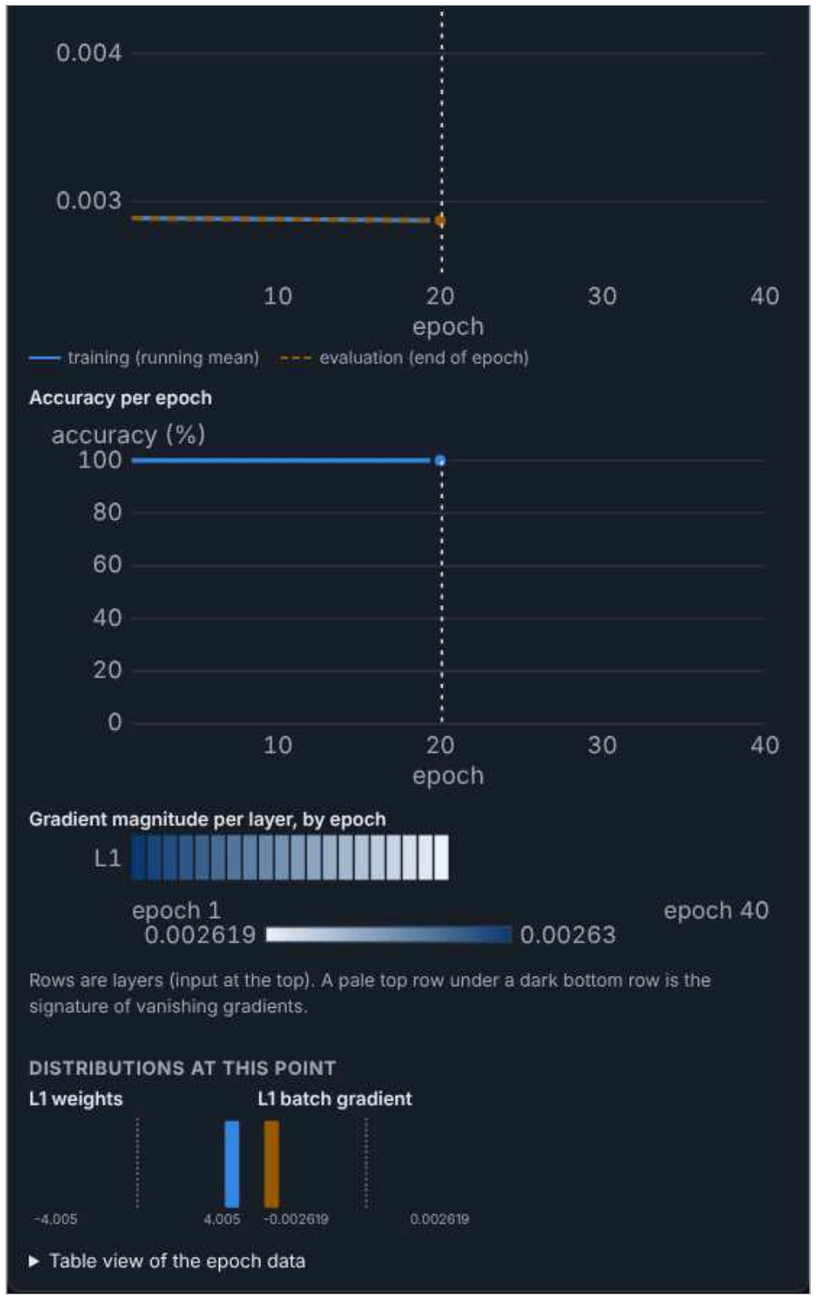


Simulated deuteranopia

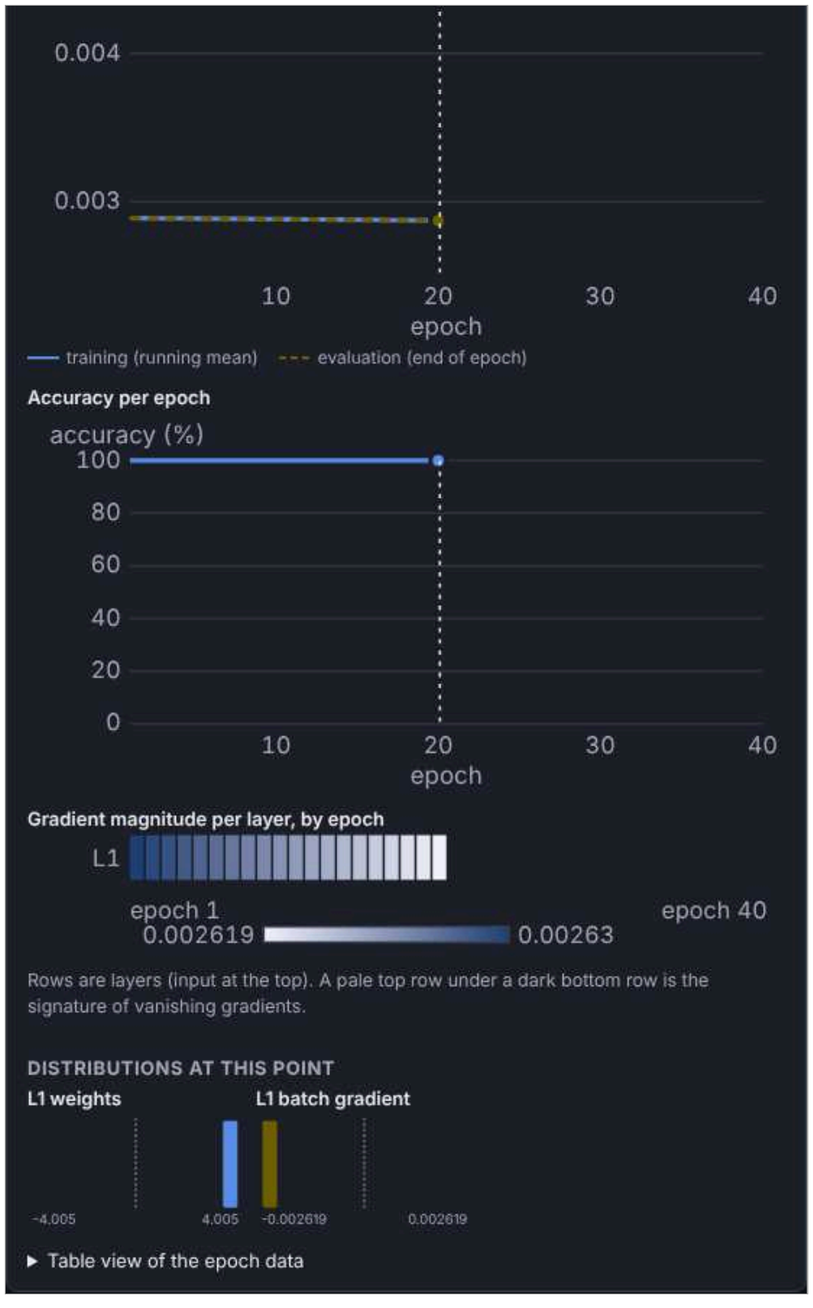


2. Activation functions — dark theme, backward phase (part 2 of 2)

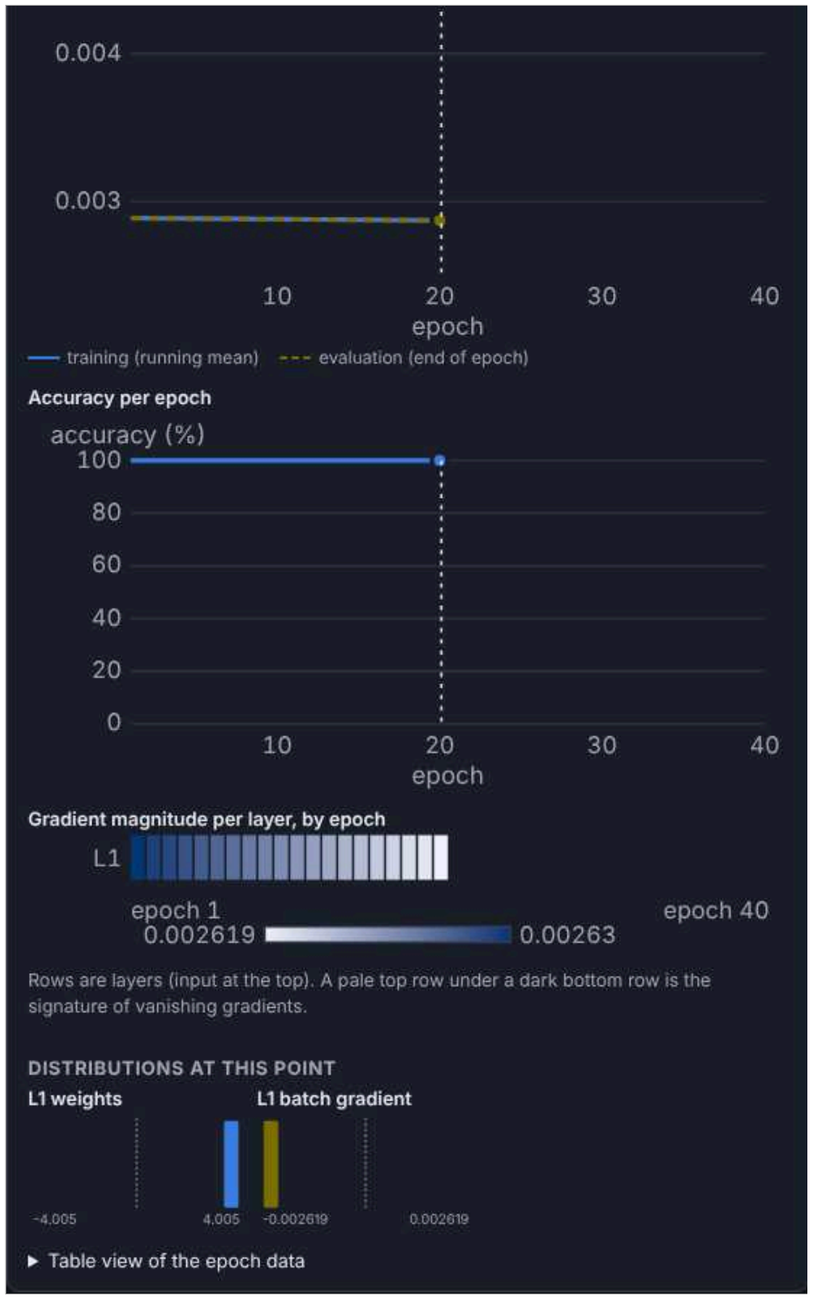
As rendered



Simulated protanopia

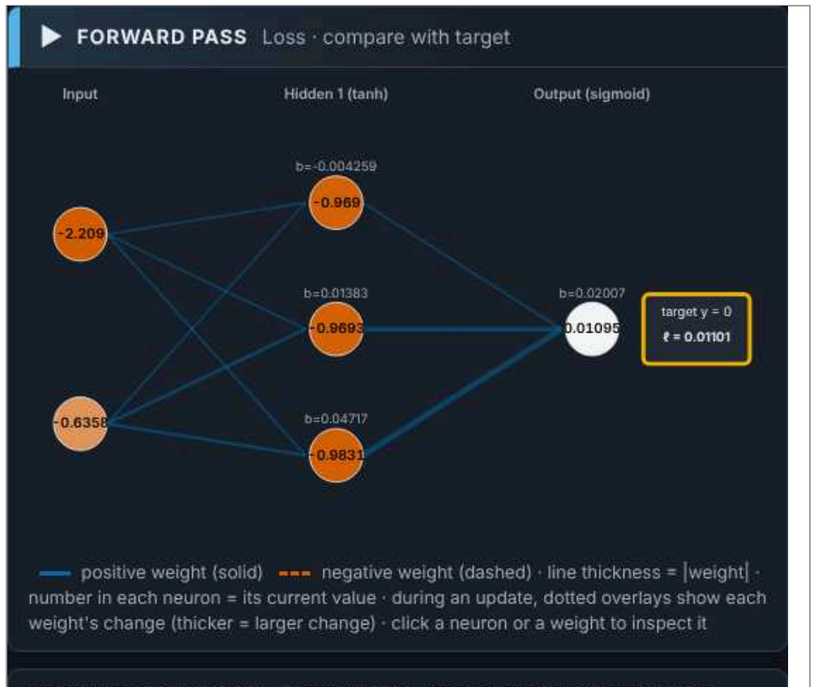


Simulated deuteranopia

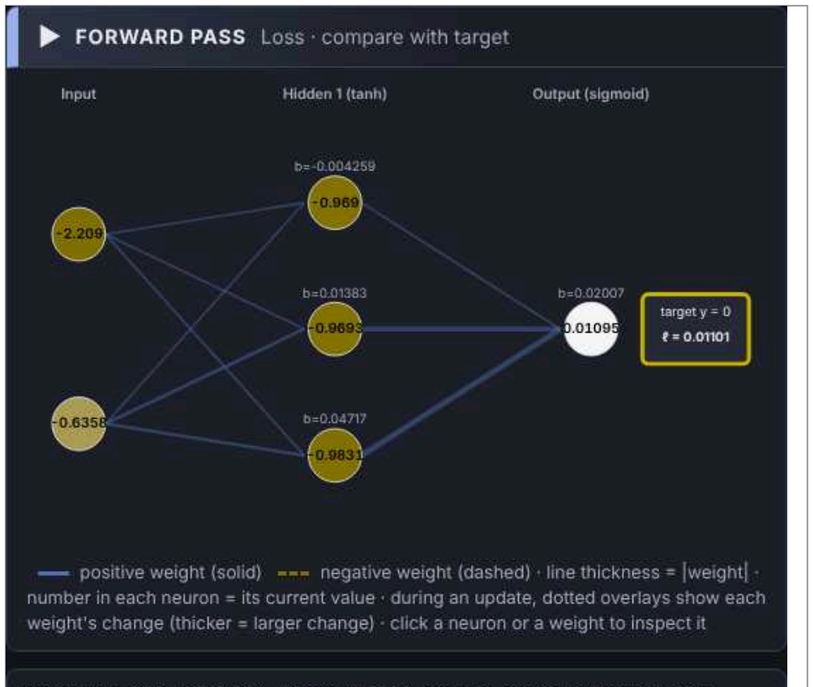


3. Forward propagation — dark theme, forward phase (part 1 of 2)

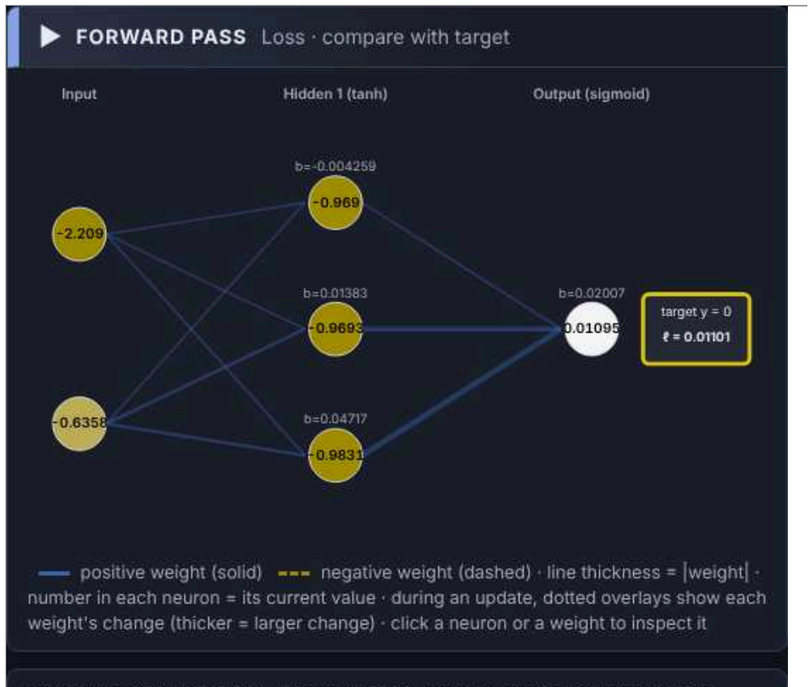
As rendered



Simulated protanopia

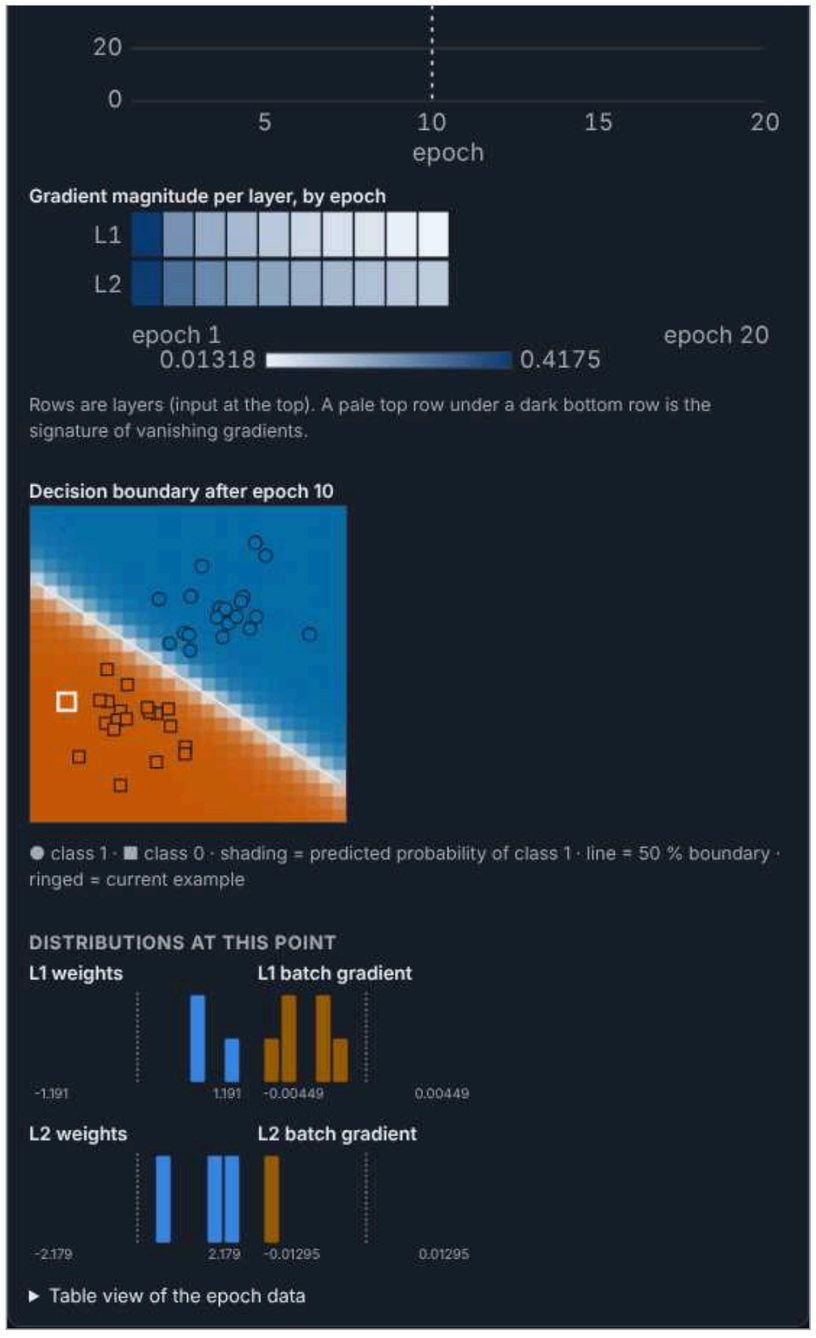


Simulated deuteranopia

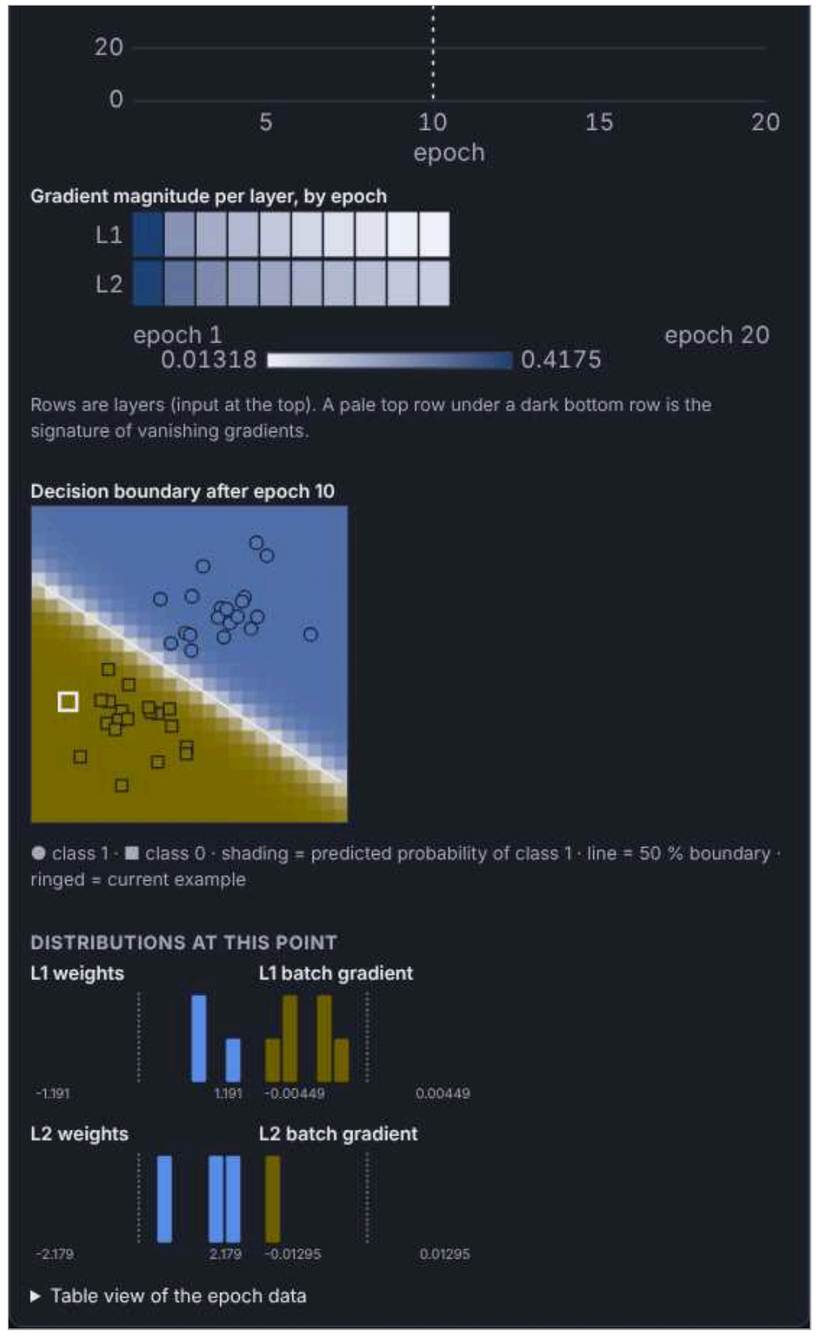


3. Forward propagation — dark theme, forward phase (part 2 of 2)

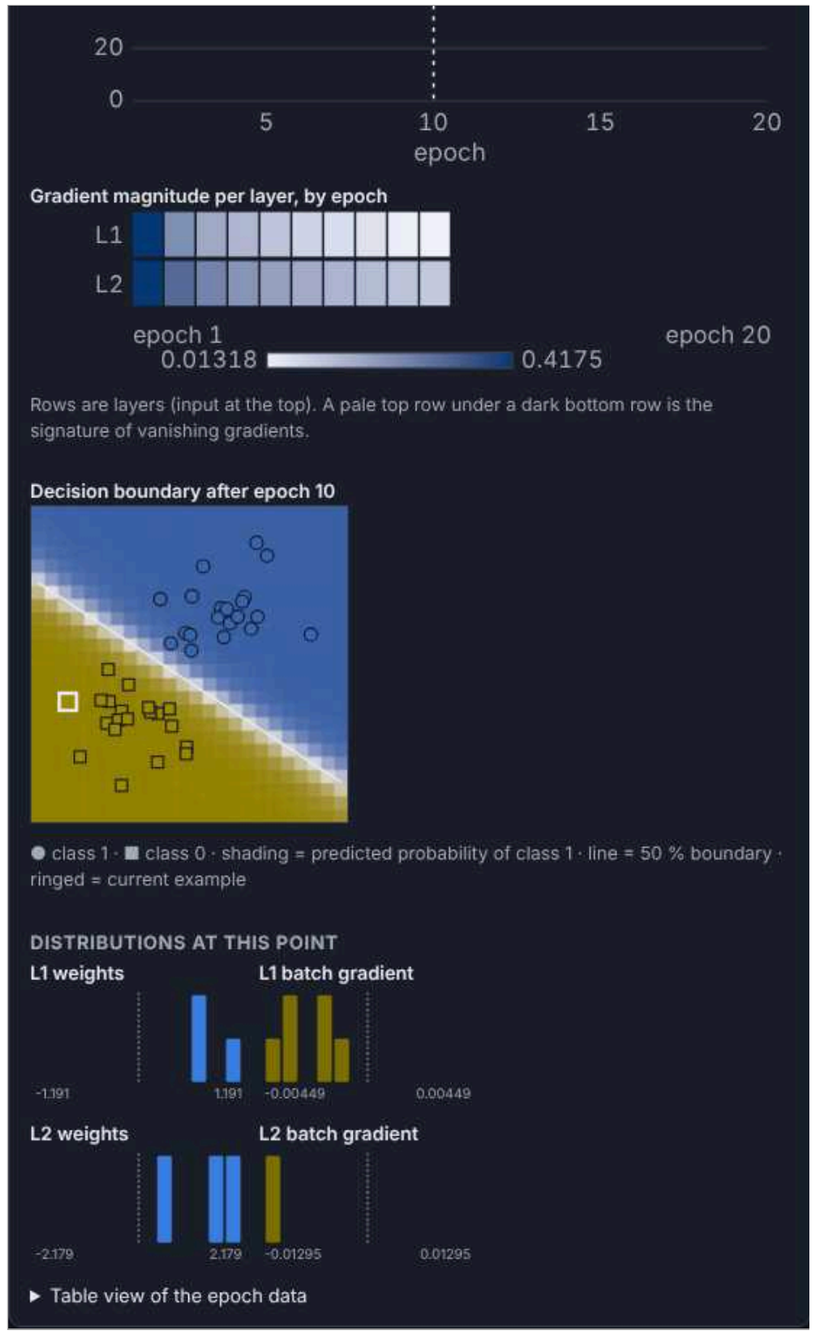
As rendered



Simulated protanopia

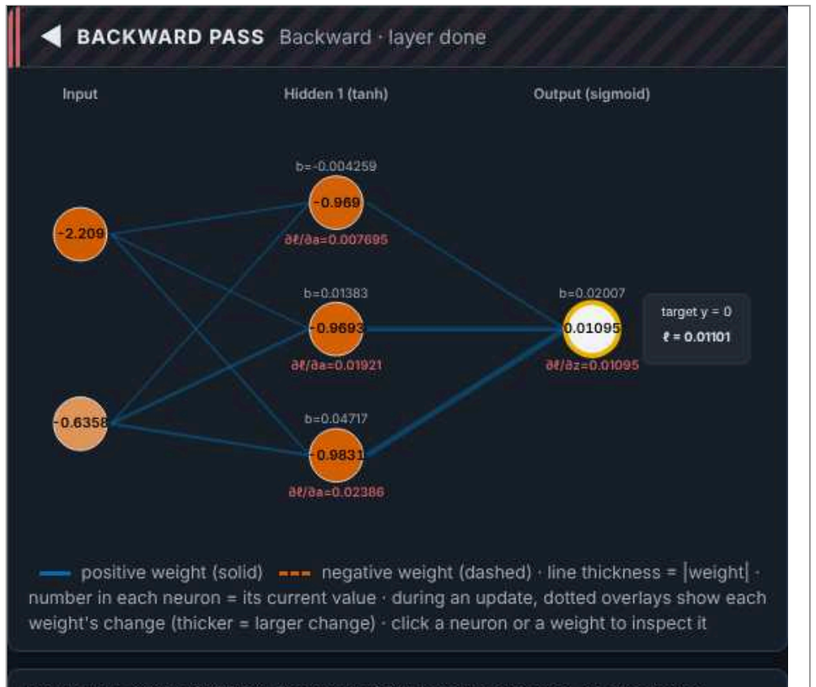


Simulated deuteranopia

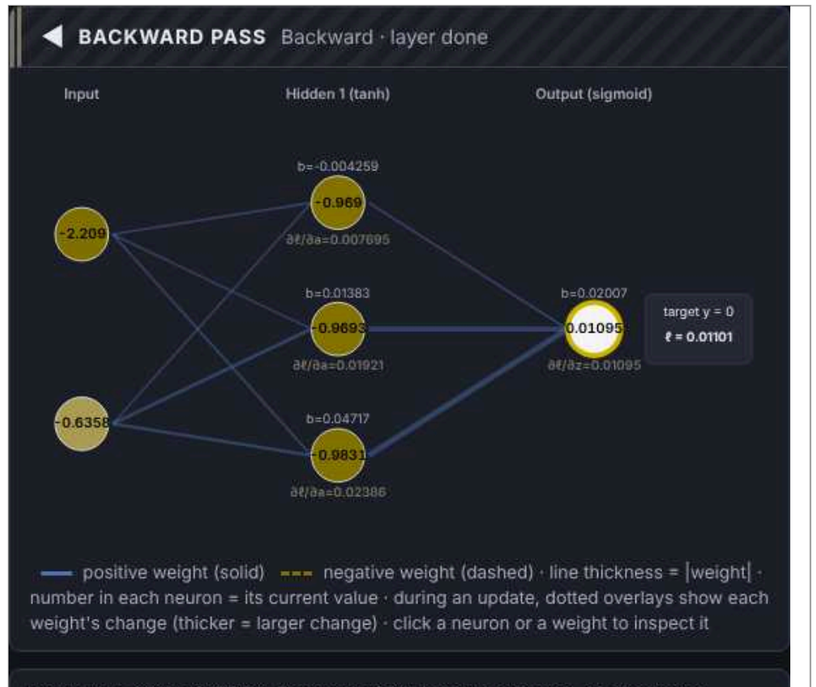


3. Forward propagation — dark theme, backward phase (part 1 of 2)

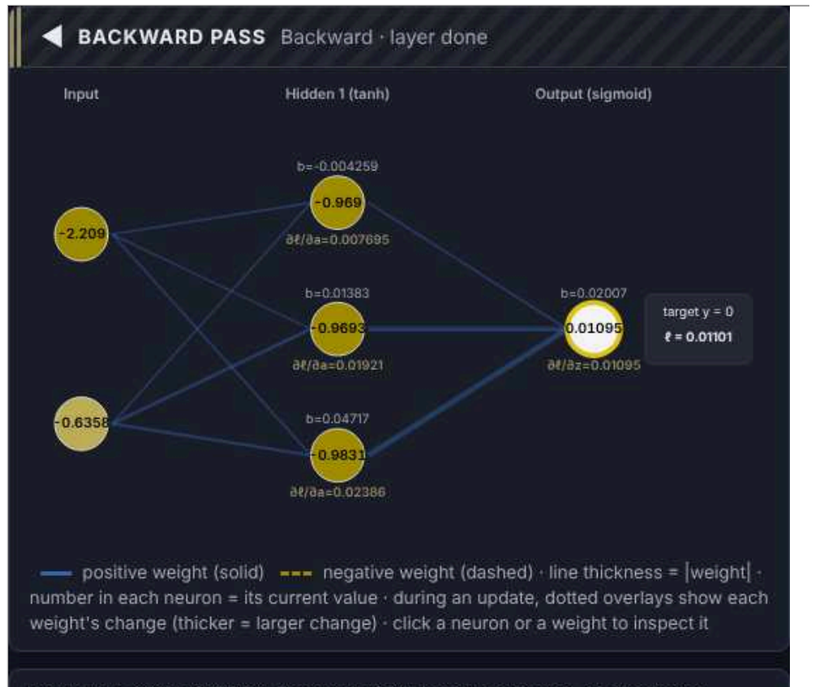
As rendered



Simulated protanopia

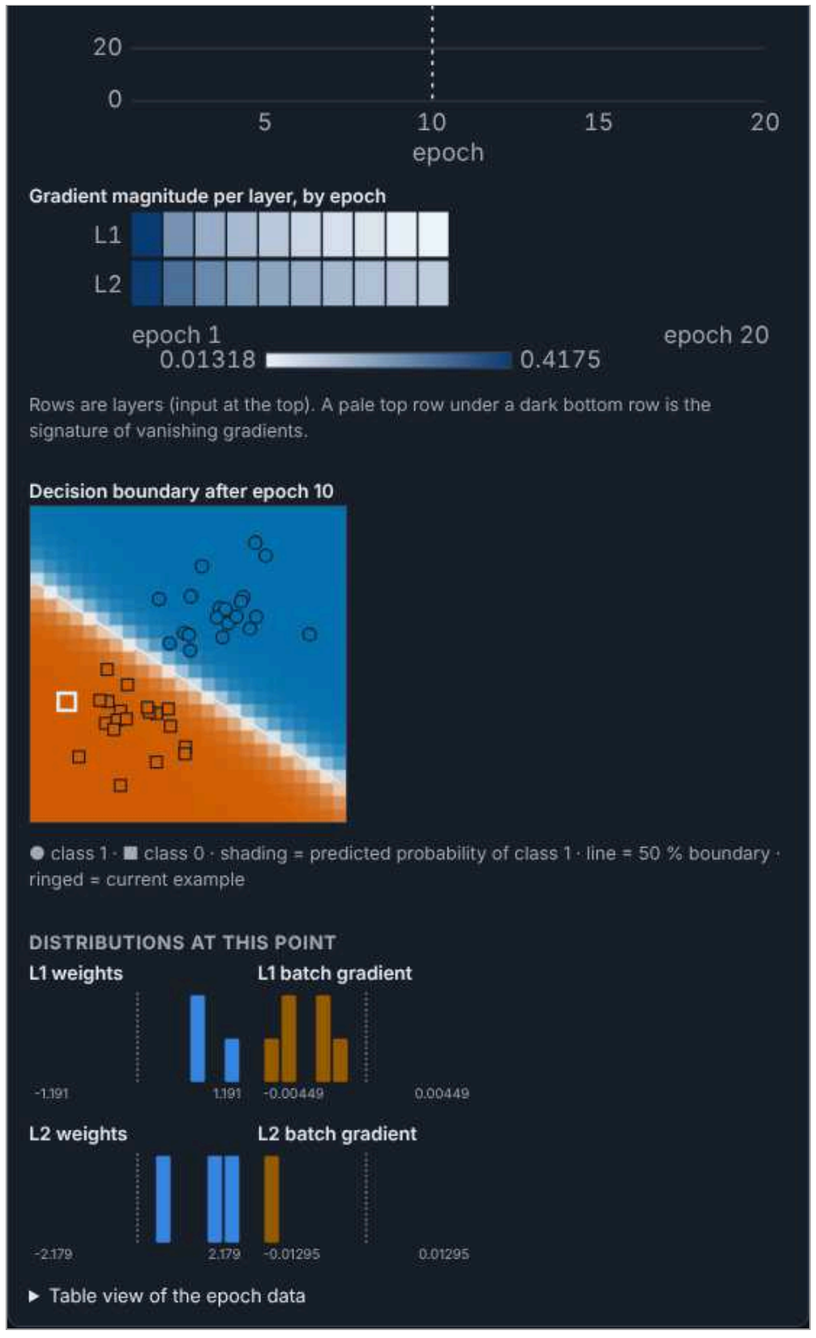


Simulated deuteranopia

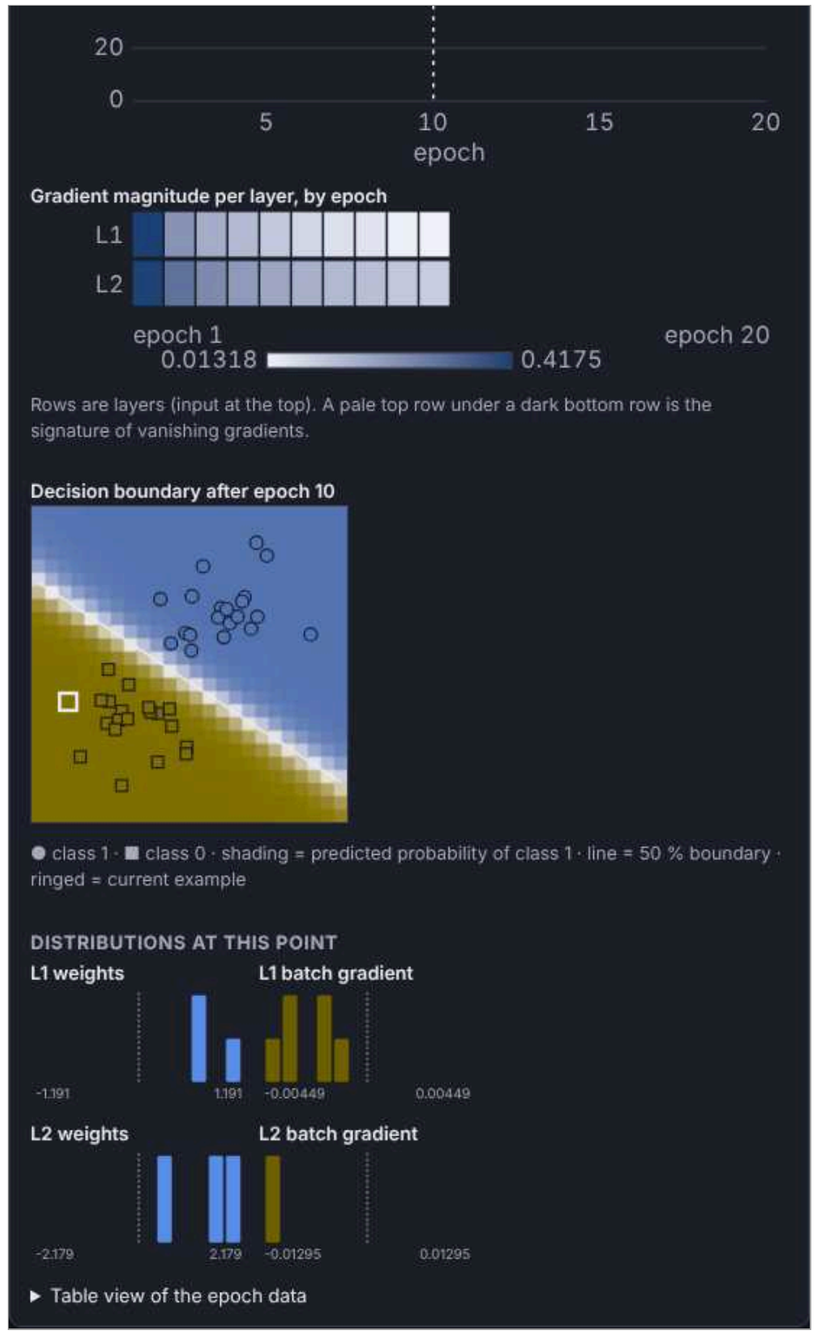


3. Forward propagation — dark theme, backward phase (part 2 of 2)

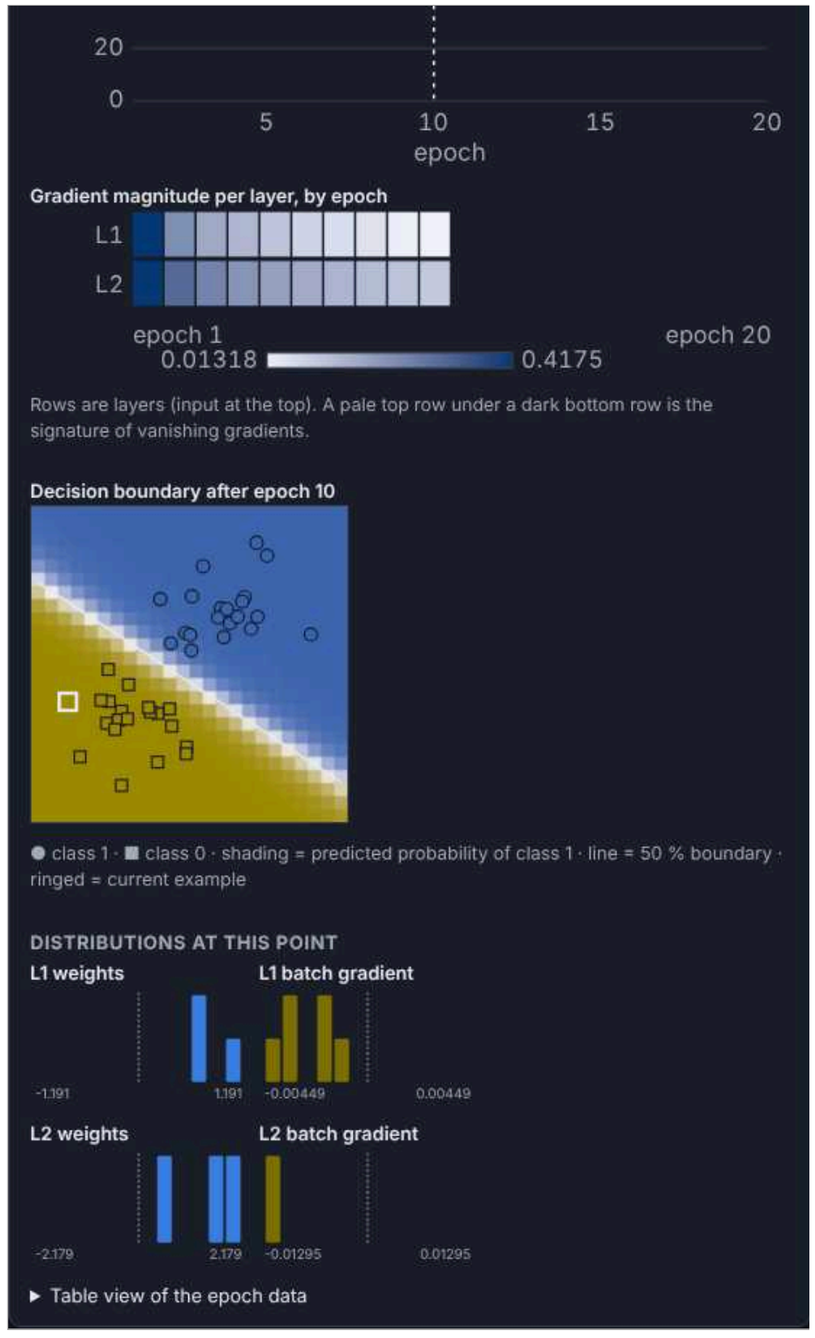
As rendered



Simulated protanopia

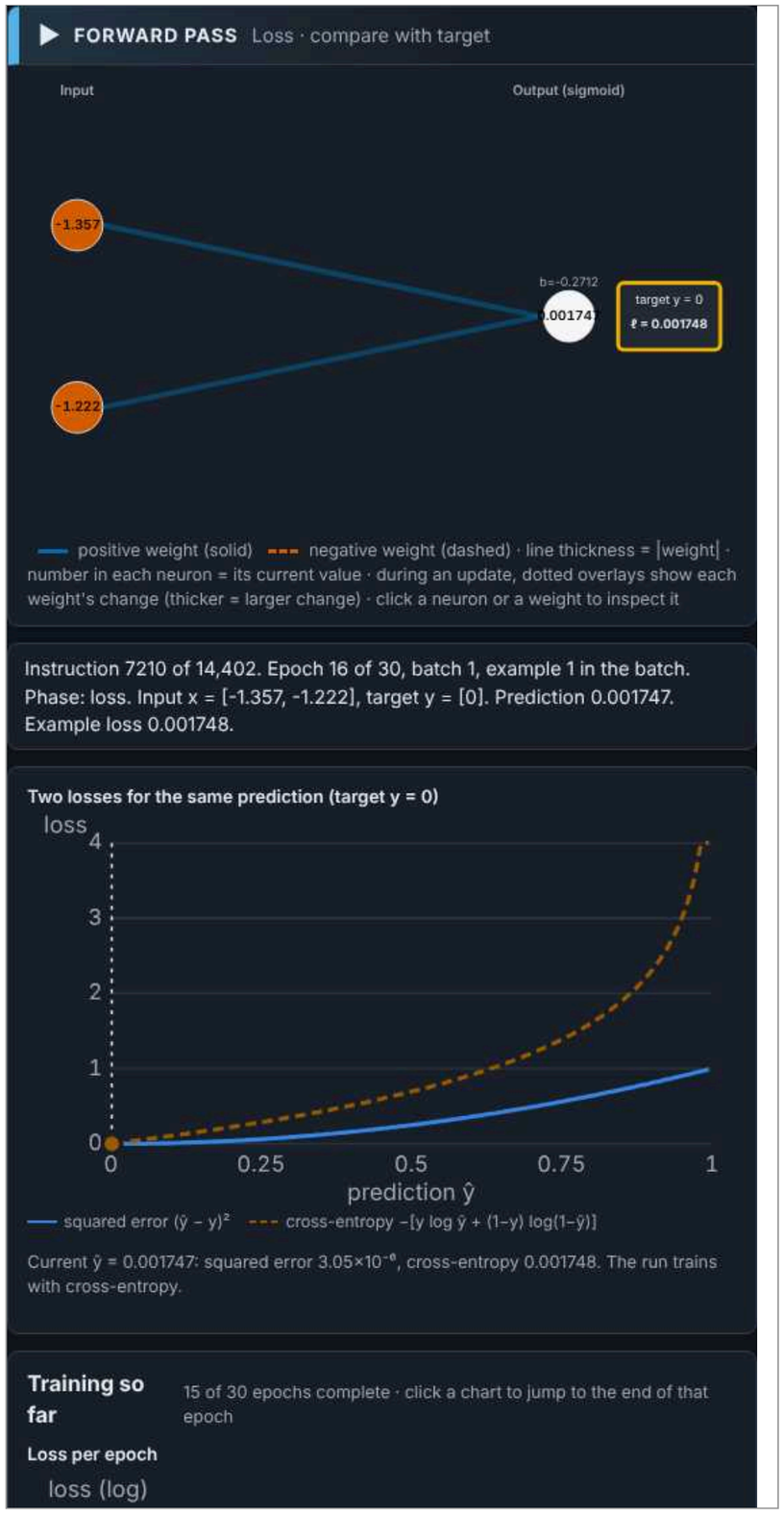


Simulated deuteranopia

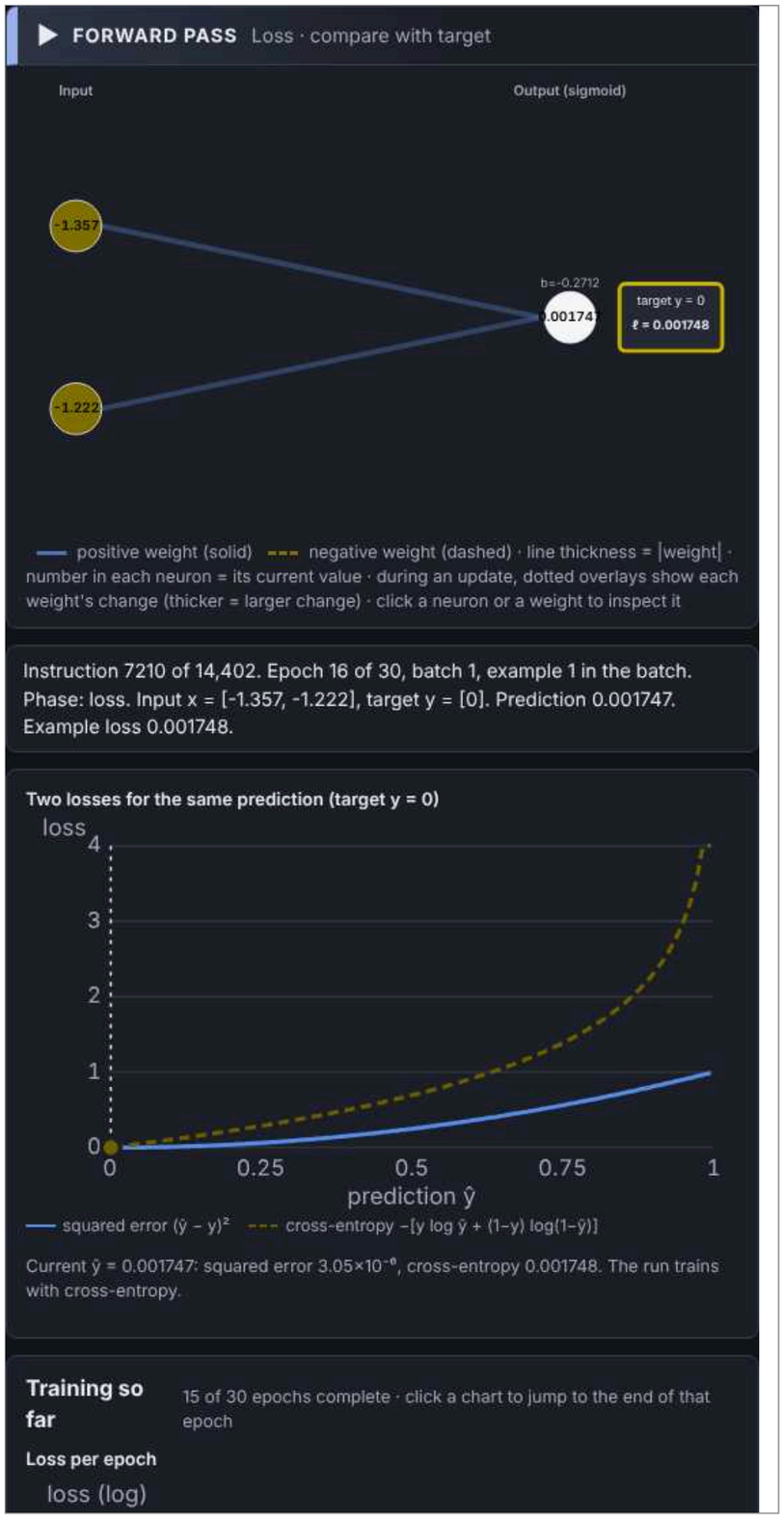


4. Loss functions — dark theme, forward phase (part 1 of 3)

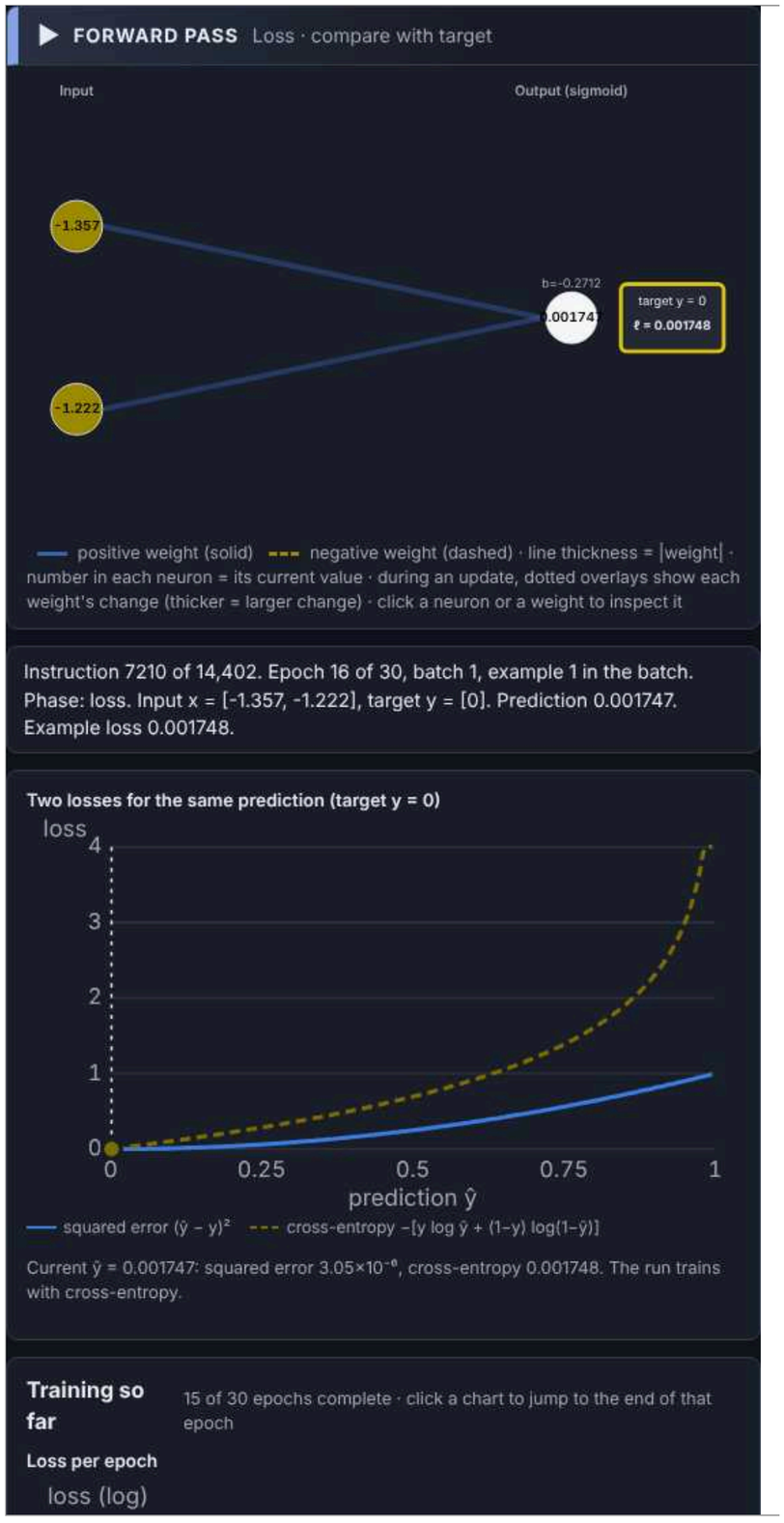
As rendered



Simulated protanopia

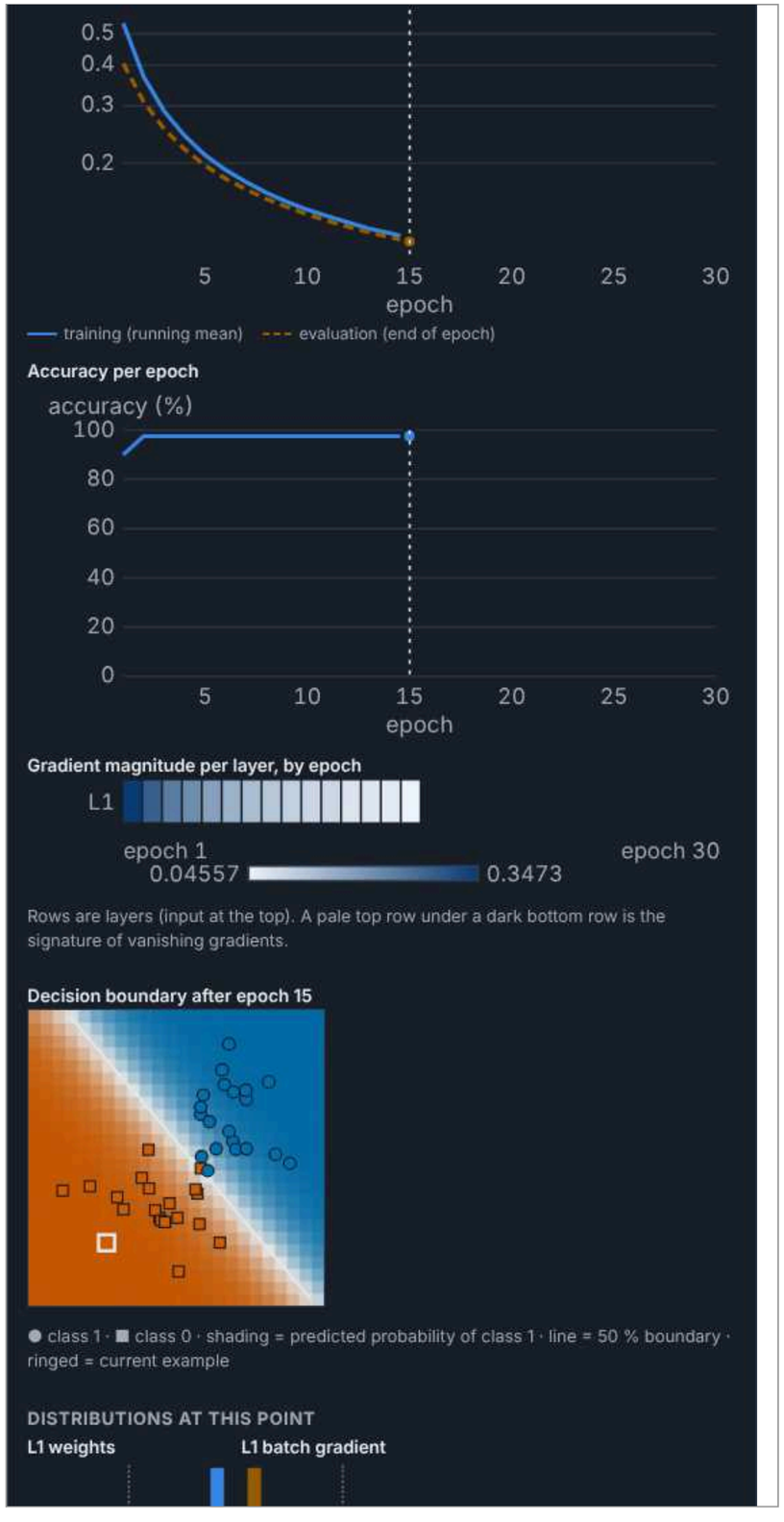


Simulated deuteranopia

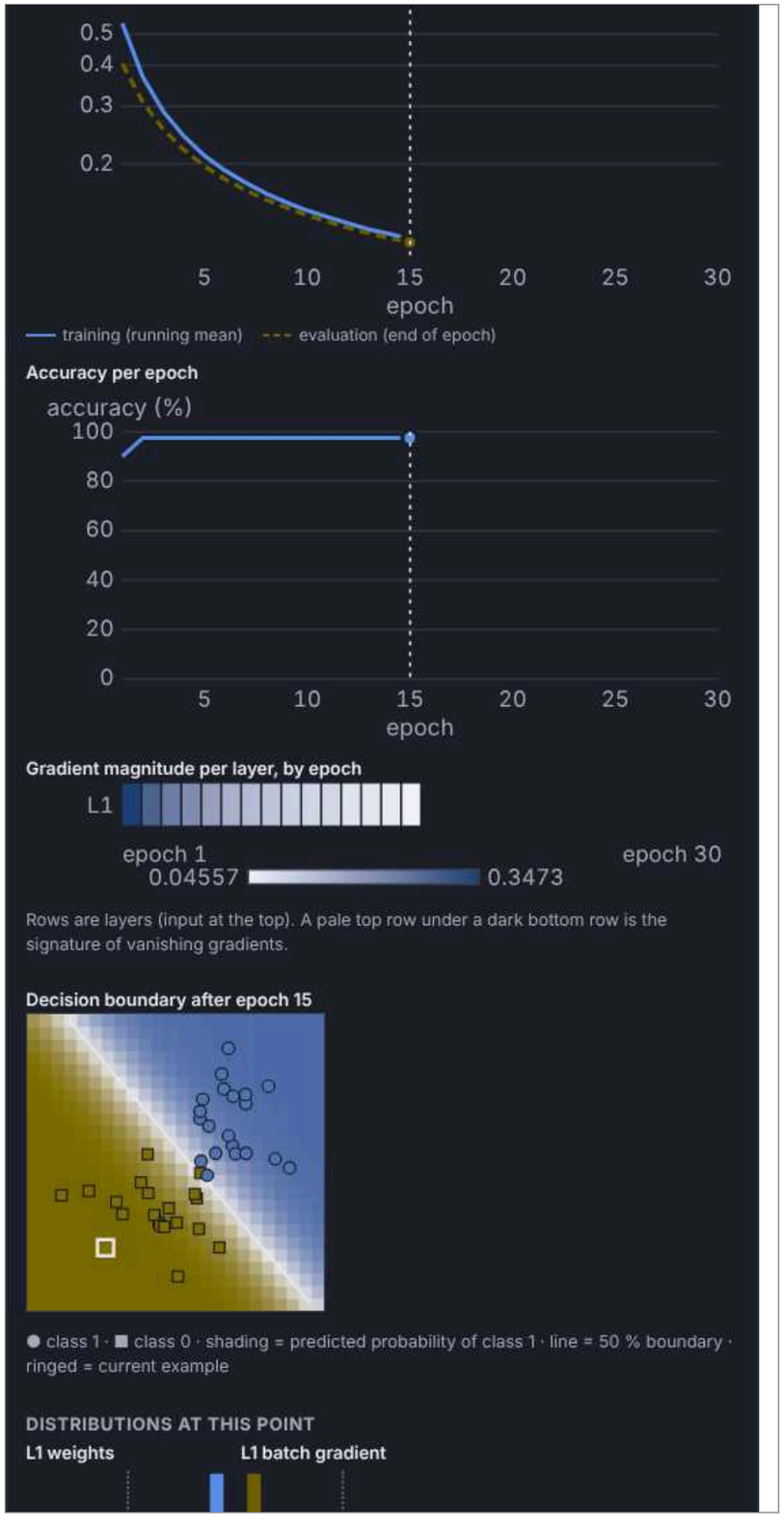


4. Loss functions — dark theme, forward phase (part 2 of 3)

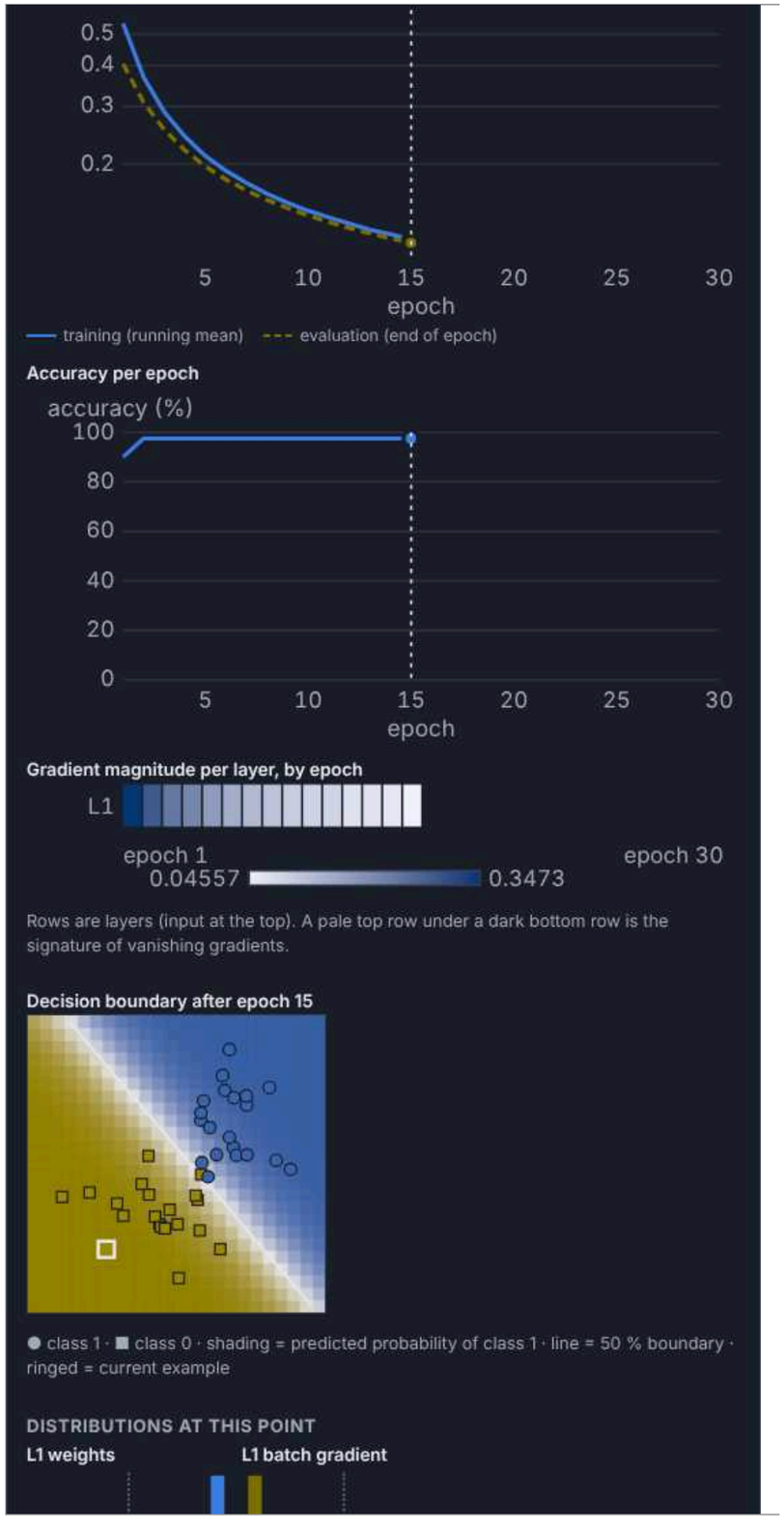
As rendered



Simulated protanopia



Simulated deuteranopia



4. Loss functions — dark theme, forward phase (part 3 of 3)

As rendered



Simulated protanopia

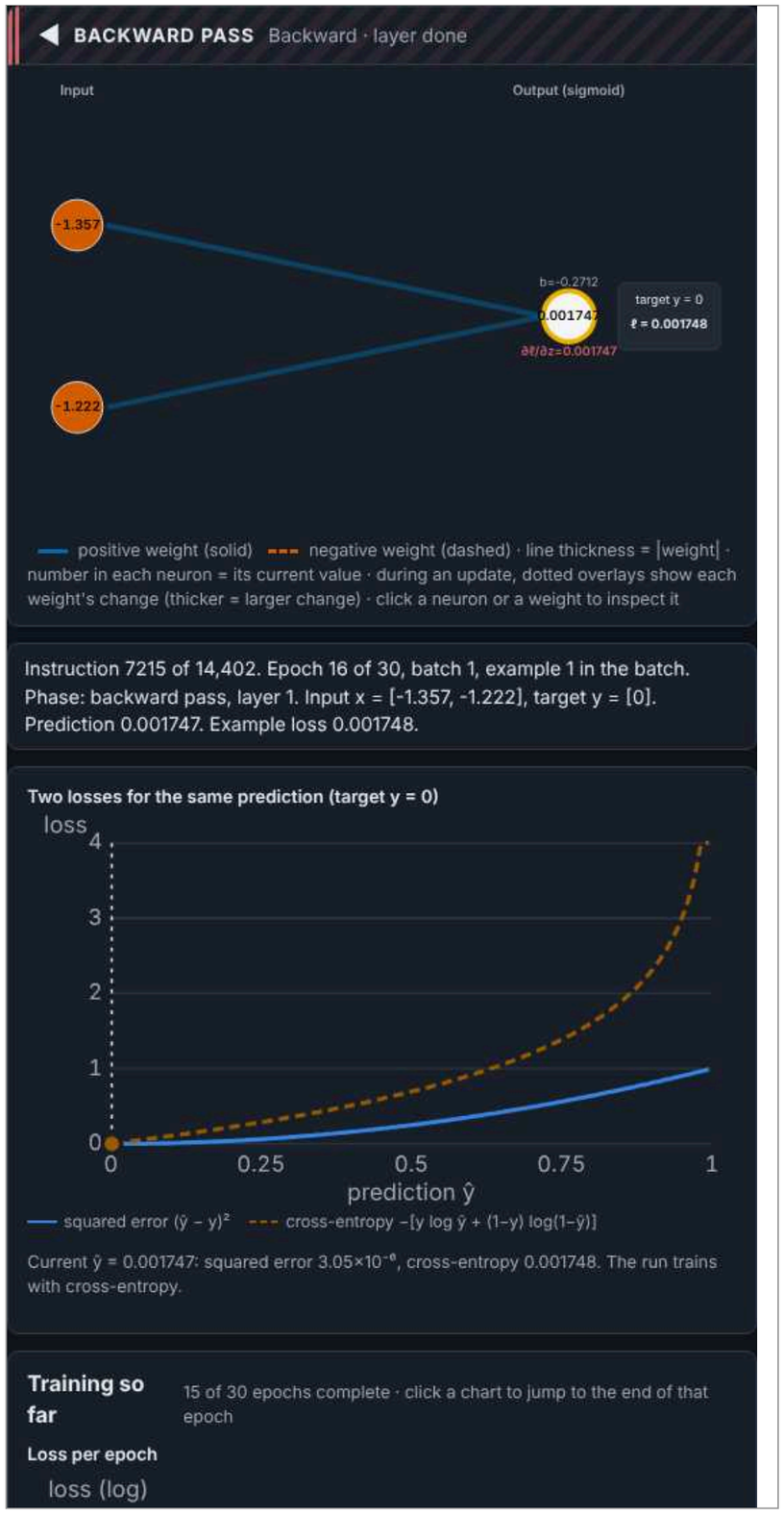


Simulated deuteranopia

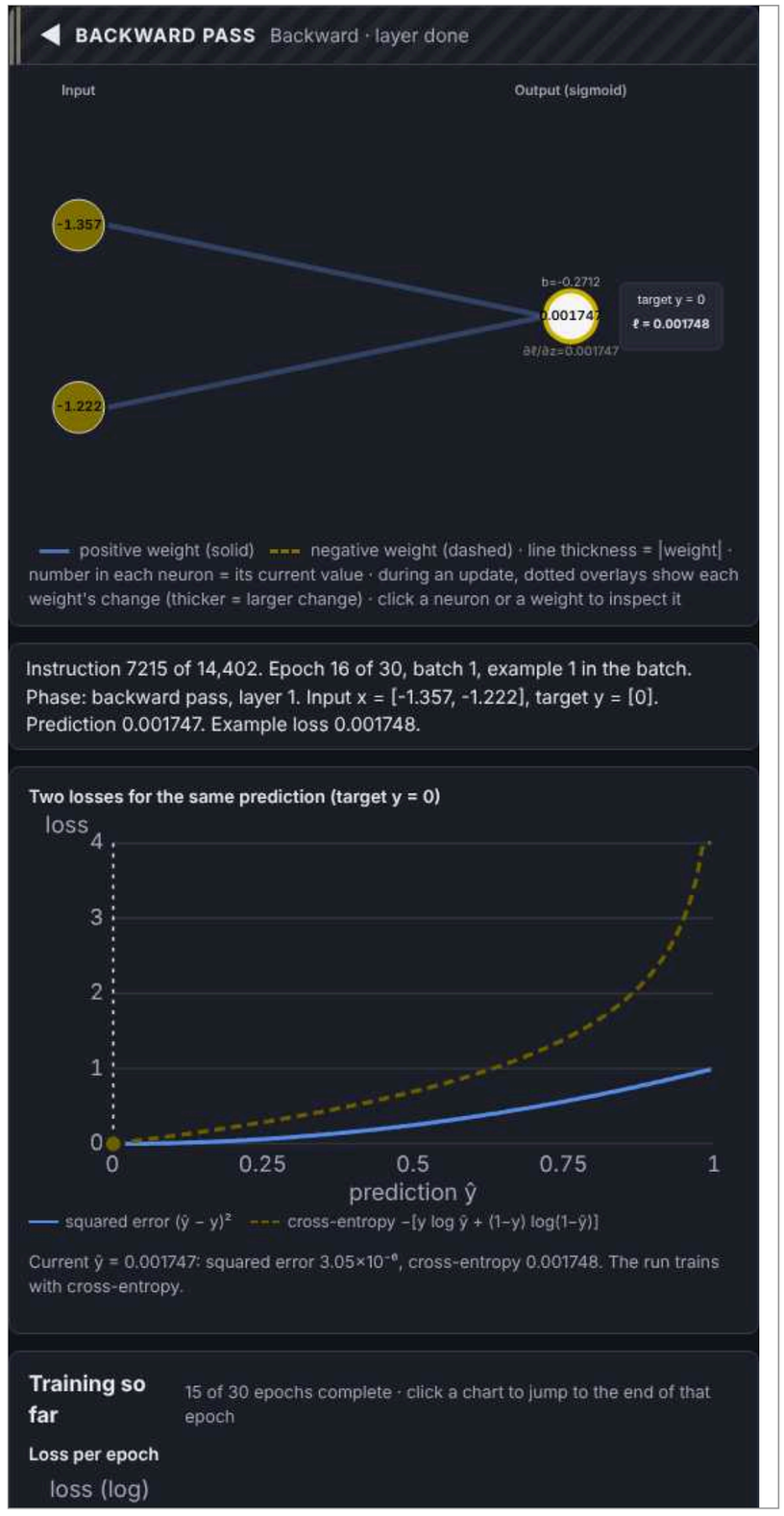


4. Loss functions — dark theme, backward phase (part 1 of 3)

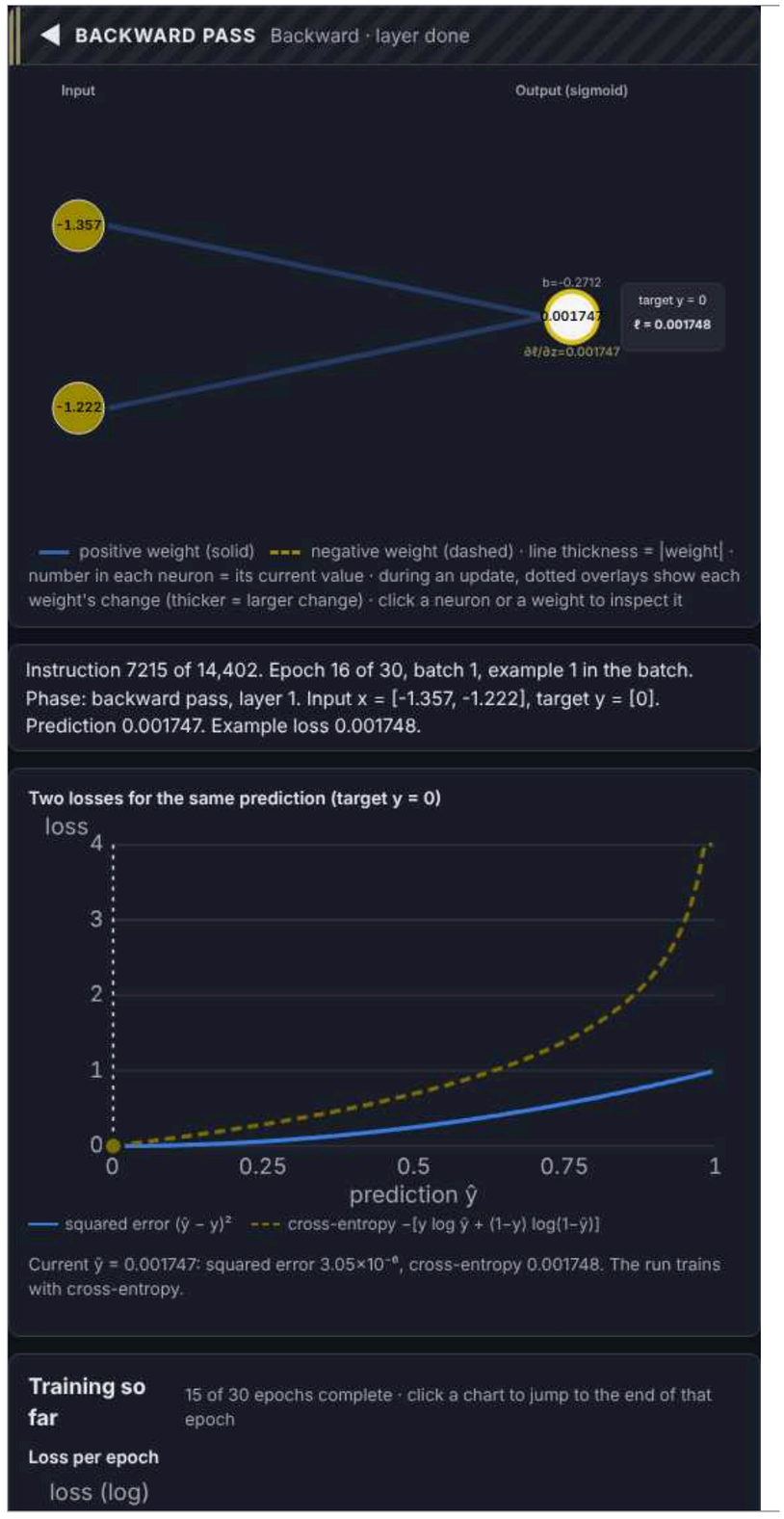
As rendered



Simulated protanopia

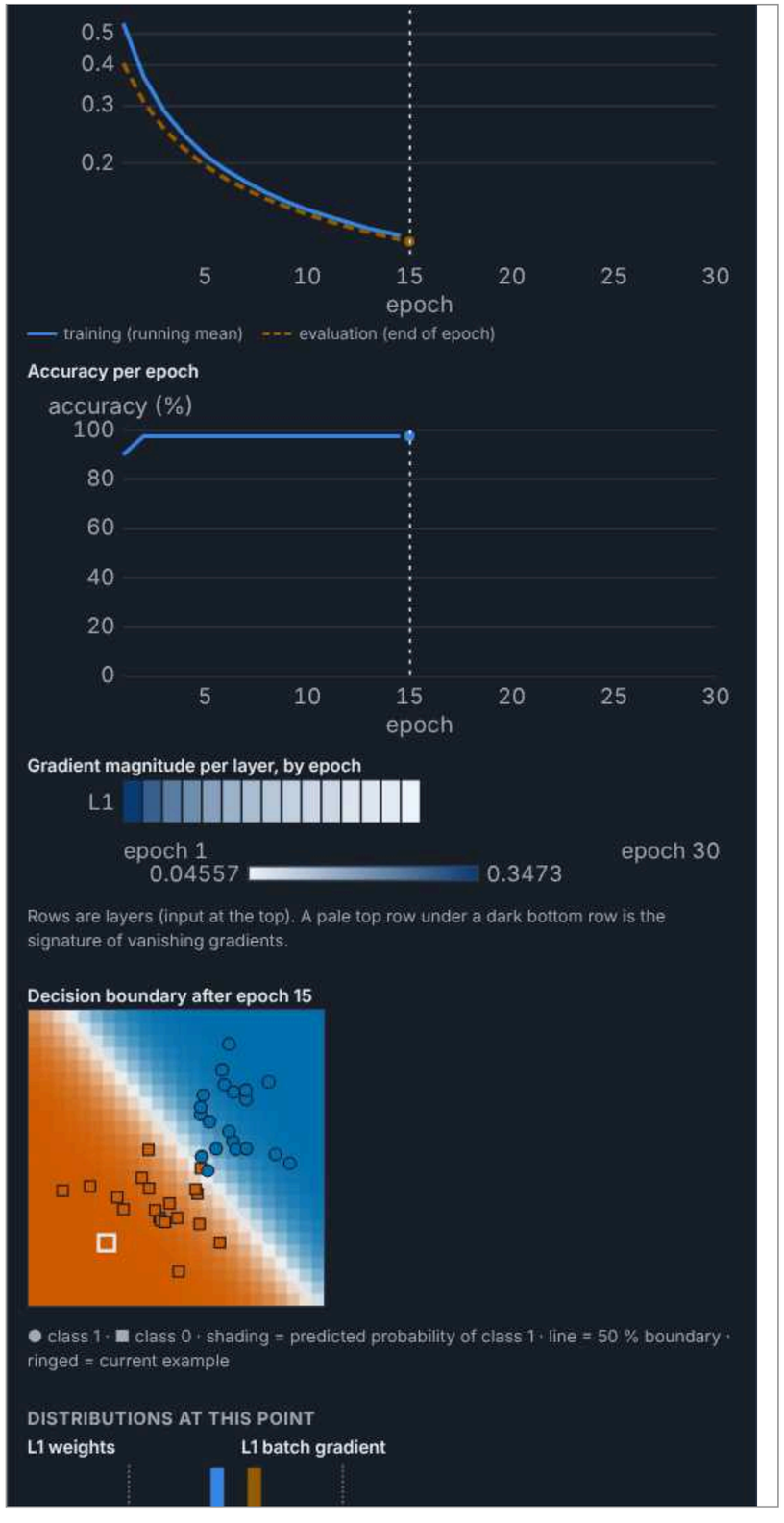


Simulated deuteranopia

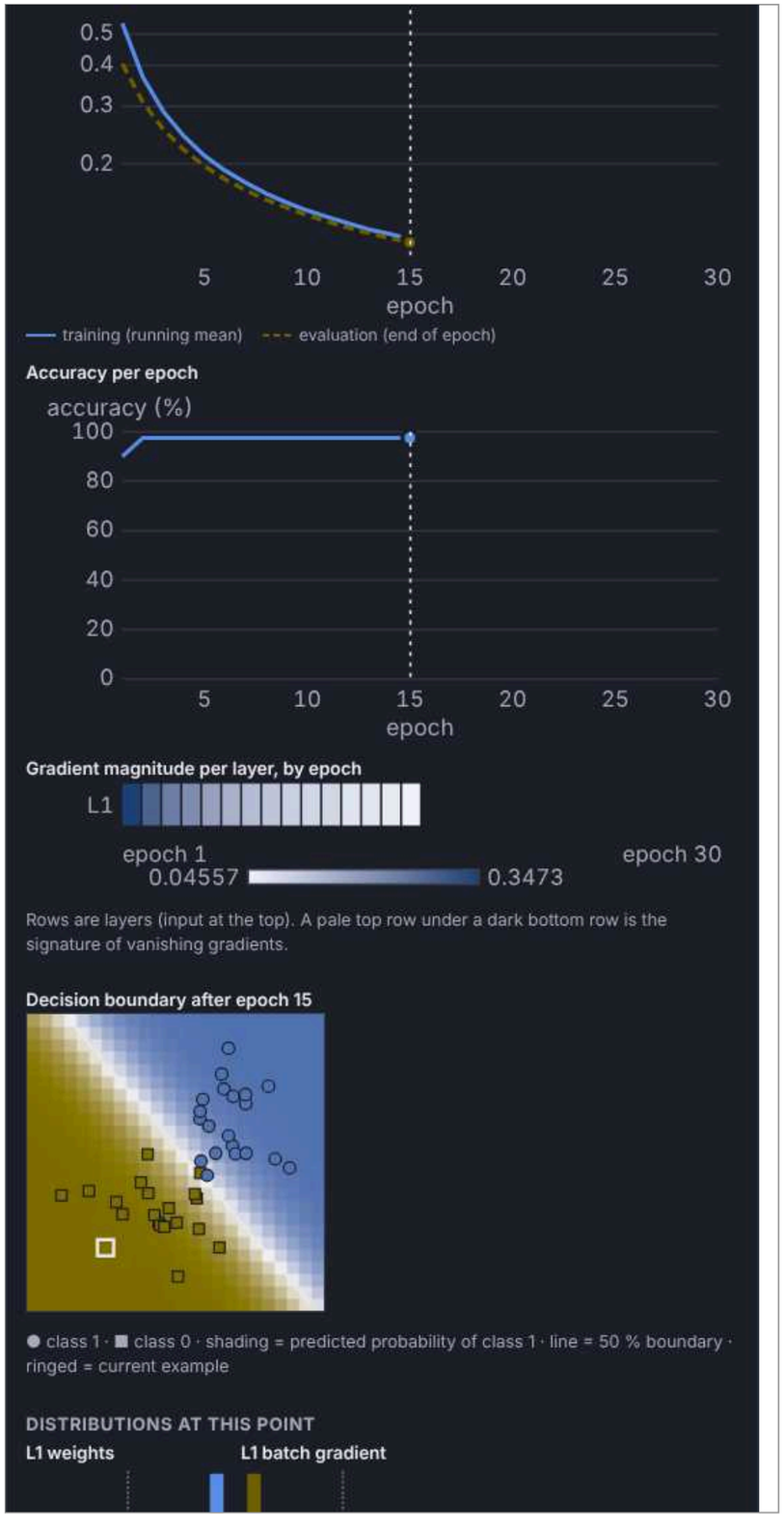


4. Loss functions — dark theme, backward phase (part 2 of 3)

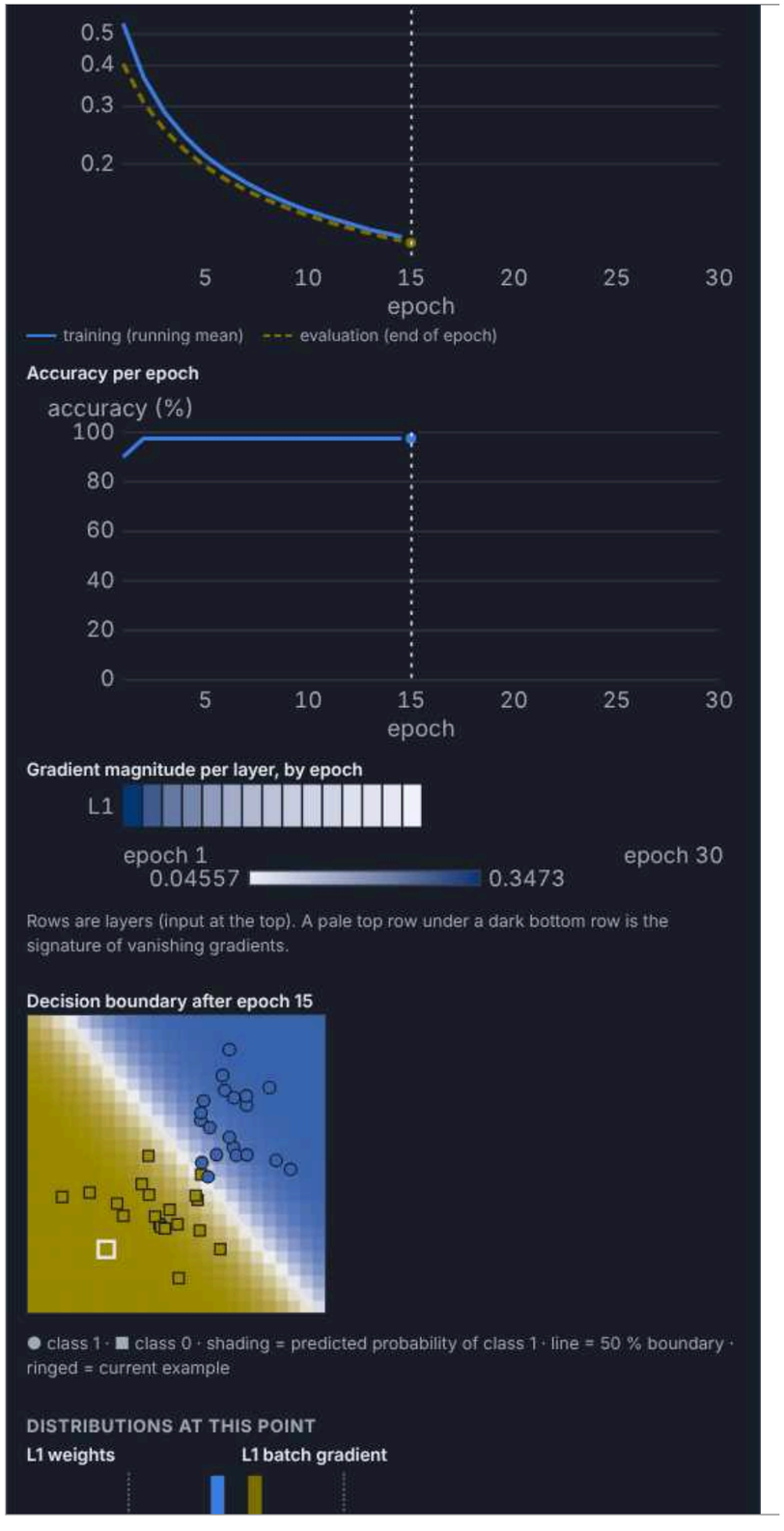
As rendered



Simulated protanopia



Simulated deuteranopia



4. Loss functions — dark theme, backward phase (part 3 of 3)

As rendered



Simulated protanopia

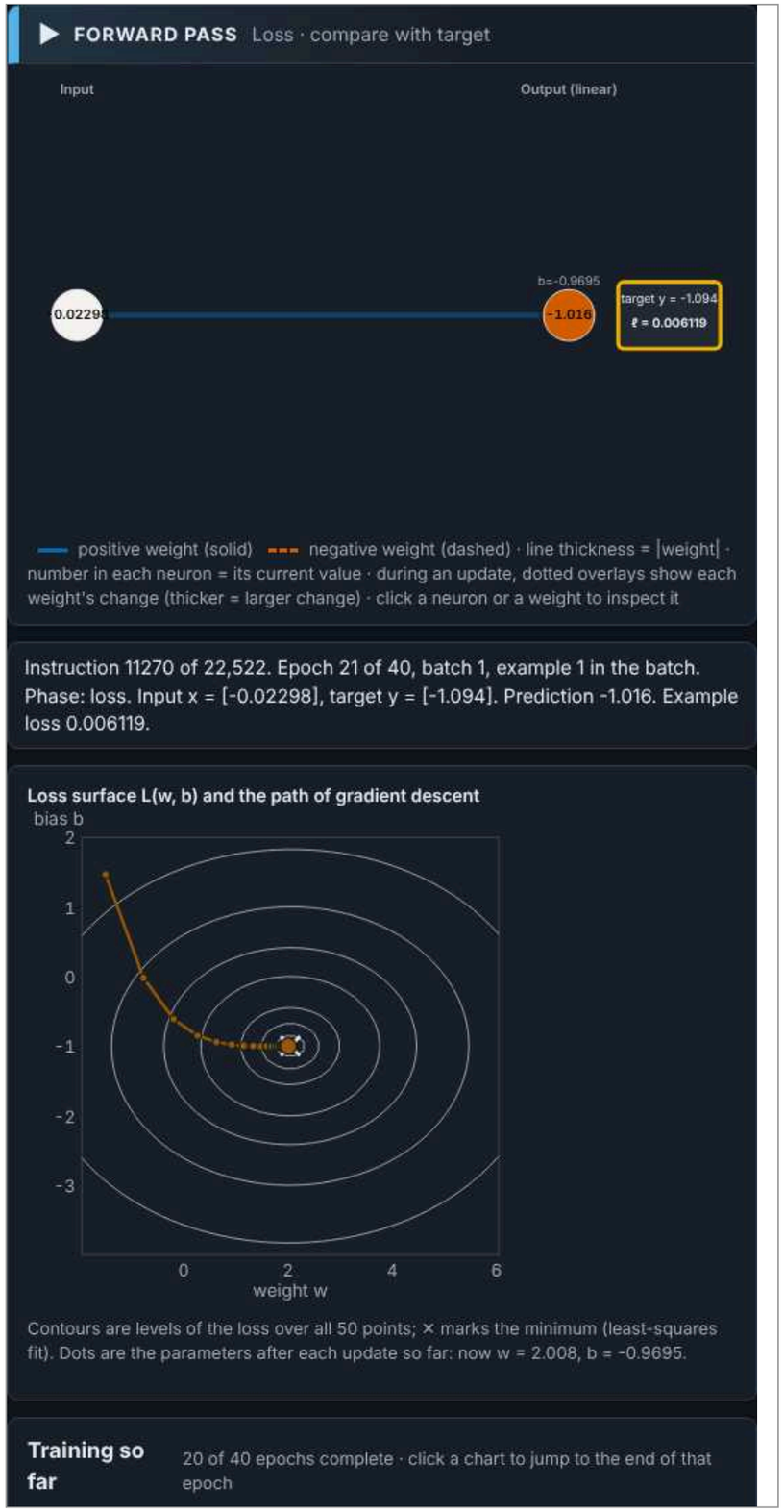


Simulated deuteranopia

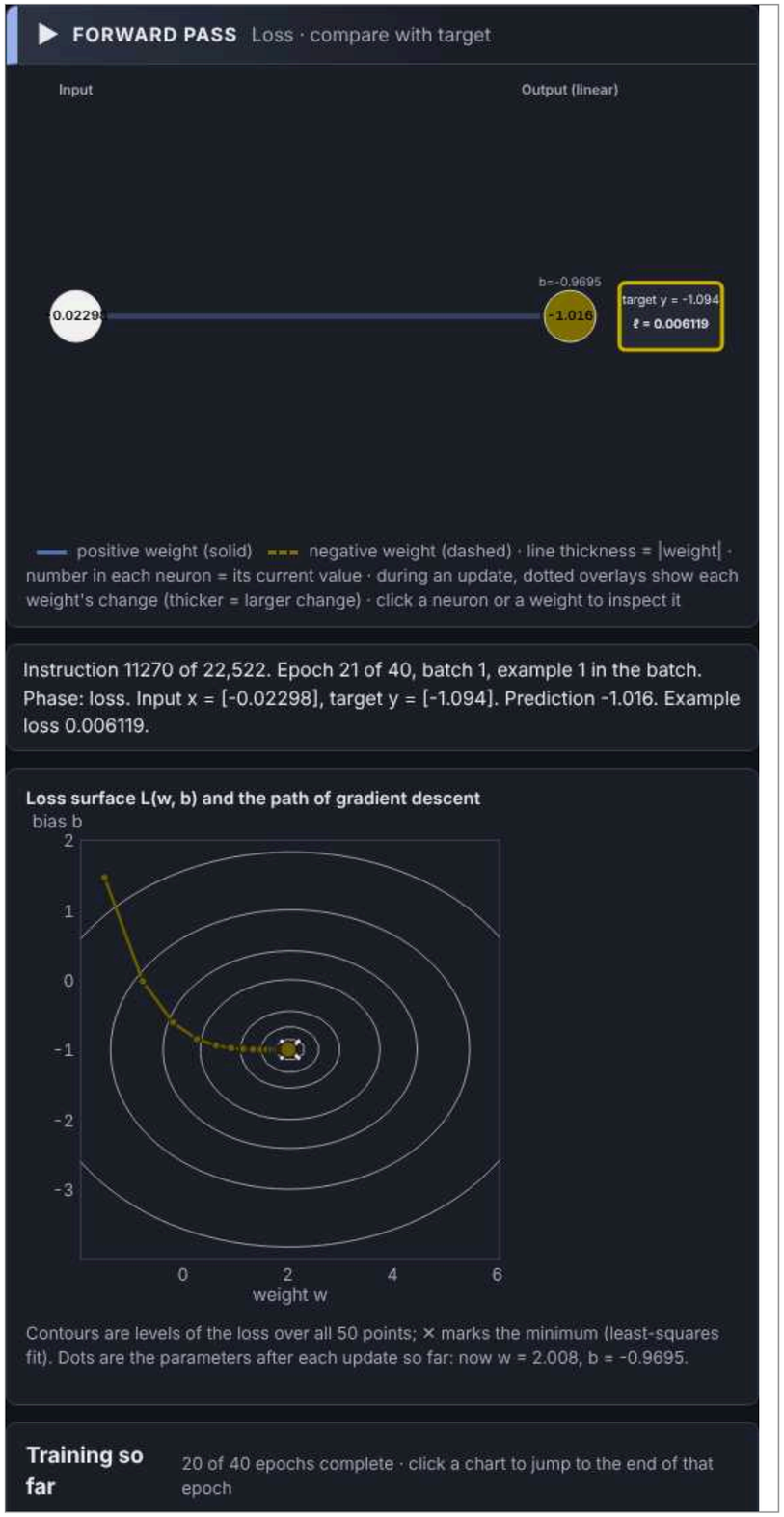


5. Gradient descent — dark theme, forward phase (part 1 of 2)

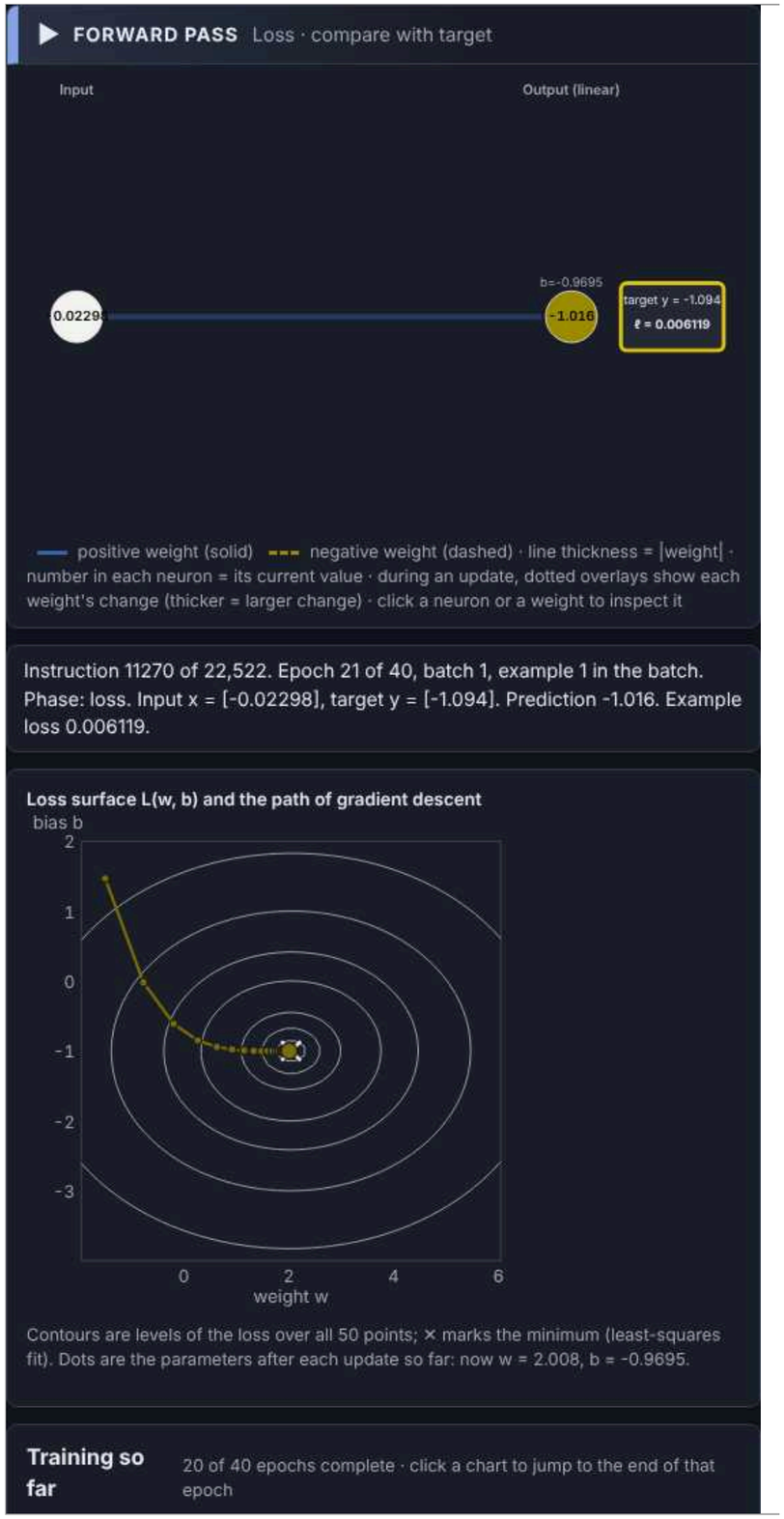
As rendered



Simulated protanopia

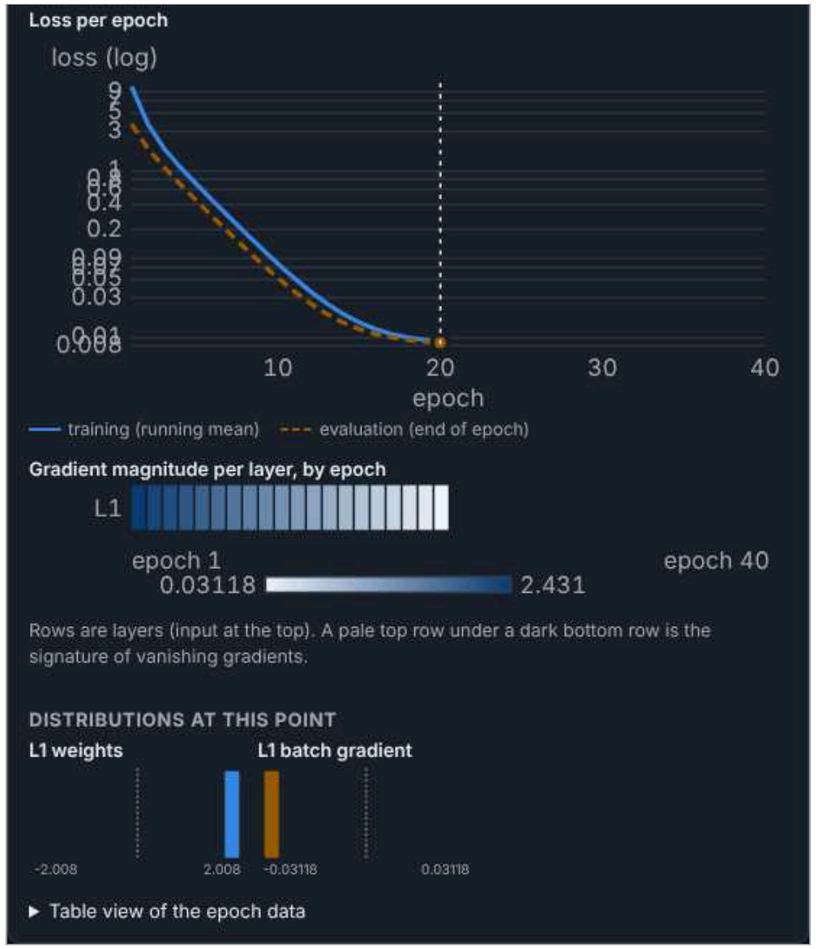


Simulated deuteranopia

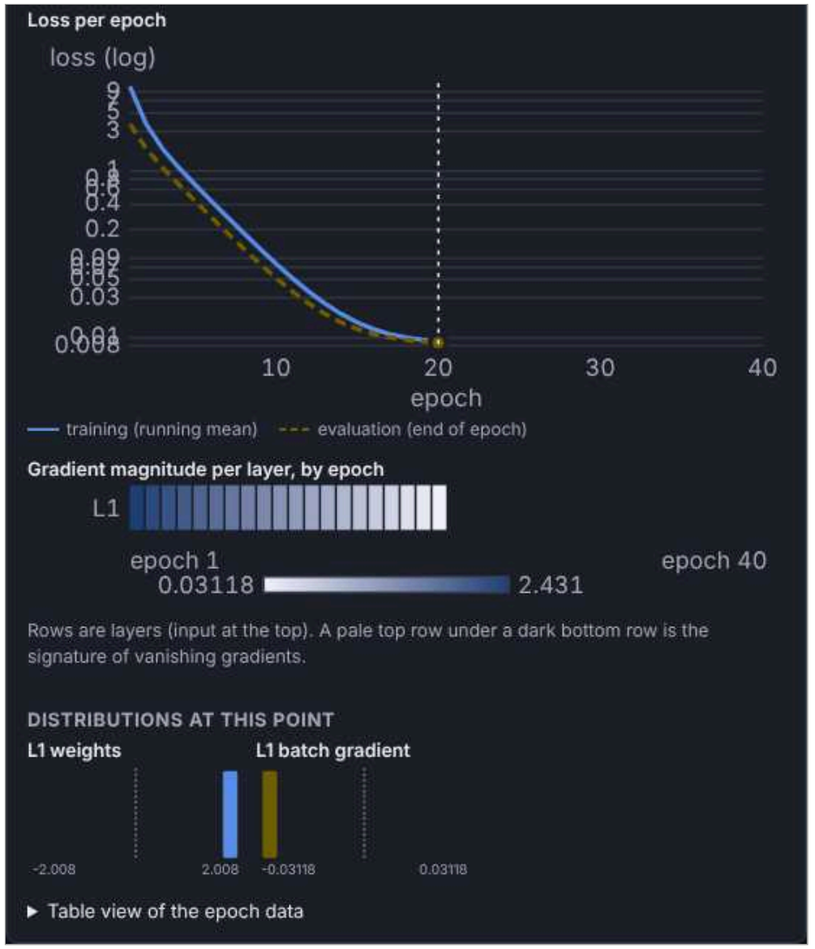


5. Gradient descent — dark theme, forward phase (part 2 of 2)

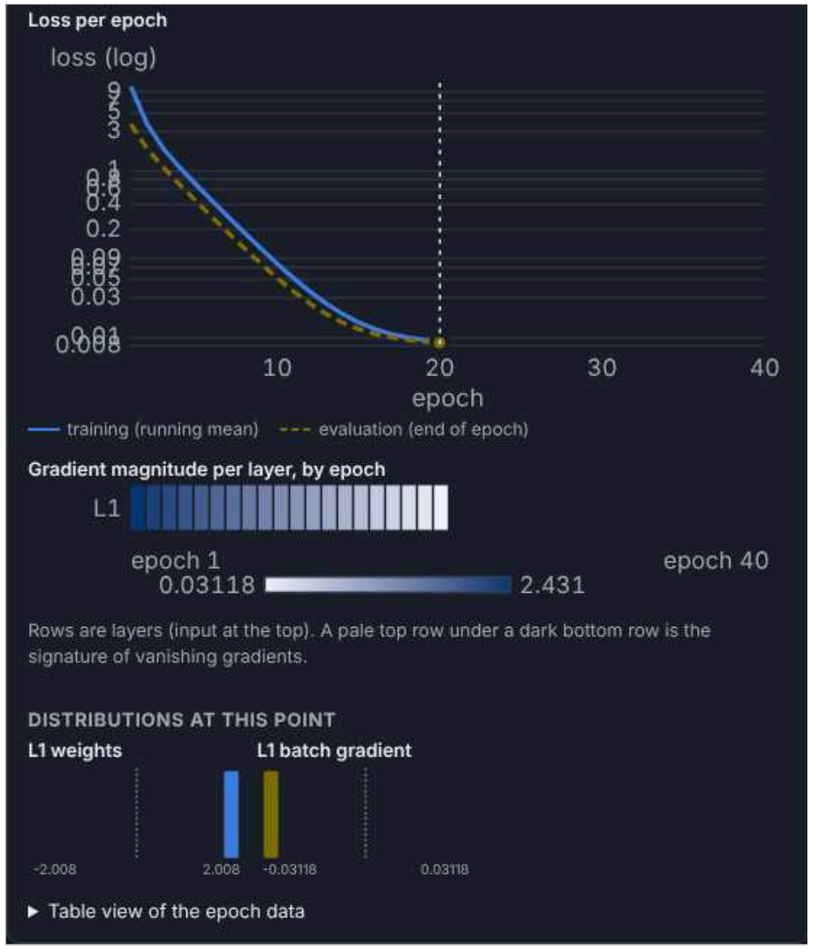
As rendered



Simulated protanopia

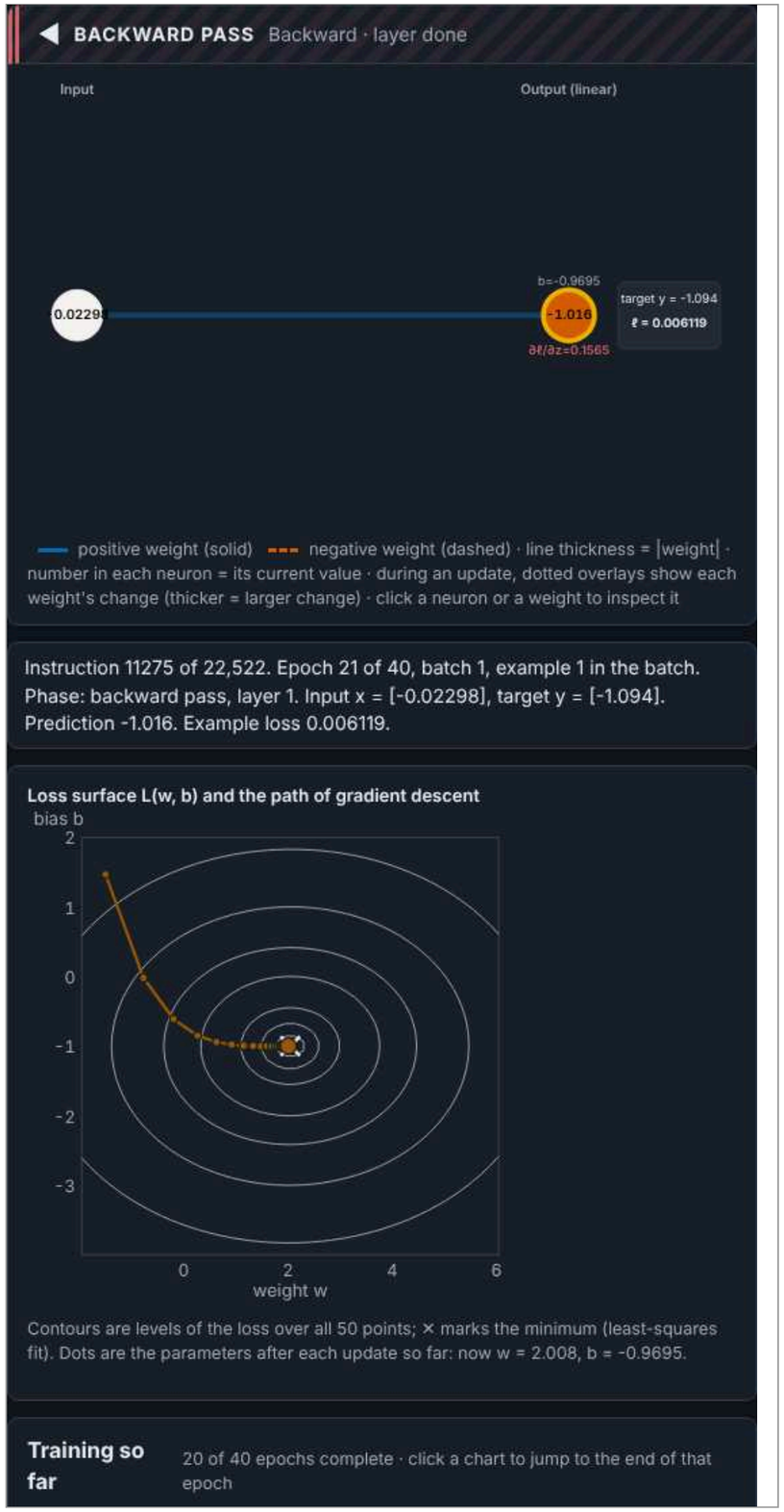


Simulated deuteranopia

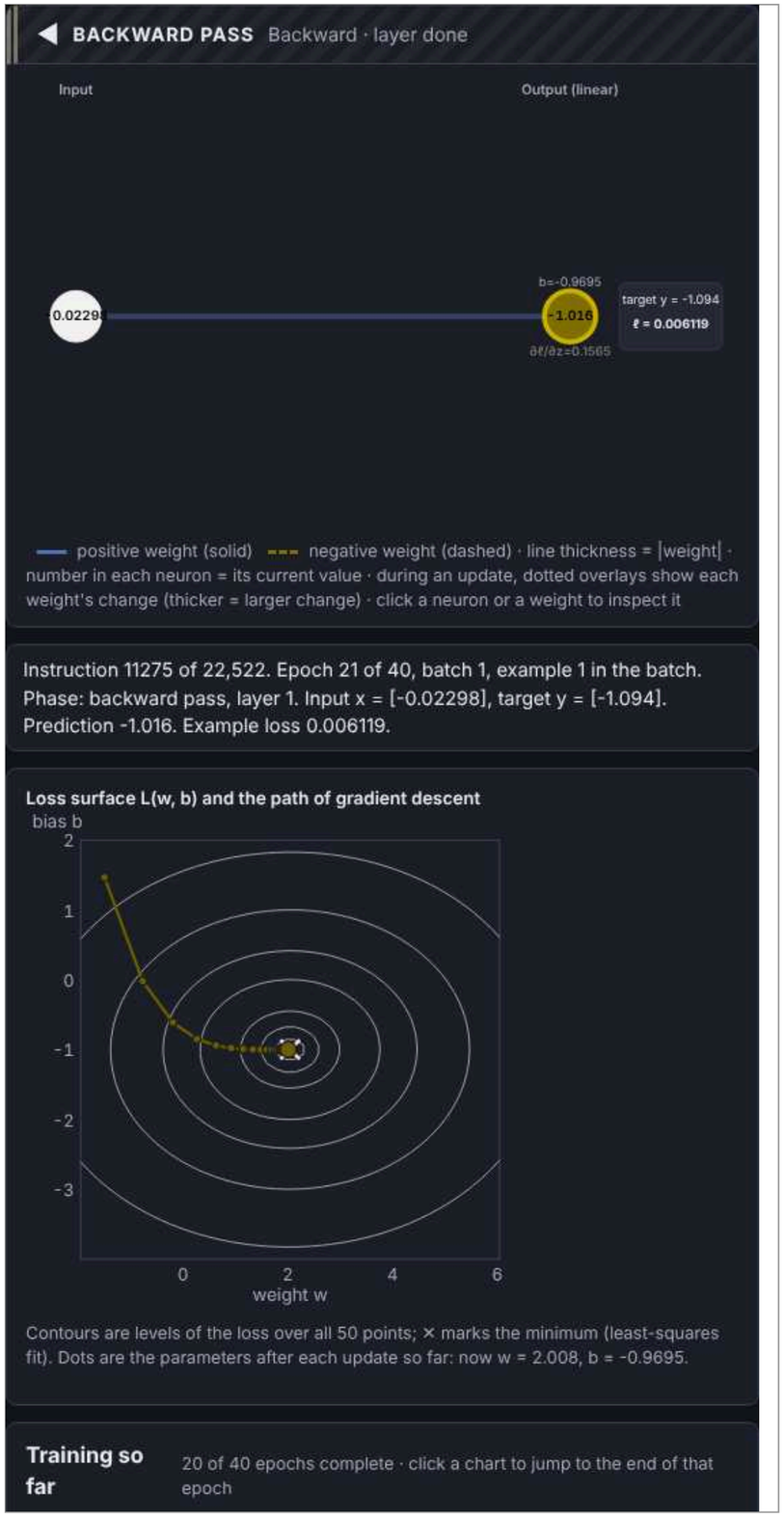


5. Gradient descent — dark theme, backward phase (part 1 of 2)

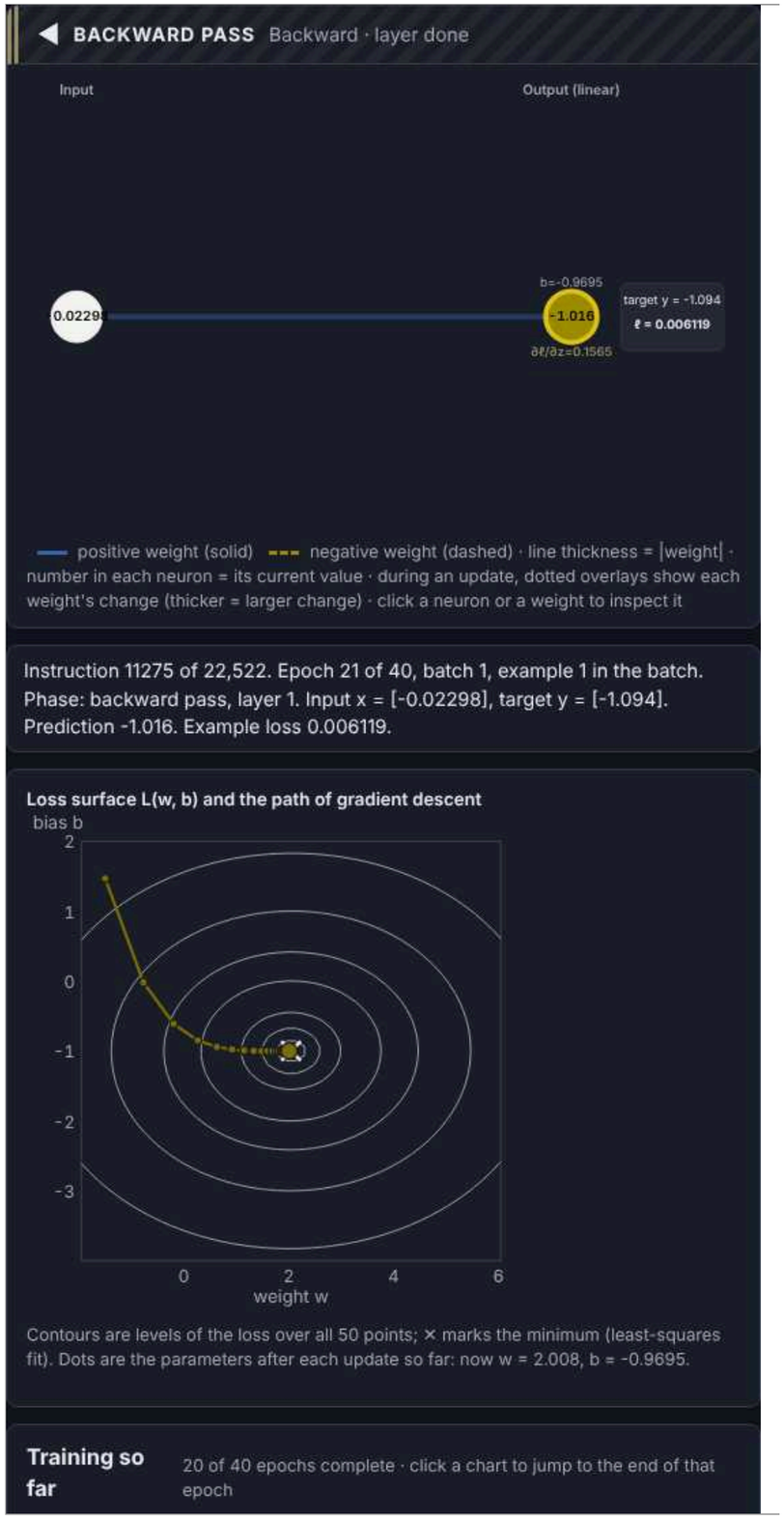
As rendered



Simulated protanopia

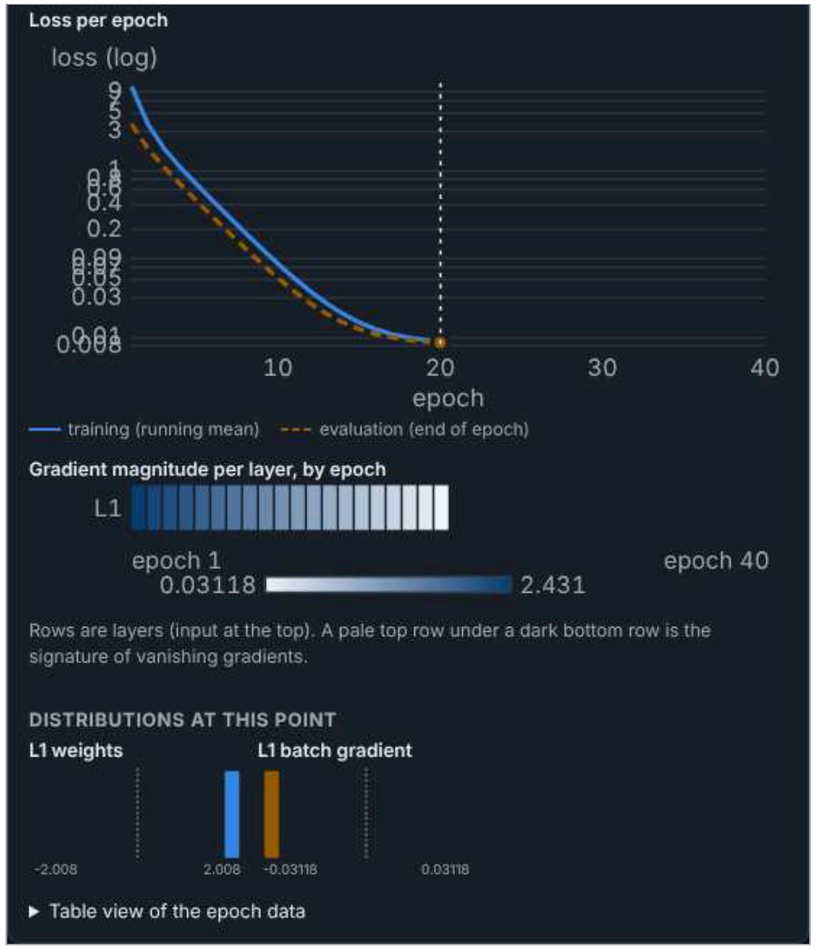


Simulated deuteranopia

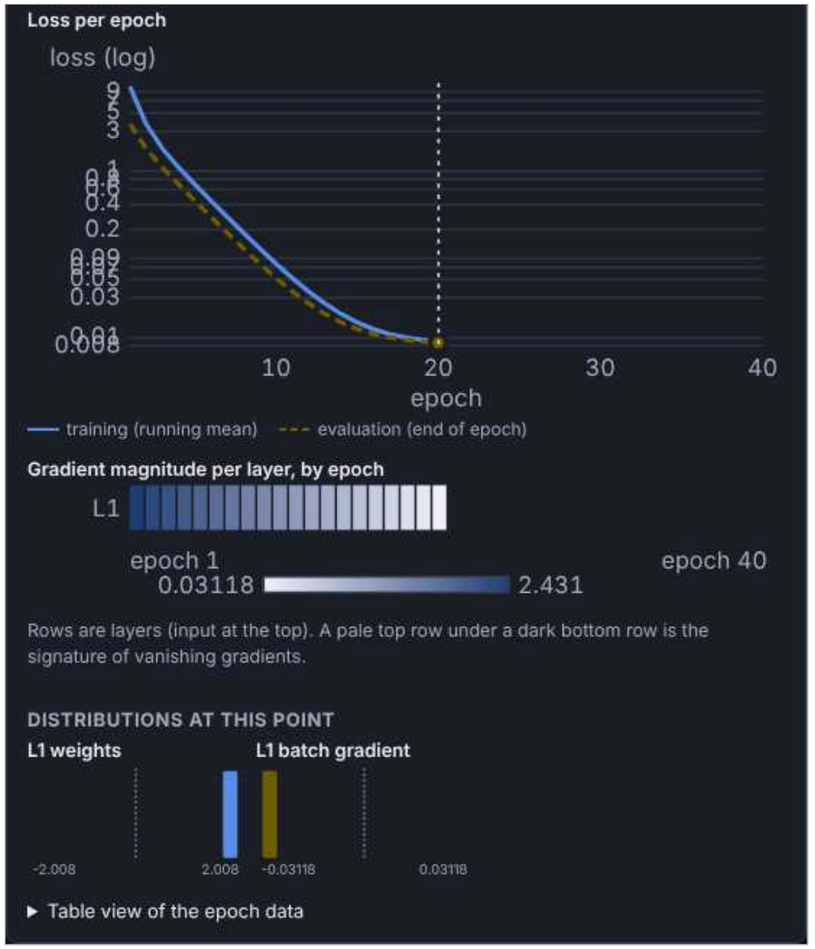


5. Gradient descent — dark theme, backward phase (part 2 of 2)

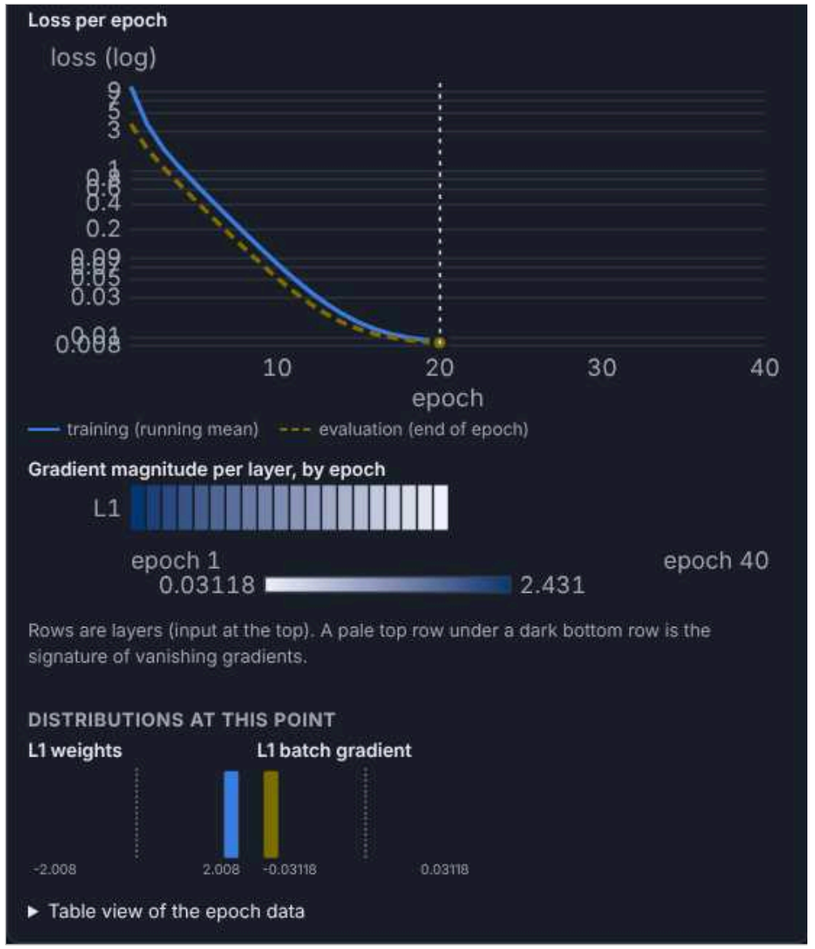
As rendered



Simulated protanopia

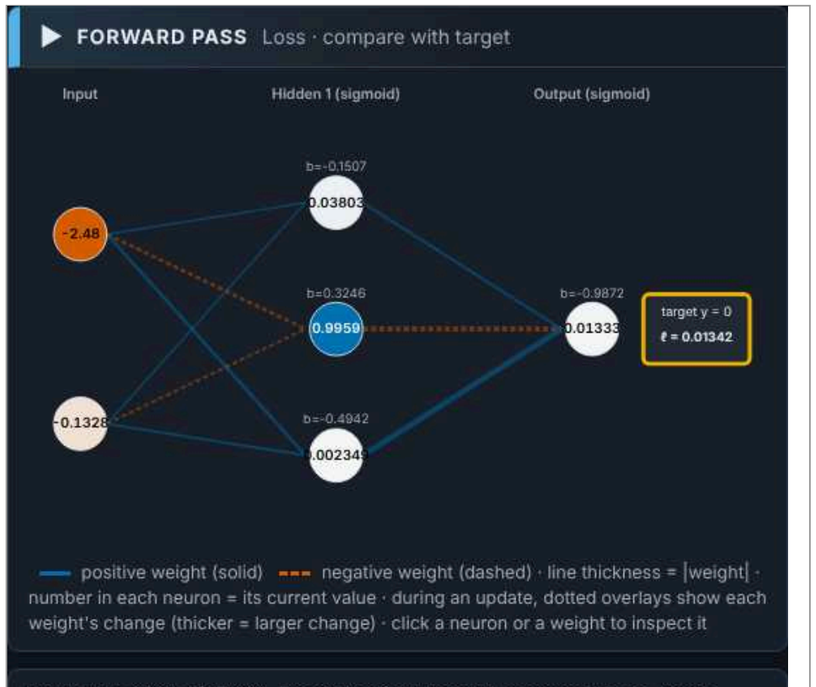


Simulated deuteranopia

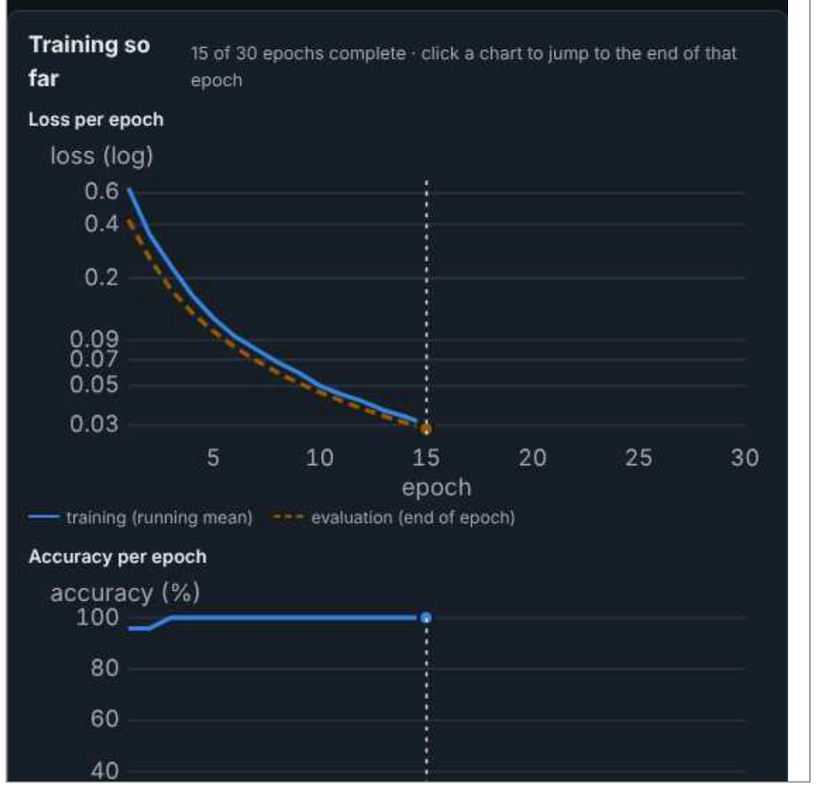


6. Back-propagation — dark theme, forward phase (part 1 of 2)

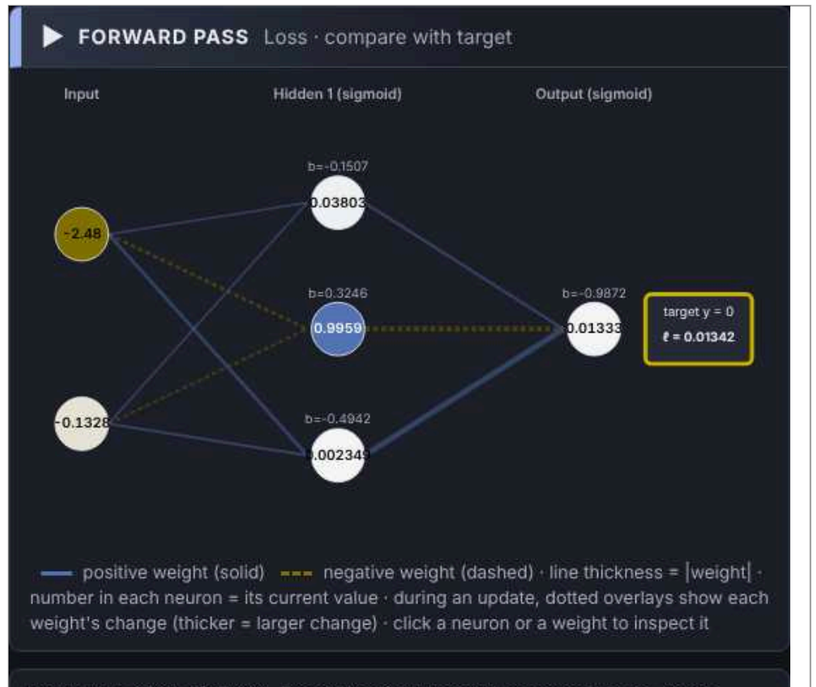
As rendered



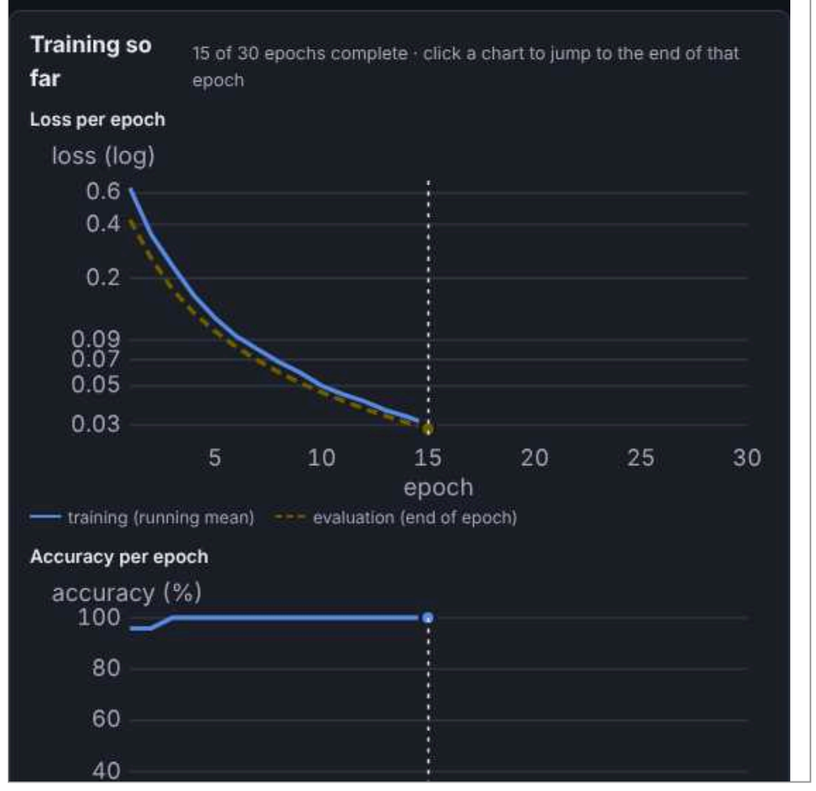
Instruction 8626 of 17,224. Epoch 16 of 30, batch 1, example 1 in the batch. Phase: loss. Input x = [-2.48, -0.1328], target y = [0]. Prediction 0.01333. Example loss 0.01342.



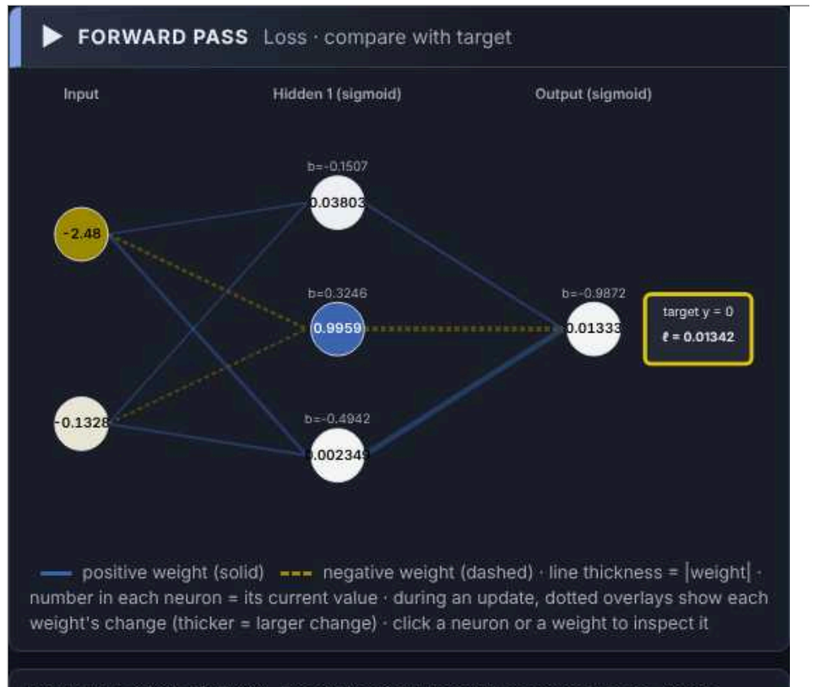
Simulated protanopia



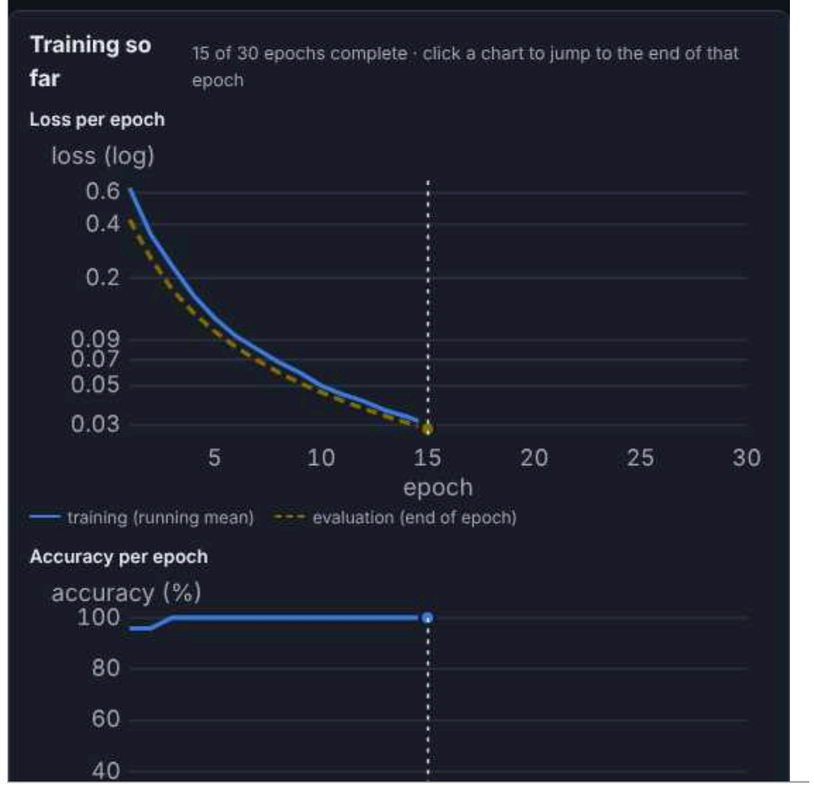
Instruction 8626 of 17,224. Epoch 16 of 30, batch 1, example 1 in the batch. Phase: loss. Input x = [-2.48, -0.1328], target y = [0]. Prediction 0.01333. Example loss 0.01342.



Simulated deuteranopia

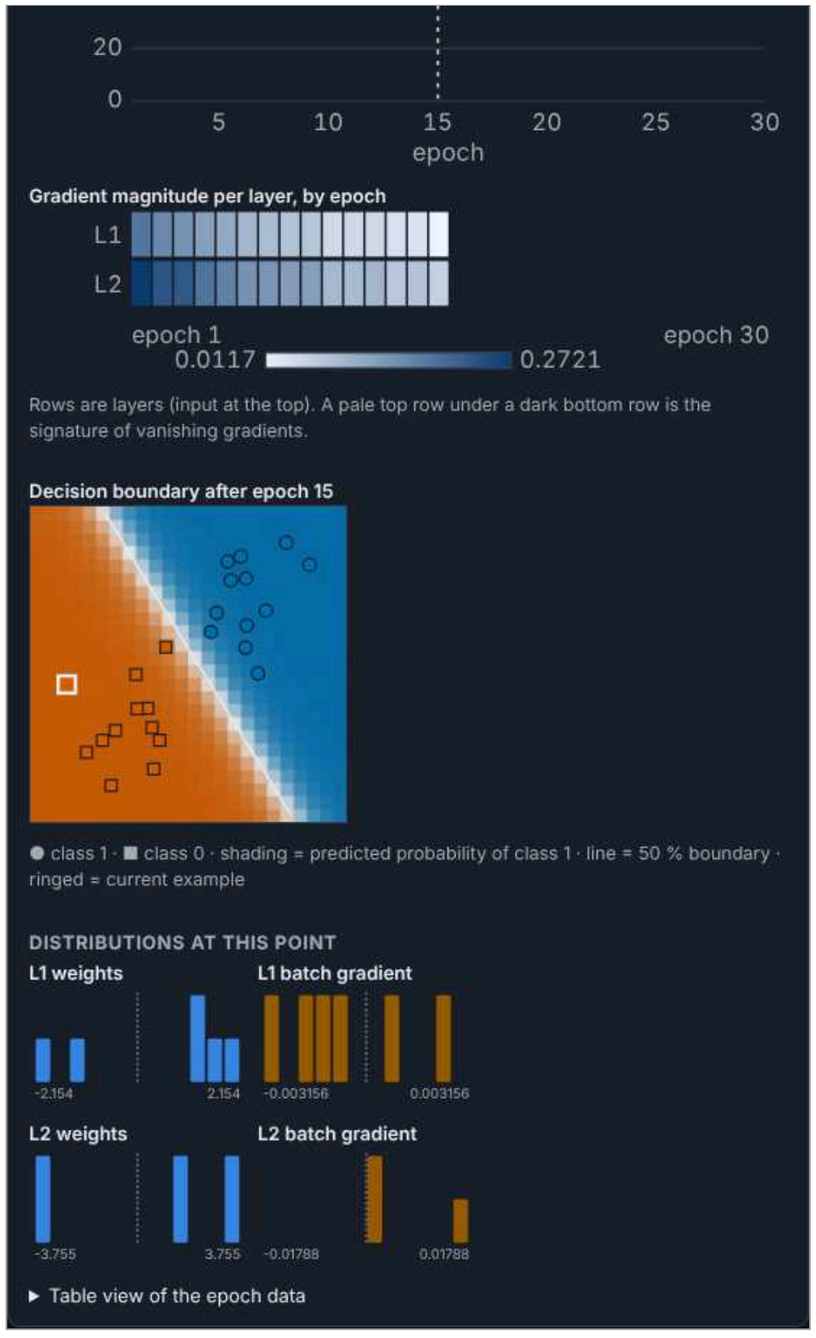


Instruction 8626 of 17,224. Epoch 16 of 30, batch 1, example 1 in the batch. Phase: loss. Input x = [-2.48, -0.1328], target y = [0]. Prediction 0.01333. Example loss 0.01342.

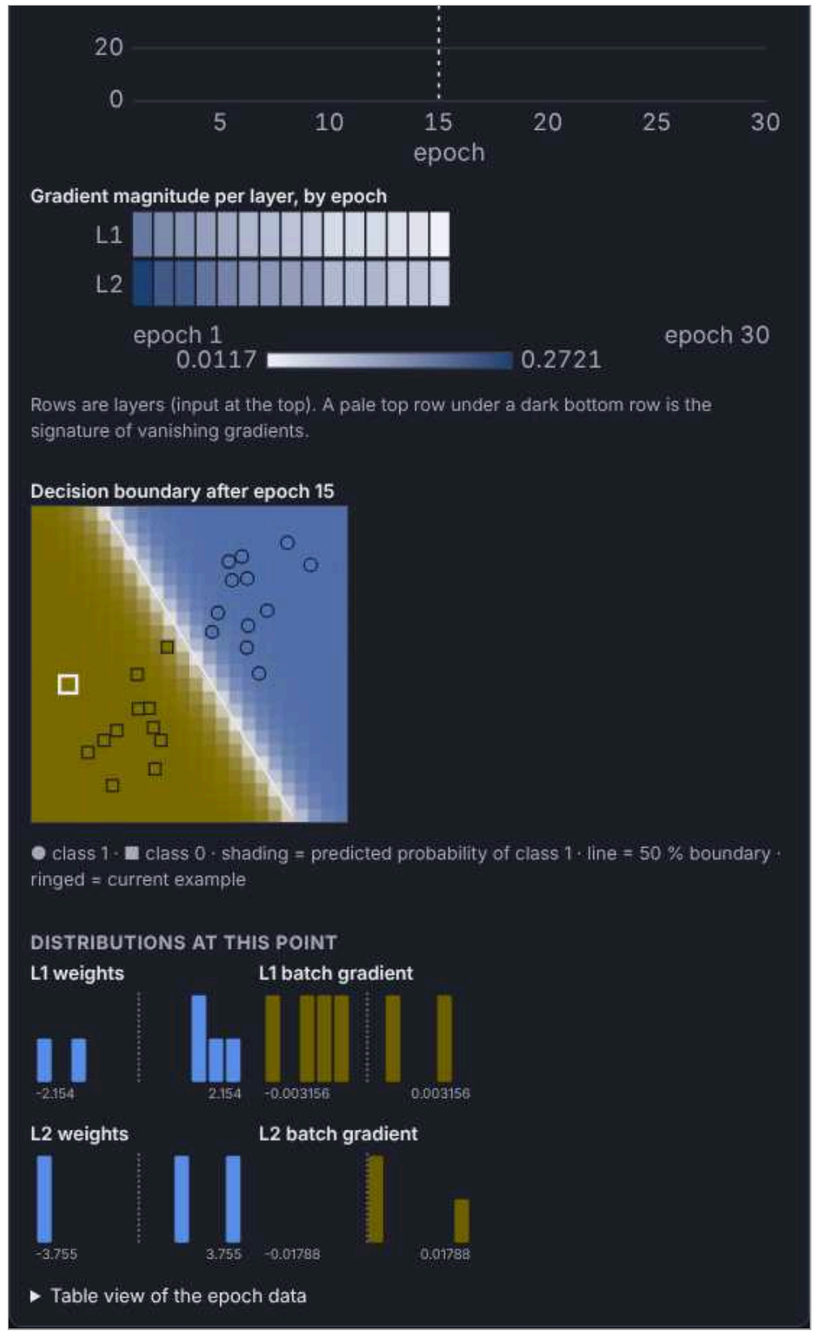


6. Back-propagation — dark theme, forward phase (part 2 of 2)

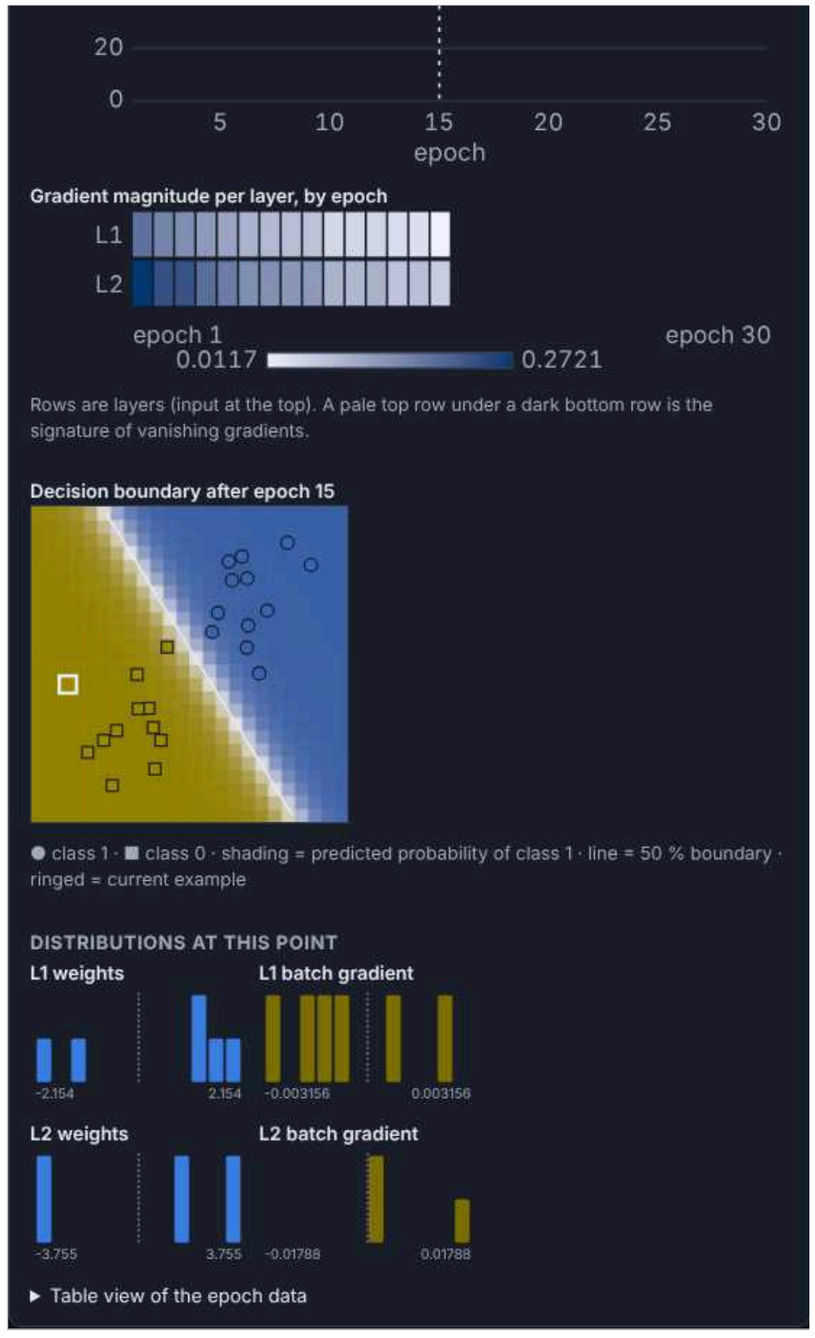
As rendered



Simulated protanopia

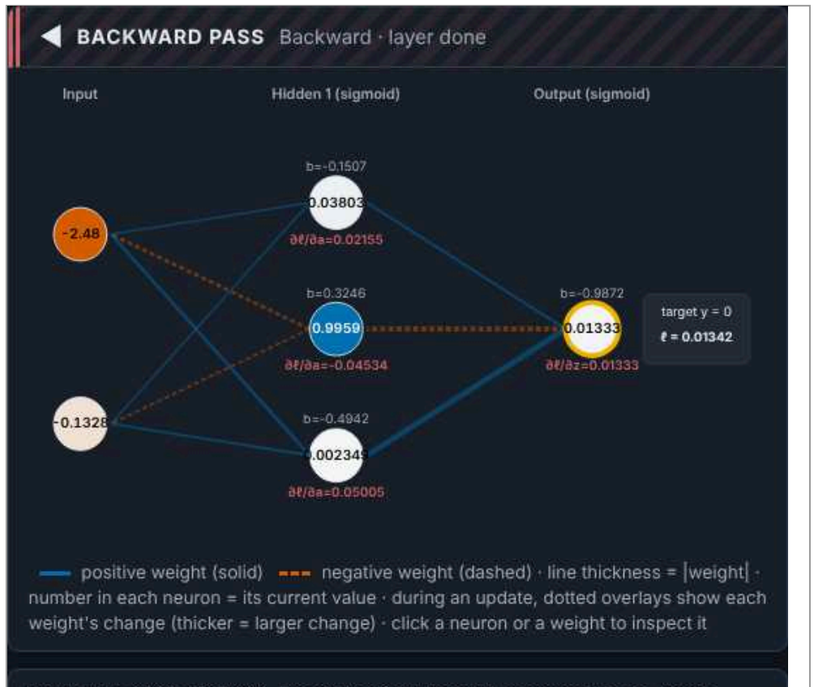


Simulated deuteranopia

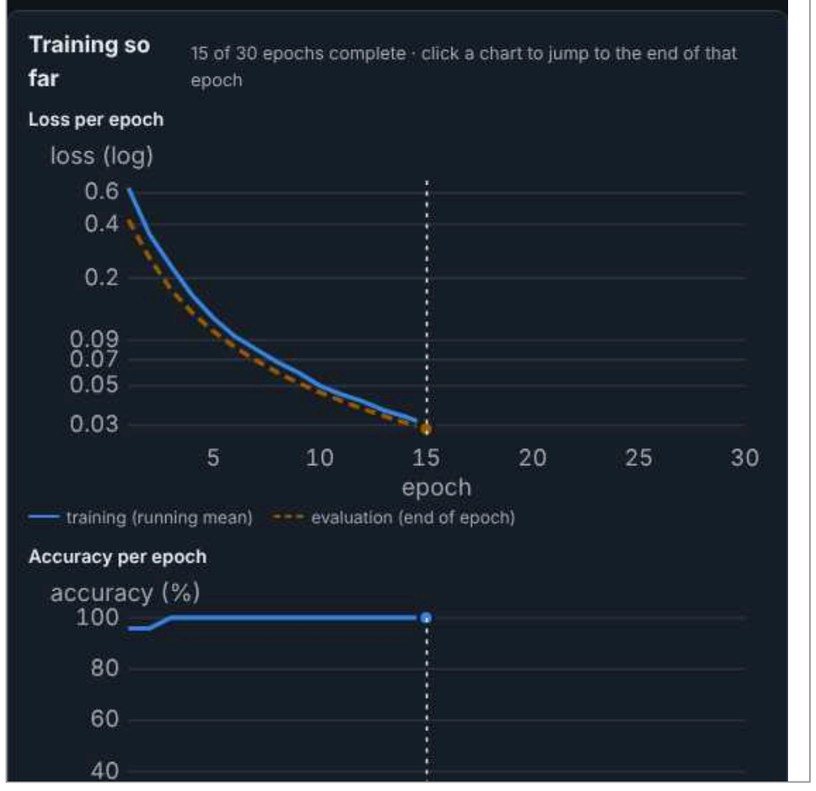


6. Back-propagation — dark theme, backward phase (part 1 of 2)

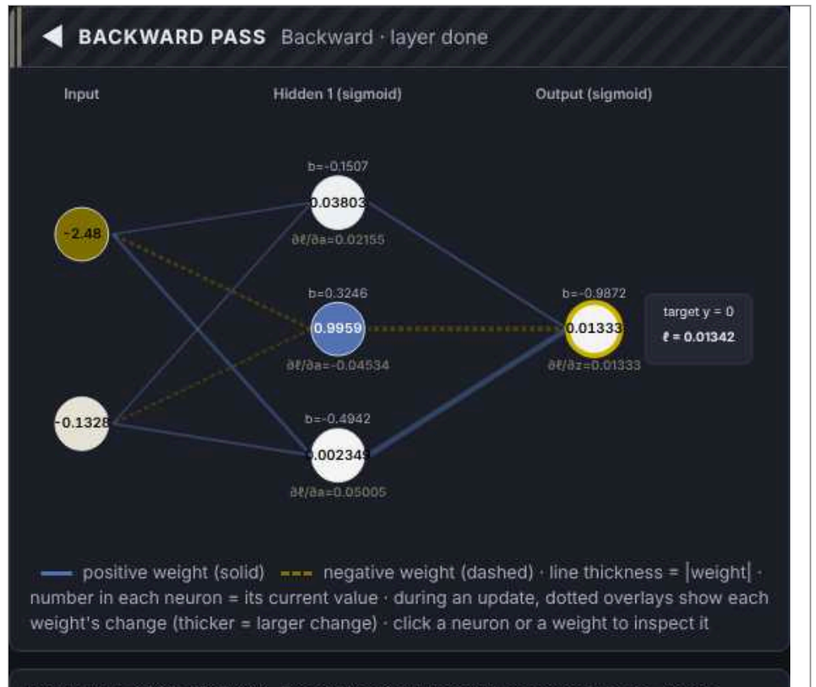
As rendered



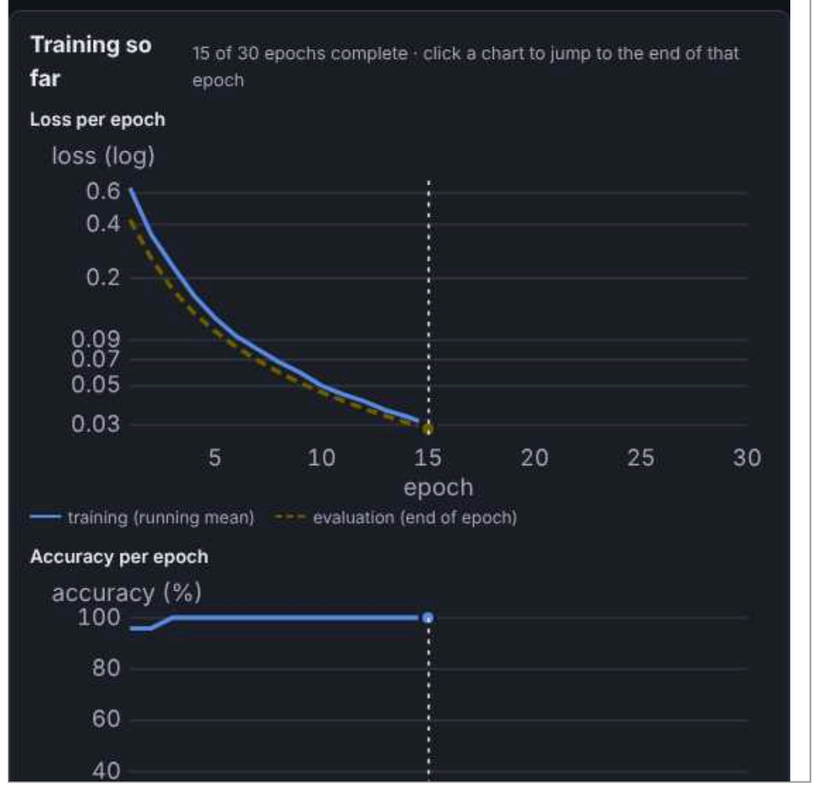
Instruction 8632 of 17,224. Epoch 16 of 30, batch 1, example 1 in the batch. Phase: backward pass, layer 2. Input x = [-2.48, -0.1328], target y = [0]. Prediction 0.01333. Example loss 0.01342.



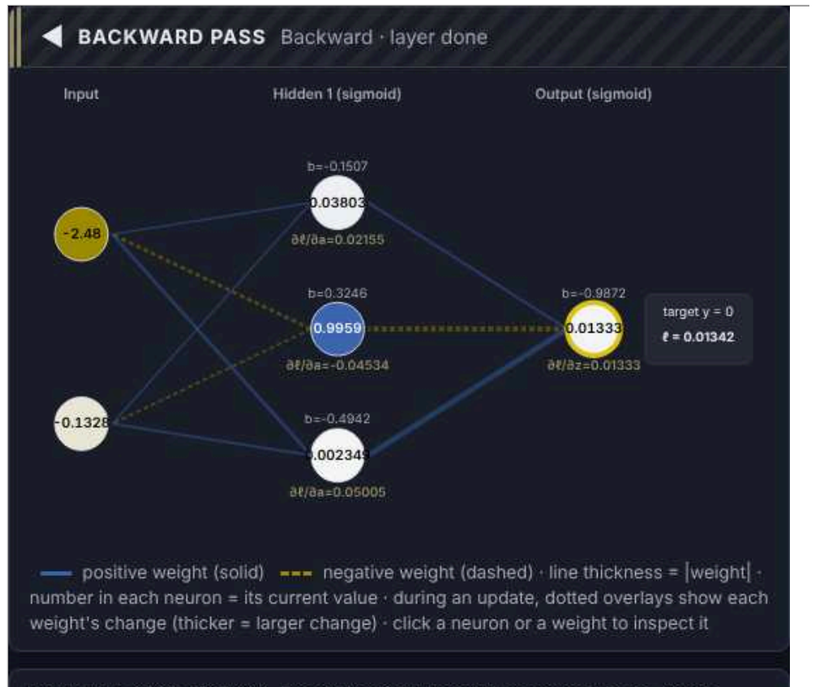
Simulated protanopia



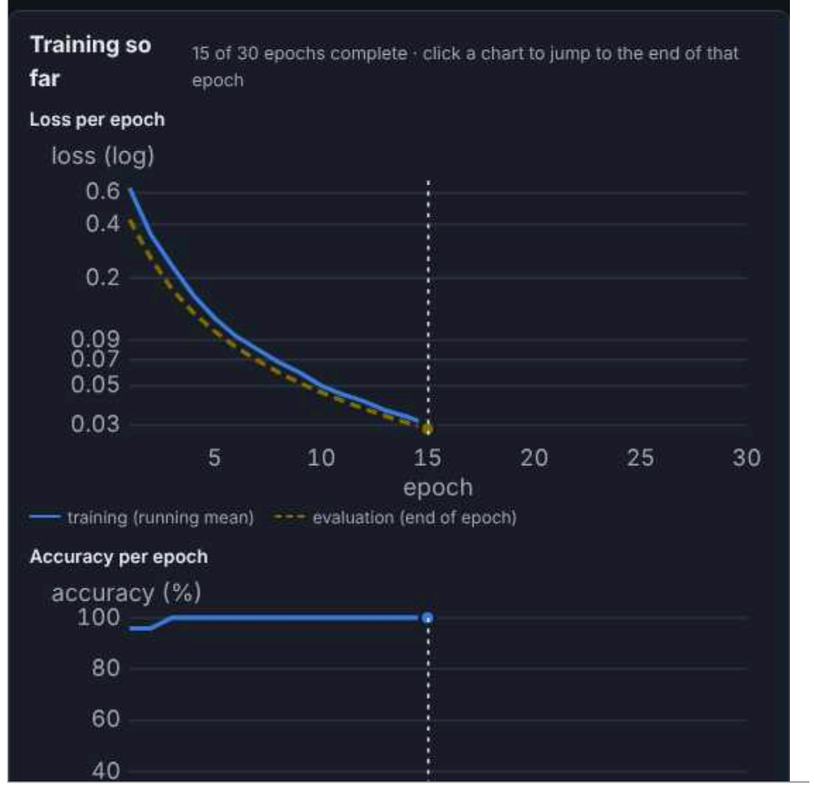
Instruction 8632 of 17,224. Epoch 16 of 30, batch 1, example 1 in the batch. Phase: backward pass, layer 2. Input x = [-2.48, -0.1328], target y = [0]. Prediction 0.01333. Example loss 0.01342.



Simulated deuteranopia

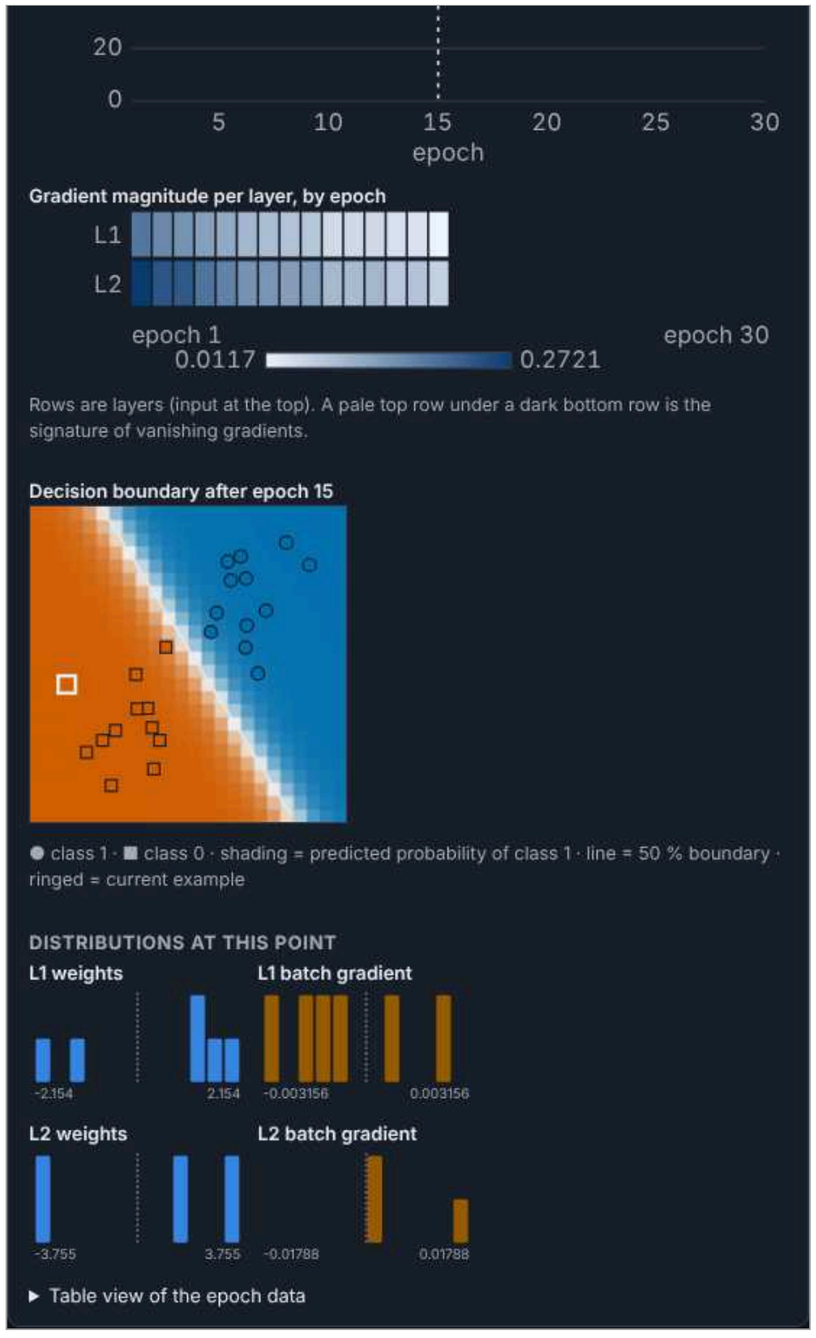


Instruction 8632 of 17,224. Epoch 16 of 30, batch 1, example 1 in the batch. Phase: backward pass, layer 2. Input x = [-2.48, -0.1328], target y = [0]. Prediction 0.01333. Example loss 0.01342.

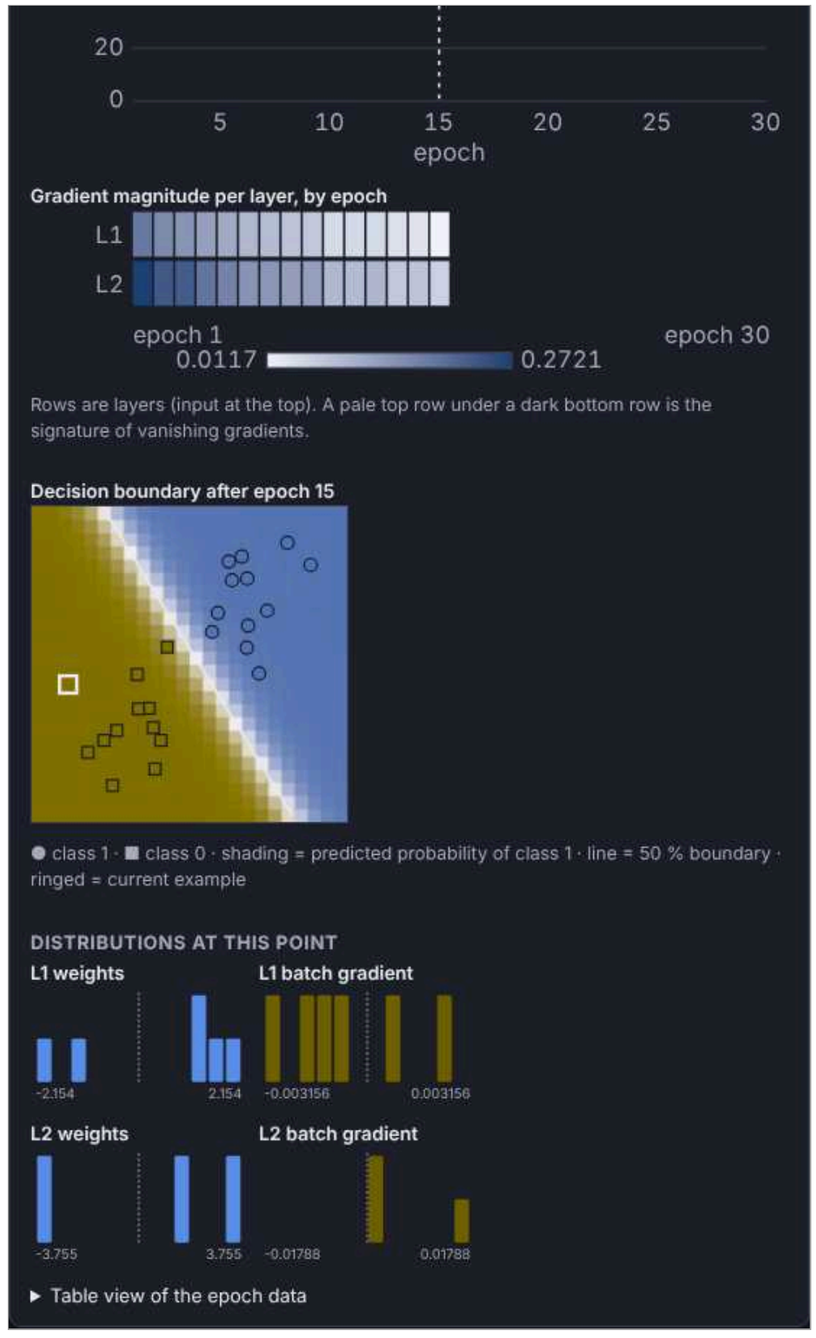


6. Back-propagation — dark theme, backward phase (part 2 of 2)

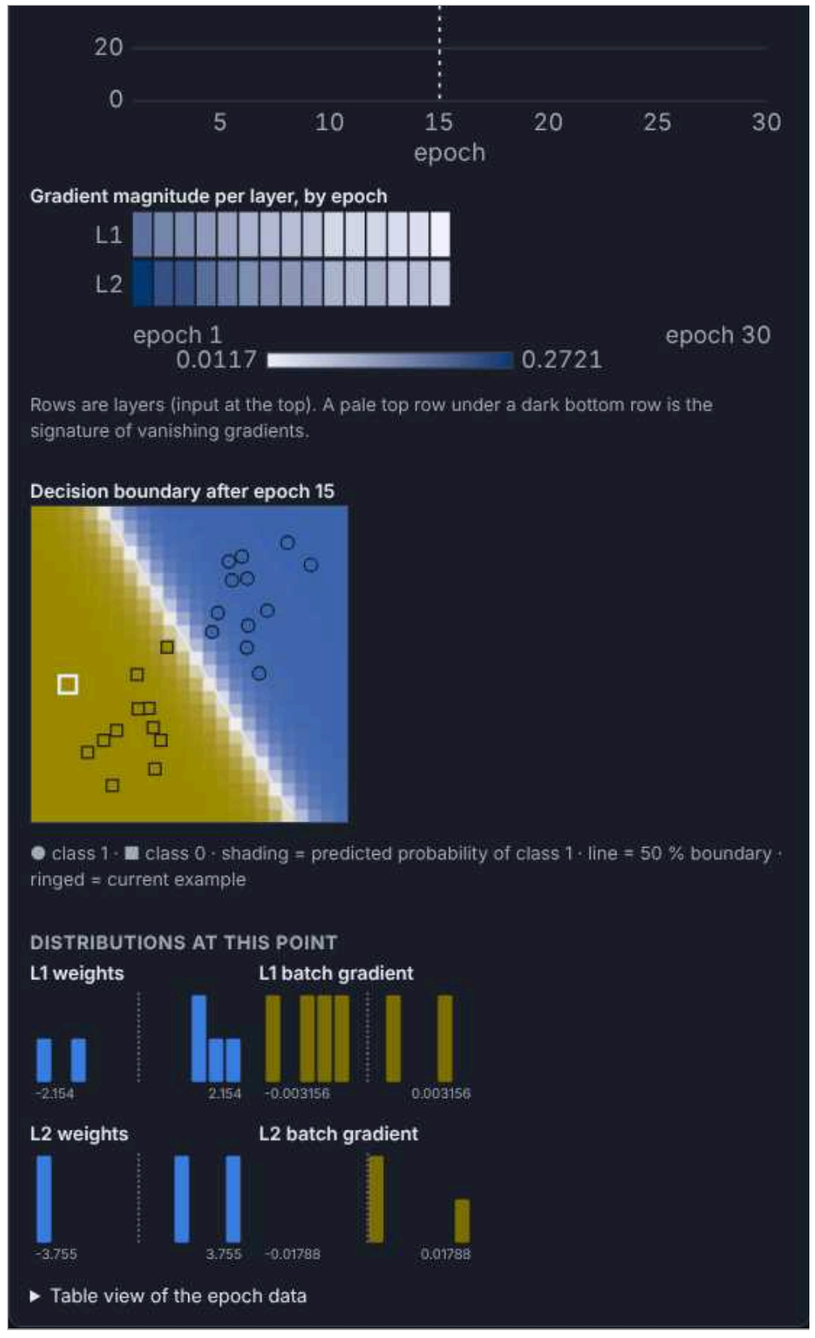
As rendered



Simulated protanopia

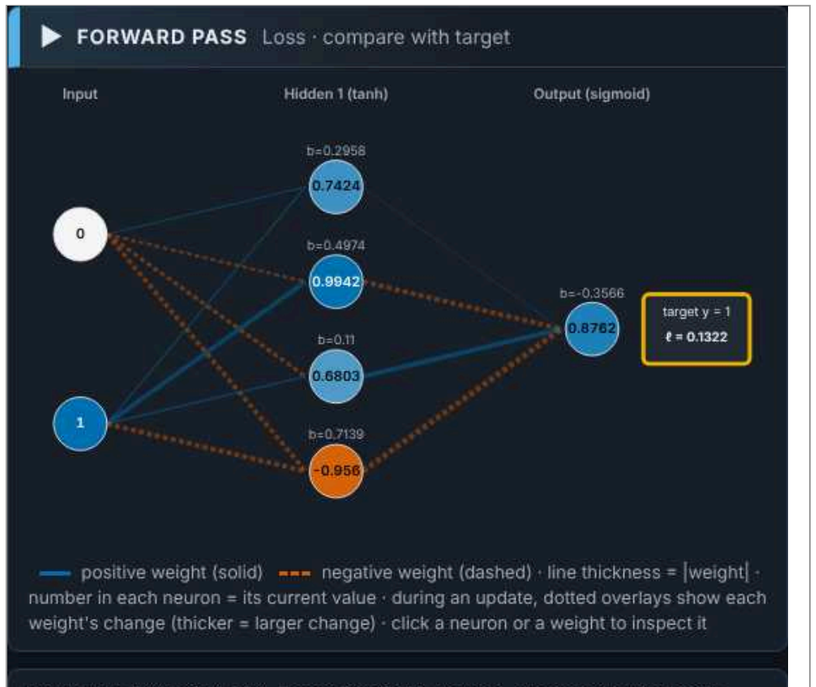


Simulated deuteranopia

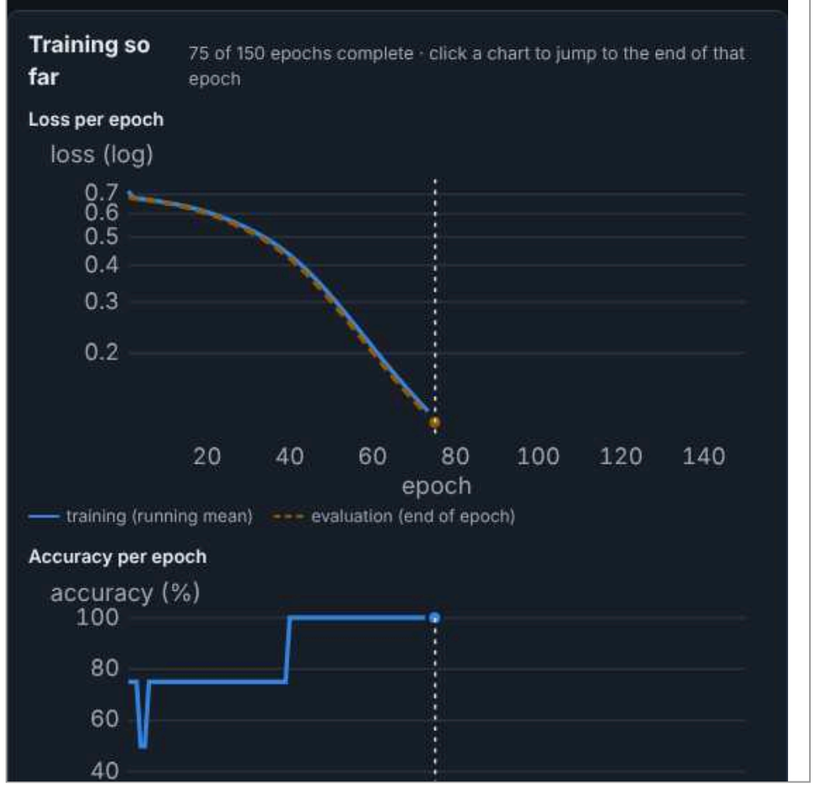


7. XOR needs a hidden layer — dark theme, forward phase (part 1 of 2)

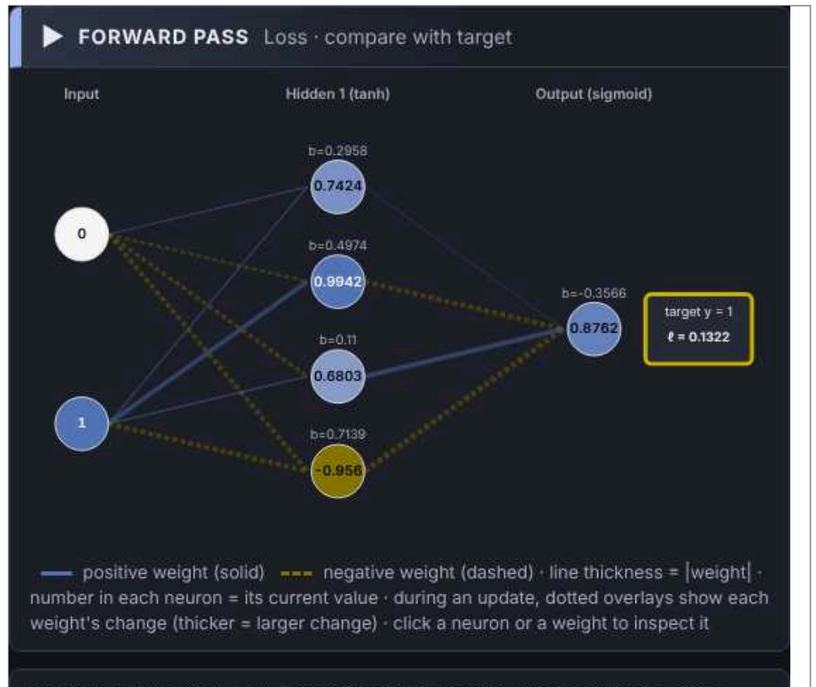
As rendered



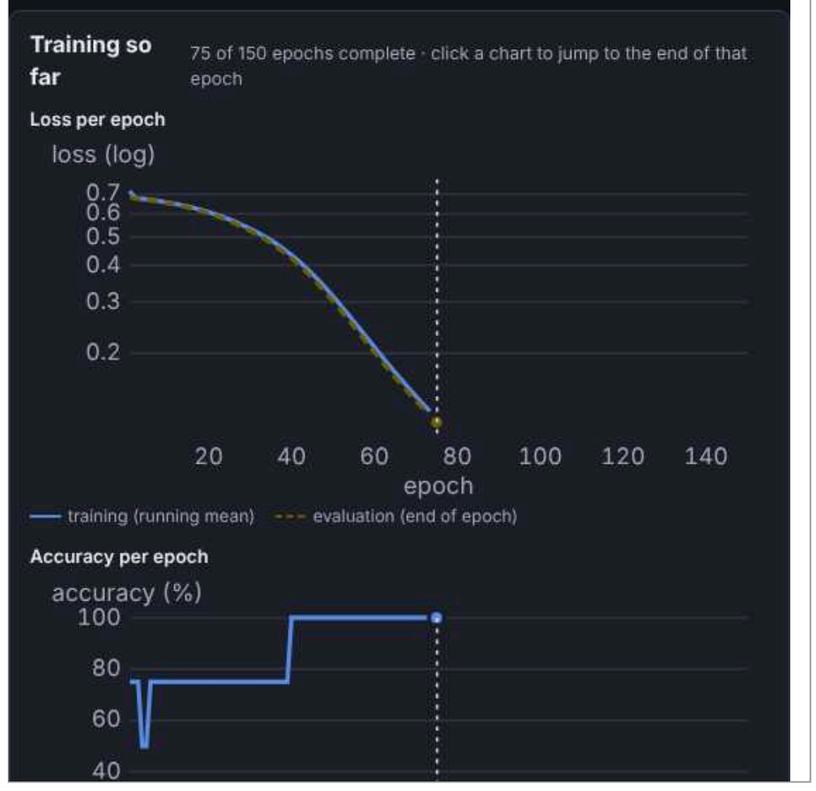
Instruction 7441 of 14,854. Epoch 76 of 150, batch 1, example 1 in the batch. Phase: loss. Input x = [0, 1], target y = [1]. Prediction 0.8762. Example loss 0.1322.



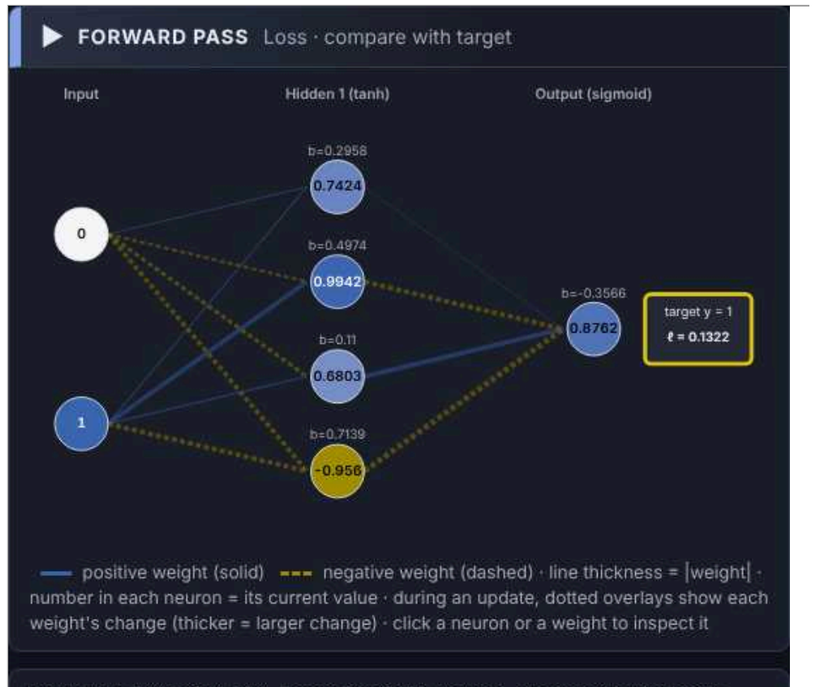
Simulated protanopia



Instruction 7441 of 14,854. Epoch 76 of 150, batch 1, example 1 in the batch. Phase: loss. Input x = [0, 1], target y = [1]. Prediction 0.8762. Example loss 0.1322.



Simulated deuteranopia

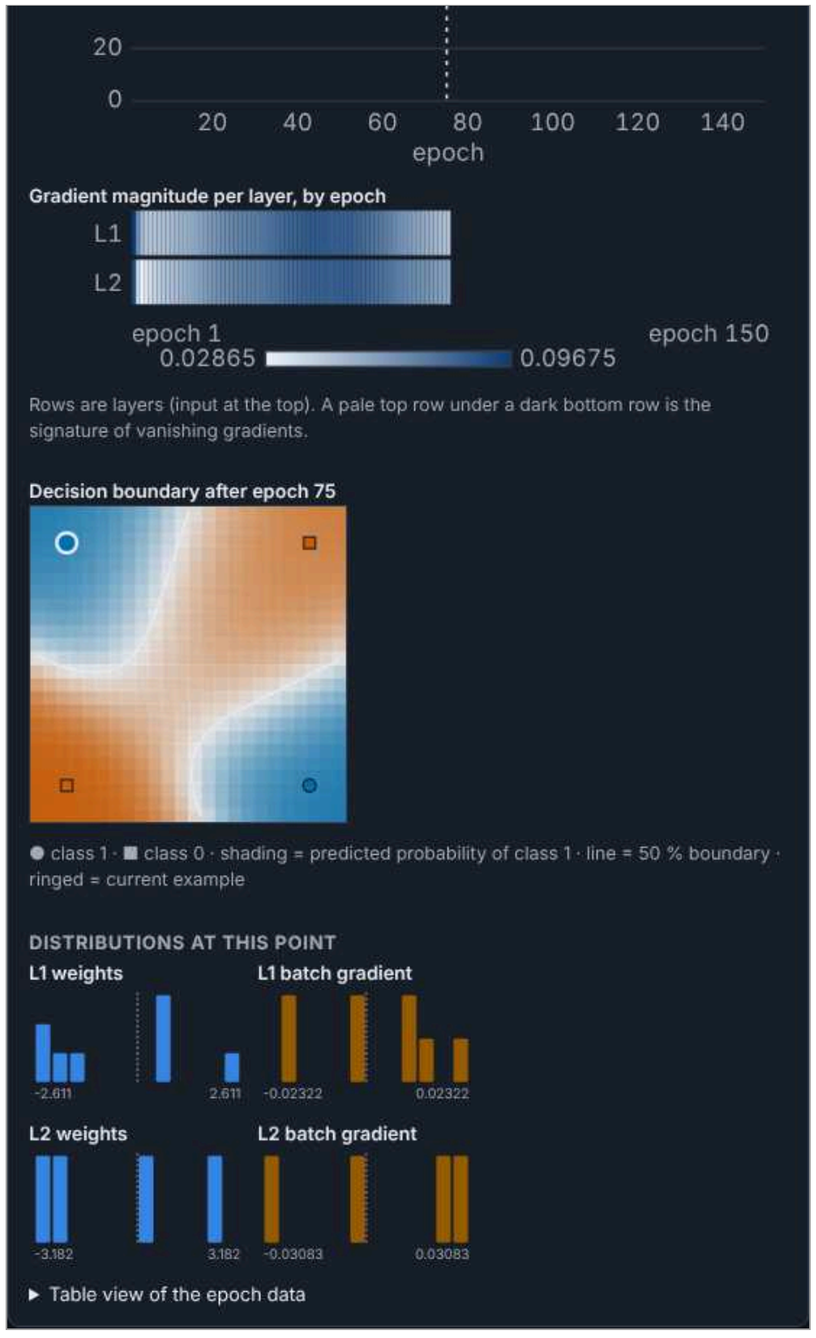


Instruction 7441 of 14,854. Epoch 76 of 150, batch 1, example 1 in the batch. Phase: loss. Input x = [0, 1], target y = [1]. Prediction 0.8762. Example loss 0.1322.

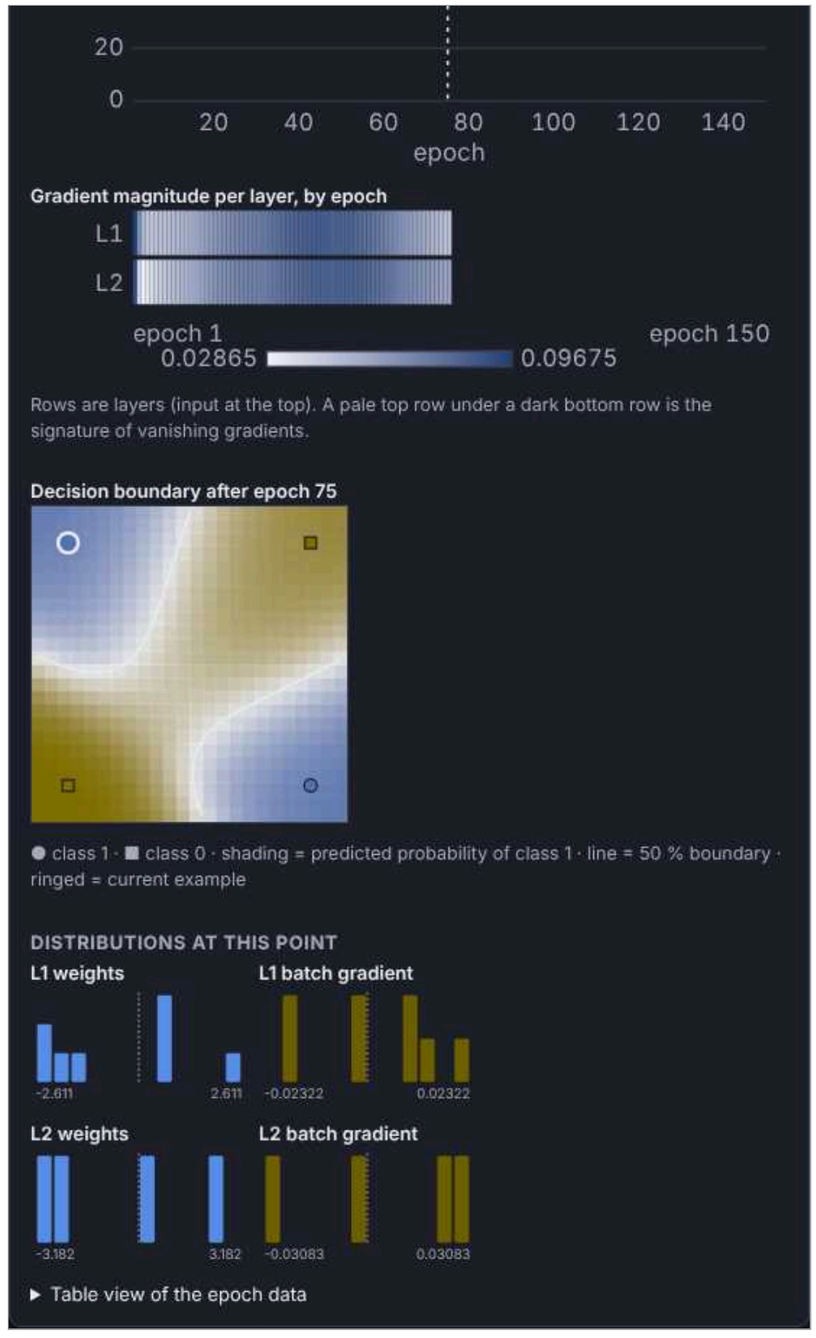


7. XOR needs a hidden layer — dark theme, forward phase (part 2 of 2)

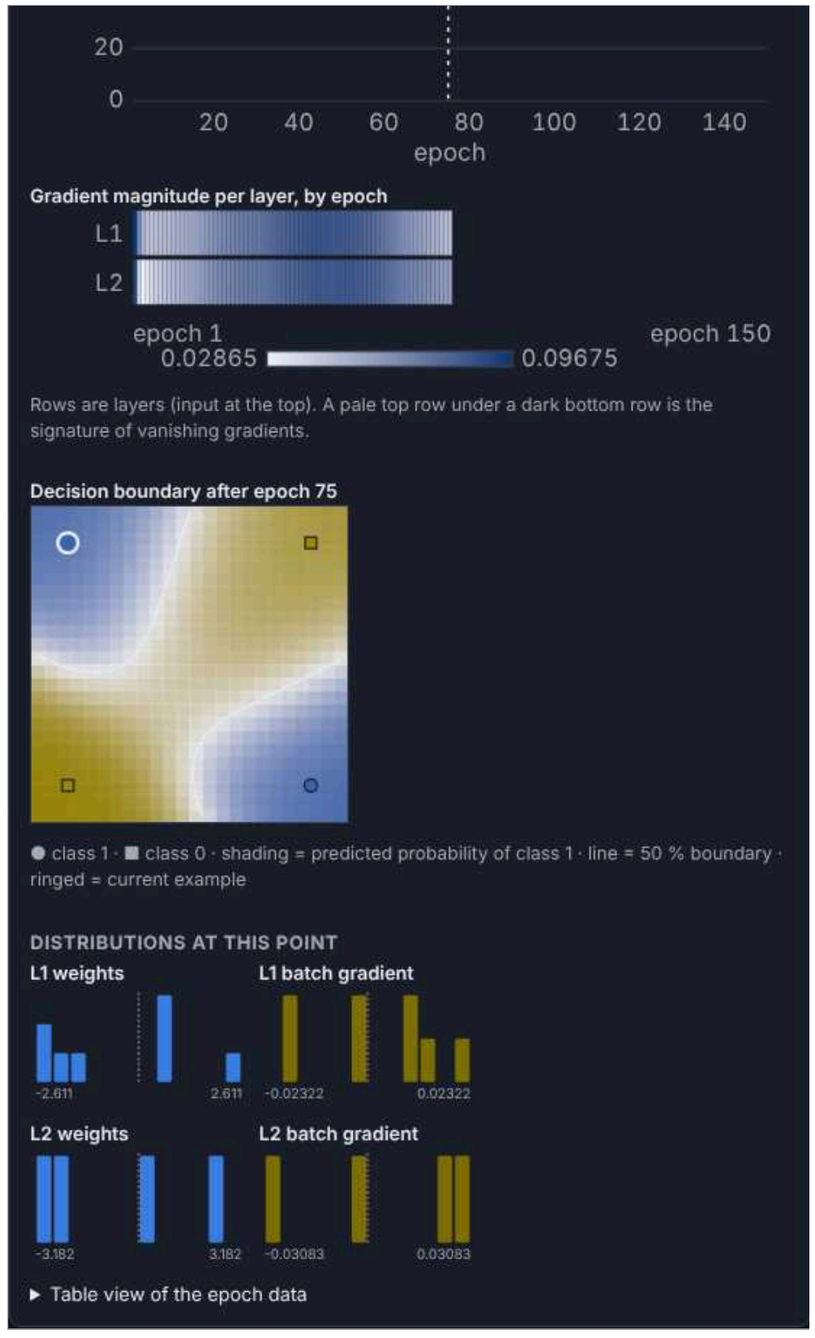
As rendered



Simulated protanopia

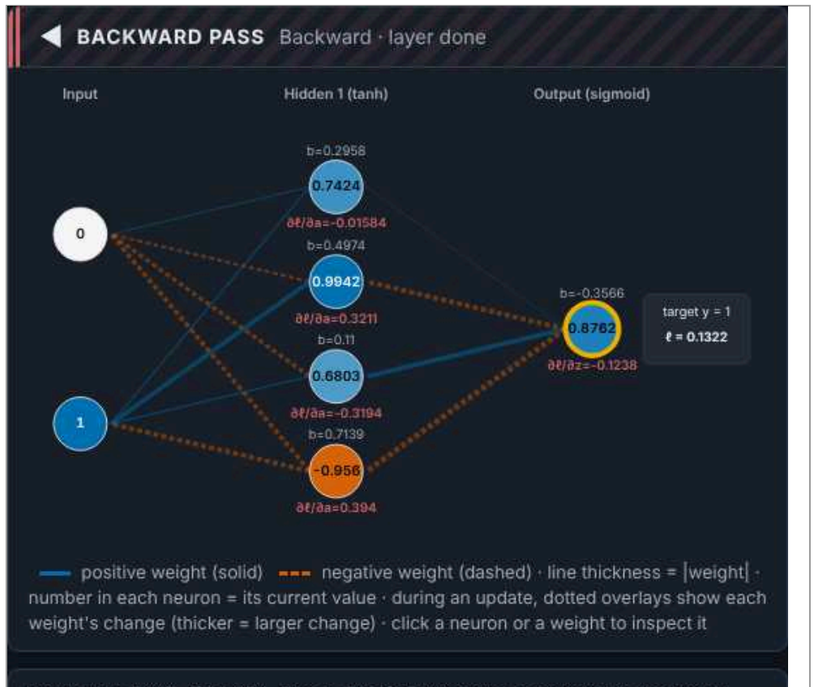


Simulated deuteranopia

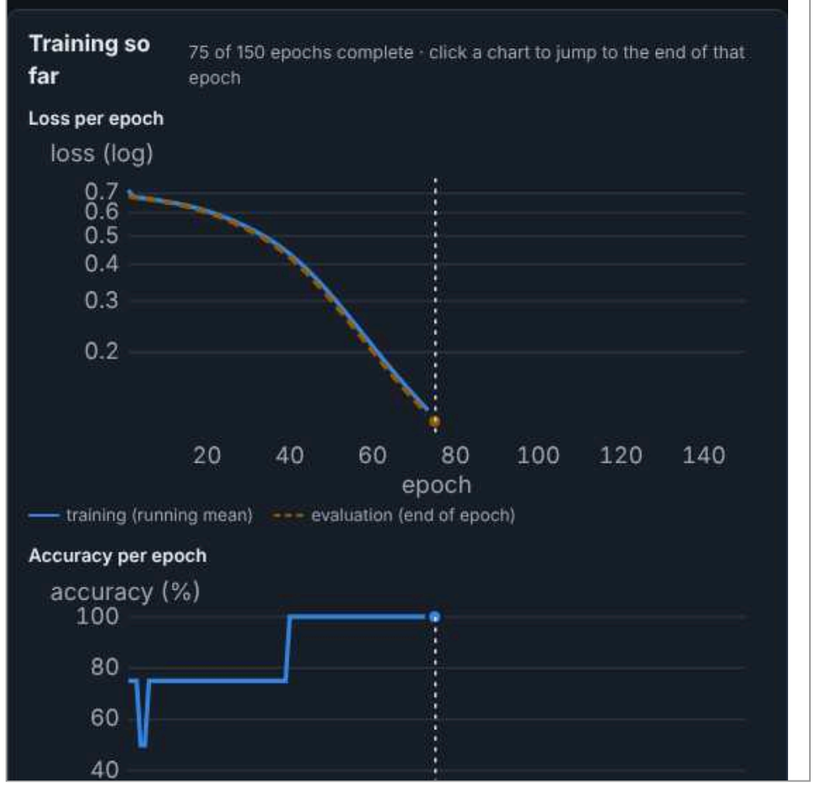


7. XOR needs a hidden layer — dark theme, backward phase (part 1 of 2)

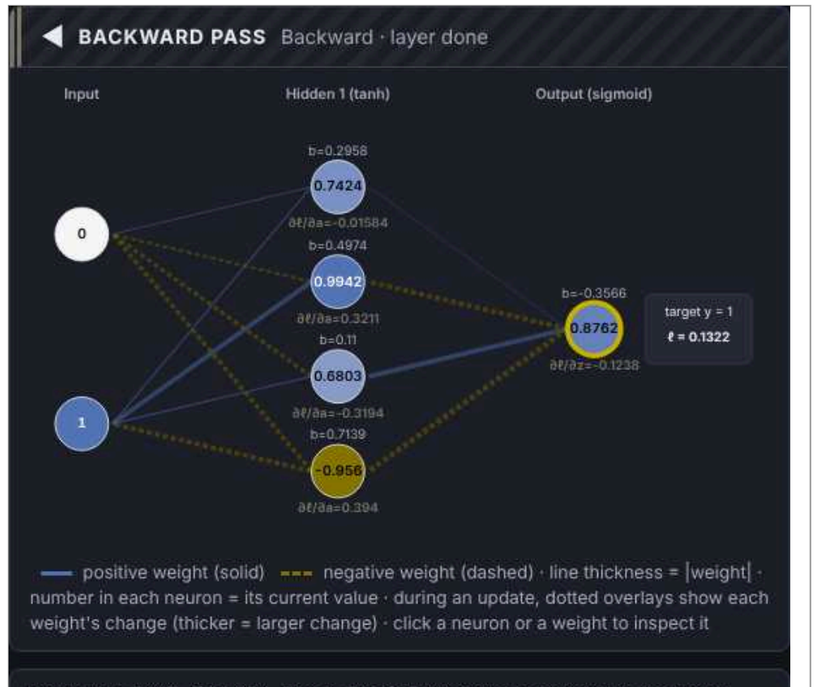
As rendered



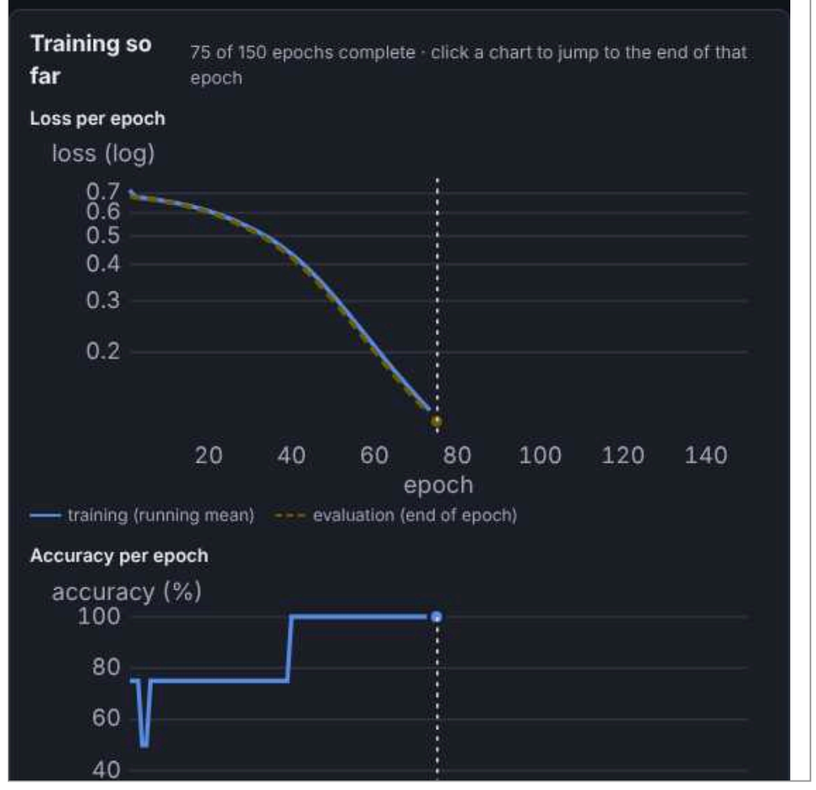
Instruction 7447 of 14,854. Epoch 76 of 150, batch 1, example 1 in the batch. Phase: backward pass, layer 2. Input x = [0, 1], target y = [1]. Prediction 0.8762. Example loss 0.1322.



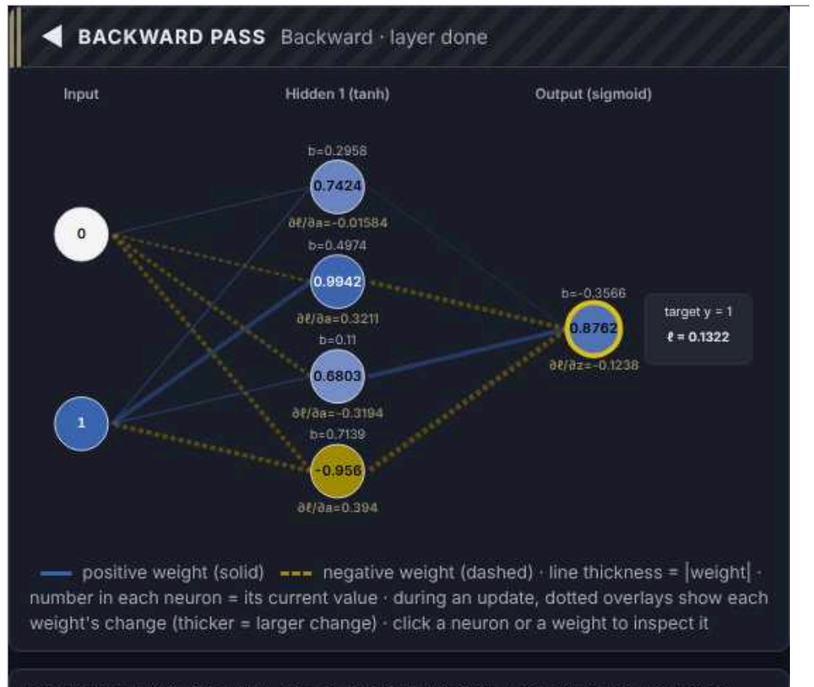
Simulated protanopia



Instruction 7447 of 14,854. Epoch 76 of 150, batch 1, example 1 in the batch. Phase: backward pass, layer 2. Input x = [0, 1], target y = [1]. Prediction 0.8762. Example loss 0.1322.



Simulated deuteranopia

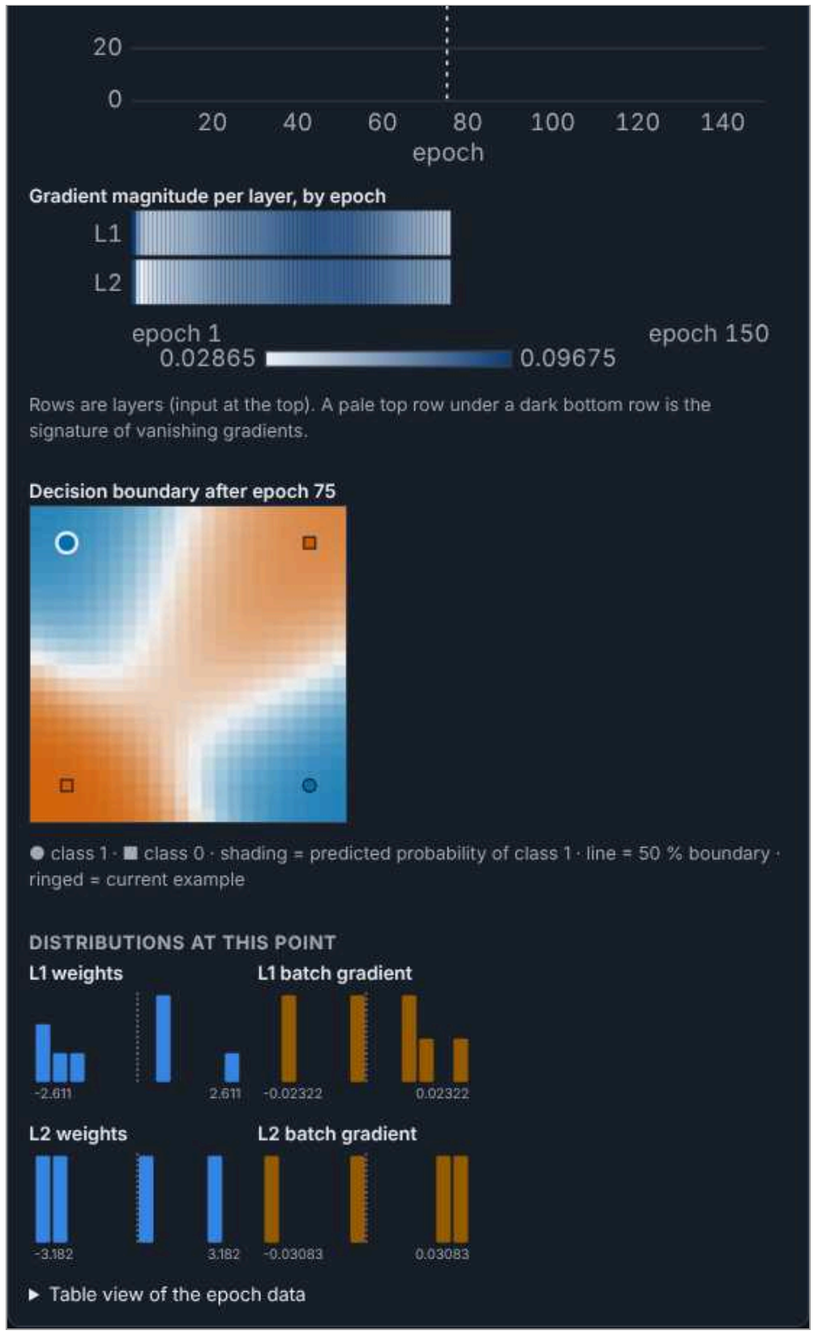


Instruction 7447 of 14,854. Epoch 76 of 150, batch 1, example 1 in the batch. Phase: backward pass, layer 2. Input x = [0, 1], target y = [1]. Prediction 0.8762. Example loss 0.1322.

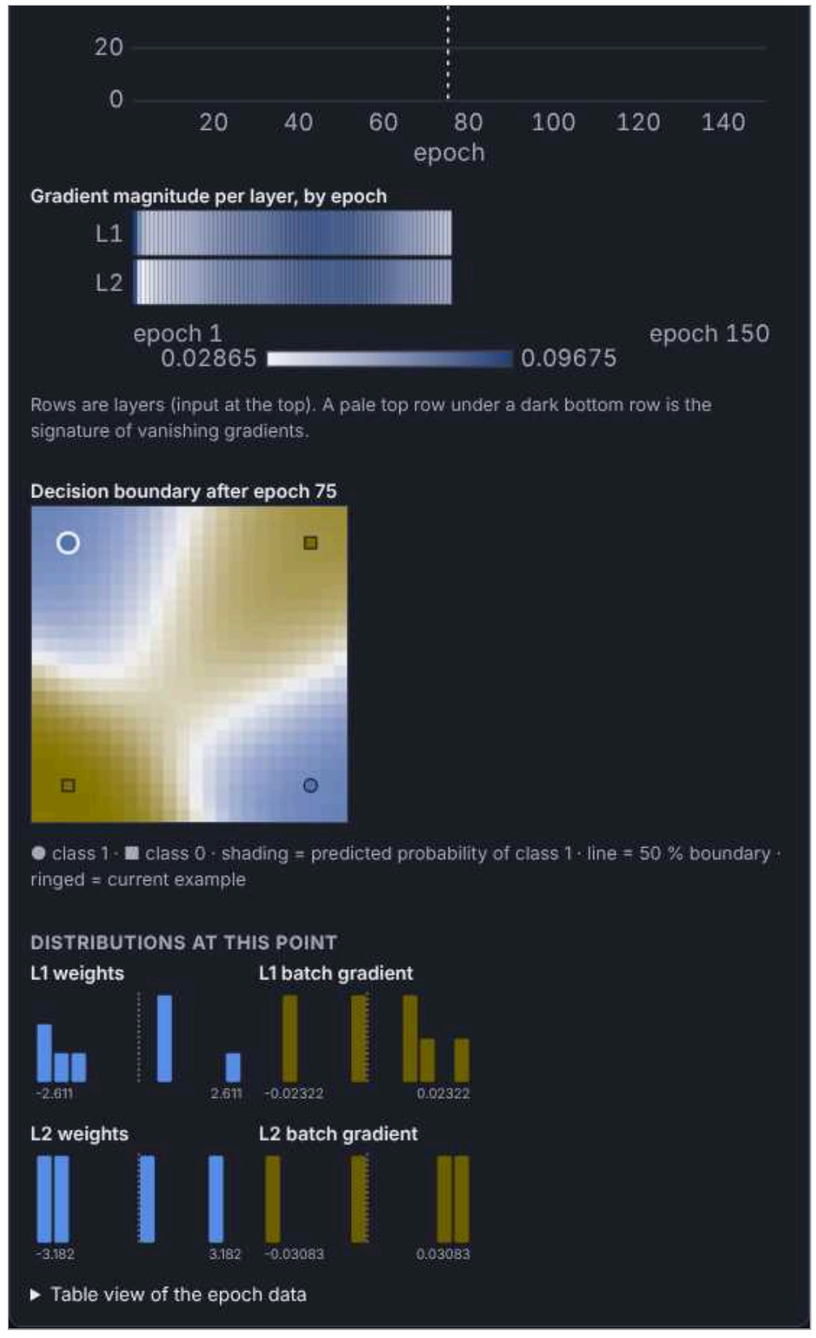


7. XOR needs a hidden layer — dark theme, backward phase (part 2 of 2)

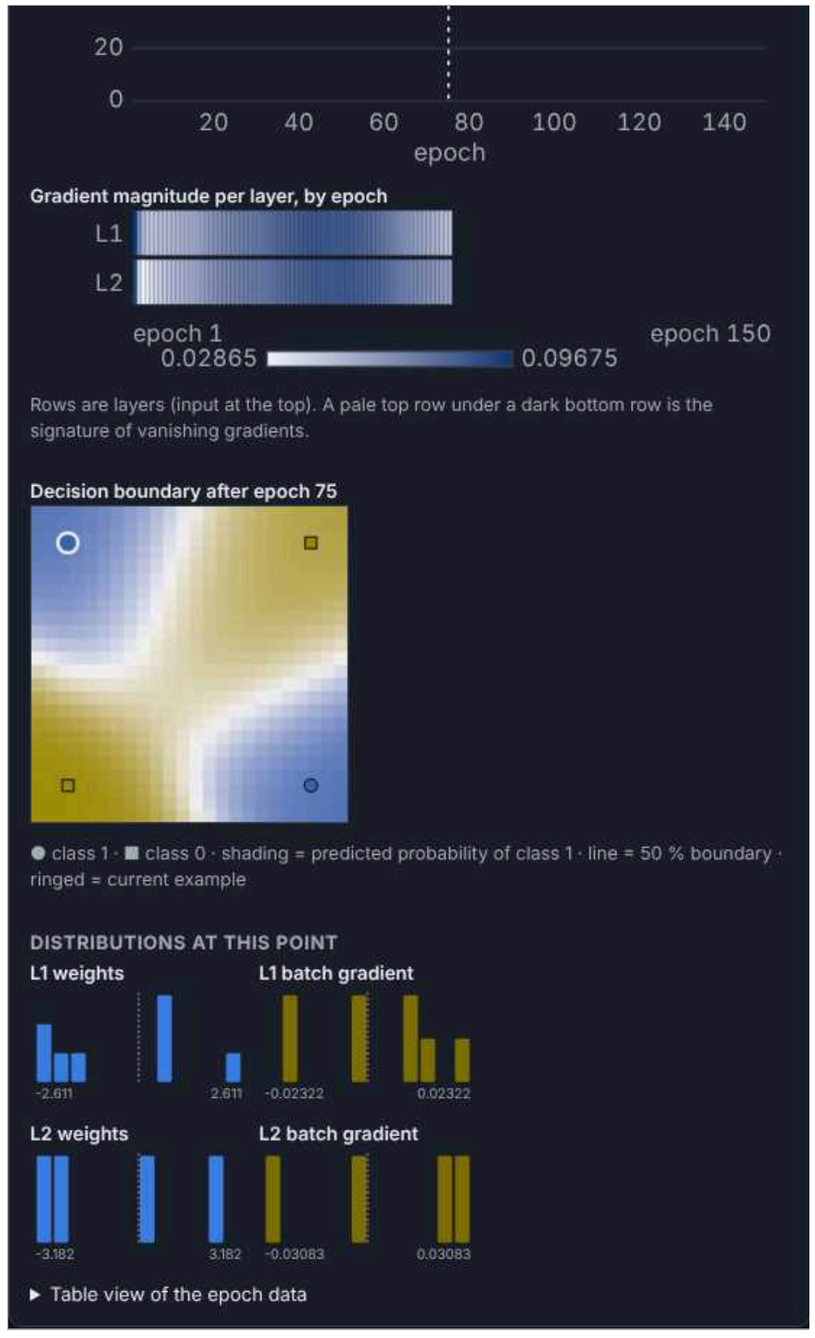
As rendered



Simulated protanopia

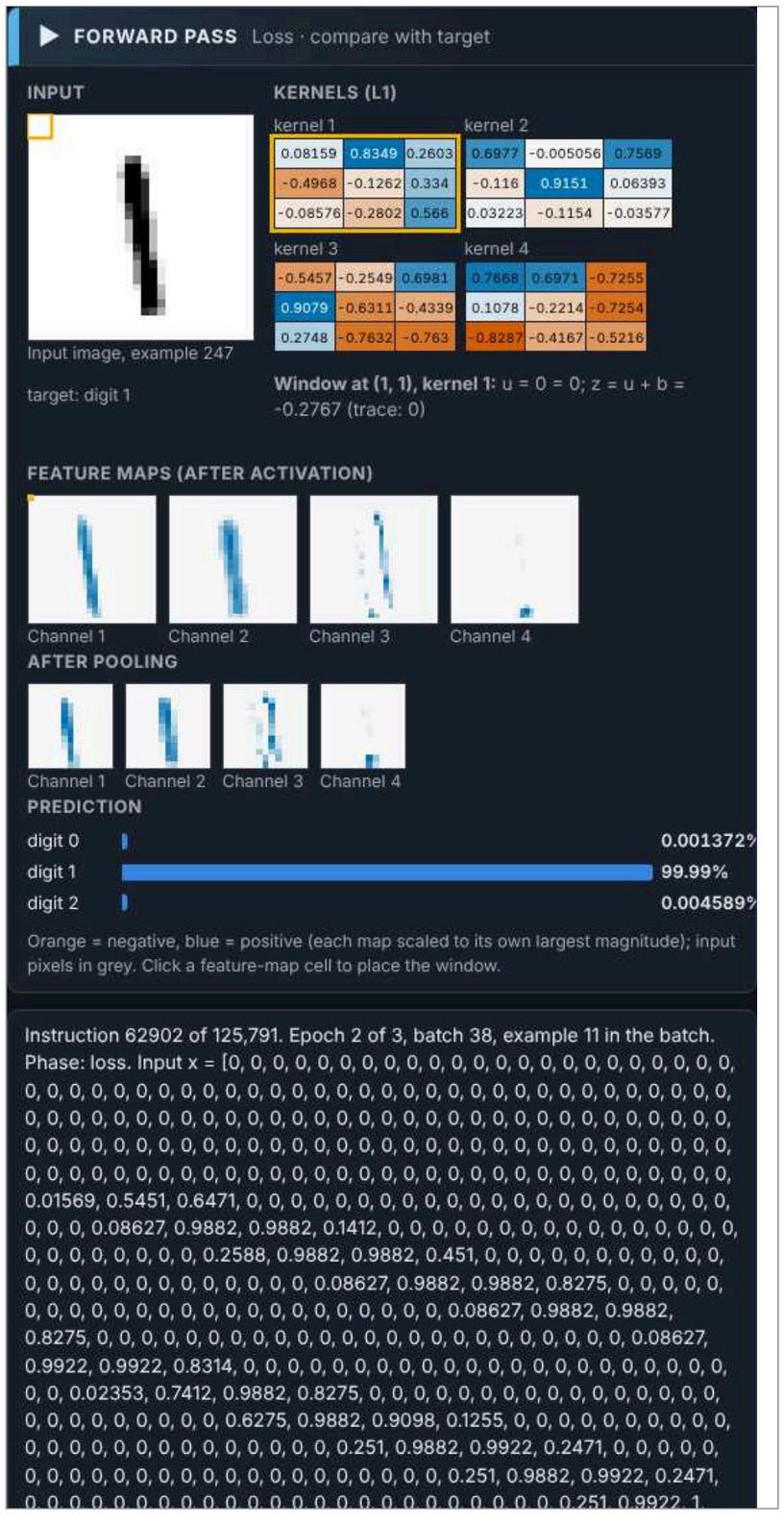


Simulated deuteranopia



8. A CNN on handwritten digits — dark theme, forward phase (part 1 of 3)

As rendered


FORWARD PASS Loss · compare with target
INPUT
KERNELS (L1)
kernel 1
0.08159 0.8349 0.2603
-0.4968 -0.1262 0.334
-0.08576 -0.2802 0.566
kernel 2
0.6977 -0.005056 0.7569
-0.116 0.9151 0.06393
0.03223 -0.1154 -0.03577
kernel 3
-0.5457 -0.2549 0.6981
0.9079 -0.6311 -0.4339
0.2748 -0.7632 -0.763
kernel 4
0.7668 0.6971 -0.7255
0.1078 -0.2214 -0.7254
-0.8287 -0.4167 -0.5216
Input image, example 247
target: digit 1
Window at (1, 1), kernel 1: u = 0 = 0; z = u + b = -0.2767 (trace: 0)
FEATURE MAPS (AFTER ACTIVATION)
Channel 1
Channel 2
Channel 3
Channel 4
AFTER POOLING
Channel 1
Channel 2
Channel 3
Channel 4
PREDICTION
digit 0 0.001372%
digit 1 99.99%
digit 2 0.004589%
Orange = negative, blue = positive (each map scaled to its own largest magnitude); input pixels in grey. Click a feature-map cell to place the window.
Instruction 62902 of 125,791. Epoch 2 of 3, batch 38, example 11 in the batch.
Phase: loss. Input x = [0, 0, 0,

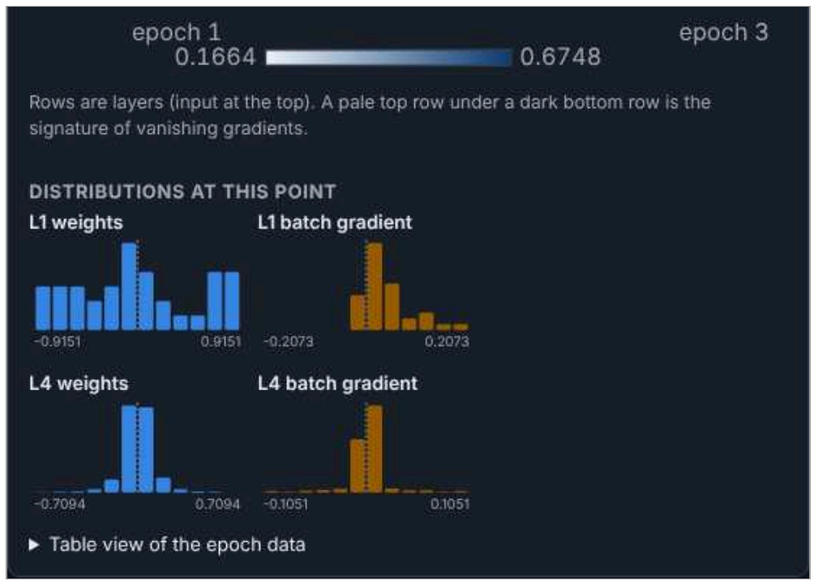
8. A CNN on handwritten digits — dark theme, forward phase (part 2 of 3)

As rendered

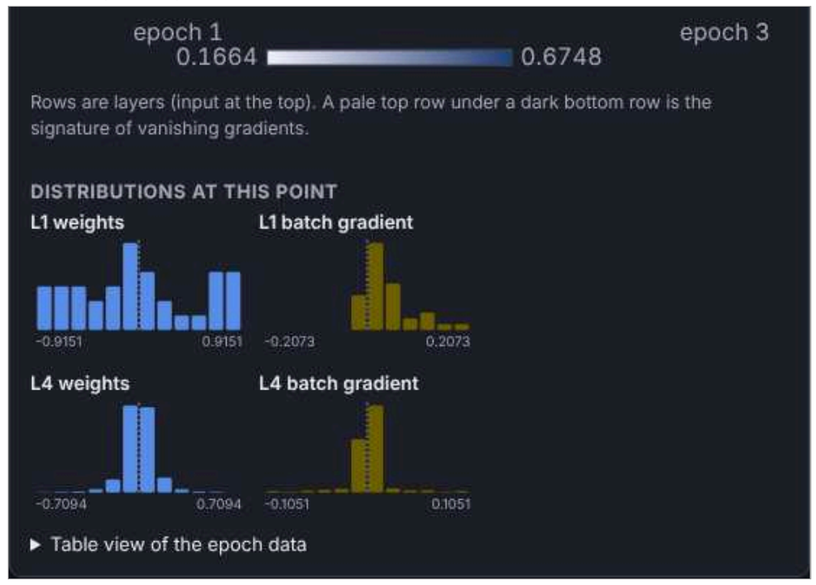
0.2471, 0, 0, 0

8. A CNN on handwritten digits — dark theme, forward phase (part 3 of 3)

As rendered



Simulated protanopia



Simulated deuteranopia

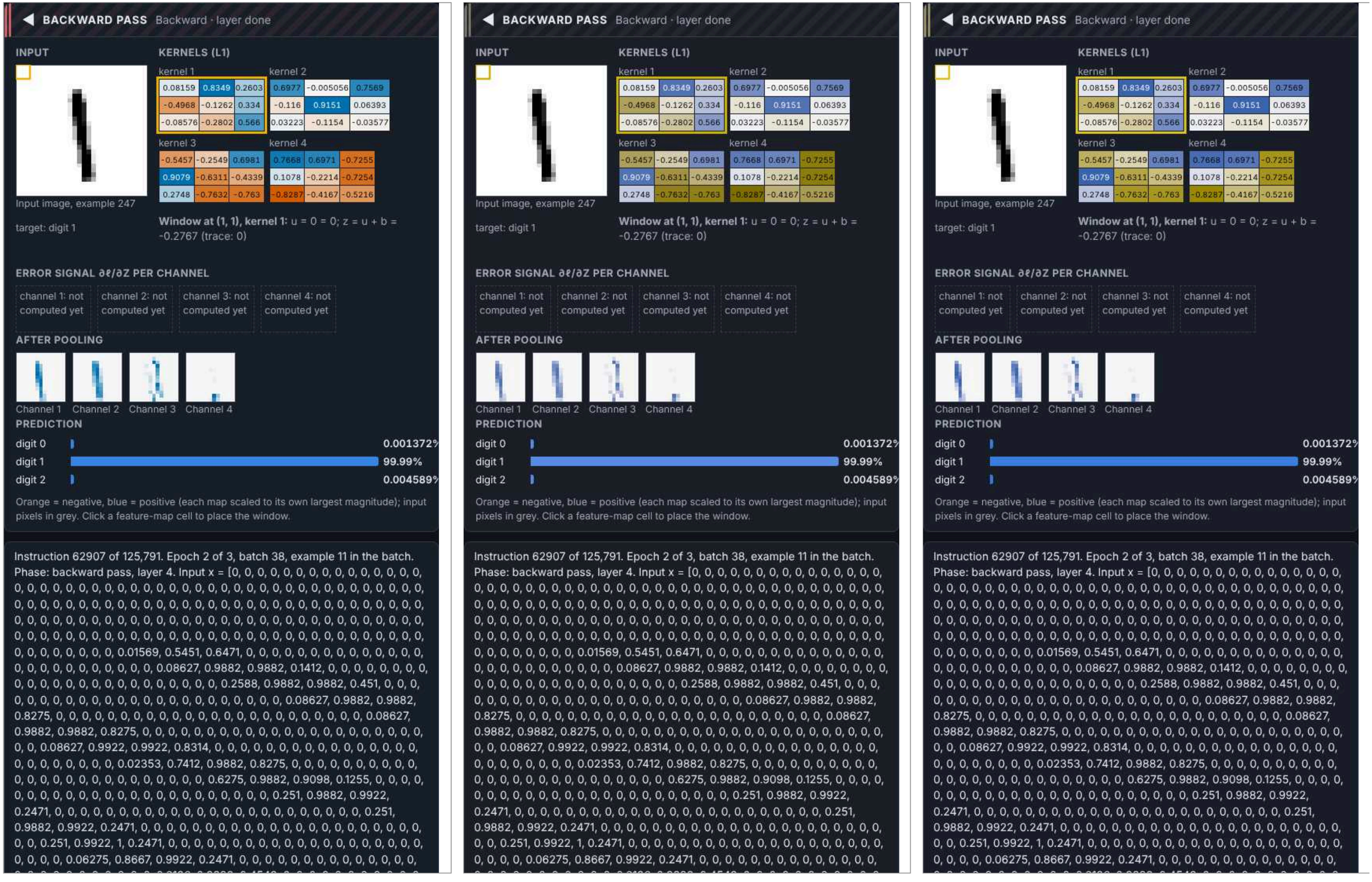


8. A CNN on handwritten digits — dark theme, backward phase (part 1 of 3)

As rendered

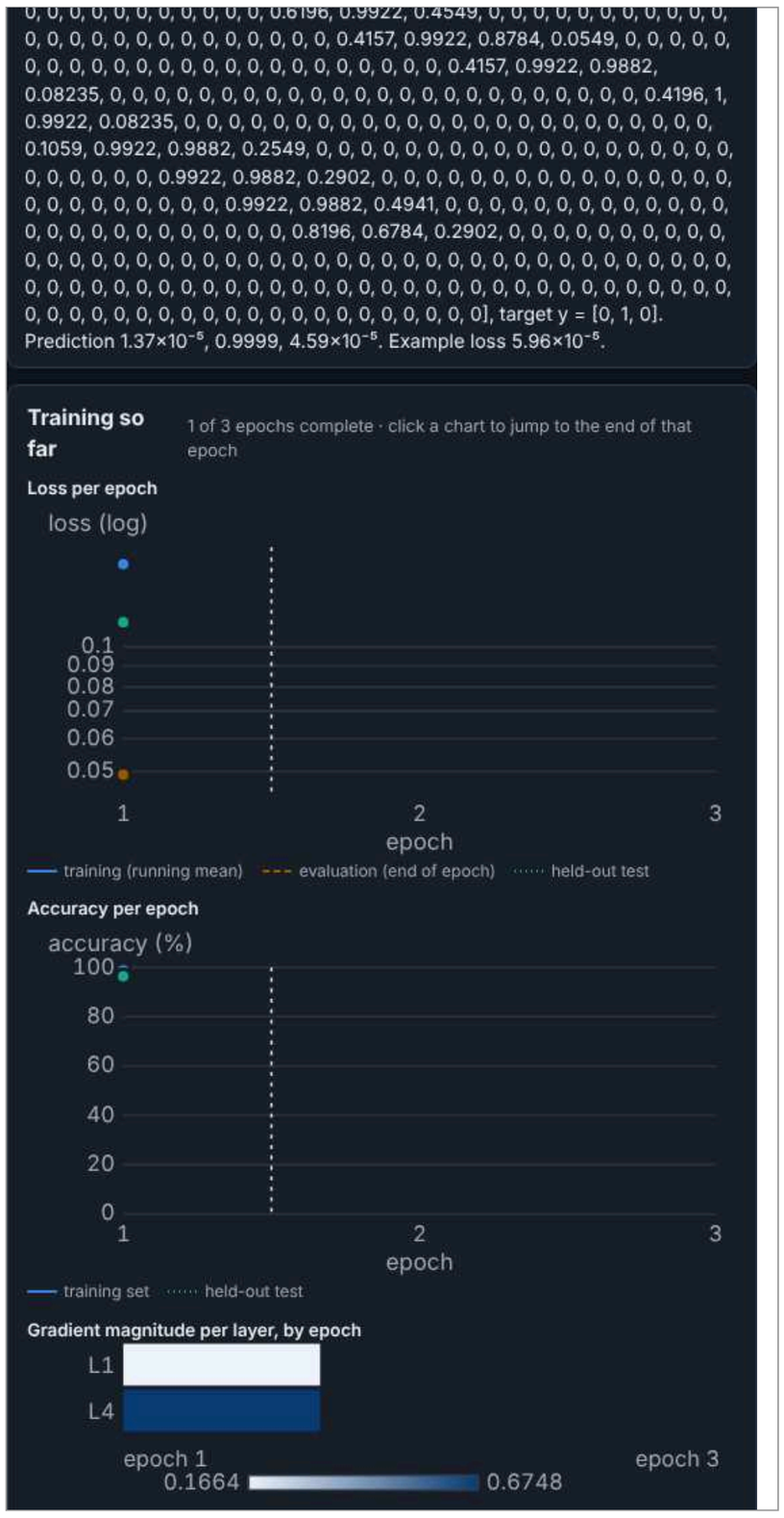
Simulated protanopia

Simulated deuteranopia



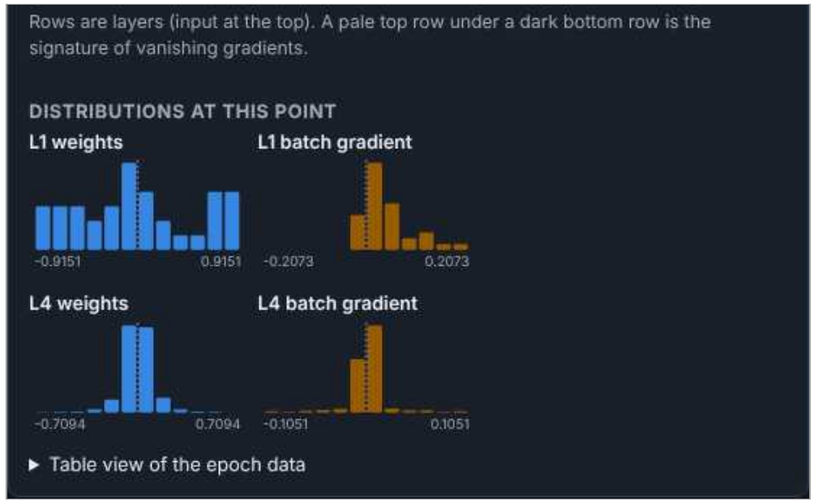
8. A CNN on handwritten digits — dark theme, backward phase (part 2 of 3)

As rendered

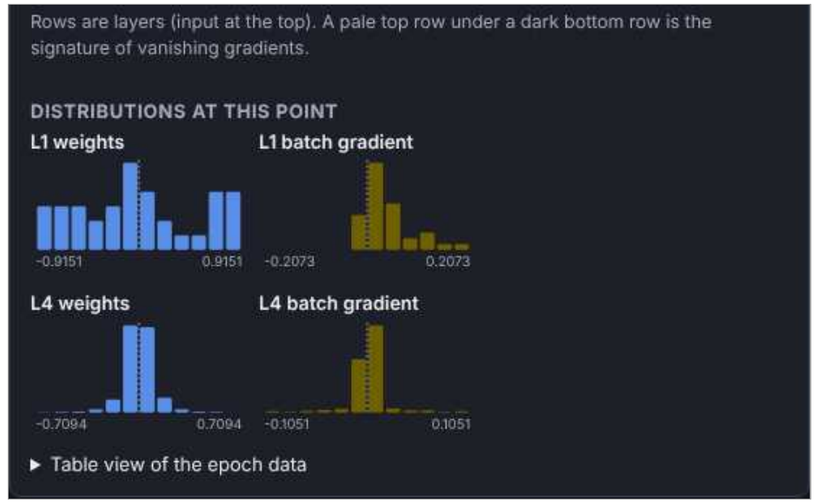

0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0.6196, 0.9922, 0.4549, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0

8. A CNN on handwritten digits — dark theme, backward phase (part 3 of 3)

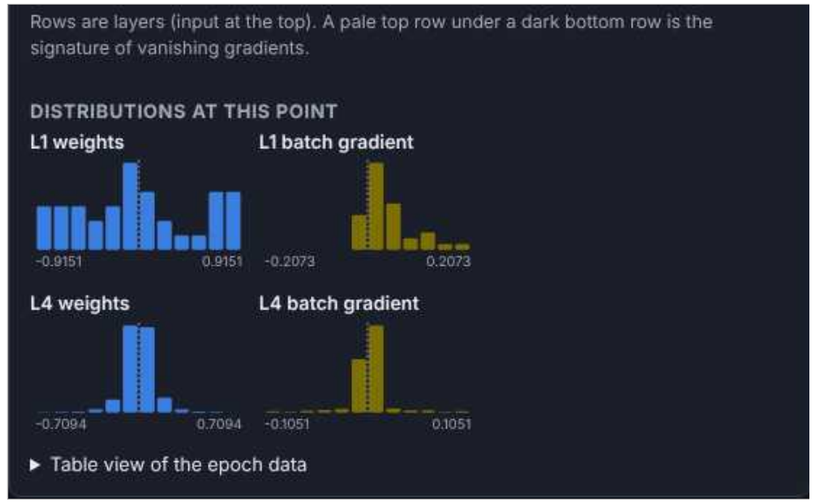
As rendered



Simulated protanopia

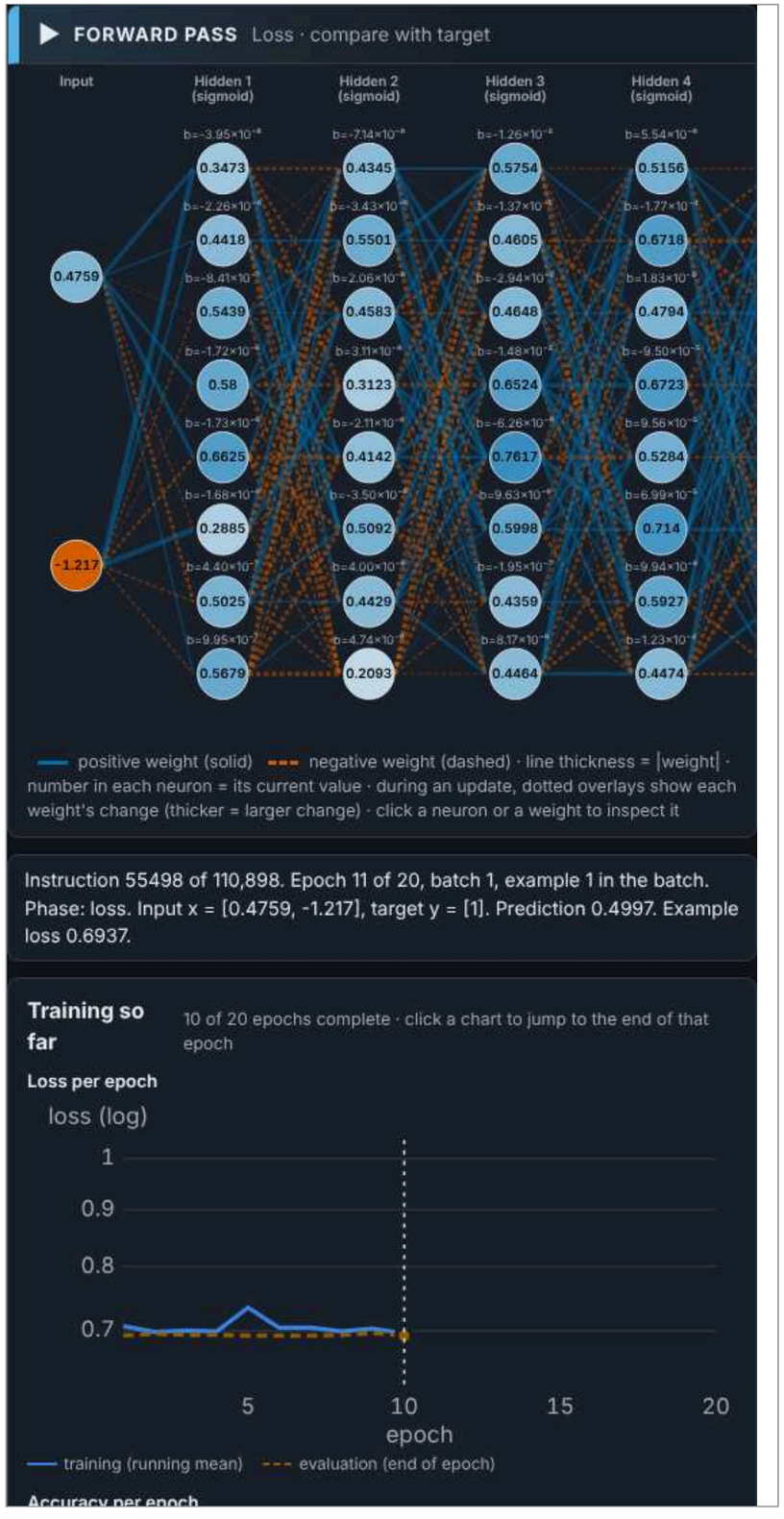


Simulated deuteranopia

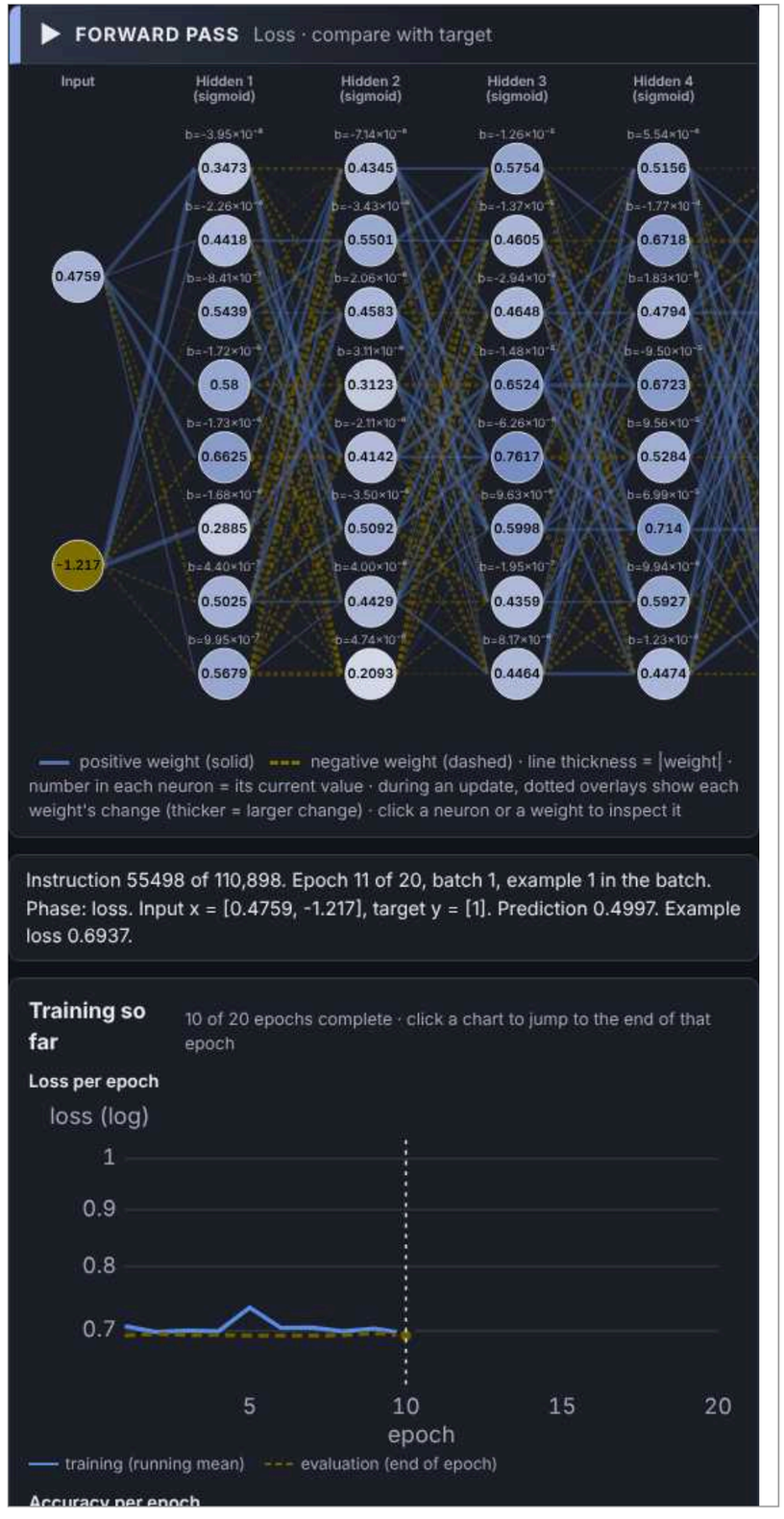


D1. Vanishing gradients — dark theme, forward phase (part 1 of 3)

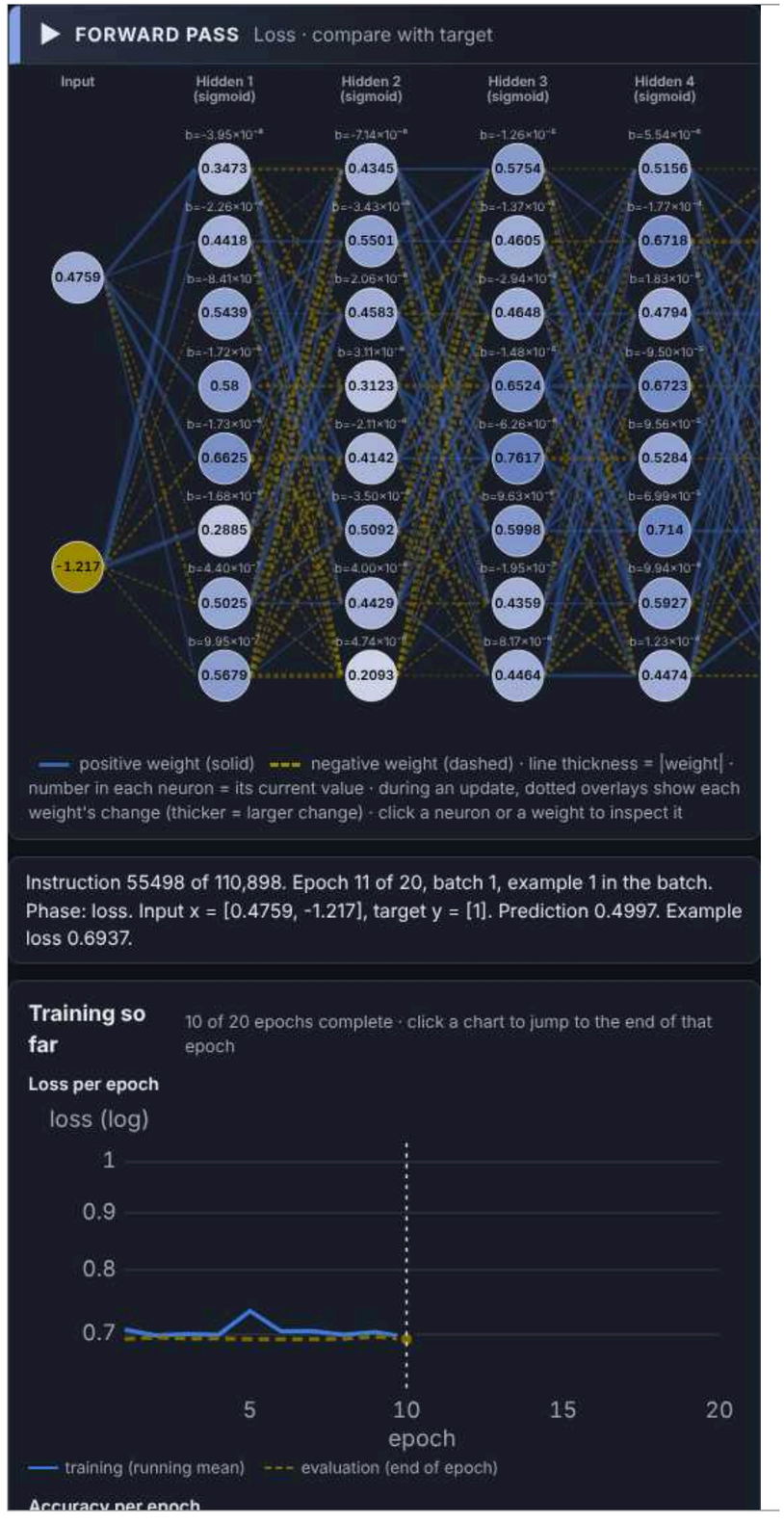
As rendered



Simulated protanopia

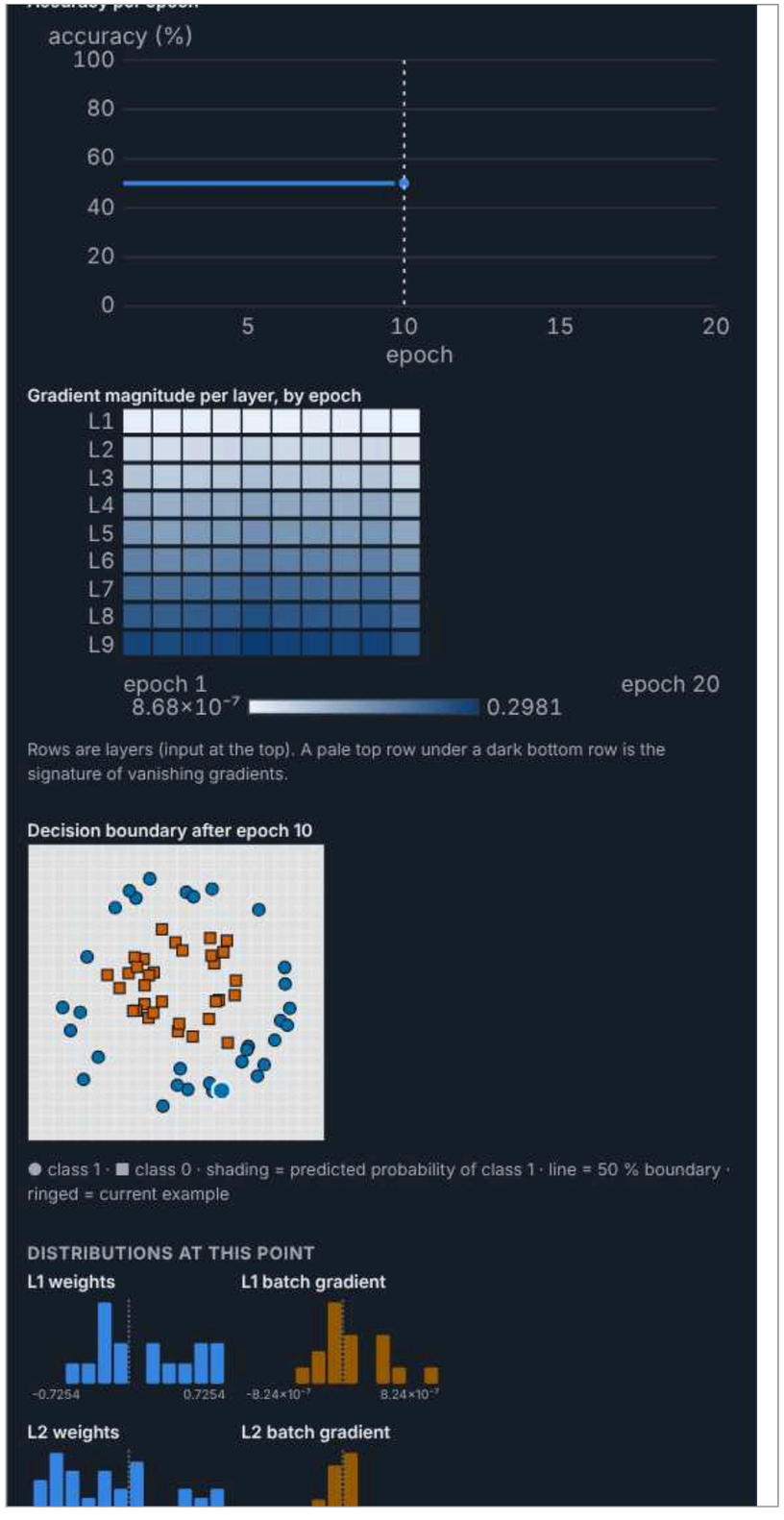


Simulated deuteranopia

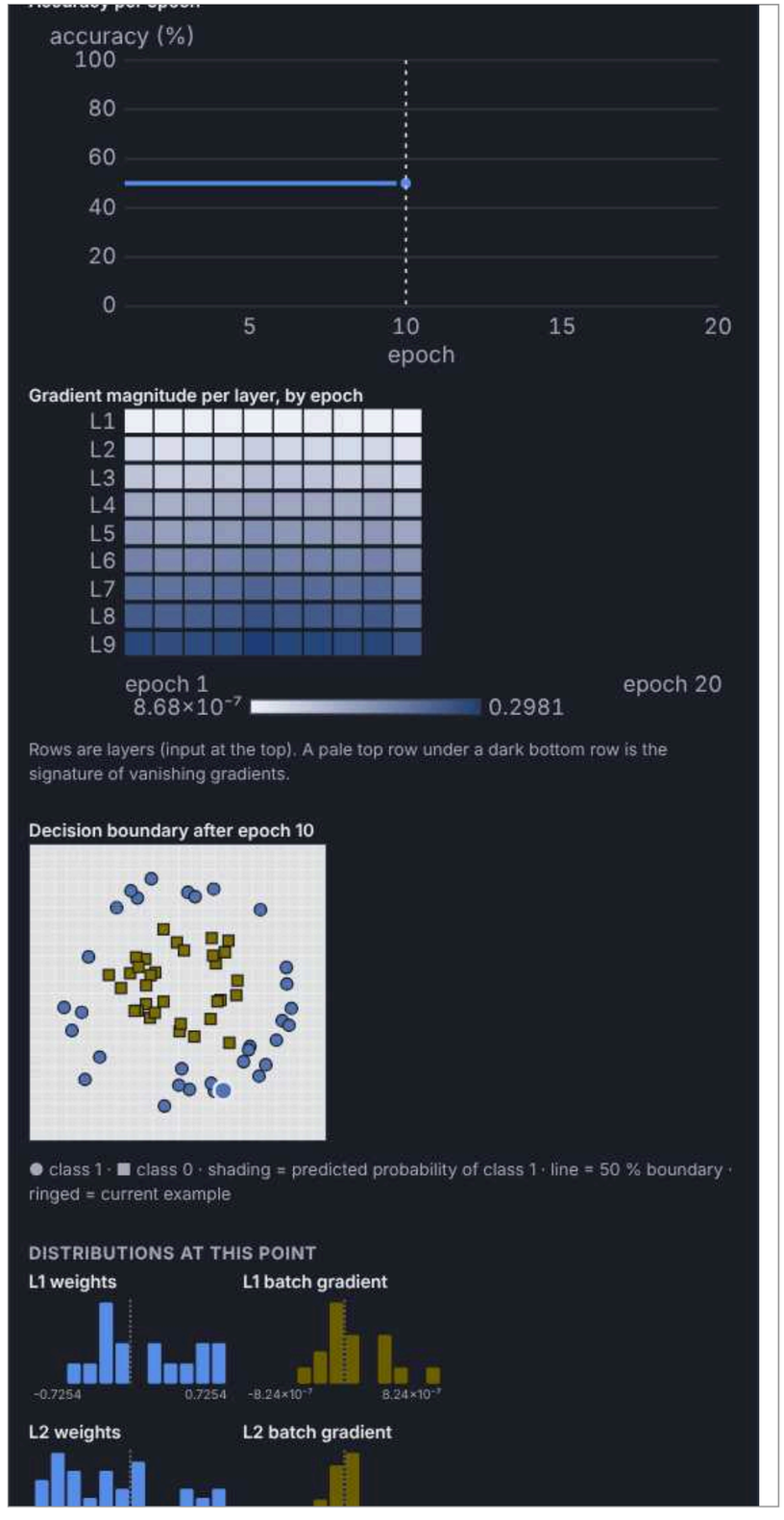


D1. Vanishing gradients — dark theme, forward phase (part 2 of 3)

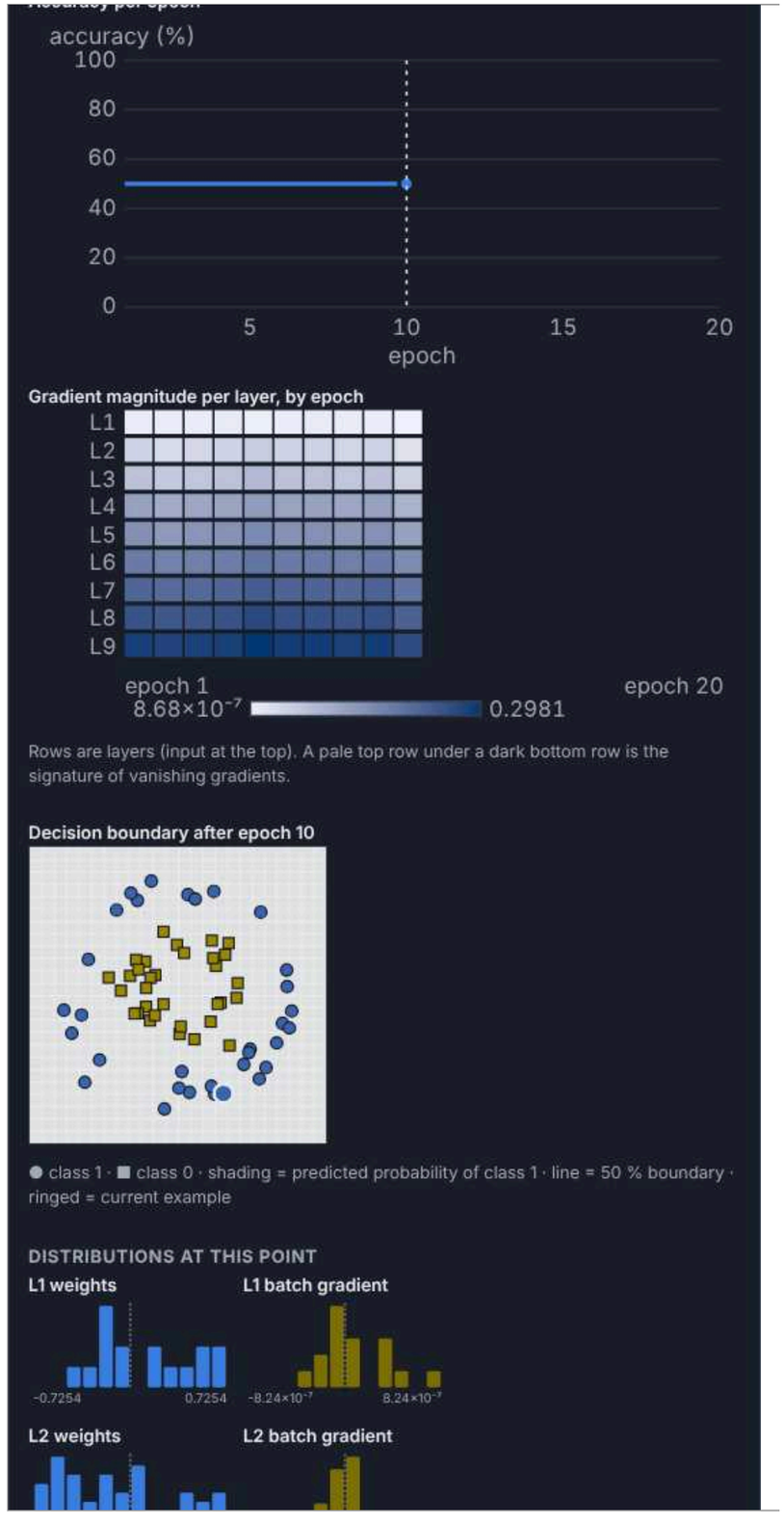
As rendered



Simulated protanopia

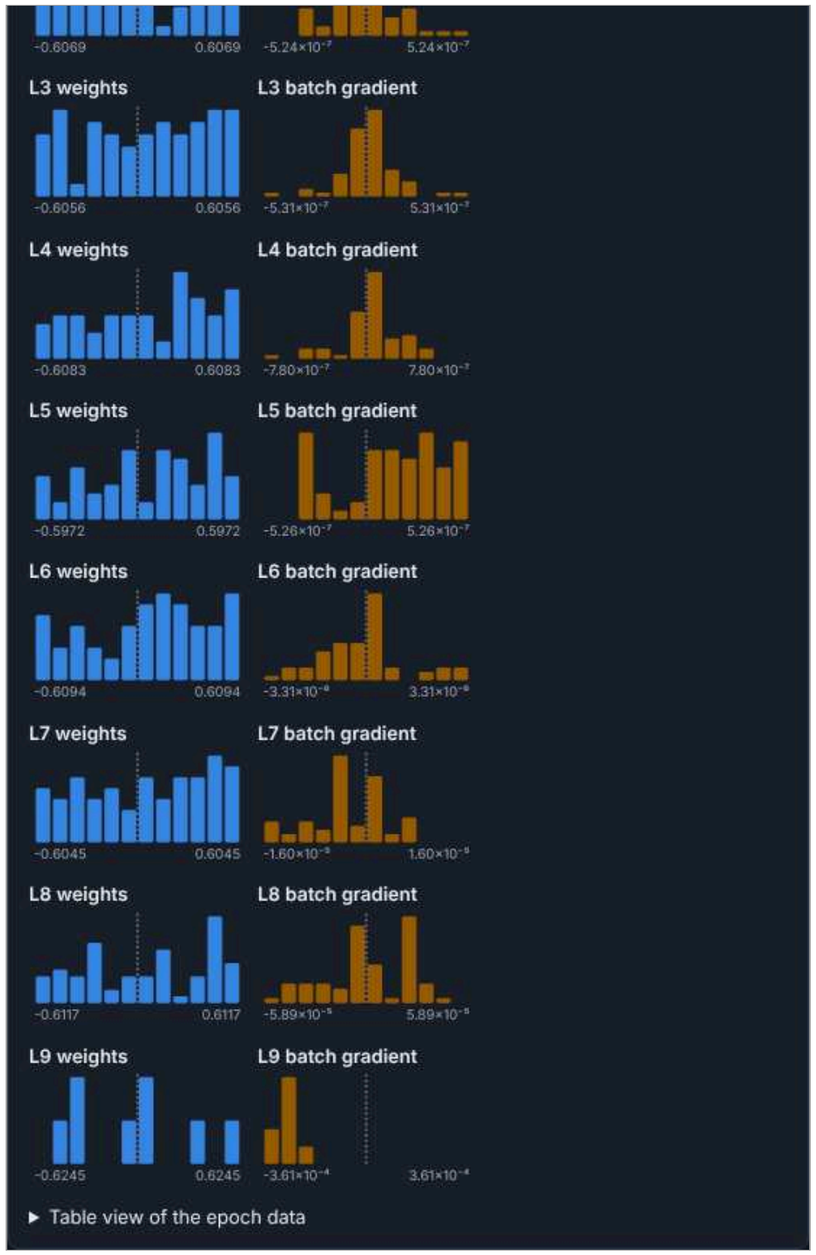


Simulated deuteranopia

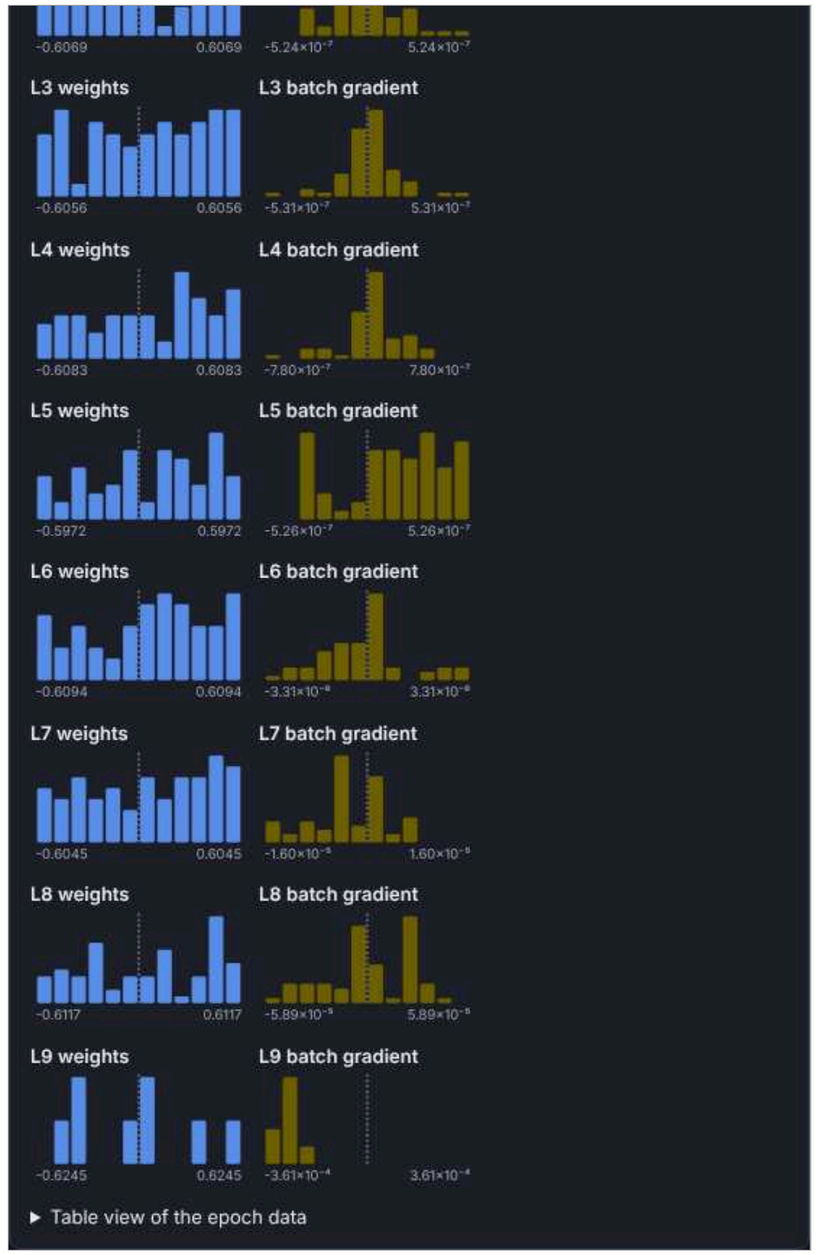


D1. Vanishing gradients — dark theme, forward phase (part 3 of 3)

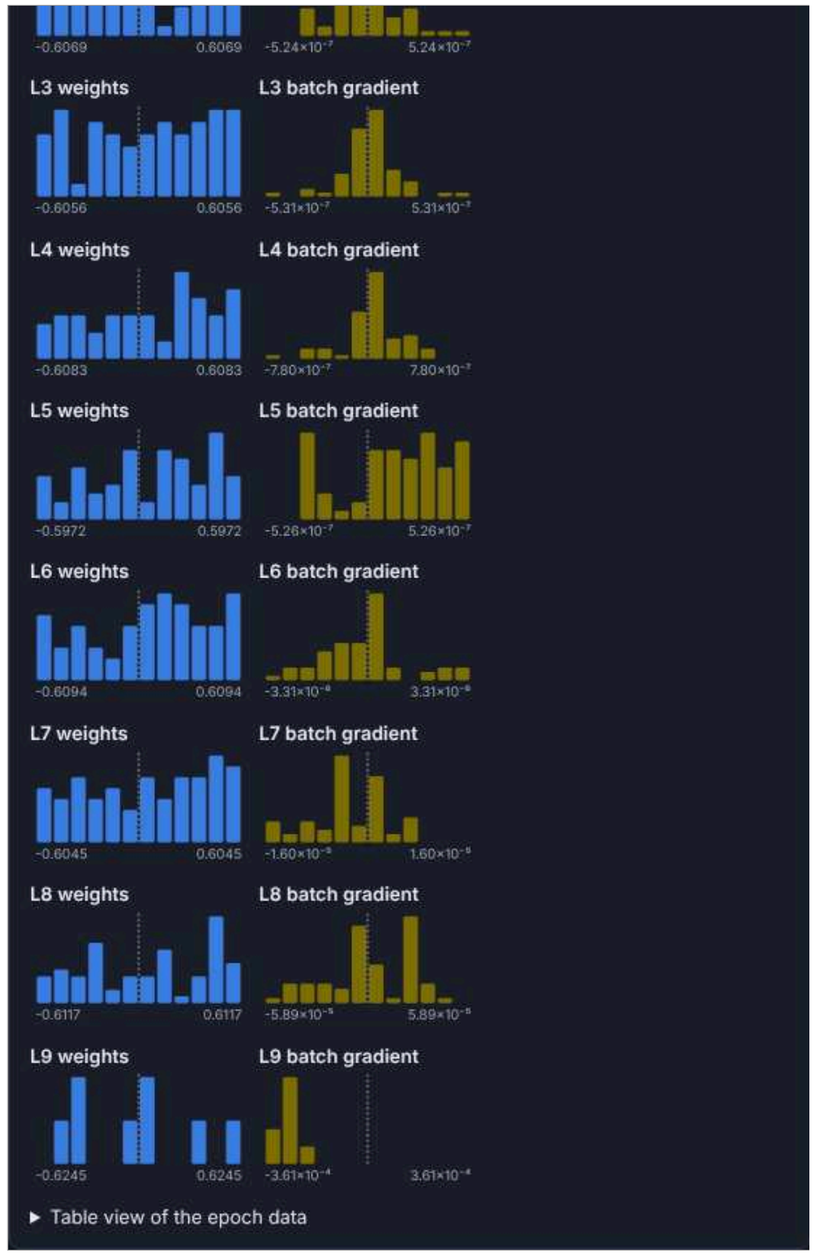
As rendered



Simulated protanopia

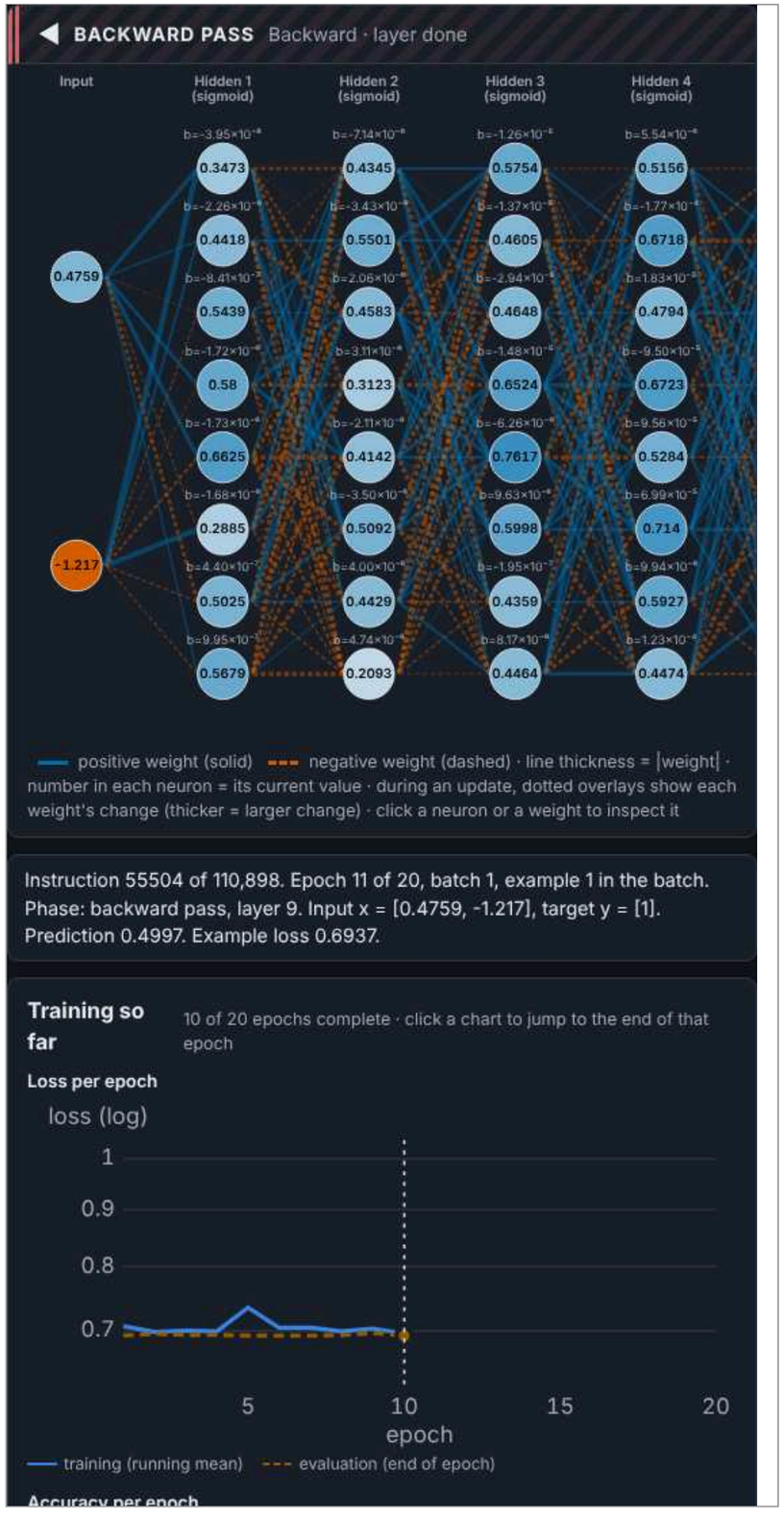


Simulated deuteranopia

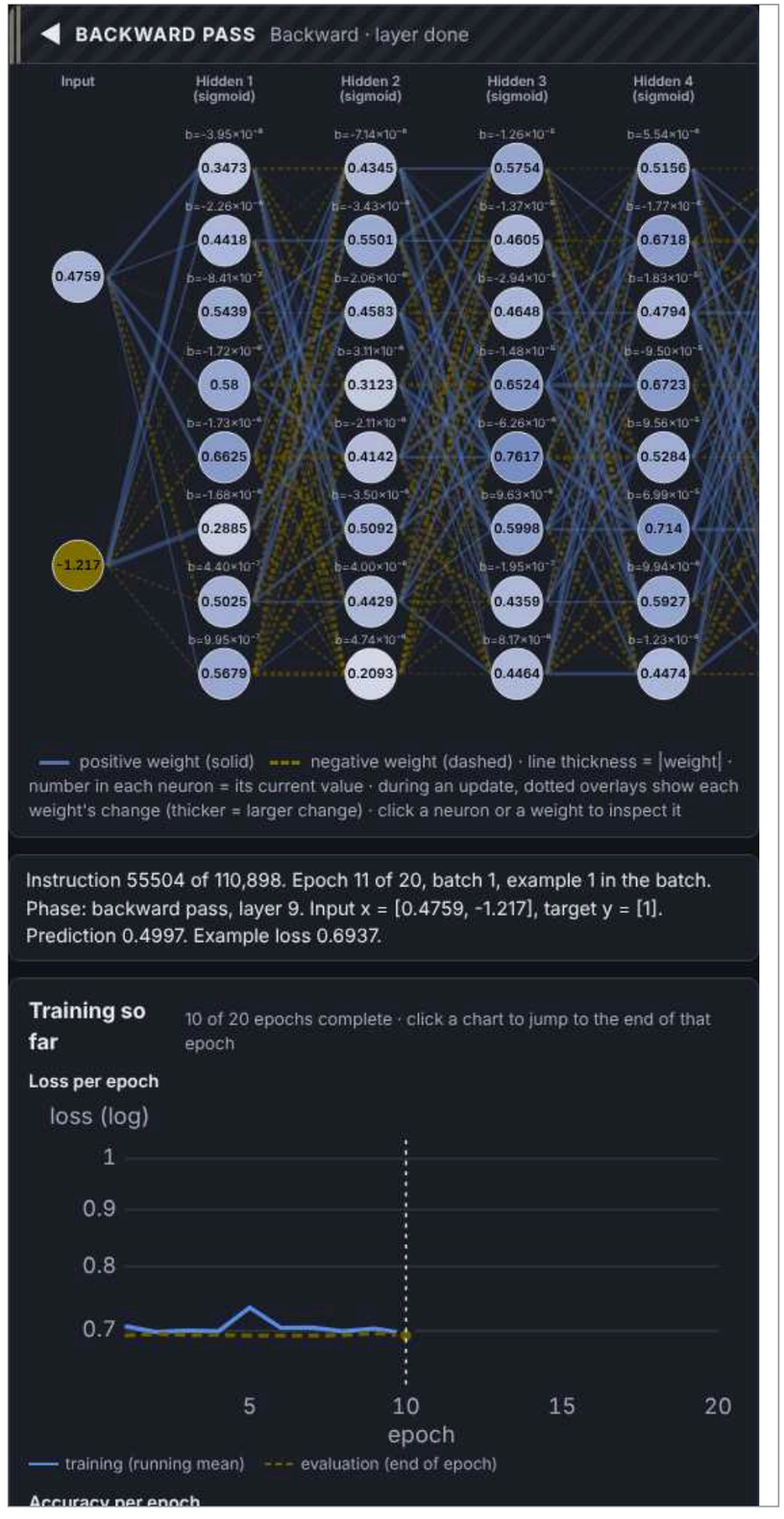


D1. Vanishing gradients — dark theme, backward phase (part 1 of 3)

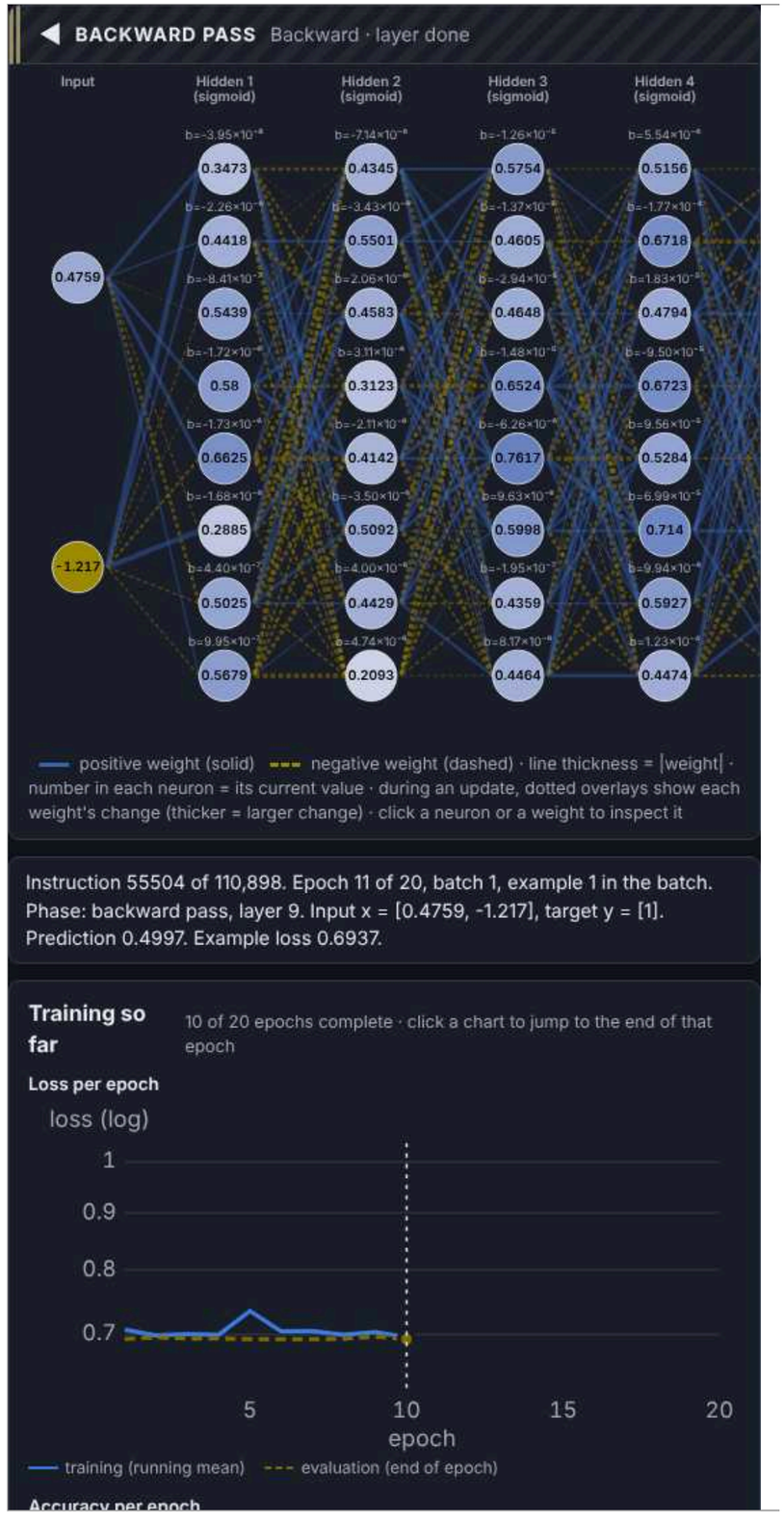
As rendered



Simulated protanopia

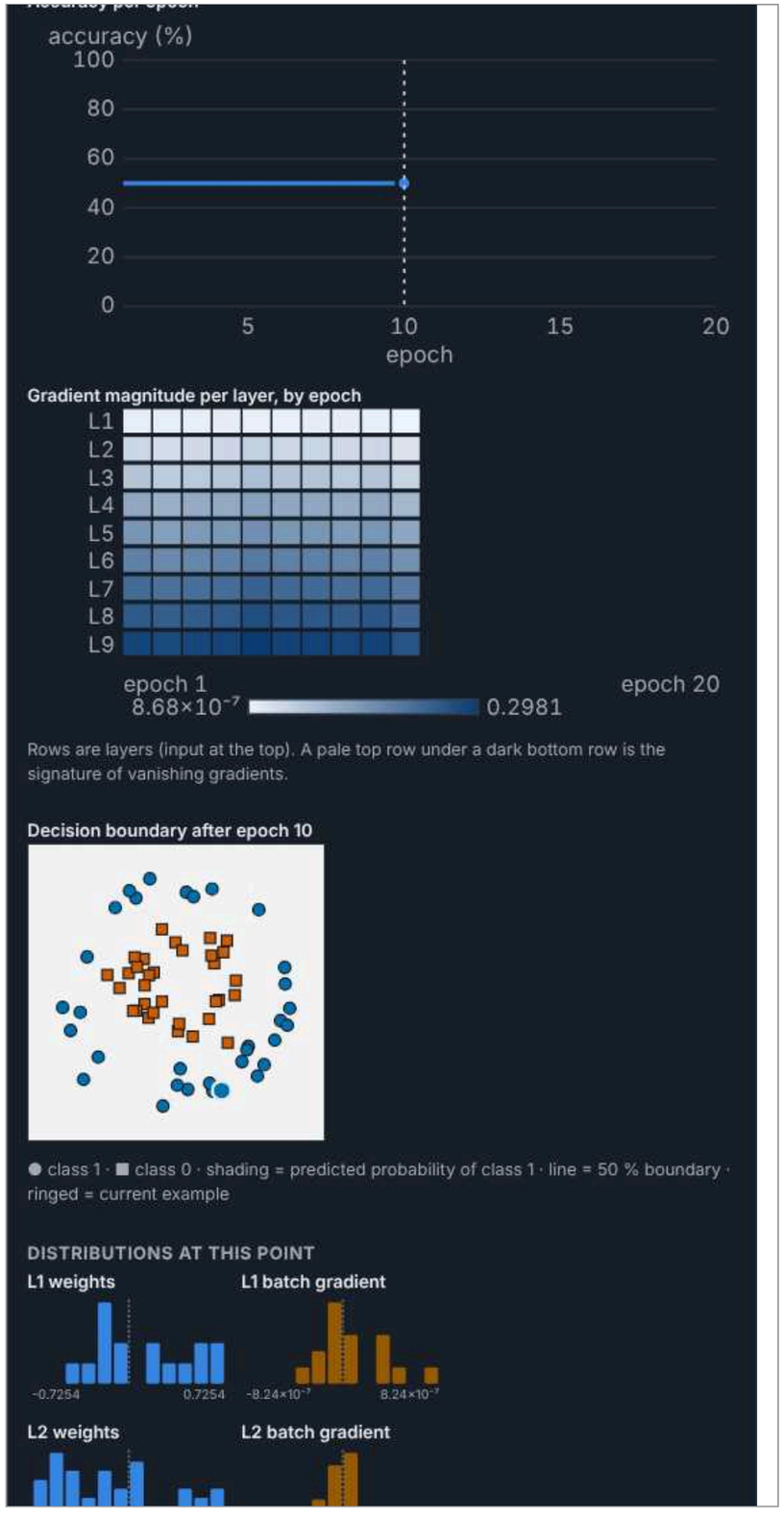


Simulated deuteranopia

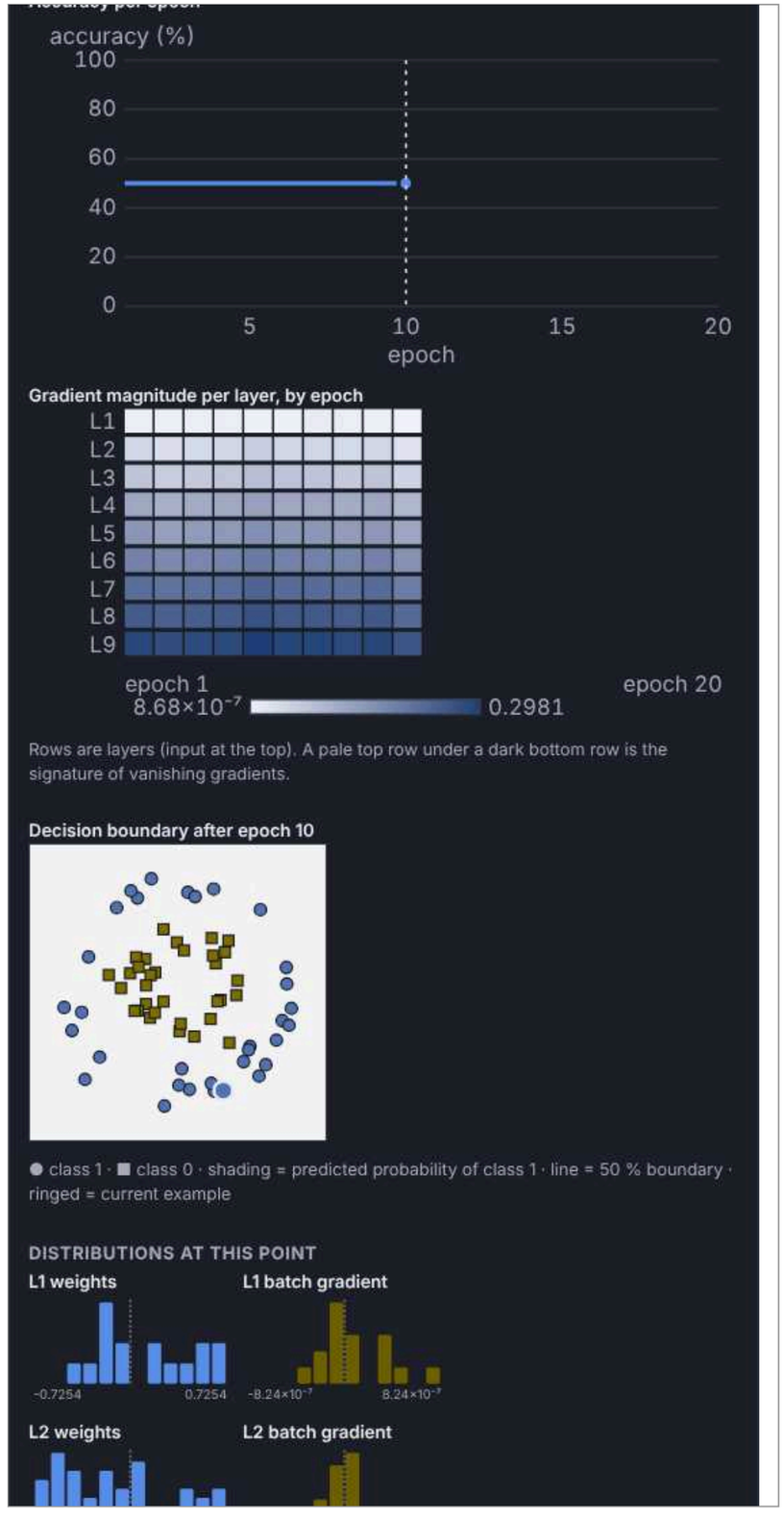


D1. Vanishing gradients — dark theme, backward phase (part 2 of 3)

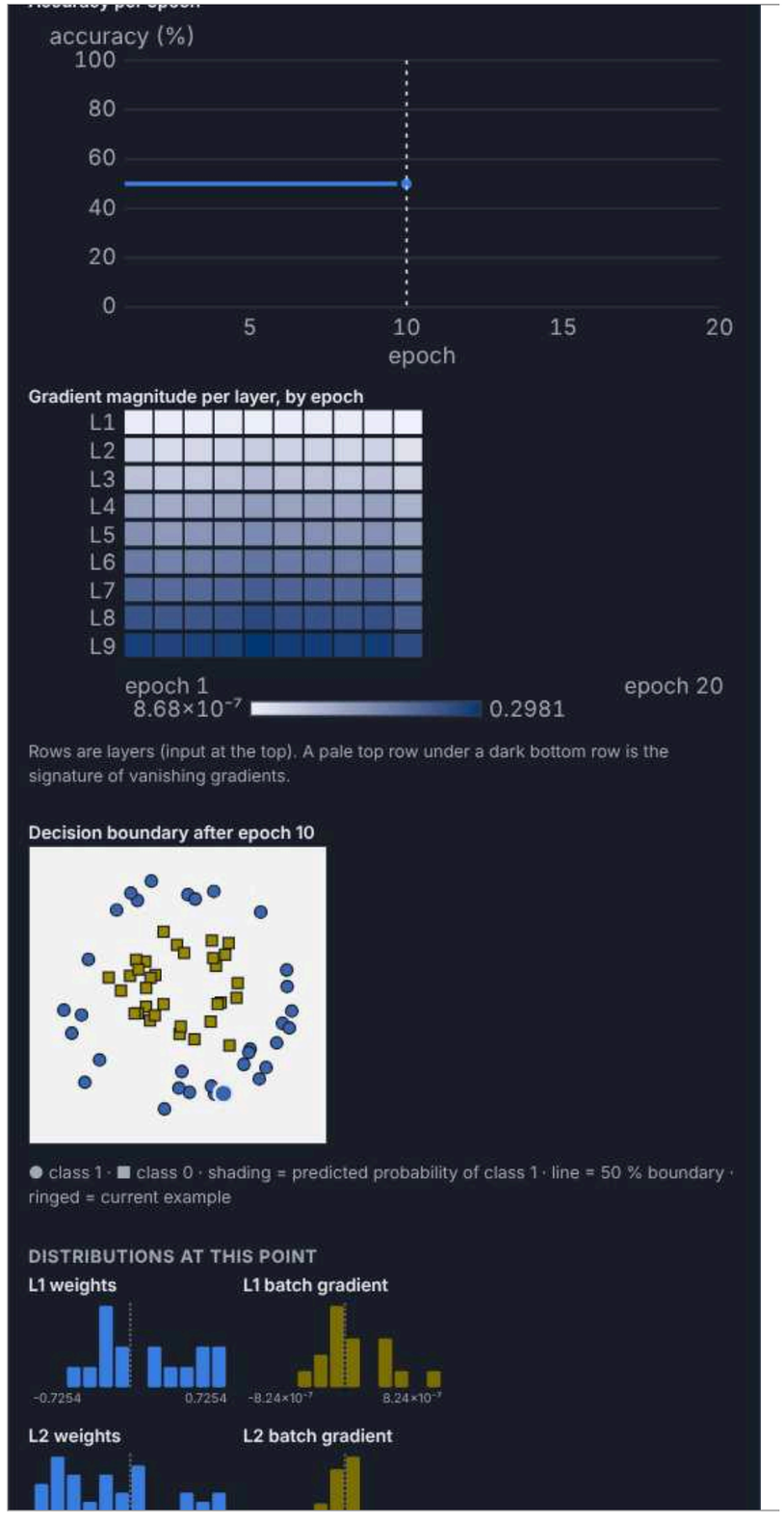
As rendered



Simulated protanopia

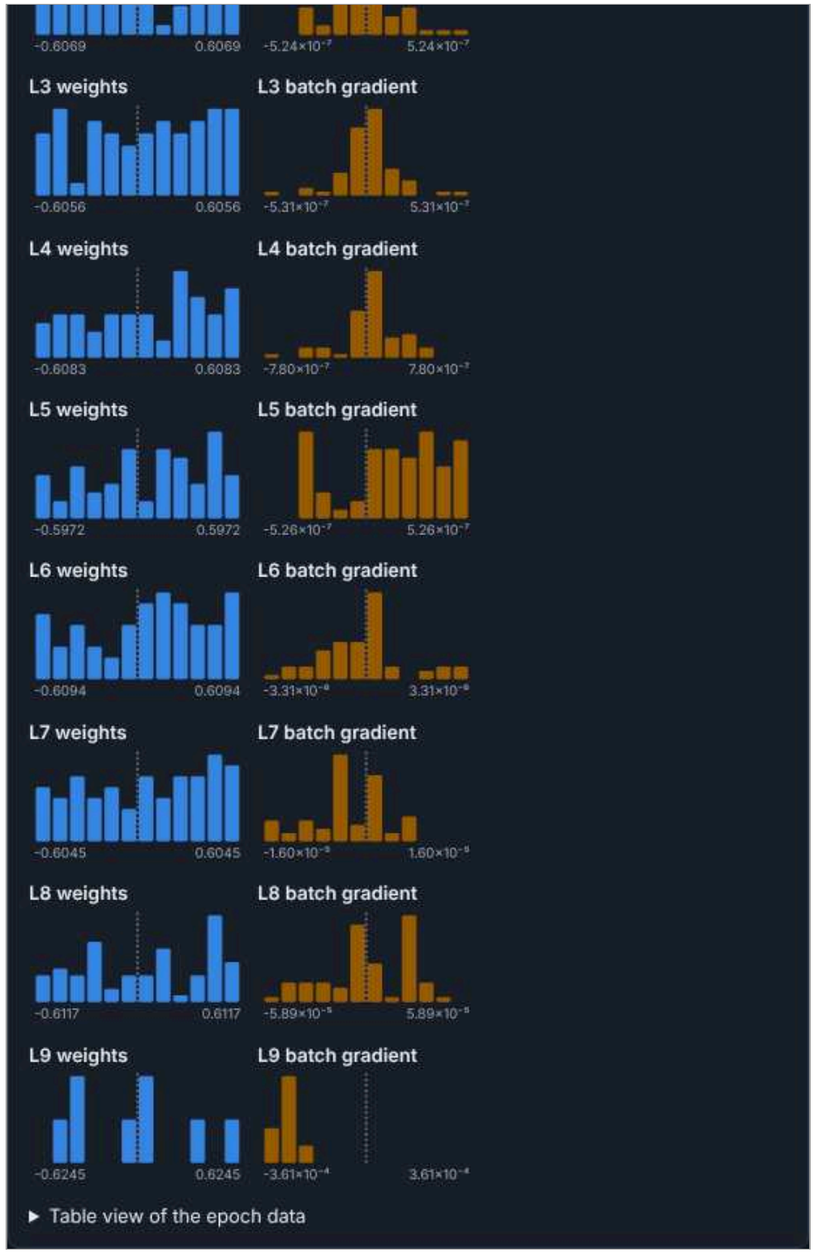


Simulated deuteranopia

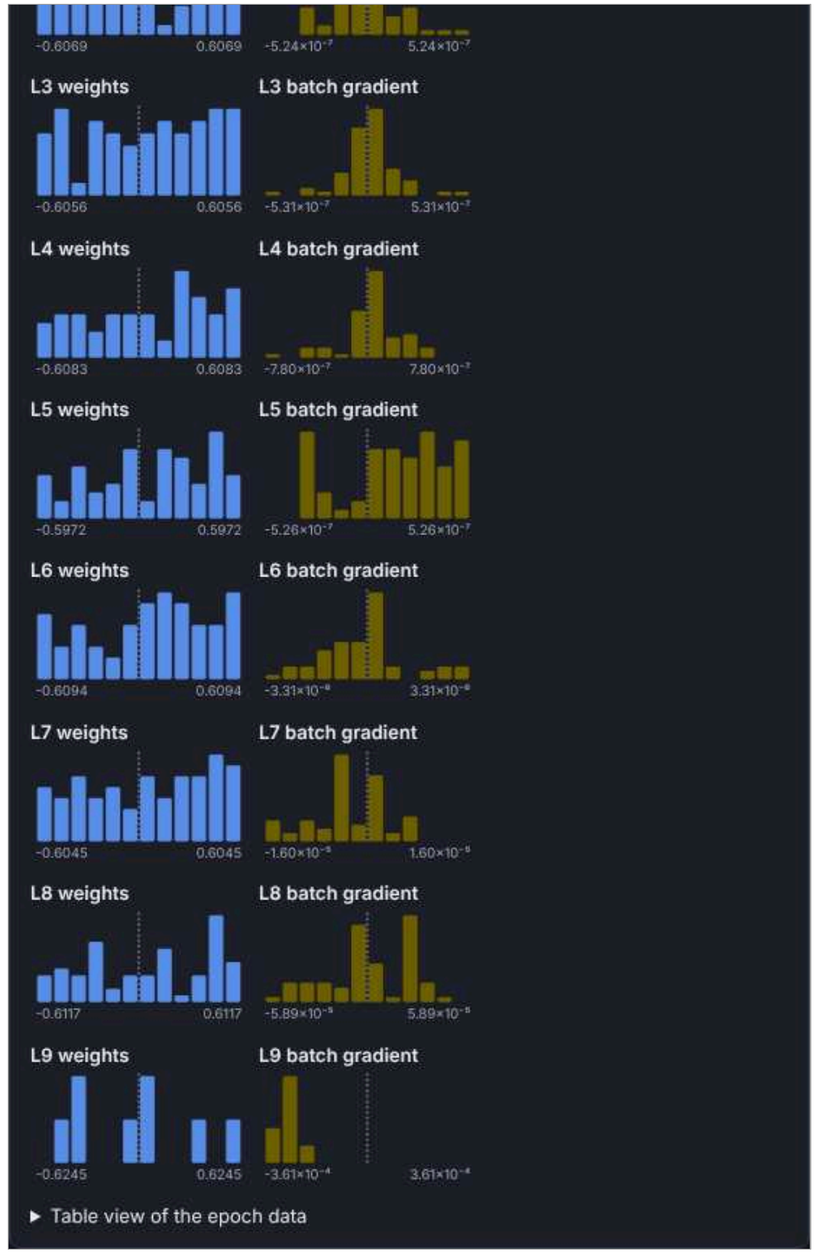


D1. Vanishing gradients — dark theme, backward phase (part 3 of 3)

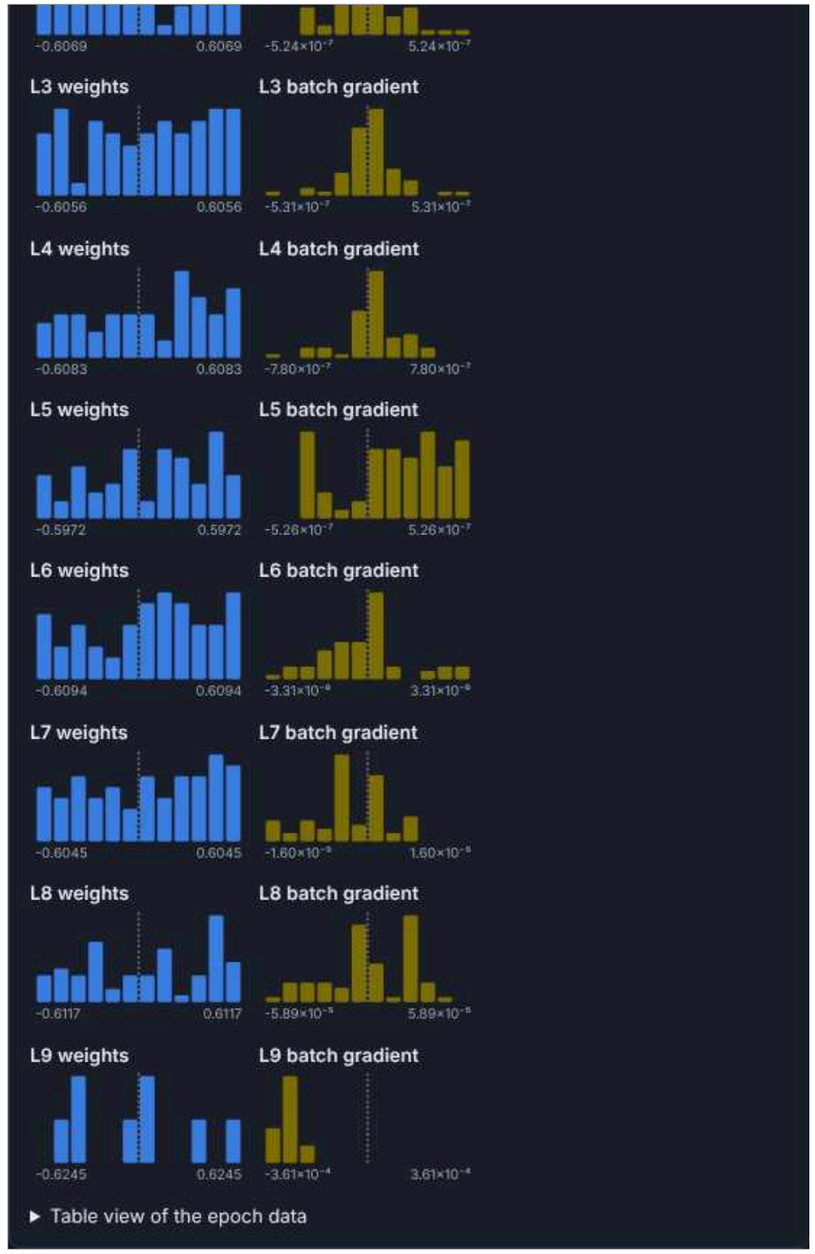
As rendered



Simulated protanopia

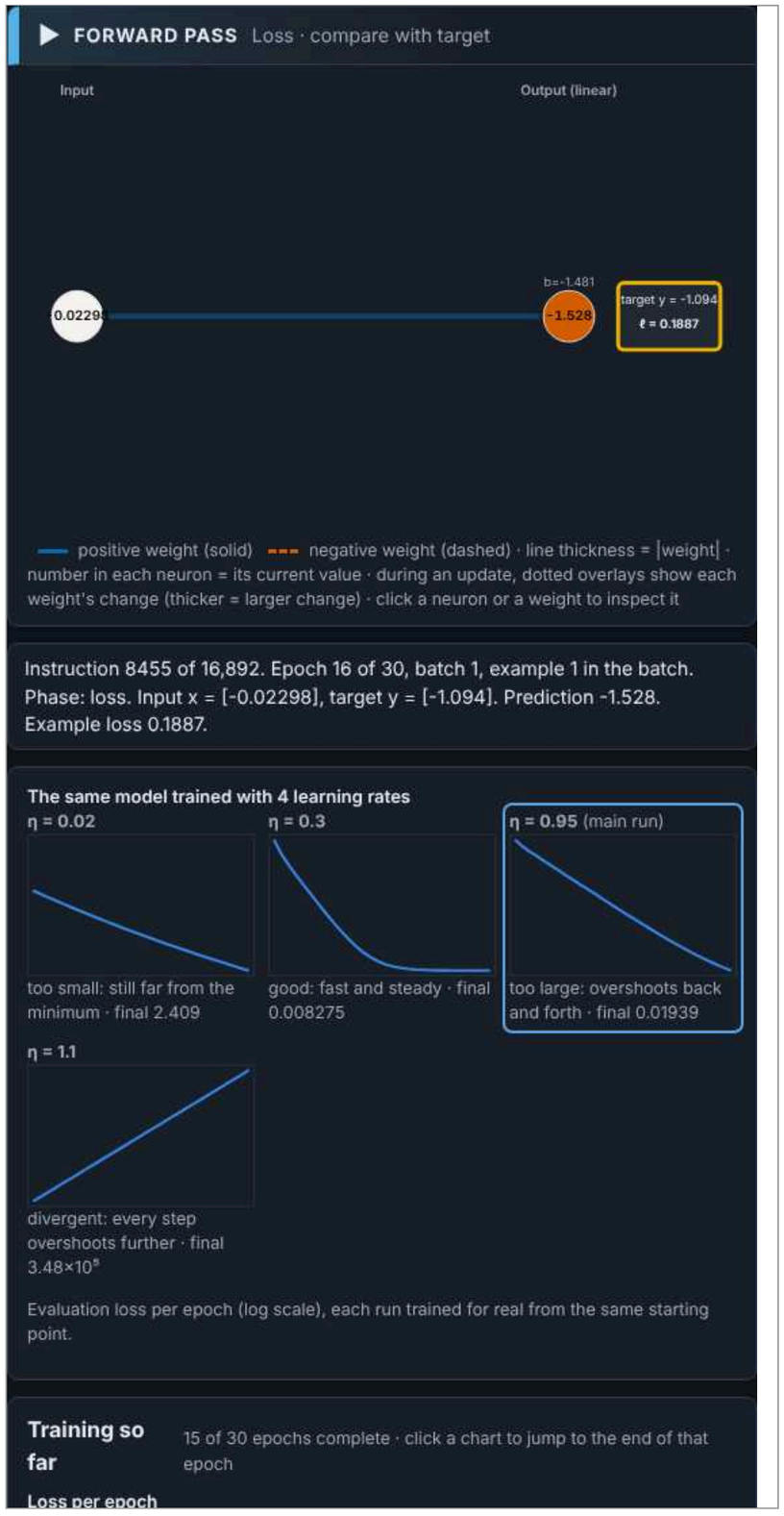


Simulated deuteranopia

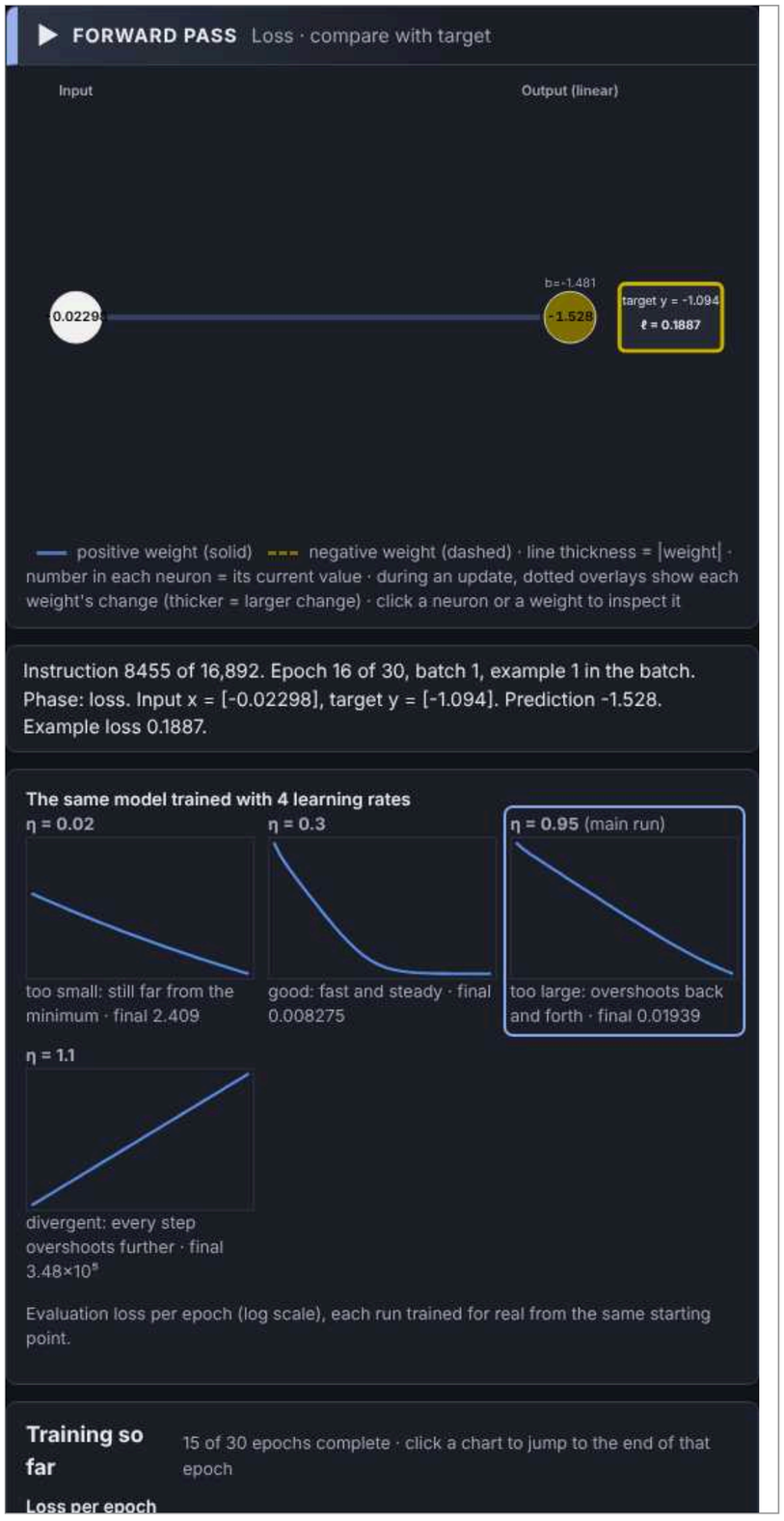


D2. Learning-rate pathology — dark theme, forward phase (part 1 of 2)

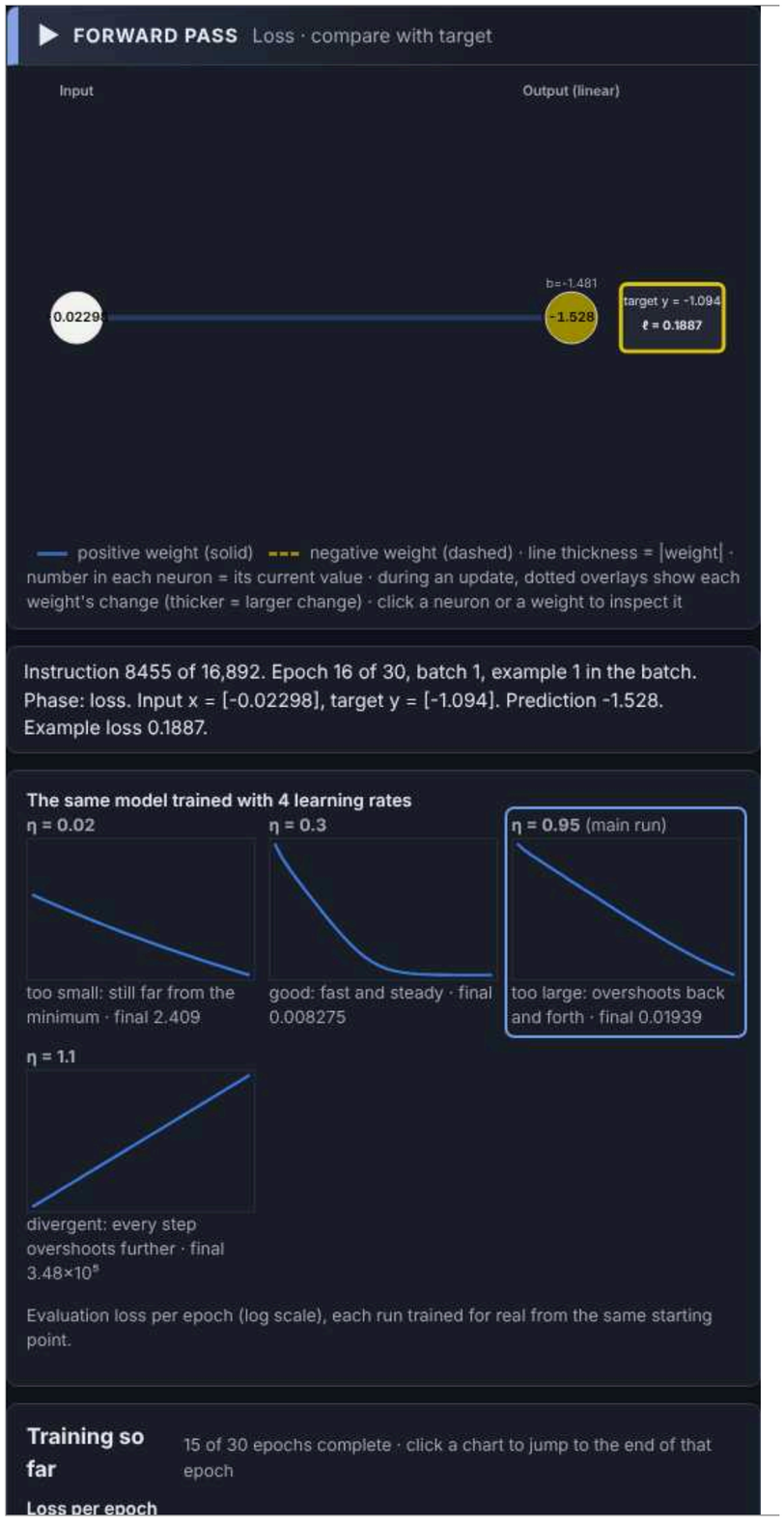
As rendered



Simulated protanopia

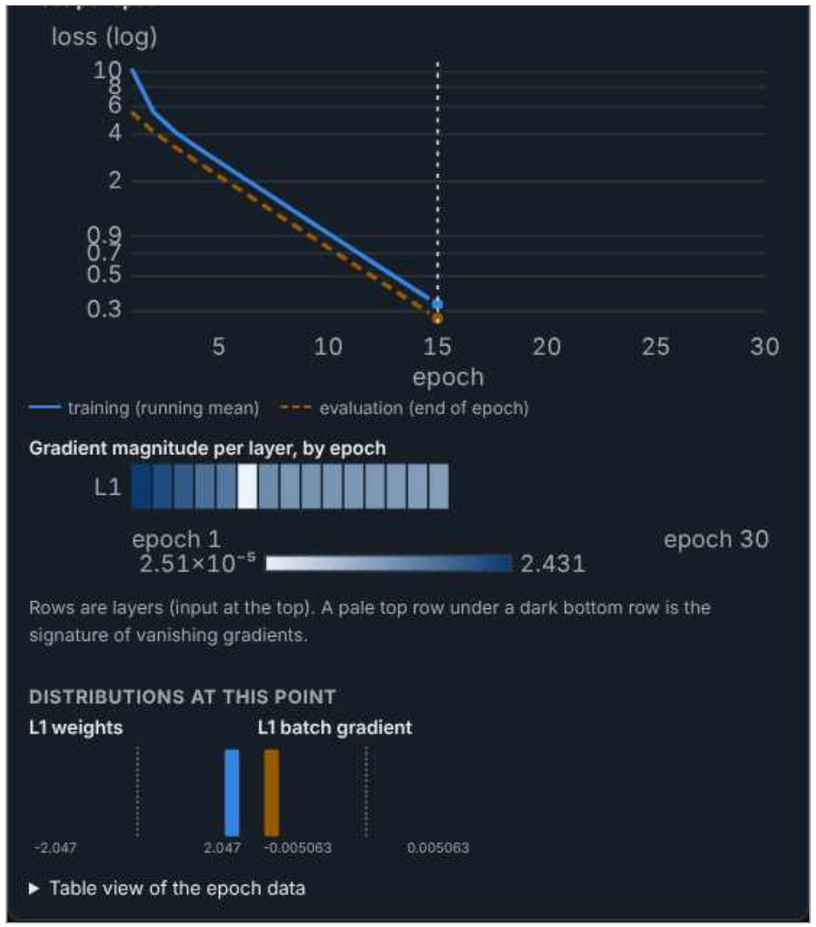


Simulated deuteranopia

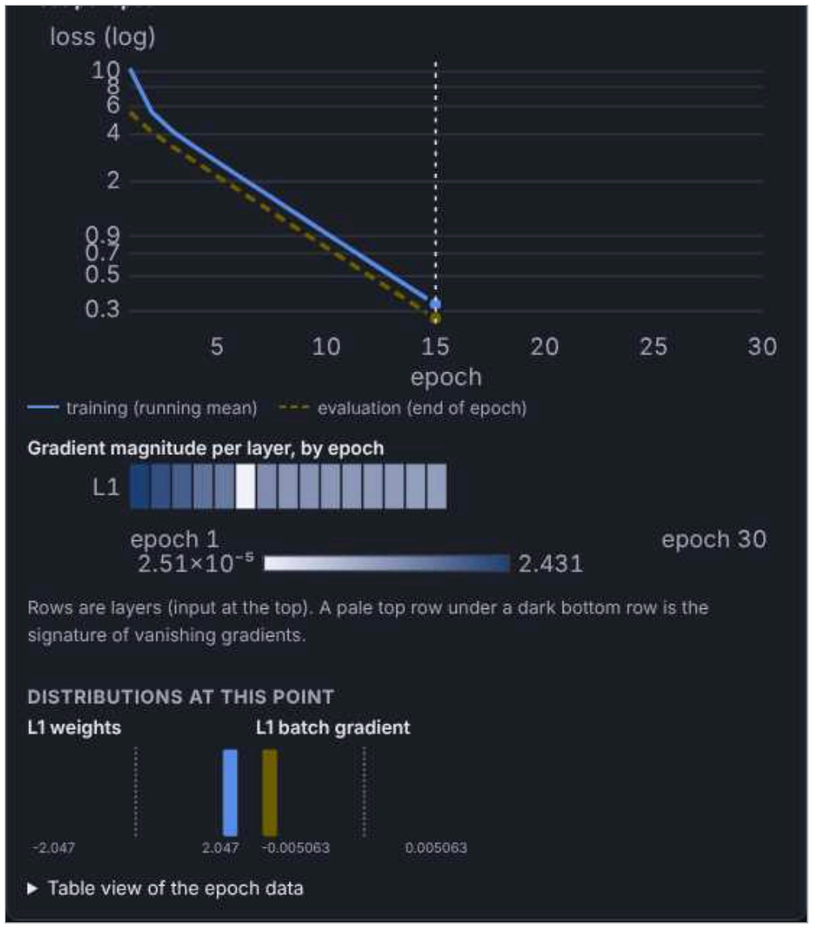


D2. Learning-rate pathology — dark theme, forward phase (part 2 of 2)

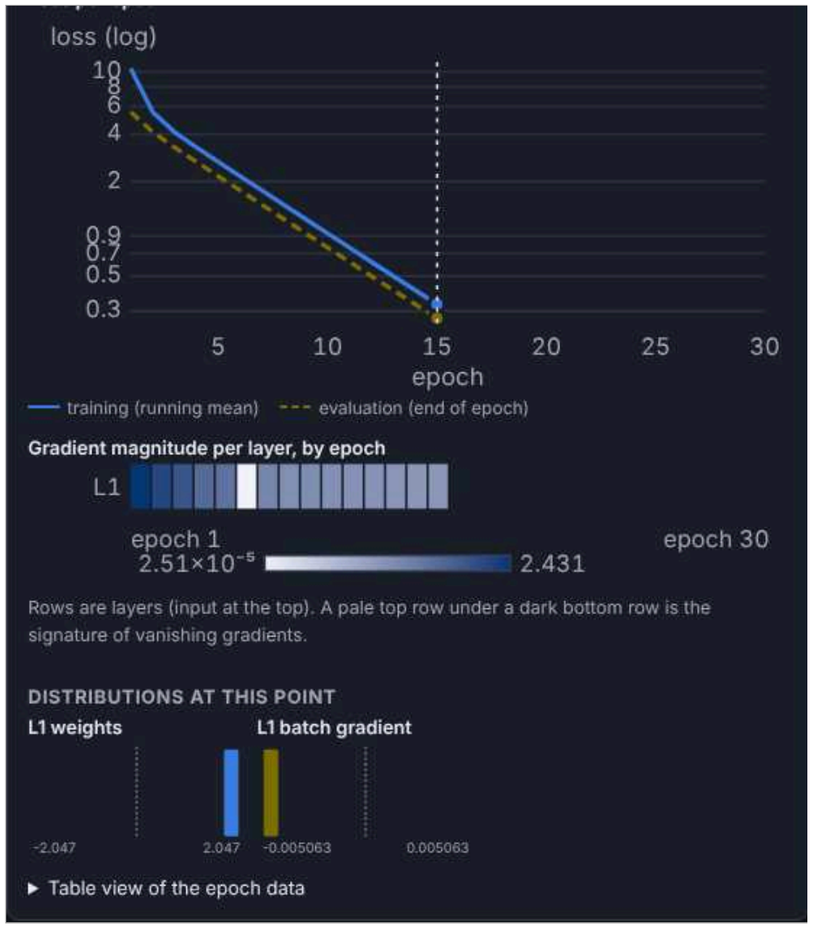
As rendered



Simulated protanopia

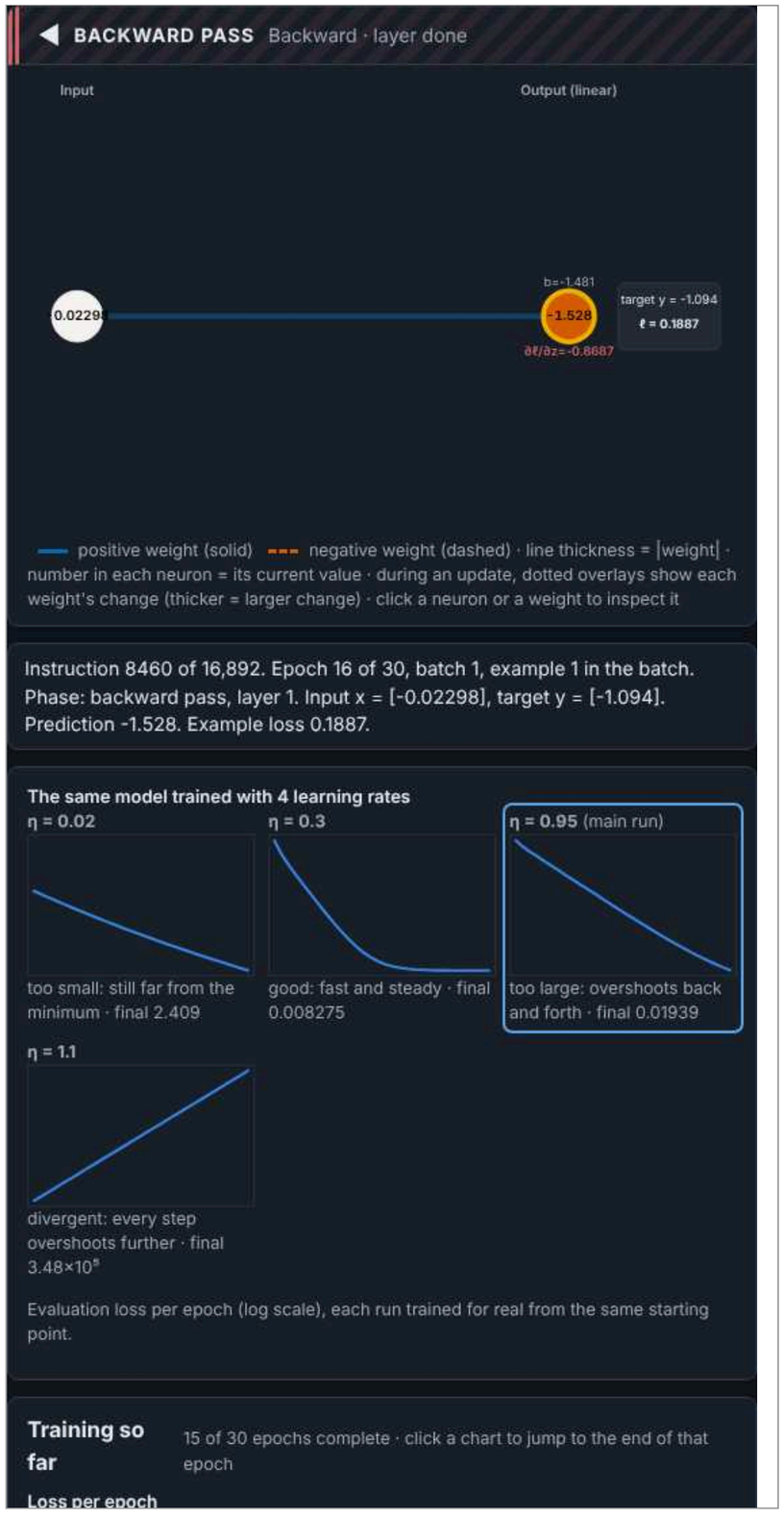


Simulated deuteranopia

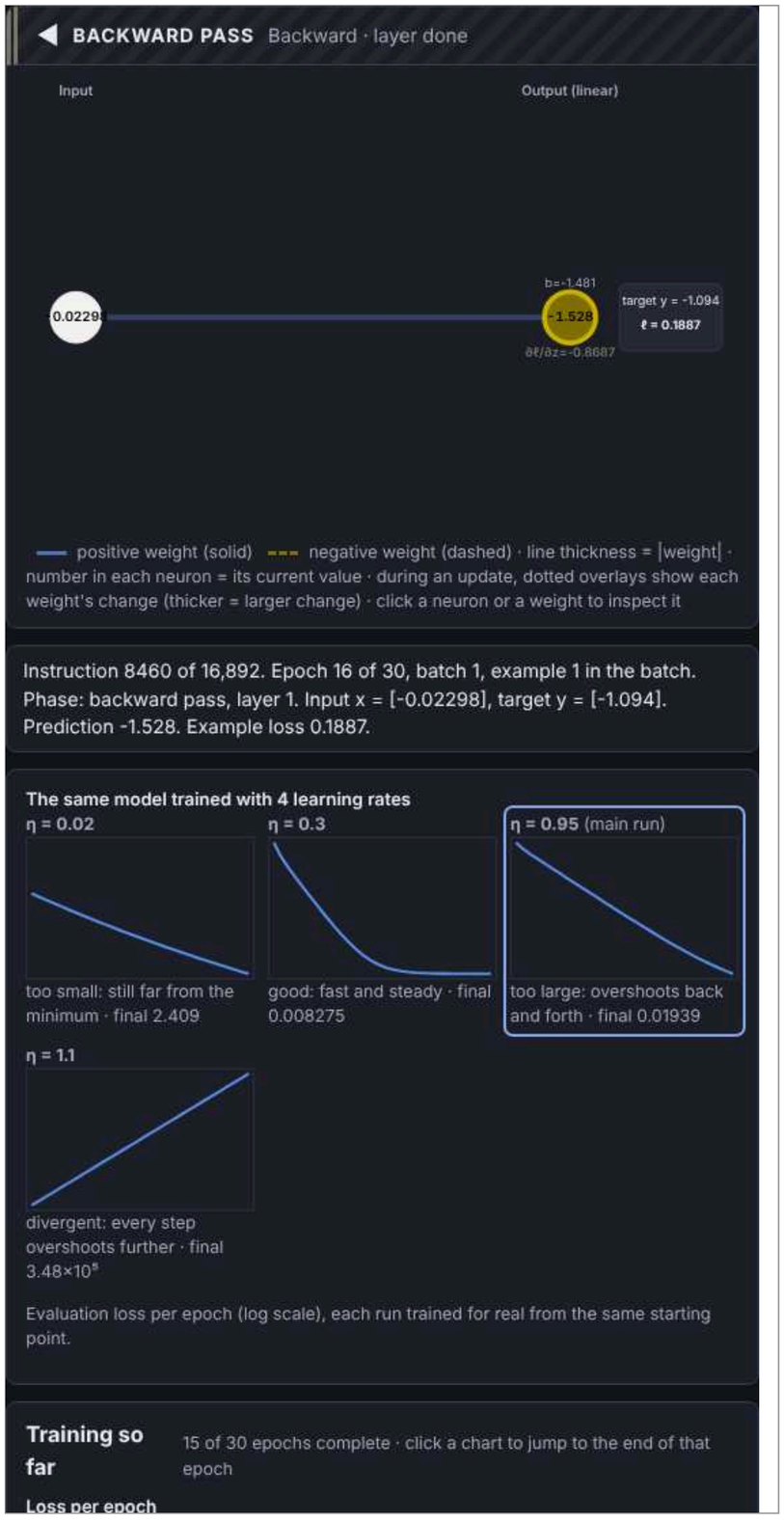


D2. Learning-rate pathology — dark theme, backward phase (part 1 of 2)

As rendered



Simulated protanopia

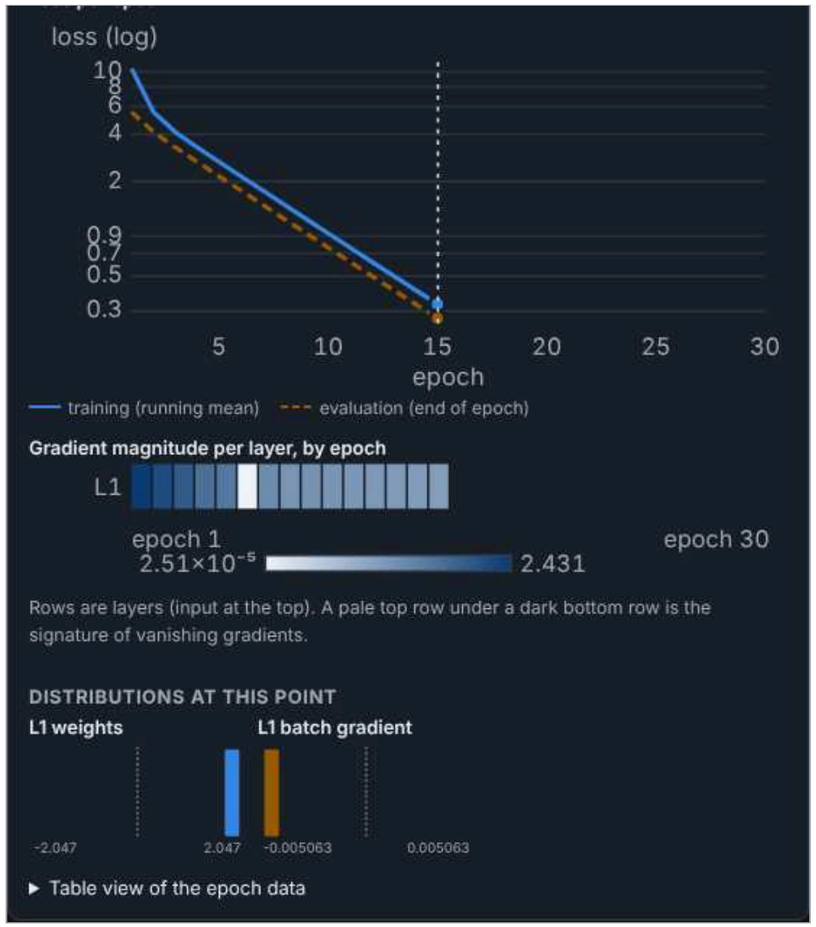


Simulated deuteranopia

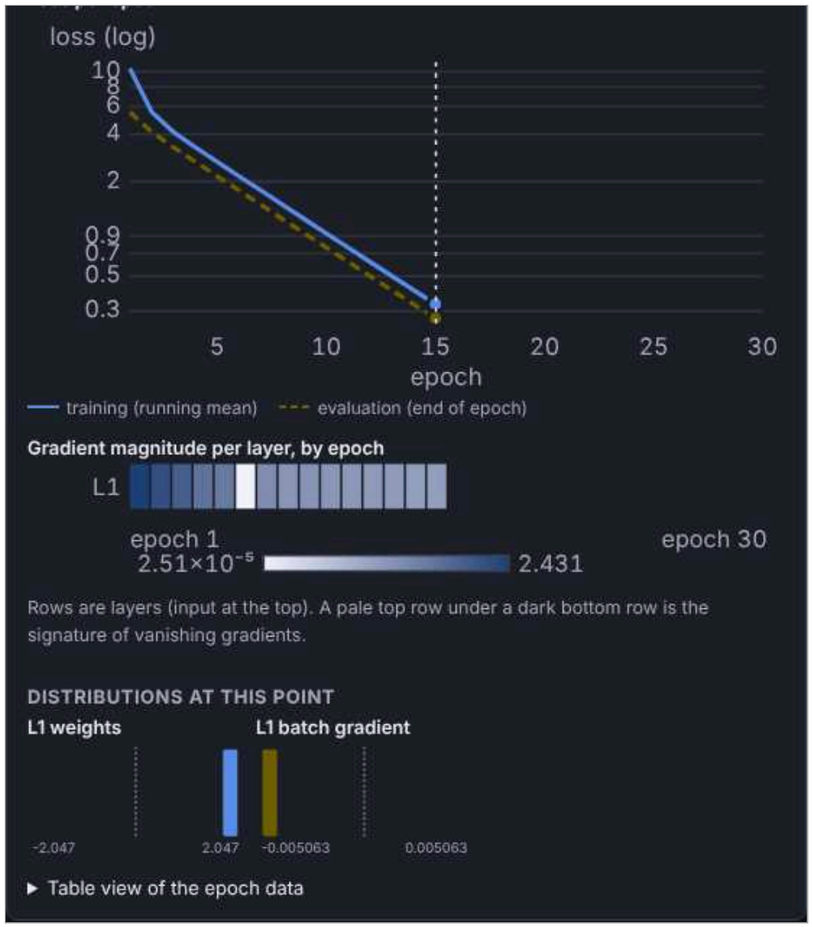


D2. Learning-rate pathology — dark theme, backward phase (part 2 of 2)

As rendered



Simulated protanopia



Simulated deuteranopia

